Geopolitical Risks and International Business

Measuring the politics of the world economy, and acting on it

Thierry Warin

17 October 2026

Table of contents

Preface 1

1 What Is Geopolitical Risk? Counting the Threat Before the War 3

I The system 19

2 The International Economic Order: Who Wrote the Rules, and What Happens When Their Share Shrinks 21

3 Anarchy and the Balance of Power: Who Sets the Terms When No One Governs 35

4 Trade, Conflict and Geoeconomic Fragmentation: From Commercial Peace to Rival Blocs 49

II Who governs 65

5 International Institutions: Who Finances, Insures and Referees the Firm 67

6 Private Authority, Standards and Global Regulation: Who Writes the Rules Firms Obey 81

III From judgement to number 97

7 Measuring Risk: How a Political Judgement Becomes a Number 99

IV The channels 115

8 Cyber Geopolitics: Restraint, Rivals and Weaponised Networks 117

9 Economic Diplomacy Redefined: Two-Level Games, Embassies and the Firm 131

10 Economic Sanctions: Do They Work, and Whom Do They Hit? 145

11 Space Geopolitics: The Orbital Commons and the New Space Race 159

12 From Indicators to Decisions: What a Multinational Can Know About Geopolitical Risk 173

References 187

Preface

Every multinational now says that geopolitics matters to it. Far fewer can say *how much*, *through which channel*, or *what they would do differently if the number changed*. The gap is not one of attention — boards discuss sanctions, export controls, wars and elections every quarter — but of measurement. Geopolitical risk reaches a firm through its market access, its inputs, its finance, its people and assets, and its data and operations, and each of those channels can be observed and counted. Most firms do not count them, and the numbers they buy are rarely examined.

This book is about that gap. It treats geopolitical risk as something to be argued about with evidence: what a concept means, how it has been turned into data, what the data show, and what they cannot show. Each chapter takes one idea that has shaped how scholars and practitioners think about the political environment of international business — the balance of power, the commercial peace, the authority of institutions and standards, the measurement of risk from text, the weaponisation of networks, sanctions, the new contest in orbit — and follows it from its origins to the numbers built on it, and from the numbers to the decisions a multinational actually takes.

The numbers disagree with the stories told about them. Indices are revised, codebooks change, countries are renamed and recoded, and the same concept measured two ways gives two answers. The chapters dwell on those disagreements, because they are where an analyst earns their keep.

The argument runs in five movements. Chapter 1 asks what is being counted when we speak of geopolitical risk. Part I describes the system a firm operates in: an economic order written by a few powers, an anarchic distribution of power, and a trading system fragmenting along geopolitical lines. Part II asks who governs that system — public institutions, old and new, and the private authority of standards. Part III shows how political judgement becomes a number, and what is lost on the way. Part IV follows the four channels through which geopolitical risk now reaches the firm most directly: cyberspace, economic diplomacy, sanctions and outer space. Chapter 12 returns to the firm, and to the step that no indicator takes for it — the decision.

Contents

- Chapter 1 · What Is Geopolitical Risk? Counting the Threat Before the War

Part — The system

- Chapter 2 · The International Economic Order: Who Wrote the Rules, and What Happens When Their Share Shrinks
- Chapter 3 · Anarchy and the Balance of Power: Who Sets the Terms When No One Governs
- Chapter 4 · Trade, Conflict and Geoeconomic Fragmentation: From Commercial Peace to Rival Blocs

Part — Who governs

- Chapter 5 · International Institutions: Who Finances, Insures and Referees the Firm
- Chapter 6 · Private Authority, Standards and Global Regulation: Who Writes the Rules Firms Obey

Part — From judgement to number

- Chapter 7 · Measuring Risk: How a Political Judgement Becomes a Number

Part — The channels

- Chapter 8 · Cyber Geopolitics: Restraint, Rivals and Weaponised Networks
- Chapter 9 · Economic Diplomacy Redefined: Two-Level Games, Embassies and the Firm
- Chapter 10 · Economic Sanctions: Do They Work, and Whom Do They Hit?
- Chapter 11 · Space Geopolitics: The Orbital Commons and the New Space Race
- Chapter 12 · From Indicators to Decisions: What a Multinational Can Know About Geopolitical Risk

How to read it

The chapters can be read in order or on their own; cross-references point back where an argument depends on an earlier one. Definitions are set apart, so that a reader looking for what a term means can find it; boxes headed *From the archive* recount an episode that made an argument concrete; each chapter closes with a short review of the literature for readers who want to go further. Every source is listed in the references, and every data source is public.

Thierry Warin · HEC Montréal · Source: github.com/warint/geopolitics-geoeconomics · 17 October 2026

Chapter 1

What Is Geopolitical Risk? Counting the Threat Before the War

1.1 A question about counting

Every risk function in every multinational firm already has a view about geopolitics. It sits in the annual report, in the paragraph on principal risks; it sits in the treasury's hedging policy, in the procurement team's list of single-source suppliers, in the country manager's sense of which visas will be hard to renew next year. What it rarely has is a *number* — and when it does, the number usually comes from somewhere else, built by someone else, for a purpose the firm never examined.

This book begins with a deliberately narrow question: before a firm can manage geopolitical risk, what exactly is it counting? The question sounds pedantic. It is not. A board that is told "geopolitical risk is at its highest level in forty years" has been told something precise about a particular index, constructed from particular newspapers, normalised to a particular base period. Whether that statement should change a capital-expenditure plan in Monterrey or a supplier contract in Penang depends entirely on what was counted, and the person presenting the slide often does not know.

The answer this chapter builds towards is that geopolitical risk is not one thing to be measured but a family of things, each visible to some instruments and invisible to others. We start with the oldest distinction in the subject, between risk and uncertainty; separate three kinds of risk that are routinely confused; sort the instruments that measure them into six families; and then take one instrument apart — the Geopolitical Risk index of Caldara and Iacoviello (2022), the most widely used number in the field, read alongside the monthly file its authors publish. The chapter closes on what that index cannot tell a firm, because a measure whose limits cannot be stated should not be used.

1.2 Risk, uncertainty, and the ground in between

The modern vocabulary starts in 1921, with an economist trying to explain where profit comes from.

From the archive · Knight, 1921

Frank Knight's *Risk, Uncertainty and Profit* began as a doctoral dissertation at Cornell and was published in 1921. Its problem was a theoretical embarrassment: in a perfectly competitive economy with full information, competition should drive profit to zero, yet profit plainly exists. Knight's answer was that some of what businessmen face can be reduced to known odds and therefore insured, pooled or priced into costs — and some cannot.

The first he called *risk*: a situation in which the distribution of outcomes is known, either a priori, as with dice, or statistically, from the experience of many similar cases, as with fire insurance. The second he called *uncertainty*: a situation unique enough that no such distribution can be formed, and in which the decision-maker must act on judgement. Risk, being insurable, becomes a cost like any other. Uncertainty cannot be shifted to an insurer, and it is the bearing of uncertainty — the willingness to commit resources on a judgement no one can verify in advance — that profit rewards (Knight 1921).

Knight's treatment was graded rather than binary: what matters is whether instances can be grouped into classes homogeneous enough to support an estimate. That is precisely the question a geopolitical risk analyst faces every morning.

Risk (in Knight's sense) — a situation in which the probabilities of outcomes can be estimated, from theory or from the experience of sufficiently similar cases, so that the exposure can be priced, pooled or insured.

Uncertainty (Knightian uncertainty) — a situation in which no reliable probability distribution over outcomes can be formed, because the case is too unlike any class of past cases. It cannot be insured away; it can only be judged.

Geopolitics lives in the ground between the two. A shipping insurer pricing piracy off the Horn of Africa is close to Knightian risk: there are thousands of transits, hundreds of incidents, and a market for war-risk premiums that updates as the incidents accumulate. A board asking whether a particular strait will be blockaded within five years is close to Knightian uncertainty: there have been blockades, but never this one, by this state, against this coalition, with these weapons.

Much of geopolitical risk analysis, and much of this book, is an attempt to move questions from the second category towards the first, and to be candid about how far they move. Every index discussed in the chapters that follow is a device for grouping unlike events into classes so that something can be counted. Grouping is never neutral. When the Geopolitical Risk index counts an article about a nuclear test and an article about a terrorist bombing as two instances of the same thing, it has made a modelling choice about what is homogeneous. When the Correlates of War project averages steel output and population into one number for "power" (Chapter 3), it has made another. The discipline is not to avoid these choices — no measurement is possible without them — but to see them, state them, and ask what they do to the conclusion.

There is a practical corollary for the firm. Knightian risk can be transferred: political-risk insurance, war-risk cover on cargo, credit insurance on export receivables are all markets for it. Knightian uncertainty cannot be transferred, only managed — by flexibility, by diversification, by being willing to pay for the ability to change one's mind. Much of what firms now call "resilience" is, in Knight's terms, an admission that the relevant exposures are uncertain rather than risky, and that the premium has to be paid in redundancy because there is no one to pay it to.

The best-known corporate answer to uncertainty of this kind was not a number at all. In the late 1960s and early 1970s a planning group at Royal Dutch Shell stopped trying to forecast the price of oil and began to write *scenarios*: internally consistent stories of how the world might develop, so that managers would rehearse decisions against several futures rather than bet on one. Pierre Wack, who led the work, described its purpose as changing the mental models of the people who decide (Wack 1985). Indices and scenarios are complements: an index tells the firm which story the world is currently writing about; a scenario tells it what it would do if the story came true.

1.3 Three kinds of risk, and the layer that comes from outside

The second distinction is between three phrases that firms, consultancies and journalists use almost interchangeably: political risk, country risk and geopolitical risk. They answer different questions, attach to different units of analysis, and call for different instruments.

Political risk — the possibility that the government of a host country changes the rules applying to a foreign firm in a way that reduces the value of its investment: expropriation, forced renegotiation, discriminatory taxation, licence withdrawal, restrictions on repatriation. Its natural unit is the firm in a host country.

Country risk — the possibility that a country as a whole fails to pay, convert or transfer: sovereign default, capital controls, currency inconvertibility, a banking crisis. Its natural unit is the country, and its natural user is the lender.

Geopolitical risk — the possibility that relations *between* states and other political actors move against the firm: war, terrorism, sanctions, export controls, the formation of blocs, state-sponsored cyber operations. Its natural unit is the international system or a pair of states, and it frequently originates outside any country in which the firm operates.

Political risk is the oldest of the three in the international business literature, and its archetypes are dramatic. When Mexico nationalised the assets of foreign oil companies in 1938, or Iran nationalised the Anglo-Iranian Oil Company in 1951, the threat came from the host government and fell on identifiable foreign firms. The scholarly literature that grew up in the 1960s and 1970s, as decolonisation multiplied the number of host governments and expropriations became frequent, was preoccupied with this kind of event. Its classic synthesis is Kobrin (1979), who complained that "political risk" was used to mean too many things — sometimes the political event itself, sometimes the instability of a regime, sometimes the consequences for a firm's operations — and that the literature rarely connected the first to the last.

Kobrin's lasting contribution was to insist that risk is a property of the relationship between an environment and a particular firm, not of the environment alone. Political instability matters to a foreign investor only through specific *contingencies* that affect specific operations; a coup that changes nothing about the mining code may be irrelevant to the mining company, while a quiet amendment to a tax treaty may be decisive. From this followed the distinction between **macro** risk, which

affects all foreign firms in a country, and **micro** risk, which falls on a sector or a single firm. Kobrin's own count of forced divestment in developing countries bore the distinction out: expropriation in the 1960s and 1970s was, for the most part, selective — aimed at particular industries rather than at foreign capital as such (Kobrin 1980).

Why particular industries? Raymond Vernon's answer was that the bargain between a foreign investor and a host government changes over the life of the investment (Vernon 1971). Before the investment is made, the firm holds the cards: capital, technology and the choice of where to go. Once the mine is dug, the capital is sunk, the technology has been learned locally, and the host's need for the firm declines while its ability to capture the rent rises. The bargain *obsolesces*. The insight has outlived the era of nationalisation, because it describes any relationship in which one party's commitment is irreversible and the other's is not.

Country risk grew from a different root — the bank lending of the 1970s to developing-country governments, and the debt crisis that followed. Its question is whether a country will service its obligations and let money move; its instruments are sovereign ratings, debt-sustainability analysis and the spreads on sovereign bonds and credit default swaps.

Geopolitical risk adds a third layer that neither earlier tradition was designed to see: risk that originates *outside* the host country. A Canadian firm operating in Vietnam can be hurt by a decision taken in Washington about Chinese inputs, by a decision in Beijing about rare-earth exports, by a decision in Brussels about which payment systems may be used for which counterparties, or by a naval incident in a sea lane neither Canada nor Vietnam controls. The host government may be perfectly stable and perfectly friendly; the threat travels through the firm's suppliers, customers, banks and networks.

The three are not mutually exclusive. Take a Canadian bank with a subsidiary in Moscow in February 2022. It faces *political* risk when the host state restricts the ability of foreign firms to sell their assets and leave; *country* risk when capital controls trap the subsidiary's earnings; and *geopolitical* risk in its purest form when its own government, acting with its allies, forbids transactions on which the subsidiary depends. All three arrive in the same quarter, through different channels, and call for different responses — a write-down, a liquidity plan, a compliance review. A risk function with only one of the three words in its vocabulary sees a third of the problem.

1.4 Six ways of counting

If geopolitical risk is a family of phenomena, it is measured by a family of instruments, and the instruments disagree. It helps to sort them by *what they count*, because what an indicator counts determines what it cannot see.

Family	What it counts	Examples in this book	Strength	Blind spot
Event-based	things that happened, coded from reports	cyber incidents (Ch. 8), sanctions (Ch. 10)	concrete, datable	only what was reported and coded
Text-based	how much the press or firms talk about a risk	GPR (this chapter), EPU (Ch. 7)	timely, long, high-frequency	measures attention, not exposure
Expert-based	scores assigned by analysts or surveys	governance indicators, commercial ratings	reaches latent concepts	opaque, sticky, perception-driven
Structural	slow-moving features of the system	capabilities (Ch. 3), trade dependence (Ch. 4)	explains *why*	rarely says *when*
Market-based	prices that embed risk	sovereign CDS spreads, implied volatility	forward-looking, daily	mixes risk with risk appetite
Firm-level	risk as perceived by firms themselves	earnings-call text (Ch. 7), disclosures	speaks to the decision	self-reported, listed firms only

A threat never carried out leaves no *event*. *Text-based* indicators — the Geopolitical Risk index, the Economic Policy Uncertainty index of Baker, Bloom, and Davis (2016), the firm-level political-risk measure of Hassan et al. (2019) —

measure *attention*, related to but not the same as the risk any reader faces. *Expert-based* indicators, such as the Worldwide Governance Indicators (Kaufmann, Kraay, and Mastruzzi 2011), pay for their reach in opacity. A *market* spread can widen because danger rose or because investors grew more fearful of the same danger.

No family is enough on its own, and the argument for **triangulating** across several is not that more is better. It is that the families fail in different ways, so their disagreements carry information. When the text is calm and the market is not, someone may be pricing something the press has not yet seen. A single index can only tell its own story; three indices that quarrel force the analyst to say which one to believe, and why.

Even two instruments from the *same* family quarrel. Figure 1.1 sets the Geopolitical Risk index beside the global Economic Policy Uncertainty index of Davis (2016), a GDP-weighted average of national policy-uncertainty indices built, like GPR, by counting newspaper articles. Over the 355 months from January 1997 to July 2026 their correlation is 0.26; averaged by year it rises only to 0.39. The months that define each series barely overlap. In October 2001 geopolitical risk stood at 513 and policy uncertainty at 170. In May 2020, in the first pandemic spring, the order was reversed: policy uncertainty at 410, geopolitical risk at 69. In April 2025, when the United States announced sweeping tariffs, policy uncertainty reached 621, the highest value in its history, while geopolitical risk stood at 141. Only in March 2022 and March 2026, with wars that were also economic-policy events, did both indices spike together.

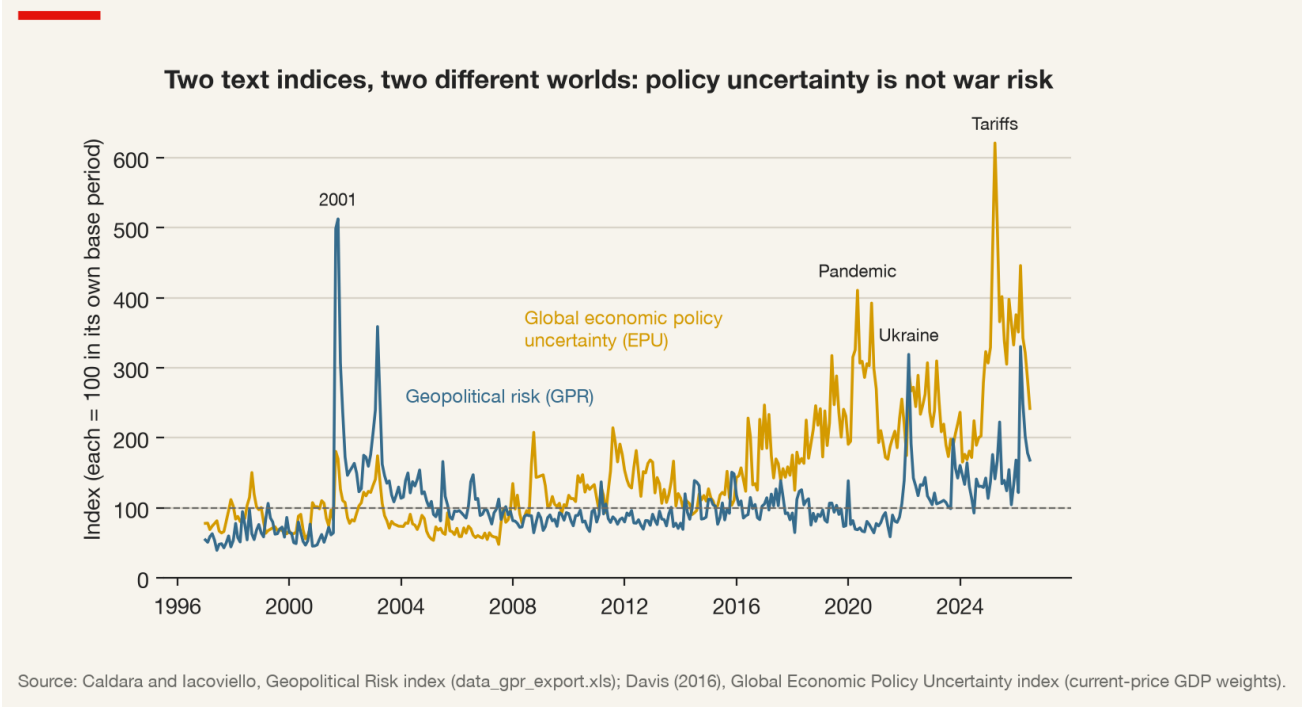


Figure 1.1: Two text indices, two different worlds: global economic policy uncertainty (EPU) and geopolitical risk (GPR), monthly, 1997–2026; monthly correlation 0.26. Source: Caldara and Iacoviello, Geopolitical Risk index; Davis (2016), Global Economic Policy Uncertainty index.

The lesson is not that one index is right. It is that "uncertainty" in the newspapers comes in kinds, and a firm's exposure to each is different. A tariff shock reaches a manufacturer through market access and inputs within weeks; a terrorist attack in New York reached airlines, insurers and tourism through assets and people, and most manufacturers hardly at all. The upward drift of policy uncertainty since 2016 is also a reminder that much of what firms now experience as "geopolitics" — tariffs, export controls, investment screening — arrives through economic policy, and is counted by the instrument that does not carry the word.

1.5 How a newspaper archive becomes an index

The rest of the chapter takes one text-based instrument apart. Dario Caldara and Matteo Iacoviello, economists at the Federal Reserve Board, published the Geopolitical Risk index in the *American Economic Review* in 2022, after circulating it for several years as a working paper. It has become the default series for anyone — central bank, investment bank, consultancy or researcher — who needs a number for geopolitical risk. That ubiquity is the reason to understand its construction exactly.

1.5.1 The definition

Geopolitical risk, in the authors' words, is "the threat, realization, and escalation of adverse events associated with wars, terrorism, and any tensions among states and political actors that affect the peaceful course of international relations" (Caldara and Iacoviello 2022).

Read it for what is in and what is out. *In* are threats as well as realisations — the index is meant to rise when war becomes more likely, not only when it starts. *In* are non-state actors: "political actors", not only states, which is how terrorism enters. *In* are tensions short of war. *Out* are domestic politics, elections, regulation and economic policy, which belong to other indices. The definition is about relations between actors, not about what a host government does to a firm.

Geopolitical Risk (GPR) index — a monthly text-based index, constructed by Caldara and Iacoviello (2022), equal to the share of articles in a fixed set of major newspapers that discuss adverse geopolitical events and threats, rescaled so that its average over a base period equals 100.

1.5.2 Four steps

The construction has four steps, and each is a decision that could have been taken differently.

The corpus. The recent index, from 1985, is built from the electronic archives of ten leading newspapers published in the United States, the United Kingdom and Canada. The historical index, from 1900, uses three American newspapers whose archives reach that far back. This is the anglophone press of three allied countries, and the historical series is the American press alone.

The dictionary. An article counts as a GPR article if it contains words from one of eight categories: war threats, peace threats, military build-ups, nuclear threats, terror threats, the beginning of war, the escalation of war, and terror acts. In most categories the search requires a combination — a word for the threat and a word that anchors the article to geopolitics — so that an article about a "price war" does not qualify. Designing such a dictionary is a trade-off between precision (only relevant articles) and recall (all relevant articles).

The share. For each month, the number of GPR articles is divided by the total number of articles published. The division matters more than it looks. The authors' own file shows that the monthly count of articles in the ten recent newspapers was about 11,600 in January 1985, about 33,400 in January 2000, and under 14,000 in September 2026. A raw count would have tripled and then halved with no change in the world. The share cares only about what fraction of the newspapers' attention went to war and terror.

The normalisation. Finally the share is rescaled so that its average over 1985–2019 equals 100; the historical index is normalised to its own average over 1900–2019. A reading of 300 means that the share of articles discussing geopolitical threats and acts was three times its base-period average. It does not mean that risk was three times higher in any sense a probability could express; it means that attention was.

The misconception to drop now. A high GPR reading does not mean that risk is high *for a particular firm*. It means that the anglophone press was writing a great deal about geopolitical tension that month. The second statement is what the index measures; the first is an inference that requires everything else in this book.

1.5.3 Forty years at a glance

Figure 1.2 draws the recent index month by month from January 1985 to September 2026, the latest month in the file. It rarely strays far from its average. The median month stands at 93; only 3.4 per cent of months exceed 200; eight months exceed 300, and they belong to five episodes — the Gulf War of 1990–91, the attacks of September 2001 and the war in Afghanistan that followed, the invasion of Iraq in 2003, the Russian invasion of Ukraine in 2022, and the war with Iran in 2026. Geopolitical risk, as the press records it, is a series of spikes separated by long stretches of near-normality, and the longest calm in the series — the late 1990s, when the annual average fell to 51 in 1997 and 57 in 2000, about half the base — sits just before its highest peak.

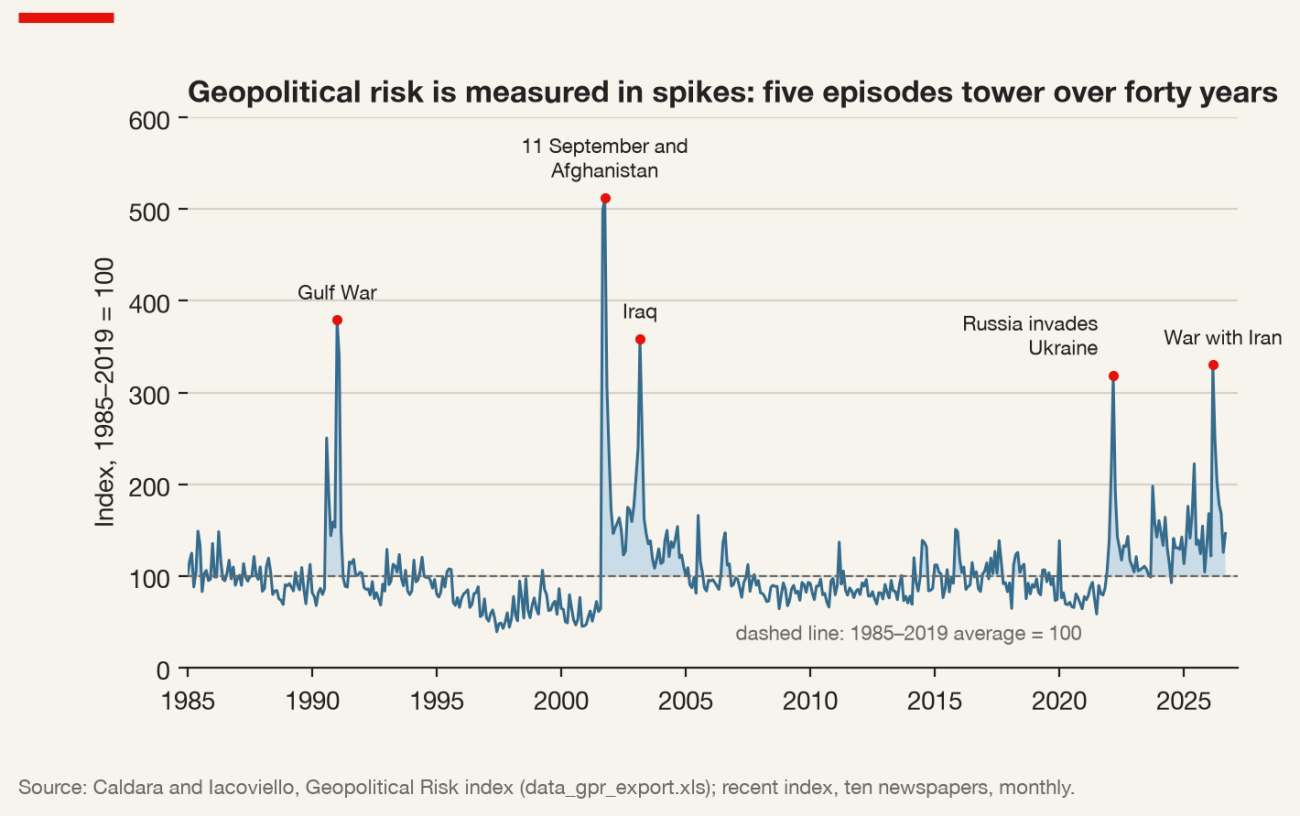


Figure 1.2: Geopolitical risk is measured in spikes. The recent Geopolitical Risk index, monthly, January 1985–September 2026 (1985–2019 average = 100); shading marks months above the average. Source: Caldara and Iacoviello, Geopolitical Risk index.

The second feature is less visible in any single month and more important for a firm: since 2022 the index has not come back down. Its annual average was 77 in 2020 and 82 in 2021; it was 158 in 2022, 122 in 2023, 134 in 2024, 145 in 2025 and 188 in the first nine months of 2026. After the invasion of Iraq the index was back at its base by 2005 (annual average 102) and stayed near it for fifteen years; since 2022 each new spike has started from a higher floor. That is a change in level, and it matters because the real-options argument below turns on whether uncertainty is expected to *resolve*. A firm can wait out a spike; it cannot wait out a plateau.

Table 1.2 lists the ten highest months, with the two sub-indices introduced below. Eight of the ten belong to the two wars against Iraq and to 2001; two belong to the 2020s. The headline blends threats and acts: March 2022 and March 2026 stand at almost the same level, 319 and 330, but one was mostly threat and the other mostly act.

Table 1.2: The ten highest months of the recent Geopolitical Risk index, 1985–2026, with the Threats and Acts sub-indices (each 1985–2019 average = 100). Source: Caldara and Iacoviello, Geopolitical Risk index, monthly file, vintage of October 2026; episodes added by the author.

Month	GPR	Threats	Acts	Episode
October 2001	513	261	854	War in Afghanistan begins
September 2001	499	260	841	Attacks of 11 September
January 1991	379	360	480	Gulf War air campaign
March 2003	359	362	429	Invasion of Iraq
February 1991	339	254	490	Gulf War ground campaign
March 2026	330	319	450	War with Iran
March 2022	319	404	251	Russian invasion of Ukraine
November 2001	307	150	510	War in Afghanistan
April 2003	256	213	332	Iraq: fall of Baghdad
August 1990	250	354	128	Iraq invades Kuwait

1.5.4 Threats and acts

The first five categories form the **GPR Threats** sub-index; the last three — the beginning of war, the escalation of war, terror acts — form **GPR Acts**. The split is the most useful thing in the paper for a firm, because the two need not move together. Threats can build for months and subside without anything happening; acts can arrive with almost no prior threat.

Figure 1.3 shows two patterns, and a third. In the autumn of 2001 acts leap from 75 in August to 841 in September and 854 in October, while threats rise only to 261: an attack with almost no public prelude. Over the winter of 2021–22 the order is reversed: threats climb from 122 in November to 321 in February and 404 in March — the highest threat reading in the recent series, debated in public for weeks — while acts stay below their average until the invasion and peak at 251. The third panel is the war of the United States and Israel against Iran, which began at the end of February 2026. Here the two moved together: in March acts jumped to 450, their highest since 2001, and threats to 319. Across the recent series the two sub-indices correlate at only 0.59.

The three shapes call for different preparation. A shock of the 2001 kind cannot be anticipated from the index; the only defence is a plan that already exists — insurance in force, a second supplier qualified — when the act arrives. A shock of the 2022 kind offers weeks in which threats accumulate in public, and a firm that has decided in advance what it will do at a given level of threat can use them. A shock of the 2026 kind, preceded by months of elevated but inconclusive threat, is the hardest to act on, because an index that has been high for a while teaches its readers to discount it.

The authors report that threats, and not only realised acts, depress economic activity. The implication for a firm is blunt: it does not need the war to pay its cost. If what moves investment is the threat, the cost is borne in the months of tension, by firms that delay, hedge and wait, whether or not the feared event ever occurs.

From the archive · The Cuban missile crisis, 1962

In mid-October 1962, photographs taken by American U-2 reconnaissance aircraft showed Soviet medium-range ballistic missile sites under construction in Cuba. On 22 October President Kennedy announced the discovery in a televised address and a naval “quarantine” of the island. For the next six days the two nuclear powers negotiated in public and in private while ships approached the quarantine line. On 28 October Khrushchev announced that the missiles would be dismantled and returned to the Soviet Union; the United States pledged not to invade Cuba and, privately, agreed to withdraw its own Jupiter missiles from Turkey.

No war began. In the categories of the GPR dictionary, the crisis is almost entirely *threats*: nuclear threats, military build-ups, war threats. In the historical series, October 1962 stands at well over twice the 1900–2019 average — the index doing exactly what its definition promises, registering the threat of adverse events and not only their realisation.

The episode also shows the limits of the clock. The risk was greatest at a moment, not over a month: the crisis lasted thirteen days. A monthly index records it as one high observation, and an annual one nearly erases it — averaged over 1962, the historical index is only about a quarter above its long-run mean. A firm deciding whether to load a ship that week needed a finer clock.

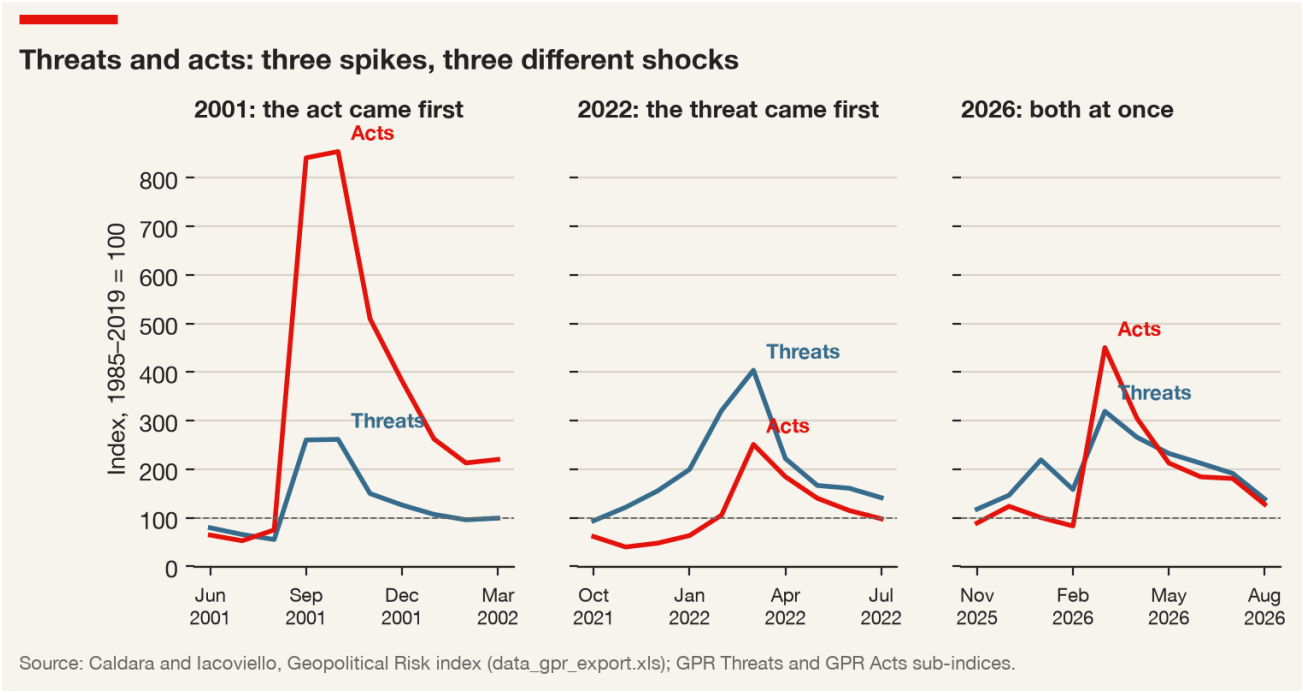


Figure 1.3: Threats and acts: three spikes, three different shocks. GPR Threats and GPR Acts sub-indices, monthly, around the attacks of September 2001, the Russian invasion of Ukraine in February 2022 and the war with Iran from February 2026; dashed line: 1985–2019 average = 100. Source: Caldara and Iacoviello, Geopolitical Risk index.

1.5.5 Country-specific indices

For 44 countries, the authors also publish country-specific series: the same search, restricted to articles that also mention the country, in columns labelled `GPRC_` and a three-letter ISO code. They are not normalised to 100; they are the percentage of articles that both discuss geopolitical risk and mention the country. A country can be the *source* of tension, its *victim*, its *mediator* or merely its *location*, and the count does not distinguish them. In March 2022 the series for Russia and for Ukraine stood at almost the same level, close to nine per cent of all articles: to the index, the invader and the invaded were two equally tense places.

The list is itself informative. It includes Ukraine, Israel, Taiwan and Venezuela and most large economies; it does not include Iran, Iraq, Pakistan or North Korea, and Africa is represented by Egypt, Tunisia and South Africa alone. The selection reflects where economic and financial data are rich enough for the comparisons the series were built for, not where geopolitical risk is greatest.

Figure 1.4 ranks the twenty countries with the highest readings over the twelve months to September 2026, against their own average over 1985–2019. Three features stand out. The first is the weight of the newspapers' home country. The United States heads the ranking at 4.5 per cent of all articles, almost double its long-run average of 2.3 and more than twice the next country, because American newspapers write about American wars and threats; the United Kingdom comes third. The country series measure geopolitical risk *as seen from* Washington, London and Toronto.

The second is the size of the departures from the past. Measured against its 1985–2019 average, Venezuela's reading over the last twelve months is almost fourteen times higher, Ukraine's nearly thirteen, Taiwan's almost six, Israel's five and Denmark's almost five. These ratios, which compare a country with itself, are the most useful thing the country series offer a firm: a subsidiary in Copenhagen does not need to know that Denmark ranks twenty-sixth, but it may need to know that its host is being written about in the language of threat at five times the usual rate.

The third is what the ranking cannot show. In March 2026 the series that jumped most were those of the United States (8.7 per cent of all articles), Israel (6.4), the United Kingdom, Russia and Saudi Arabia. Iran, the country at the centre of that

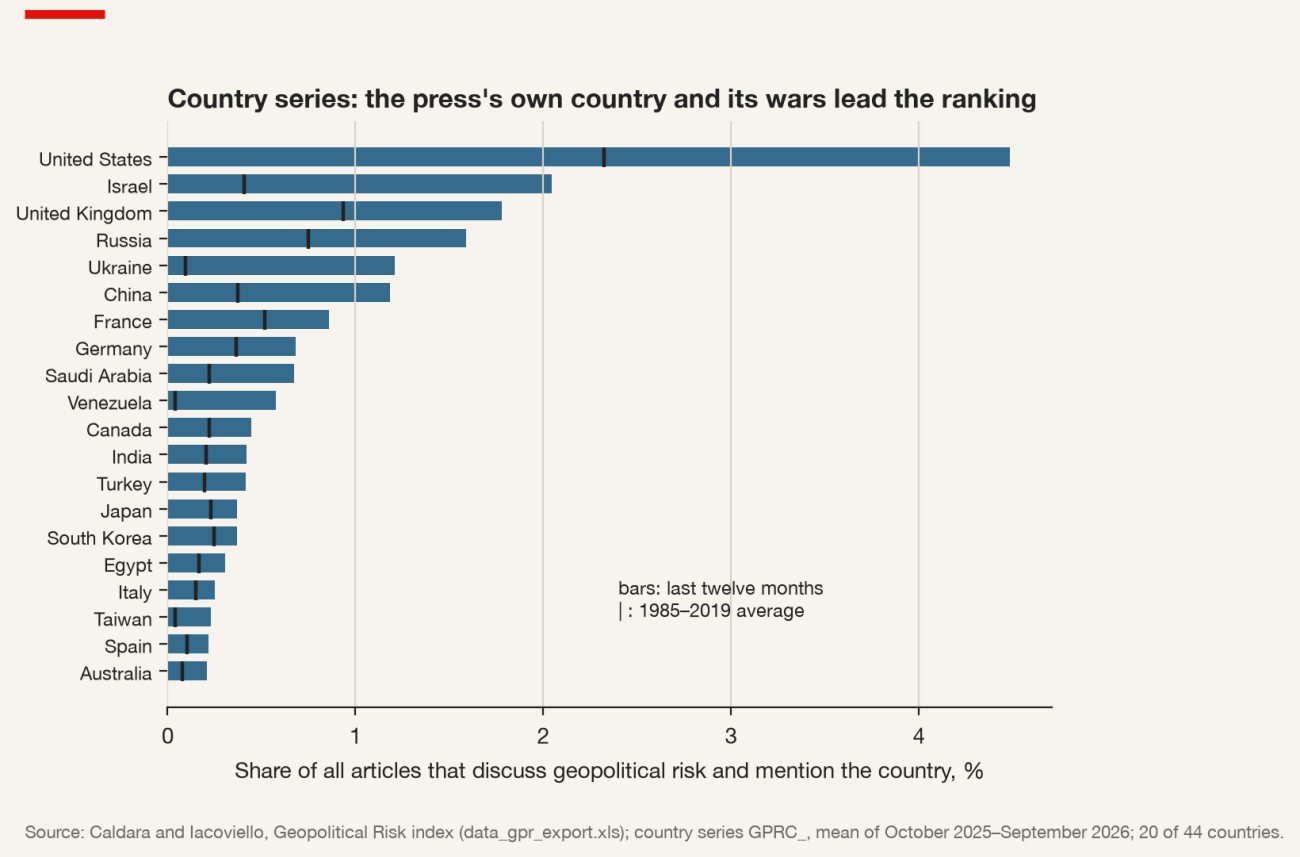


Figure 1.4: Country series: the press's own country and its wars lead the ranking. Share of all articles that discuss geopolitical risk and mention the country, October 2025–September 2026 (bars) and 1985–2019 (ticks); twenty of 44 countries. Source: Caldara and Iacoviello, Geopolitical Risk index.

month's war, has no series at all: it is not among the 44. A firm relying on the country series would have seen that war only through Iran's neighbours and adversaries.

1.6 Does it measure something real?

A text-based index is a hypothesis about the world: that the frequency of certain words in certain newspapers tracks a phenomenon of interest. The paper tests the hypothesis in three ways, and reading them is good training in how to evaluate an index one did not build.

The first is **narrative**. The index should spike when, on any reasonable reading of history, geopolitical tension spiked: the two World Wars, Korea, the Cuban missile crisis, the Gulf War, 11 September 2001, Iraq in 2003, Ukraine in 2022. It does. In the historical series the highest month is September 1939, followed closely by June 1944 and August 1914; in the recent series it is October 2001, followed by September 2001 and January 1991. This is a necessary test and a weak one.

Figure 1.5 shows the narrative test at the scale of the century. The two world wars dominate everything: the historical index reached 472 in August 1914, 484 in September 1939 and 473 in June 1944, and averaged 213 across the whole of the 1940s. Measured on this scale, the events that dominate the recent series shrink. October 2001 stands at 304, March 2026 at 252 and March 2022 at 167, below the Cuban missile crisis at 228. The average since 2020 is about 100. Against total war the present looks ordinary; against 1985–2019 it looks exceptional. The base period is a choice of what to compare the present with.

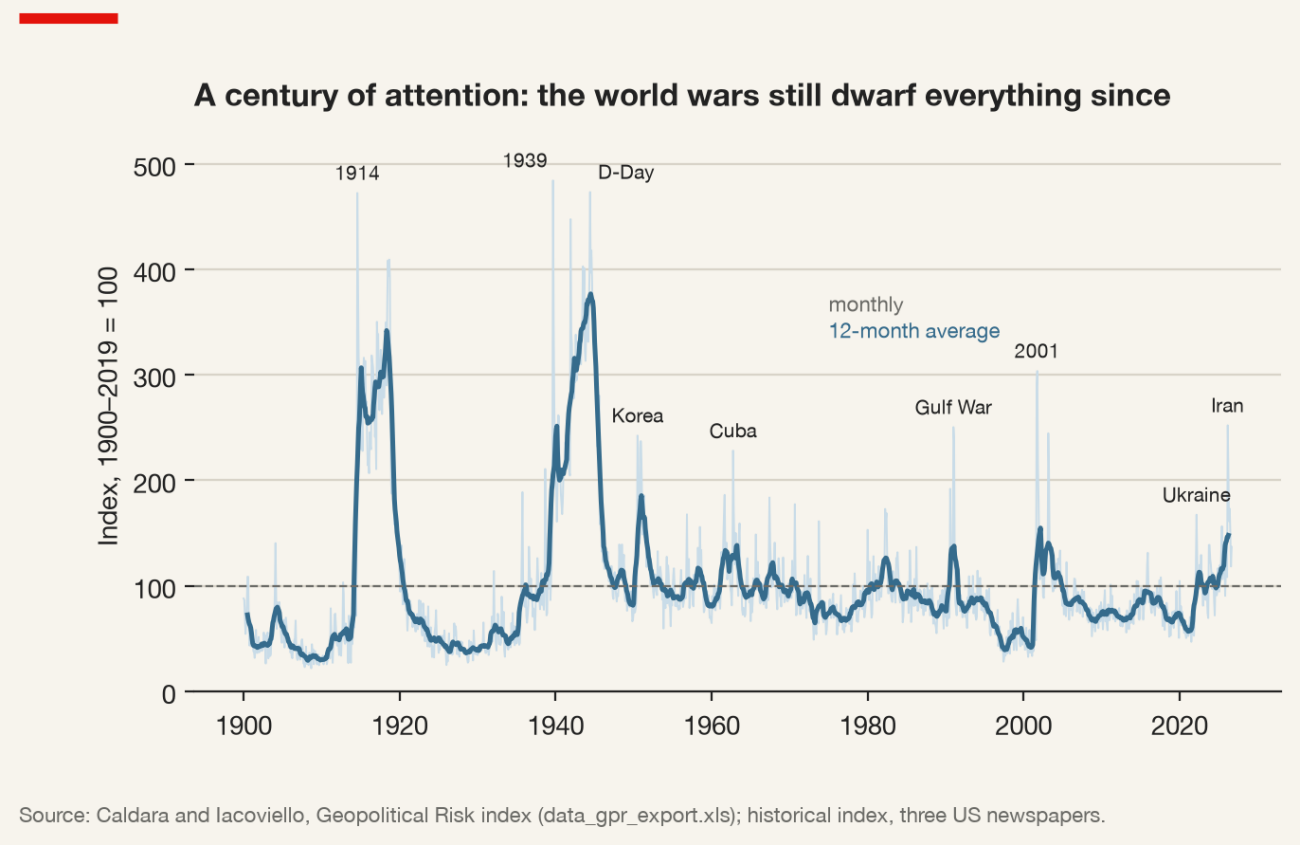


Figure 1.5: A century of attention: the world wars still dwarf everything since. The historical Geopolitical Risk index, monthly (light) and 12-month centred average (dark), 1900–2026 (1900–2019 average = 100). Source: Caldara and Iacoviello, Geopolitical Risk index, historical series from three American newspapers.

The historical file also reports the share of articles in each of the eight dictionary categories, and the composition tells a story the headline hides (Table 1.3). In the decade of the First World War, articles about the beginning and the escalation of war made up almost five per cent of everything the three newspapers printed; terror was all but absent. By the 2000s,

terror acts were the largest single category, at 1.2 per cent of all articles, up from almost nothing in the 1910s, while articles about wars beginning had fallen to 0.4 per cent. Nuclear threats appear in the 1940s, peak in the 1960s and recede. In the 2020s so far, military build-ups are the largest category at 1.8 per cent — higher than in either world-war decade — and war threats have doubled from their 2000s level. The same reading meant a war in Europe in one decade, bombings in another, and rearmament in a third.

Table 1.3: What the index counts has changed: share of all articles in the three historical newspapers in each dictionary category, per cent, decade averages of monthly shares; an article can fall into several categories. Source: Caldara and Iacoviello, Geopolitical Risk index, historical category shares (SHAREH_CAT_1–8).

Decade	War threats	Peace threats	Military build-ups	Nuclear threats	Terror threats	Beginning of war	Escalation of war	Terror acts	Any category
1910s	0.91	0.14	1.13	0.00	0.01	2.38	2.23	0.01	6.17
1940s	0.95	0.22	1.25	0.27	0.02	2.13	3.47	0.10	7.68
1960s	0.61	0.20	1.04	0.94	0.04	0.49	0.78	0.31	3.78
1980s	0.35	0.09	1.05	0.81	0.27	0.41	0.56	0.63	3.40
2000s	0.26	0.06	0.72	0.50	0.39	0.40	0.56	1.23	3.28
2020s (to Sept. 2026)	0.56	0.04	1.79	0.44	0.14	1.08	0.44	0.42	3.60

The second is **audit**. The authors had samples of flagged articles read by people, to estimate how often the algorithm counted an article a human would not consider to be about geopolitical risk, and how often it missed one a human would. A text measure whose false-positive rate is unknown is a measure whose movements cannot be interpreted.

The published file also carries alternative versions of the index — one with a narrower search, one with a broader search that requires the threat word and the geopolitical anchor only to appear in the same article rather than close together, and one that drops the excluded words used to screen out false positives. The narrow version moves almost in lockstep with the headline index, with a monthly correlation above 0.9; the broad one much less, at about 0.64, and its highest month is March 2022 rather than the autumn of 2001. "The" geopolitical risk index is one point in a space of defensible choices.

Table 1.4 sets the four versions side by side. The disagreement is not only about levels but about *rankings* of the episodes that matter. On the narrow version, March 2022 (406) stands higher than March 2026 (350); on the version without excluded words it stands higher still, at 390 against 265; on the headline index the order is reversed, 319 against 330. A board told that the spring of 2026 was "the most dangerous month since the invasion of Iraq" has been told something true of one of four published series and false of the other three. The narrow version exaggerates the 2001 peak, at 889; the broad one flattens the peaks and lifts the background.

Table 1.4: One concept, four published indices: alternative versions of the recent Geopolitical Risk index, monthly, January 1985–September 2026 (each 1985–2019 average = 100). Source: Caldara and Iacoviello, Geopolitical Risk index, columns GPR, GPR_BASIC, GPR_NOEW and GPR_AND.

Version	Correlation with headline	Highest month (value)	March 2022	March 2026
Headline (GPR)	1.00	October 2001 (513)	319	330
Basic (narrow search)	0.93	September 2001 (889)	406	350
Excluded words kept	0.84	September 2001 (511)	390	265
Broad (AND search)	0.64	March 2022 (284)	284	176

The third is **economic content**. If the index measures something real, it should be related to economic outcomes in the

ways theory predicts. The paper finds that higher geopolitical risk predicts lower investment and employment, and more so in industries more exposed to it.

None of the three tests establishes that the index measures *risk* in Knight's sense — a probability of adverse outcomes. What they establish is that it measures *attention to geopolitical adversity*, that this attention is concentrated in the right months, that the counted articles are mostly about the right things, and that the attention is associated with economic decisions. That is a considerable achievement. It is also narrower than the word "risk" in the name suggests, and the gap between the two is where the misuse of the index lives.

1.7 Why waiting has value

Why should a firm's investment fall when the newspapers write about war, if the war has not yet begun and may never begin? The answer comes from the theory of irreversible investment under uncertainty.

The real-options channel — when an investment is costly to reverse and its payoff depends on information that will arrive later, the opportunity to wait has value; an increase in uncertainty raises that value and so leads firms to postpone investment, hiring and entry, even if expected returns are unchanged.

Bernanke (1983) made the argument for cyclical investment: a firm that waits for news gives up some return, but avoids having sunk capital into a project the news would have shown to be a mistake. Because a plant cannot be un-built at its cost, bad outcomes weigh more heavily in the decision than good ones, and more uncertainty about them makes waiting more attractive. Dixit and Pindyck (1994) gave the idea its general treatment: an unexercised investment opportunity is like a call option, and exercising it early destroys the option's time value.

Bloom (2009) turned the theory into a macroeconomic prediction and tested it: a spike in uncertainty should produce a rapid drop in hiring and investment, followed by a rebound as the uncertainty resolves. The pause-and-rebound pattern is the fingerprint of the real-options channel. Political uncertainty works the same way: Julio and Yook (2012) find that corporate investment falls before national elections, when the identity of the next government is unknown, and recovers after them. A geopolitical threat raises uncertainty about future states of the world without yet changing the present one — the purest form of the shock the model describes — and Caldara and Iacoviello's finding that threats depress investment is, read this way, Bernanke's 1983 argument applied to a new source of uncertainty.

The firm-level implication is concrete. The decisions most exposed to geopolitical threats are large, lumpy and hard to reverse: a greenfield plant, an acquisition, a take-or-pay supply contract. A firm that wants to keep investing through a period of tension is paying to convert the first kind of decision into the second — leasing rather than building, dual-sourcing rather than committing, staging an investment in tranches with exit points.

Consider a Canadian fertiliser exporter in March 2022, with the Threats sub-index at the highest level since 1985. "Monitoring the situation" is not a response. A response names a decision — hedge the currency, delay the warehouse in Europe, sign longer contracts with customers who have lost a Russian or Belarusian supplier — and the threshold that would trigger it. Some decisions are worth taking *because* of the uncertainty: a longer contract with a stranded customer converts a volatile opportunity into committed revenue. Others are worth postponing for the same reason: the warehouse is irreversible, and its value depends on how the war ends. The index can be an input to the threshold that separates the two. It cannot be the threshold, because it says nothing about the exporter in particular.

1.8 From the system to the profit and loss account

A firm needs something more specific than an aggregate: the routes by which a change in relations between states reaches *its* costs, revenues and assets. This book uses five, and they recur in every chapter.

Channel	How it reaches the firm	An example	Where it returns
Market access	tariffs, quotas, bloc-based trade diversion, procurement bans	a market closes to firms from a rival bloc	Chapter 4

Channel	How it reaches the firm	An example	Where it returns
Inputs	a critical input or service controlled by a foreign state or a single provider	a critical service from one satellite operator	Chapter 11
Finance and payments	sanctions on banks, asset freezes, exclusion from payment networks	a correspondent bank exits a jurisdiction	Chapter 10
Assets and people	conflict, expropriation, evacuation, detention	staff evacuated, a plant abandoned	Chapters 3, 5
Data and operations	state-sponsored cyber operations, data-localisation demands	operations halted by an intrusion	Chapter 8

The same geopolitical event reaches different firms through different channels, and therefore with different magnitudes and timing. A war in a grain-exporting region reaches a food processor through *inputs*, a shipping line through *assets and people*, a bank through *finance and payments*, and a software firm through *data and operations* when the conflict spills into cyberspace. A global index records one event. A firm's exposure is a vector, and the index is a scalar.

The channels also say where the evidence sits inside the firm: market access in the geographic breakdown of revenue; inputs in the supply-chain map; finance in the treasury's list of banks, currencies and payment routes; assets in the fixed-asset register and the list of staff posted abroad; data with the chief information security officer. An assessment that does not ask each of these offices a question is an assessment written from the newspaper.

1.9 What the index cannot see

Every measure has a boundary beyond which it stops licensing conclusions, and the boundary is part of the measure. The GPR index has at least four.

The first is that it measures **attention in the anglophone press**. The editors of American, British and Canadian newspapers decide what is news for their readers, and their readers are not in Abidjan or Shenzhen. A violent dispute that dominates the regional press of West Africa can leave the index untouched; a crisis involving the United States can dominate it. That is legitimate to measure, and it is not the risk faced by a subsidiary elsewhere.

The second is that it is **global**. One number per month for the world cannot distinguish a risk that falls on the Baltic from one that falls on the Taiwan Strait, and for most firms that distinction is the whole question. The country series help, but only for 44 countries, and only in the sense of "mentioned together with tension".

The third is that **attention fades while risk persists**. Newspapers cover change. A frozen conflict — a ceasefire line, a sanctions regime in its tenth year — can be a constant and material risk for firms near it while generating ever fewer articles. On the index, a persistent risk looks like a declining one. Firms often hold assets for longer than editors hold attention.

The fourth is that the index is **not a forecast**. It rises with events and threats as they are reported. Its predictive content for investment arises because firms *respond* to the same reported threats; that is a statement about behaviour, not about the probability of war.

A fifth follows from the construction. The denominator is all articles: if newspapers devote their space to an election or a pandemic, the geopolitical share can fall with no change in geopolitics. And the numerator depends on a fixed vocabulary; a new kind of threat — a cyber operation against a power grid, a satellite jammed in orbit — counts only insofar as journalists describe it in the words of war and terror.

1.10 How a concept becomes data

Every measure in this book passes through the same sequence. A concept — geopolitical risk, defined in words — becomes an *operational definition*: the eight categories, the ten newspapers, the share, the base period. The operational definition becomes a *dataset*: a file with one row per month and columns for the recent index, its components, the historical index, the country series, the article counts and the alternative dictionaries. The dataset becomes a *figure*. And the figure becomes an *argument*, the moment someone puts it on a slide in front of a board.

Every link is a place where meaning can be lost. Between concept and operational definition, everything the dictionary does not capture is dropped. Between operational definition and dataset, the series is revised: the file is updated monthly and the authors refine it, so a figure drawn from the current file will differ, in places, from the one printed in the journal. Between dataset and figure, choices of frequency and smoothing change what the eye sees. The recent index stood above 500 in October 2001; its average over the whole of 2001 is about 170. A geopolitical shock is a monthly phenomenon, and a yearly average buries it. And between figure and argument, the slide's title replaces the paper's caveats.

1.11 A short review of the literature

The distinction between measurable risk and unmeasurable uncertainty is due to Knight (1921), and it remains the vocabulary of the field. In international business, political risk became a subject in its own right in the 1960s and 1970s, when expropriations were frequent: Vernon (1971) explained why the bargain between investor and host decays once capital is sunk; Kobrin (1979) defined political risk as a property of the firm–environment relationship and distinguished macro from micro risk; Kobrin (1980) showed that forced divestment was selective rather than wholesale. Later work moved to measurement: Henisz (2000) built an index of political constraints on policy change, and Jensen (2003) showed that democracies attract more foreign direct investment. The expert-based tradition is summarised by the Worldwide Governance Indicators (Kaufmann, Kraay, and Mastruzzi 2011). Bremmer and Keat (2009) made the practitioner's case for political knowledge as an investment input, and Wack (1985) described how one firm learned to plan under uncertainty it could not price.

The economics of why uncertainty matters for investment runs from Bernanke (1983) through the real-options synthesis of Dixit and Pindyck (1994) to Bloom (2009), who showed that uncertainty shocks produce a sharp fall and rebound in activity; Julio and Yook (2012) documented the same mechanism around elections.

The measurement of uncertainty from text is the most recent strand. Baker, Bloom, and Davis (2016) built the Economic Policy Uncertainty index from newspaper articles and validated it by human audit — the template Caldara and Iacoviello (2022) followed for geopolitical risk, adding the split between threats and acts, a historical series to 1900, and country-specific indices. Hassan et al. (2019) moved the method to the firm, measuring each company's political risk from its own earnings calls — the measure examined in Chapter 7.

The through-line is a shift of the unit of analysis: from the host country, to the system, and back to the firm. The indices have become better at counting what is said. Whether what is said is what a firm is exposed to remains the open question.

1.12 What a count of articles does not license

The Geopolitical Risk index does what it says. It measures, month by month and for more than a century, how much of the attention of a fixed set of anglophone newspapers went to war, terror and the threat of both; it separates the threat from the act; and it moves with investment in the way the theory of irreversible decisions predicts. That makes it the natural starting point for anyone who wants to put a number on geopolitical risk. It does not make it the end point.

What the index does not license is the inference from the system to the firm. A high reading says that the press of three allied countries was preoccupied with geopolitical adversity; it does not say that a given subsidiary, supplier or payment route was exposed to it. A falling reading says that the press has moved on; it does not say that the frozen conflict next to a firm's plant has thawed. Whether a spike reaches the firm depends on the channel and on the firm's position in it, none of which a count of articles can observe.

This is not a criticism of Caldara and Iacoviello. It is a statement about the division of labour between a measure and a judgement, and Knight's distinction returns here in practical form. The part of geopolitical risk that can be counted is

real, and the index captures much of its timing. The part that cannot — the particular blockade, the particular sanction, the particular expropriation — has to be judged, by someone who knows where the firm's exposures lie. The most useful sentence to write beside any chart of the index is a sentence about what the chart cannot see: a supplier, a subsidiary, a payment route, named, with the channel through which the system would reach it.

The chapters that follow can be read as a sequence of instruments built to see what this one misses: the distribution of power (Chapter 3), the trade ties through which conflict travels (Chapter 4), the institutions and standards that govern access to finance and markets (Chapters 5 and 6), the firm's own words (Chapter 7), and the networks through which states now reach firms directly. Each has its own blind spot. Read against one another, they come closer to what a board needs to know than any one of them can alone.

Part I

The system

Chapter 2

The International Economic Order: Who Wrote the Rules, and What Happens When Their Share Shrinks

2.1 The rules a firm does not notice

A multinational firm lives inside a set of rules it rarely thinks about. When a Canadian manufacturer ships components to a plant in Mexico, assembles them, and sells the result in Germany, it assumes a great deal: that the tariff it pays in Germany is the same as its competitors pay; that a dispute over that tariff will be settled by procedure rather than by retaliation; that the dollars it earns can be converted and moved; that the patent on the component will be respected in the country where it is assembled. None of these assumptions is a law of nature. Each was written, by particular states, at particular moments, in their own interest, and each is held in place by something — power, shared interest, or institutions with momentum of their own.

This chapter is about that set of rules: the international economic order. Its question is simple to state. Who wrote the rules multinationals play by, and what happens to the rules when the writers' share of the world shrinks? The answer has three parts. The first is historical: how the post-1945 order was built, in layers, by the United States and its allies. The second is economic, and comes from Richard Baldwin's *The Great Convergence*, read against the World Bank's own national-accounts data: how a fall in the cost of moving ideas after about 1990 moved industry and income from the countries that wrote the rules to a handful that did not. The third is political: why that convergence produced a backlash, and what the backlash means for a firm whose supply chain was built on the old assumptions.

2.2 What an order is

The word "order" is used loosely, so it is worth fixing.

International order — a set of rules, institutions and shared expectations governing relations between states, sustained by some distribution of power among them.

International economic order — the part of the international order that governs trade, money and investment: tariff commitments and their exceptions, exchange-rate and capital-account arrangements, the treatment of foreign investors and their property, and the procedures by which disputes about all of these are settled.

Three questions describe any economic order, and they are worth keeping in view for the rest of the book. *Who wrote the rules?* — which states designed the institutions, and in whose interest. *What do the rules say?* — how open they are to trade, to capital and to investment, and how disputes are resolved. *What holds them in place?* — whether it is the power of a dominant state, a common interest among many, or the inertia of institutions that are costly to abandon. The first question is about history, the second about law, and the third about politics. A firm's exposure to geopolitical risk is largely a matter of the third: rules that are held in place by one state's power are as durable as that power and that state's willingness to use it.

The third question has a classic answer in political economy, and a classic rejoinder. Kindleberger (1973), writing about the Great Depression, argued that the interwar collapse of trade and finance happened because no state was both able and willing to stabilise the system — Britain was able no longer, and the United States was not yet willing. An open world economy, on this view, needs a stabiliser: a state that keeps its market open in a slump, lends counter-cyclically, and acts as lender of last resort. Krasner (1976) generalised the point into what became known as hegemonic stability theory: openness

is most likely when a single state is economically dominant, because the dominant state gains most from openness and has the means to enforce it. The rejoinder came from Keohane (1984), whose *After Hegemony* argued that institutions, once established, can sustain cooperation even as the hegemon that created them declines, because they reduce the costs of making and monitoring agreements and states have reasons to keep them. Whether the post-1945 order outlives the relative decline of its authors is the empirical test of that debate, and it is being run now.

Hegemonic stability — the proposition that an open international economy is most likely to be created and maintained when a single dominant state has both the capacity and the interest to supply its rules and absorb the costs of enforcing them.

2.3 The post-1945 order, in four layers

The order that multinationals take for granted was assembled in layers over half a century. It is easier to understand as a stack than as a single design.

From the archive · Bretton Woods, 1944

In July 1944, while the war in Europe and the Pacific was still being fought, delegates of forty-four allied nations met at the Mount Washington Hotel in Bretton Woods, New Hampshire. The two dominant figures were John Maynard Keynes for the United Kingdom and Harry Dexter White for the United States Treasury, and the conference largely ratified a compromise their governments had negotiated in advance. The British wanted a system that would protect a debtor country's freedom to pursue full employment; the Americans, who would be the world's creditor, wanted a system in which their money would not be used to finance others' deficits indefinitely.

The conference created the International Monetary Fund, to oversee a system of fixed but adjustable exchange rates and to lend to countries facing temporary balance-of-payments difficulties, and the International Bank for Reconstruction and Development, the core of what became the World Bank. Currencies were pegged to the dollar; the dollar was convertible into gold at a fixed price for foreign official holders. Capital controls were not merely tolerated but expected: the founders had watched short-term capital flows destabilise the interwar economy and did not intend to repeat the experience (Eichengreen 2019).

What is striking, read from the twenty-first century, is how much of the design reflected the balance of power in the room. The United States held most of the world's monetary gold and was the only major economy undamaged by the war. The institution's voting shares, its headquarters in Washington, and the central role of the dollar all followed. The rules were multilateral in form and American in their foundations — which is the observation that every later debate about the order starts from.

The second layer was trade. The planned International Trade Organization, which was to have stood beside the Fund and the Bank, was never established: its charter was negotiated but not ratified by the United States. What survived was a provisional agreement, the **General Agreement on Tariffs and Trade** of 1947, which turned out to be one of the most durable provisional arrangements in history. The GATT rested on two principles. **Most-favoured-nation** treatment required each member to extend to all members any tariff concession it gave to one; **national treatment** required that imported goods, once across the border, be treated no worse than domestic ones. On that basis, successive negotiating rounds cut tariffs on manufactured goods among the industrial countries.

The record of those rounds, which Table 2.1 reproduces from the WTO's own history of the GATT years, is worth reading row by row, because it shows the order growing in two directions at once. It grew in *membership*: twenty-three countries negotiated in Geneva in 1947, sixty-two in the Kennedy Round of the 1960s, one hundred and twenty-three in the Uruguay Round. And it grew in *scope*: the first five rounds dealt with tariffs alone; the Kennedy Round added anti-dumping; the Tokyo Round reached non-tariff measures and a set of plurilateral "codes" that only some members signed; the Uruguay Round took in services, intellectual property, agriculture, textiles and dispute settlement, and created a new organisation to administer the result.

Table 2.1: The GATT negotiating rounds, 1947–1994: more countries, and more of the economy, at each round. Source: World Trade Organization, *Understanding the WTO: The GATT years* (World Trade Organization, n.d.).

Years	Place / name	Subjects covered	Countries
1947	Geneva	Tariffs	23
1949	Annecy	Tariffs	13
1951	Torquay	Tariffs	38
1956	Geneva	Tariffs	26
1960–61	Geneva (Dillon Round)	Tariffs	26
1964–67	Geneva (Kennedy Round)	Tariffs, anti-dumping	62
1973–79	Geneva (Tokyo Round)	Tariffs, non-tariff measures, "framework" agreements	102
1986–94	Geneva (Uruguay Round)	Tariffs, non-tariff measures, rules, services, intellectual property, dispute settlement, textiles, agriculture, creation of the WTO	123

Two features of the table matter for what follows. The first is that the countries that joined late joined an agenda already set. A developing economy that acceded in the 1980s or 1990s did not negotiate the most-favoured-nation principle, the treatment of anti-dumping duties, or the structure of tariff bindings; it accepted them as the price of entry, and bargained over its own schedule. The second is that the last round moved the order behind the border, into the domestic regulation of services, patents and investment, where political resistance is strongest. Even before then, the WTO's own history records, the rule-writers had turned in the recessions of the 1970s and early 1980s to bilateral market-sharing arrangements and agricultural subsidies the GATT could not discipline (World Trade Organization, n.d.).

The third layer is not an institution but a bargain, and the political scientist John Ruggie gave it its name.

Embedded liberalism — the post-war compromise in which states committed to multilateral openness in trade while retaining the domestic policy autonomy — macroeconomic management, social insurance, capital controls — needed to make that openness politically sustainable at home (Ruggie 1982).

Ruggie's point was that the post-war order was not the liberalism of the nineteenth century restored. It was liberalism *embedded* in a domestic social settlement: governments would open their markets, but they would also cushion their citizens from the dislocations openness caused, and they would keep the instruments to do so. Free trade was managed; capital was controlled; full employment was a government responsibility. The bargain held for roughly a quarter of a century, and its unravelling is the background to most of what follows.

The fourth layer is the neoliberal turn and the creation of the World Trade Organization. The monetary pillar of Bretton Woods broke first.

From the archive · The end of gold convertibility, 1971

On 15 August 1971, President Nixon announced that the United States would suspend the convertibility of the dollar into gold for foreign official holders, and imposed a temporary surcharge on imports. The announcement was made without consulting the other members of the IMF. A realignment of exchange rates negotiated at the Smithsonian Institution in December 1971 tried to restore a system of pegs; by 1973 the major currencies were floating against one another, and they have floated since (Eichengreen 2019).

The episode is usually told as the end of Bretton Woods, and in its monetary design it was. But it is more instructive as a demonstration of the third question — what holds the rules in place. The system had depended on the United States' willingness to subordinate its own policy to the gold parity. When that willingness ended, the rule ended, unilaterally, on a Sunday evening. The institutions survived: the IMF found new roles, and the dollar remained the centre of the system, now without the anchor. The order changed form without collapsing, which is the pattern to look for whenever someone announces the end of an order.

Through the 1980s and 1990s, capital controls in the industrial countries were dismantled, and many developing countries followed. In trade, the Uruguay Round of negotiations, launched in 1986, ended with the creation of the **World Trade**

Organization in 1995. The WTO differed from the GATT in two ways that matter for firms. Its rules reached far beyond tariffs on goods — into services, into intellectual property through the TRIPS agreement, into investment measures through TRIMs — so that the order began to govern not only what crossed borders but what happened behind them. And it had a binding dispute-settlement mechanism, with an Appellate Body, so that rulings could no longer be blocked by the losing party. China joined in 2001, after fifteen years of negotiation, on terms that bound it to many of these rules.

How should the result be characterised? Ikenberry (2011) argues that the American-led order was **easy to join and hard to overturn**: open, rule-based, and institutionalised enough that rising states would gain more by joining it than by challenging it. On this reading, the order is not merely an expression of American power but a set of institutions that bind the hegemon too, and that therefore give others a reason to accept it. Chapter 3 sets the realist reply beside this view: that institutions reflect the distribution of power and will be revised when it shifts.

2.4 Firms as authors as well as subjects

It would be a mistake to read the order as something states wrote and firms merely obeyed. At several points, firms were among its authors, and the layers that matter most to multinationals today are the ones in which their influence was greatest.

The clearest case is intellectual property. Before the Uruguay Round, patents, copyrights and trademarks were governed by conventions administered outside the trade regime, with weak enforcement and wide latitude for national policy; many developing countries did not grant patents on pharmaceuticals at all. The TRIPS agreement changed that, by making minimum standards of intellectual-property protection a condition of WTO membership, enforceable through the dispute-settlement mechanism. Sell (2003) traces how this happened: a small coalition of chief executives from American pharmaceutical, software and entertainment firms organised in the 1980s, defined the problem as one of trade rather than of intellectual property, persuaded the United States government to adopt their framing, and worked with counterparts in Europe and Japan to build a joint position. Twelve corporations, in her account, effectively drafted the core of a public international law. The episode is the strongest evidence that the post-1945 order was not only American in its foundations but, in some of its later layers, corporate.

A second case is the protection of foreign investment, which the GATT and the WTO largely left aside. Here the instrument was the bilateral investment treaty. The first was signed between West Germany and Pakistan in 1959; the Convention establishing the International Centre for Settlement of Investment Disputes, under World Bank auspices, followed in 1965; and in the 1990s the number of such treaties multiplied. Their distinctive feature is that they allow a foreign investor to bring a claim directly against a host state before an international tribunal, without needing its own government to take up the case. For a multinational, this is the layer of the order that most directly answers Kobrin's political risk (Chapter 1): it is the legal response to expropriation and discriminatory treatment. It is also the layer that has attracted the most political contestation in the 2010s, from governments that came to see investor-state arbitration as a constraint on regulation — Rodrik's trilemma, applied to a single clause.

The point for the firm is twofold. First, the rules a firm relies on are not all of a piece: some are multilateral (tariff bindings), some are bilateral (investment treaties), some are private or semi-private (standards, the subject of Chapter 6). They will fail in different ways. Second, if firms helped write some of the rules, they have a voice in how they are rewritten: the influence a firm exercises over the rules is one of the few geopolitical exposures it can partly manage from the inside.

2.5 Three costs, three unbundlings

Baldwin's book offers a different way into the same history, one that starts not from institutions but from costs. Globalisation, he argues, has been shaped by three successive constraints: the cost of moving **goods**, the cost of moving **ideas**, and the cost of moving **people** (R. Baldwin 2016). Each time one of these costs fell sharply, it "unbundled" something that geography had previously held together.

Unbundling — in Baldwin's account, the separation, made possible by a fall in a particular cost of distance, of activities that had previously had to take place in the same location: first production from consumption, then the stages of production from one another.

2.5.1 The first unbundling: goods move, ideas stay

From the nineteenth century, steam ships and railways made it cheap to move goods. Production no longer had to happen near consumption: a factory in Manchester could clothe customers in India. But the cost of moving ideas remained high, because coordinating complex production still required people to be close to one another. So industry clustered — in Britain, western Europe, North America, later Japan — where know-how could circulate cheaply within a local area. The result was the great divergence: the industrialising North pulled away from the rest of the world in income, because the knowledge that made factories productive stayed where it was invented. Trade grew, but it was trade in finished goods between nations, and the comparative advantage that governed it was a national property.

2.5.2 The second unbundling: ideas move, inside the firm

From about 1990, information and communication technology made it cheap to move ideas — to coordinate complex activities at a distance. The factory itself could now be unbundled. Stages of production that had to sit together in one place could be dispersed across countries, each stage located where its costs were lowest, and coordinated from headquarters by telecommunications and computers. A firm could combine its own technology, management and marketing with low-wage labour in a nearby developing country — for the United States, Mexico; for Germany, Poland and central Europe; for Japan, the rest of East Asia.

The crucial move in Baldwin's argument is what travelled with the stages of production. When a Japanese firm moved assembly to Thailand, it did not move only the assembly. It moved the know-how to perform it: the process engineering, the quality systems, the managerial practice. Know-how that had been confined within national borders now flowed across them — but inside the firm's own production network, under its control. This is why the second unbundling produced convergence rather than divergence. A handful of developing economies, those that joined the supply chains of northern firms, industrialised at a speed previously unknown, because they did not have to invent the know-how; they had it brought to them.

Global value chain — the international fragmentation of a production process into stages located in different countries and coordinated, usually by a lead firm, so that what crosses borders is increasingly intermediate goods, services and know-how rather than finished products.

Baldwin's empirical picture of the shift emphasises its regional character. Supply-chain trade, R. Baldwin and Lopez-Gonzalez (2015) show, is organised around a few hubs — "Factory North America", "Factory Europe", "Factory Asia" — each pairing high-technology headquarter economies with lower-wage neighbours. The second unbundling was less a globalisation than a set of regionalisations, linked by the headquarters economies.

2.5.3 The third unbundling: people, virtually

The third constraint is the cost of moving people, and specifically of face-to-face interaction. Baldwin's prospective third unbundling would come from telepresence and telerobotics: technologies that let a worker in one country perform services in another without travelling. It is a forecast rather than a finding, and the least developed part of the argument.

2.5.4 The multinational as the vehicle

For the multinational, the key point of the book is a single sentence's worth of argument: in the second unbundling, *the multinational firm is the vehicle of convergence*. The convergence of incomes between the old industrial economies and the new ones did not happen through anonymous markets in finished goods. It happened through supply chains that multinational firms built, owned or controlled, along which they moved technology and organisational capability.

That is why geopolitics now reaches firms so directly. If know-how, jobs and capabilities travel along supply chains, then supply chains are where states will intervene when they decide that the movement of know-how threatens their security. The export controls of the 2020s on advanced semiconductors and the equipment to make them are attempts to stop exactly the flow the second unbundling made cheap. The firm's supply chain has become a geopolitical object, not because firms chose to make it one, but because it carries what states care about.

2.6 The convergence in the data

Baldwin's evidence for the convergence is a picture: the share of the G7 — Canada, France, Germany, Italy, Japan, the United Kingdom and the United States — in world income, which rose for over a century and then fell sharply from around 1990. The share lost by the G7 was gained mostly by a small group of developing economies, which he calls the "Industrialising Six": China, Korea, India, Poland, Indonesia and Thailand.

The picture can be checked against the World Bank's World Development Indicators, and the check is instructive, because reproducing it requires three decisions, each of which shapes what the figure says.

The first is that **the G7 is not in the data**. The World Bank publishes the world total and many aggregates — income groups, regions, the euro area — but no G7. The analyst constructs it by summing the seven members, and the G7 that results is a political grouping turned into a statistic by someone's choice. Done with GDP in current US dollars, the result confirms Baldwin's shape and dates its turning point precisely. In the release of October 2026, the G7 produced about 66 per cent of world output in 1970. The share dipped in the 1970s, recovered through the 1980s, and peaked at about 67.5 per cent in 1993. It then fell almost without interruption: below 60 per cent by 2005, below 50 by 2010, and about 44 per cent in 2024 and 2025. In a single generation the economies that wrote the rules went from two-thirds of world output to well under half. Figure 2.1 draws the series.

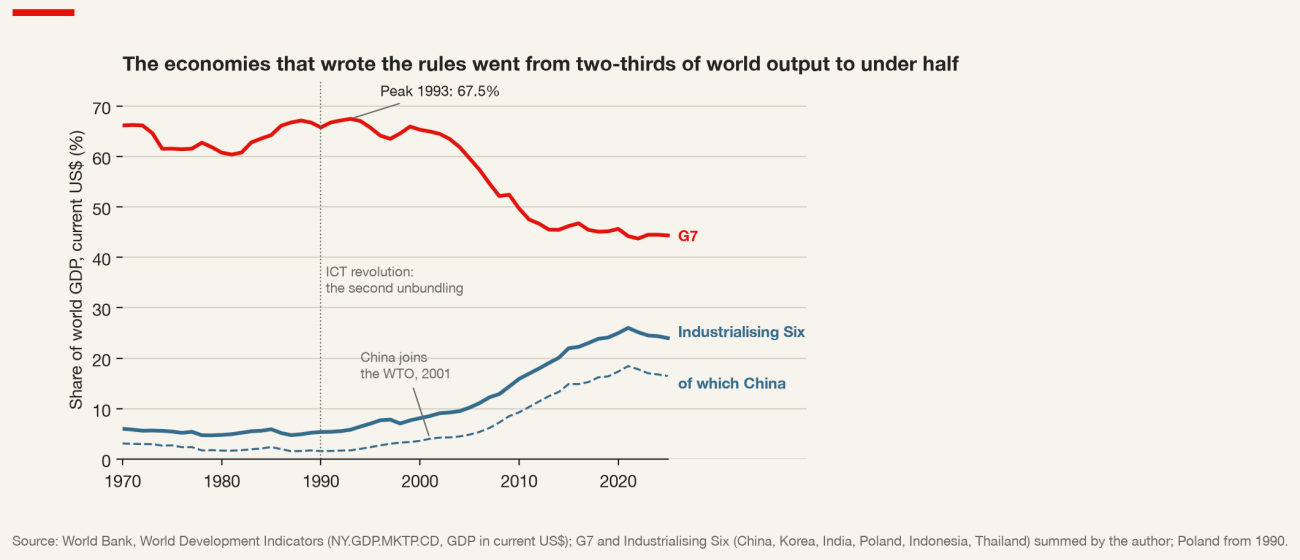


Figure 2.1: The G7's share of world GDP at market exchange rates peaked in 1993 at 67.5 per cent and has fallen to about 44 per cent; the Industrialising Six rose from about 5 to about 24 per cent, two-thirds of it China. Source: World Bank, World Development Indicators (NY.GDP.MKTP.CD); aggregates summed by the author.

Three things are easy to miss. The first is how flat the G7 line is for two decades before the fall. The 1970s dip — from 66 per cent in 1972 to about 60 per cent in 1981 — is the oil shock and the dollar's depreciation after the end of convertibility, not convergence: oil exporters' incomes rose in dollar terms, and the dollar value of everything outside the United States moved with the currency. The recovery to the 1993 peak is in large part the yen and the European currencies appreciating against the dollar after the Plaza Accord of 1985. None of this is the great convergence; it is exchange-rate noise around a stable distribution. The second is the timing of the fall. Baldwin dates the second unbundling to about 1990, but the G7 share at market rates did not fall decisively until the early 2000s — it was still 65 per cent in 2001 and below 60 only in 2005. The lag is the time it took supply-chain industrialisation to show up in national output measured in dollars, and it coincides with China's accession to the WTO. The third is that the fall has stopped. Since 2013 the G7 share has moved between about 44 and 47 per cent, and the Industrialising Six peaked at 26 per cent in 2021. At market rates the convergence has, for now, paused.

The Industrialising Six account for most of the difference. Their combined share was about 6 per cent in 1970 and a little over 5 per cent in 1990; it reached about 16 per cent in 2010 and about a quarter of world output by the early 2020s. Within the six, China dominates. Its GDP at market exchange rates was about 6 per cent of the United States' in 1990 and 41 per

cent in 2010; it reached a high of about 77 per cent in 2021 and has since fallen back, to about 64 per cent in 2024, partly because the dollar strengthened. The convergence is real, and it is not a straight line.

2.6.1 Whose decline?

A G7 total hides who is in it. Figure 2.2 takes the aggregate apart, and the result disagrees with the way the story is usually told. The usual account is about American decline: the hegemon that wrote the rules is losing the weight to enforce them. At market exchange rates, the data say something else. The United States produced about 36 per cent of world output in 1970, a figure inflated by the dollar's overvaluation under the fixed parity, but about 26 per cent in 1990 and about 26 per cent in 2025. Over the period of the great convergence, the American share is essentially unchanged. The share the G7 lost was lost by the other six. Japan's fall is the most dramatic in the panel: from 7 per cent of world output in 1970 to a peak of about 18 per cent in 1994, at the height of the yen, and down to under 4 per cent in 2025, the combined effect of three decades of slow growth, deflation and a weaker currency. The four large European economies together produced about 23 per cent of world output in 1990 and about 13 per cent in 2025; Germany's share roughly halved, France's halved, and Italy's more than halved. In 1990 the G7 outside the United States produced about 40 per cent of world output; in 2025, about 18 per cent.

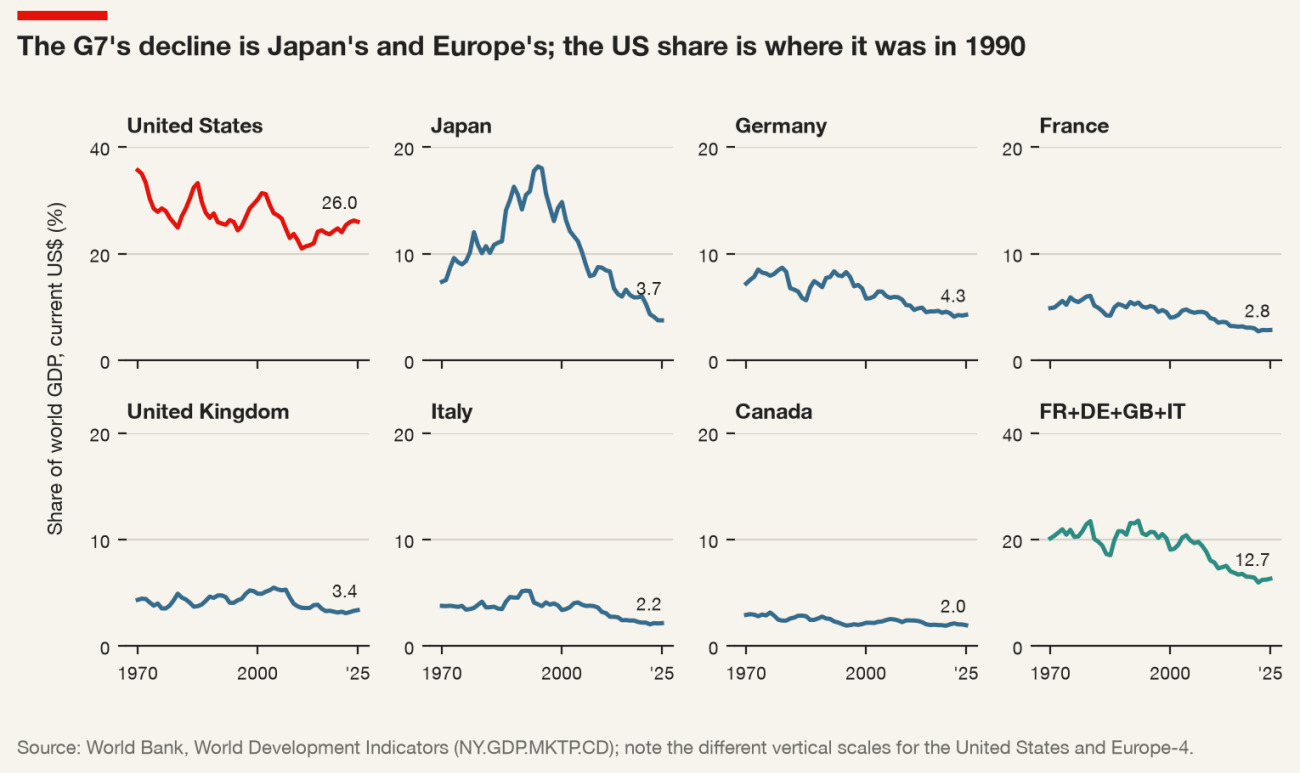


Figure 2.2: At market exchange rates the United States' share of world GDP is about where it was in 1990; the G7's lost share was lost by Japan and the four large European economies. Source: World Bank, World Development Indicators (NY.GDP.MKTP.CD).

The decomposition matters for the argument of this chapter in two ways. For hegemonic stability theory, the relevant quantity is the hegemon's share, not the club's; on that measure, at market rates, the stabiliser of the post-1945 order has not shrunk since the end of the Cold War, while its allies have. The order's problem is less that its leading author is weaker than that the coalition which shared the costs of running it — the Europeans and Japan, who financed the institutions, matched American tariff cuts and carried a large part of the votes in the Fund and the Bank — now carries much less weight. That changes the politics of burden-sharing, which is a recurring theme of the 2020s. For a multinational, the decomposition means that "the G7 market" is not a single market whose relative size is declining; it is one market, the American, that has held its weight, and several that have not. A European firm whose home market has halved as a share of world demand faces a different problem from an American one.

The second decision is the **unit of account**. Market exchange rates are the natural measure if the question is purchasing power on world markets — what an economy's output can buy abroad, how large it looks to an exporter selling in dollars. But the share then moves with the dollar. When the dollar strengthens, every other economy shrinks in dollar terms and the US share rises, without any change in what anyone produces. Part of any year-to-year movement in the series is the exchange rate, and the Chinese reversal after 2021 is partly that.

Purchasing-power parity (PPP) — a conversion of national output into a common unit using the relative prices of a common basket of goods and services rather than market exchange rates, so that an economy's output is valued at what it would cost in a reference country.

The alternative is GDP at purchasing-power parity. Because many goods and services — especially non-traded services — are cheaper in lower-income economies, converting their output at market exchange rates understates its real volume. The difference is large. The World Bank's PPP series begins only in 1990, which is to say at the start of the convergence, and in that year the G7's share at PPP was already about 46 per cent — twenty points below its share at market rates. By 2025 it was about 28 per cent. On this measure China's economy passed that of the United States in 2014, and by 2025 was about a third larger; at market rates it has never passed it.

Which measure is right depends on the decision. If a board is deciding where the purchasing power for its *exported* product sits — how many dollars or euros a market can pay — market exchange rates are the relevant measure. If it is deciding where real production capacity and local markets for locally made goods are, PPP is closer. A firm that shows its board only one of the two has made a strategic choice without stating it.

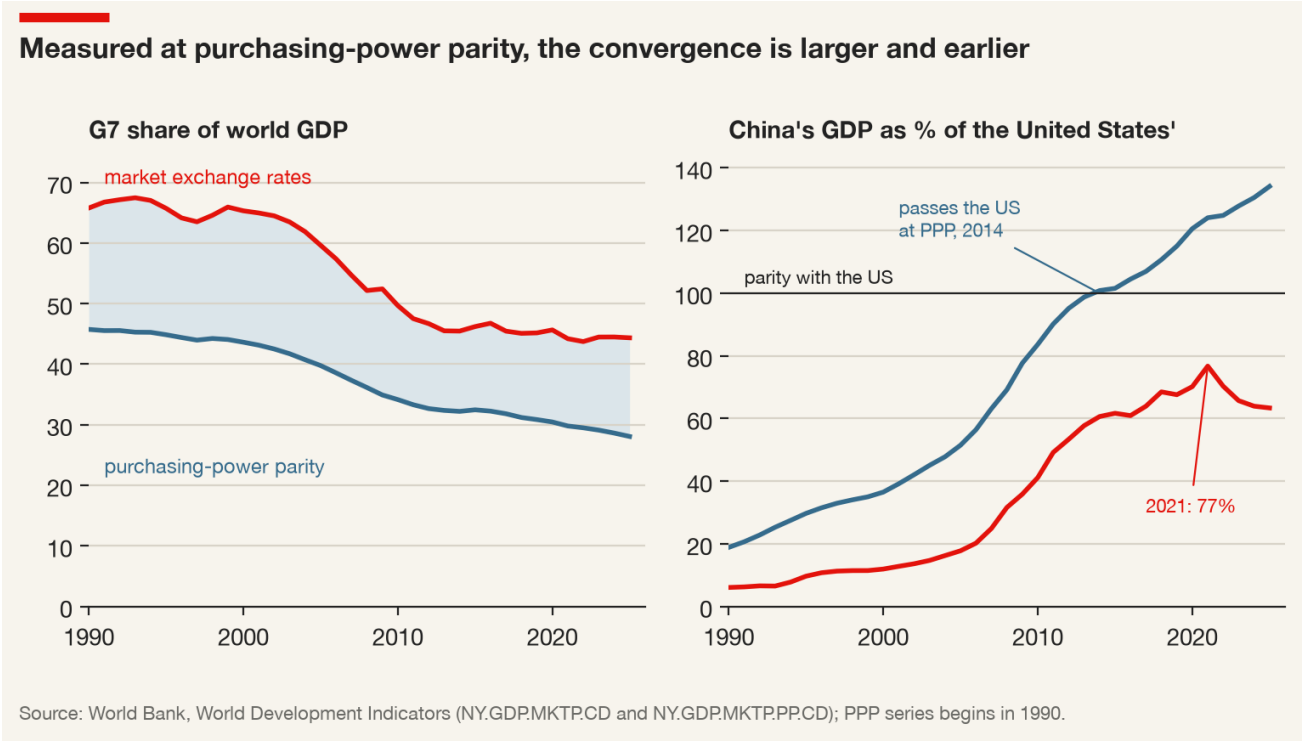


Figure 2.3: At purchasing-power parity, the G7's share of world output was already 46 per cent in 1990 and is 28 per cent in 2025; China's output passed that of the United States in 2014, while at market rates it peaked at 77 per cent of US output in 2021. Source: World Bank, World Development Indicators (NY.GDP.MKTP.CD and NY.GDP.MKTP.PP.CD).

Figure 2.3 sets the two measures side by side, and Table 2.2 gives the benchmark years. The two measures agree on the direction and disagree on almost everything else. At PPP the G7 was never a two-thirds club in the period the data cover; it was already below half when the convergence began, and the Industrialising Six overtook it in 2020 — something that has not happened, and on current trends will not soon happen, at market rates. The China–United States comparison is the sharpest case. At PPP the crossing came in 2014 and the gap has widened every year since, to a Chinese economy about 134 per cent of the American in 2025; at market rates China rose to 77 per cent of US output in 2021 and has fallen back to

63. Two respectable measures tell one story of a power transition already completed and another of a transition that stalled. The realist argument of Chapter 3, about whether a rising state's power is approaching that of the established one, depends on which is used.

Table 2.2: Shares of world GDP at benchmark years, at market exchange rates and at purchasing-power parity (per cent). Source: World Bank, World Development Indicators (NY.GDP.MKTP.CD, NY.GDP.MKTP.PP.CD), October 2026 release; G7 and Industrialising Six summed by the author; the PPP series begins in 1990.

Year	G7, market	G7, PPP	Industrialising Six, market	Industrialising Six, PPP	United States, market	United States, PPP	China, market	China, PPP
1970	66.2	—	6.0	—	35.7	—	3.1	—
1990	65.8	45.7	5.4	11.9	25.9	20.1	1.6	3.8
2010	49.7	34.2	15.9	25.4	22.5	16.6	9.3	13.9
2024	44.5	28.6	24.4	32.7	26.2	14.7	16.8	19.1

The table also shows why the two measures should not be averaged or blended. They answer different questions, and the gap between them is itself information: it is a rough index of how cheap non-traded goods and services are in an economy relative to the reference. As economies converge in income, their price levels tend to rise towards those of richer economies and the gap narrows — so part of what the market-rate series records as convergence over a long period is the closing of the price gap, not only growth in volume. The PPP numbers are also revised whenever the International Comparison Program publishes a new round of price surveys, so an analyst who quotes "China passed the United States in 2014" is quoting a particular vintage of price data, and should say so (World Bank 2026).

The third decision is about **who is in the denominator**. The world total is a World Bank aggregate that includes economies for which data are sparse or estimated, particularly in the early decades, and not every member of the Industrialising Six is present throughout. Poland has no GDP figure in the series before 1990: for the first twenty years of the period, the Six are in fact five. A stacked chart of shares — G7, I6, rest of the world — therefore shows the "rest of the world" absorbing whatever is missing from the Six, which is an artefact of the data, not a fact about the world.

2.7 From convergence to backlash

The convergence was, by most measures of global welfare, a remarkable success. It was also politically destabilising in the countries whose share was shrinking, and the most useful framework for understanding why comes from Dani Rodrik.

Rodrik's trilemma — the proposition that a country can have at most two of three things: deep economic integration with the world economy, national sovereignty in policy-making, and democratic politics (Rodrik 2011).

The logic is that deep integration requires harmonised rules — on products, on investment, on capital flows — and those rules constrain what national governments can do. If the nation-state keeps its sovereignty and pursues deep integration, democratic choice must be constrained: governments adopt whatever policies markets reward, a condition Rodrik describes with Thomas Friedman's phrase, the "golden straitjacket". If integration and democracy are both kept, the rules must be made democratically above the state — a global federalism no one has built. If sovereignty and democracy are kept, integration must be shallow: the Bretton Woods compromise, with managed openness and room for national policy.

The trilemma is a reading of the history in the previous sections. Embedded liberalism was the third option: shallow integration that left room for democracy and sovereignty. The neoliberal turn and the WTO pushed towards deep integration — Rodrik calls it hyperglobalisation — and, for a while, the countries concerned accepted the straitjacket. The backlash of the 2010s and 2020s — Brexit, the tariff wars that began in 2018, the return of industrial policy, export controls, investment screening, "de-risking" — can be read as a sequence of choices of sovereignty and democracy over deep integration.

The economics behind the politics became clearer with evidence on the distribution of the gains. Autor, Dorn, and Hanson (2013) showed that US local labour markets more exposed to rising Chinese import competition after 2000 suffered larger falls in manufacturing employment and wages, and higher take-up of public transfers, with effects that persisted. What the "China shock" literature showed was that the losers were geographically concentrated, that adjustment was slow, and

that the compensation the embedded-liberal bargain had promised did not reach them. A bargain that is not honoured stops being a bargain.

2.7.1 Slowbalisation

The phenomenon that the press calls *slowbalisation* is visible in a single series, world trade as a percentage of world GDP, drawn in the left panel of Figure 2.4. It was about 26 per cent in 1970 and 38 per cent in 1990, and peaked at about 60 per cent in 2008; it fell to 52 per cent in 2009 and has oscillated since between the low fifties and 62 per cent, reached in 2022, a year of high commodity prices. The last year in the release jumps to well above any earlier value, but it rests on reports from 138 economies, against 183 for 2022, and should be read as provisional. Whether the flattening is a retreat from globalisation or a change in its form is contested. A retreat would show up as firms reshoring and supply chains shortening; a change of form, as trade in goods stagnating while poorly measured trade in services and data continues to grow. The series alone cannot settle the question, because its denominator, GDP, has shifted towards services, which are traded less, in exactly the period in question.

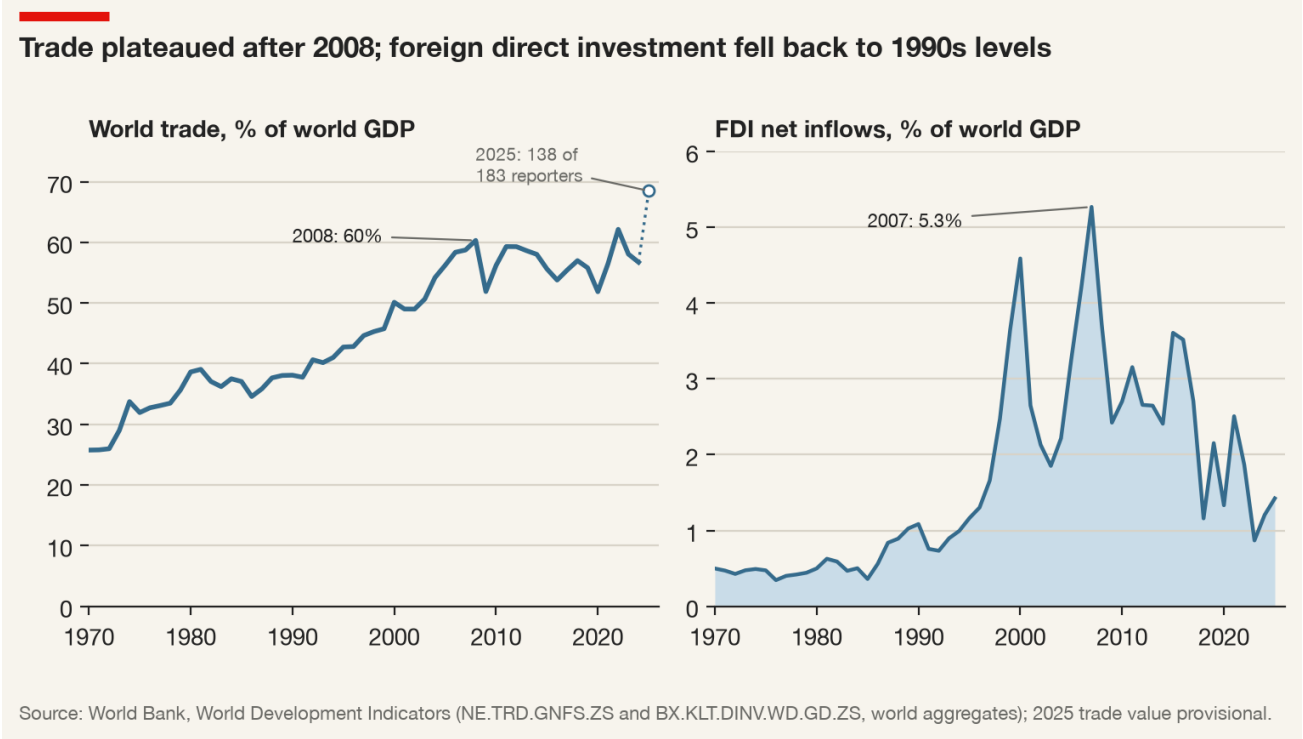


Figure 2.4: World trade rose from 26 per cent of world GDP in 1970 to 60 per cent in 2008 and has moved sideways since; net inflows of foreign direct investment peaked at 5.3 per cent of world GDP in 2007 and were near 1 per cent in 2023–2025. Source: World Bank, World Development Indicators (NE.TRD.GNFS.ZS, BX.KLT.DINV.WD.GD.ZS), world aggregates; the last trade observation is provisional.

The right panel tells a starker story, and it is the one closer to the multinational's own decisions. Net inflows of foreign direct investment — the money firms commit to owning and controlling operations abroad — were under 1 per cent of world GDP until the late 1980s, rose to 4.6 per cent in 2000 and 5.3 per cent in 2007, waves of cross-border mergers riding the second unbundling, and have since fallen back. They averaged 3.3 per cent of world GDP in 2000–2007 and 1.6 per cent from 2018, with a low of 0.9 per cent in 2023: the level of the early 1990s. The series is noisy, because it records financial flows rather than factories — the 2018 trough reflects American multinationals repatriating foreign earnings after the 2017 US tax reform (UNCTAD 2019), and much FDI passes through holding companies in low-tax jurisdictions (Damgaard, Elkjaer, and Johannesen 2024) — but the direction is not in doubt. Trade has plateaued; the investment that builds supply chains has retreated. If slowbalisation is a change of form, it is one in which firms keep trading through the networks they built while becoming much more reluctant to build new ones abroad.

2.7.2 The dispute mechanism

One institutional fact belongs in every firm's risk register. The WTO's Appellate Body has been unable to hear appeals since December 2019, when the terms of its members expired and the United States blocked the appointment of replacements. A panel ruling can now be appealed "into the void", where no body exists to hear it, and so never become binding. For a firm, a discriminatory measure by a trading partner is now more likely to be resolved by bargaining or retaliation between governments than by a legal ruling — by power rather than by procedure.

2.8 Interdependence as an instrument

The first two parts of this chapter describe an order designed to make interdependence safe. The third must add that interdependence was never only an economic fact. Albert Hirschman, writing in 1945 about Nazi Germany's trade with south-eastern Europe, showed that trade relations could be structured deliberately to create dependence: by trading with small partners for whom the trade was a large share of their total, a great power could make the threat of interrupting it a source of political influence (Hirschman 1945). The gains from trade, in his account, were also the measure of the power the trade conferred — the cost a partner would bear if it stopped.

Farrell and Newman (2019) brought the argument into the age of networks. The infrastructure of globalisation — financial messaging, dollar clearing, internet backbones, cloud platforms — is not flat; it has hubs, and the hubs sit disproportionately in a few jurisdictions. States that control the hubs can use them for surveillance, observing what flows through, and for exclusion, cutting adversaries off. Interdependence, on this reading, has been "weaponised": the very networks that the economic order built become instruments of coercion. Chapters 8, 10 and 11 develop the argument for cyber, sanctions and space. Its relevance here is that the order's openness and its vulnerability are the same thing seen from two sides.

For a multinational, the practical form of weaponised interdependence is jurisdictional reach. A firm that clears payments in dollars, stores data with a provider headquartered in one country, or uses components containing technology controlled by another, has placed part of its operations under that country's law, wherever the firm itself is incorporated and wherever the transaction takes place. The order built after 1945 made it natural to route business through American banks, platforms and technology; the same routing now carries American regulatory authority with it, and China and the European Union have begun to build their own instruments of reach. A firm's exposure is therefore a matter not only of where it sells and produces, but of which networks it runs through and whose law travels along them.

2.9 What it means for the multinational

The history and the economics lead to four implications for a firm's strategy.

The market is no longer where the rules are written. For most of the post-war period, the largest markets and the rule-makers were the same countries. The convergence separated them. A European firm now earns a growing share of its revenue in markets whose governments did not design the WTO, did not participate in Bretton Woods, and do not feel bound by the spirit — as distinct from the letter — of rules they joined late. The firm's legal environment and its commercial environment have come apart.

The supply chain is a geopolitical object. If the second unbundling moved know-how along supply chains, then supply chains are where governments now intervene to stop it: export controls, outbound investment screening, restrictions on technology transfer, requirements to locate certain stages of production in trusted jurisdictions. The question for the firm is no longer only where a stage is cheapest, but where it is permitted, and by whom.

Multilateral protection is weaker. With the Appellate Body unable to hear appeals, the guarantee that a trading partner's discriminatory measure could be challenged and removed through procedure has weakened. Firms that relied implicitly on that guarantee — exporters into markets with a history of trade remedies, for example — face disputes settled more often by negotiation and retaliation. The firm's government's bargaining power has become part of the firm's risk profile.

Measurement choices are strategy choices. At market exchange rates the G7 still produces nearly half of world output; at PPP, little more than a quarter; and within the G7 the American share has held while Japan's and Europe's have shrunk. A risk analyst's first duty is to show both measures, and the members, and say which question each answers.

Mapped onto the five channels of this book, the order's transformation runs mainly through **market access** — the tariffs, trade remedies and bloc-based diversion of Chapter 4 — and through **inputs**, where the know-how and components that the second unbundling dispersed are now subject to control. But it reaches **finance and payments** too, because the dollar-centred monetary system that survived 1971 is the platform on which financial sanctions operate.

2.10 What the figure does not license

Baldwin's figure is a powerful image, and its power is a reason for care. It does not license four conclusions that boards and commentators routinely draw from it.

A falling G7 share is **relative**. The G7 economies kept growing in absolute terms through the whole period; the denominator grew faster. A falling share does not mean the G7 market shrank, and a firm selling into the G7 did not lose customers because the line went down.

At current US dollars, the share **moves with the dollar**: a reading in any single year can say as much about currency markets as about convergence.

Convergence is **concentrated**. Most of the gain went to a handful of economies, and above all to China. "The South" did not converge; specific economies that joined specific supply chains did.

It shows **income**, not **rule-making power**. Whether a shrinking share of world GDP means a shrinking ability to set the rules is a question about the relation between economic size and political influence — the subject of Chapter 3. Hegemonic stability theory predicts that it does; institutionalists predict that rules can outlast the share. The chart cannot settle the dispute, because it does not measure the thing in dispute.

2.11 A short review of the literature

The political economy of the post-war order begins with the hegemonic-stability debate. Kindleberger (1973) attributed the interwar collapse to the absence of a stabilising power; Krasner (1976) generalised the argument to the structure of trade; Keohane (1984) replied that institutions can sustain cooperation after hegemony. Ruggie (1982) named the domestic bargain — embedded liberalism — on which post-war openness rested, and Ikenberry (2011) reinterpreted the American-led order as liberal, institutional and self-binding, and therefore "easy to join and hard to overturn". The monetary history, from Bretton Woods through the end of gold convertibility, is told by Eichengreen (2019).

The economics of the new globalisation is organised by R. Baldwin (2016) around falling costs of moving goods, ideas and people, with the multinational's supply chain as the conduit for the second unbundling; R. Baldwin and Lopez-Gonzalez (2015) map the regional structure of supply-chain trade. The distributional consequences in the rich countries are documented most influentially by Autor, Dorn, and Hanson (2013), and their political consequences are read through Rodrik (2011)'s trilemma of integration, sovereignty and democracy.

A third strand treats economic interdependence as a source of power rather than only of gains. Hirschman (1945) showed how asymmetric trade could be designed to create political dependence; Farrell and Newman (2019) extended the argument to the networked infrastructure of finance and communications, introducing "weaponised interdependence" — the strand that Chapters 8, 10 and 11 follow.

The through-line is that every account of the order answers the same question — what holds the rules in place — and that the answers differ chiefly in how much weight they give to the distribution of power. The great convergence is the event that puts that weight to the test.

2.12 Whether the rules follow the income

The international economic order was written by a small number of states, at a moment when they held most of the world's gold, most of its industry and, by the time the series begins, two-thirds of its output. The rules they wrote were multilateral in form, and in several layers they were written with the help of the firms that would live under them. Those rules made the second unbundling possible: they made it cheap and safe for a multinational to move know-how across borders inside its own production network, and the movement of know-how is what produced the convergence. The order, in this sense, financed its own relative decline — which is not a failure of design, since an open order was always going to spread industry to the places it reached.

What the data cannot say is whether the rules will follow the income. The G7's share of world output has fallen from about two-thirds to well under half at market prices, and to little more than a quarter at purchasing-power parity. The institutions the G7 built are still standing: the Fund and the Bank still lend, the WTO's schedules still bind tariffs, the dollar still clears most of the world's trade. But some of what made them work has already gone. The Appellate Body cannot sit. The embedded-liberal bargain no longer holds in the countries that wrote it. And the networks through which goods, money and data move have become instruments of the states that host their hubs.

Two readings of that evidence compete. On the hegemonic-stability reading, rules that rested on the power of their authors will be revised as that power shrinks, and the revision is already under way — in tariffs, in export controls, in the rival institutions that rising states have begun to build. On the institutionalist reading, rules that are costly to abandon and useful to everyone will outlast the share of those who wrote them, and what looks like decline is a change of form, as in 1971. The evidence of the last decade supports neither reading cleanly. The order has changed form without collapsing; it has also lost two of the things that made it an order rather than a set of habits.

For a multinational, the practical consequence is that the rules can no longer be treated as a constant in the business plan. A firm that built its supply chain on the assumption of non-discriminatory tariffs, procedural dispute settlement and neutral financial networks built it on rules that were held in place by a particular distribution of power. A falling G7 share does not tell the firm that its G7 customers are poorer, that emerging markets in general are richer, or that the rules it relies on have already changed. It tells it that the economies that wrote the rules now produce a much smaller share of the world's income than when they wrote them, and that the question of what holds the rules in place — which the next chapter answers from the realist side — has become a question about the firm's own operations.

Chapter 3

Anarchy and the Balance of Power: Who Sets the Terms When No One Governs

3.1 A system with no government

The previous chapter described the international economic order as a set of rules, written by particular states and held in place by something. This chapter asks what that something is when it is not law. Inside a country, a contract is enforced by courts, and the courts are backed by a state that holds a monopoly of legitimate force. Between countries there is no such state. There is no world government that can make a rule binding, enforce a contract between two governments, or guarantee that a state which disarms will not be invaded. That absence is the starting point of the realist tradition in international relations, and it is the reason this book cannot treat geopolitics as a branch of regulation.

The question that follows is direct. In a system with no government above states, who sets the terms of trade — and how would a multinational know that the balance is shifting? The realist answer to the first half is blunt: the terms are set, in the end, by those who hold the capabilities others need, and they are revised when the distribution of capabilities changes. The second half is a measurement problem, and the chapter answers it with two centuries of data on the distribution of material capabilities among the major powers, asking when, and on what measure, China overtook the United States.

The chapter defines anarchy; sets out the defensive and offensive realist theories; turns to power transition theory and Graham Allison's "Thucydides's Trap"; and examines how power is measured, and what the standard measure rewards and misses.

3.2 Anarchy, and what it does not mean

Anarchy — in international relations, the absence of any authority above states capable of making binding rules, enforcing agreements between them, or guaranteeing their survival. It describes the *structure* of the system, not the level of violence within it.

The word is a trap for newcomers, because in ordinary language it means chaos. In realist theory it means nothing of the kind. Most of the time, most states are at peace with most others; they trade, sign treaties, join organisations and settle disputes by negotiation. Anarchy is compatible with all of that. What it denies is that any of it is *guaranteed* by a higher authority. A treaty is kept because the parties find it in their interest to keep it, not because a court can compel them; a trading relationship continues because both sides value it, not because a sovereign above them will punish defection.

Three consequences follow, in the realist reading, and each matters for firms.

The first is **self-help**. Because no one else guarantees a state's survival, each state must ultimately provide for its own security. It may ally with others, but alliances are arrangements of convenience, and a state that relies entirely on an ally has placed its survival in someone else's hands.

The second is **uncertainty about intentions**. A state can observe another's capabilities — its army, its industry, its population — but it cannot observe its intentions with confidence, and intentions can change faster than capabilities. A peaceful neighbour with a large army is a neighbour whose government could change. Because the cost of misjudging a hostile state is so high, states tend to plan for capabilities rather than for stated intentions.

The third is concern for **relative gains**. Because today's partner may be tomorrow's adversary, states care not only about how much they gain from cooperation, but about how much more the other side gains. A trade agreement that makes both countries richer but makes the rival richer faster shifts the balance of capabilities, and in an anarchic system that shift can be a threat.

Relative gains — the concern of a state with its gains from cooperation *relative to* those of its partner, rather than only with its own absolute gains, arising from the possibility that the partner's gains will be converted into capabilities used against it (Grieco 1988).

Joseph Grieco's 1988 article made this the central point of difference between realists and liberal institutionalists such as Keohane (Chapter 2): institutions can solve the problem of cheating, Grieco argued, but they cannot solve the problem that cooperation may help a rival more than oneself. Powell (1991) sharpened the argument by showing that the weight a state places on relative gains is not fixed but depends on circumstances — in particular, on whether the use of force is a live possibility between the two states. Where force is unthinkable, as between Canada and the United States, relative gains matter little and cooperation is easy; where it is conceivable, relative gains matter a great deal.

That is why the concept belongs in a book on international business. Relative-gains thinking is the logic by which trade becomes a security question. A sale of advanced semiconductors to a rival, or of the equipment to make them, may be profitable for the seller and beneficial for the buyer, and still be prohibited, because it narrows a capability gap the seller's government wishes to preserve. Export controls on advanced chips are not a failure of economic reasoning; they are the application of a different calculus, in which the question is not "are we better off?" but "are we better off *relative to them*?". A firm that sells into the gap between two powers' capabilities should expect its business to be judged by the second question.

3.3 Two realisms

Realists agree that anarchy shapes behaviour. They disagree about how much power states want, and the disagreement has direct implications for what to expect from the relationship between the United States and China.

3.3.1 Defensive realism: enough is enough

Kenneth Waltz's *Theory of International Politics* (1979) is the founding text of structural, or neo-, realism. Its claim is that the behaviour of states is shaped less by their domestic characteristics — their regimes, their leaders, their ideologies — than by the structure of the system they inhabit: anarchy, plus the **distribution of capabilities** among the units (Waltz 1979). States are alike in their basic function, which is to survive, and they differ in how well equipped they are to do it.

Balance of power — the distribution of capabilities among the major states of a system, and the tendency, in realist theory, of states to respond to a concentration of power by building their own capabilities or forming coalitions against the stronger state ("balancing").

In Waltz's version, states seek an *appropriate* amount of power: enough to be secure. Seeking more is usually self-defeating, because a state that grows too strong provokes others to balance against it, and ends up less secure than before. The system therefore tends towards equilibrium. Waltz also argued that a **bipolar** system — two great powers, as in the Cold War — is more stable than a multipolar one, because with only two principal actors there are fewer relationships to manage, fewer opportunities for miscalculation, and no uncertainty about who the main threat is.

Waltz's theory sits on an older idea that is essential for understanding how conflict can arise without anyone wanting it.

Security dilemma — the situation in which measures a state takes to increase its own security, such as arming or forming alliances, reduce the security of others, who respond in kind, leaving all less secure than before even though none intended aggression (Herz 1950; Jervis 1978).

John Herz named the dilemma in 1950. Robert Jervis's 1978 article gave it its modern form by asking when it is most severe. His answer turned on two variables: whether offensive and defensive weapons and postures can be distinguished, and whether offence or defence has the advantage. Where defensive preparations look defensive and defence is strong, states can secure themselves without threatening others, and the dilemma is mild. Where they cannot be told apart and offence is favoured, every precaution looks like preparation for attack, and the dilemma is acute.

The security dilemma is the most useful single idea in this chapter for a risk analyst, because it describes escalation without intent. A government imposes an export control to protect a technological lead; the other side reads it as an attempt at containment and responds with restrictions on a critical mineral; the first reads that as coercion and tightens the control. Each step is defensive in the mind of the state that takes it. A firm caught in the sequence does not need either government to want a confrontation in order to lose a market or a supplier.

3.3.2 Offensive realism: as much as possible

John Mearsheimer's *The Tragedy of Great Power Politics* (2001) accepts Waltz's starting point but reaches a darker conclusion. It rests on five bedrock assumptions (Mearsheimer 2001):

1. The international system is **anarchic**.
2. Great powers inherently possess some **offensive military capability**.
3. States can never be certain about other states' **intentions**.
4. **Survival** is the primary goal of great powers.
5. Great powers are **rational actors**.

None of these, Mearsheimer notes, is individually alarming. Together they imply that great powers fear one another, cannot rely on anyone else for their security, and conclude that the best guarantee of survival is to be as powerful as possible relative to potential rivals. States are therefore not satisfied with an appropriate amount of power. They **maximise relative power**, and the ultimate goal is hegemony.

Global hegemony, however, is practically impossible, because of what Mearsheimer calls "the stopping power of water": large bodies of water make it extremely difficult to project land power across oceans. The realistic goal is therefore **regional hegemony** — domination of one's own region — together with preventing any other great power from dominating its region. The United States, on this account, is the only regional hegemon in modern history, having achieved dominance in the Western Hemisphere in the nineteenth century, and its foreign policy in the twentieth century can be read as a series of efforts to prevent the emergence of a hegemon in Europe or in North-East Asia.

The prediction for China follows mechanically. If China continues to grow economically, it will seek to dominate Asia as the United States dominates the Western Hemisphere, and the United States, which has spent a century preventing exactly that, will try to stop it. Mearsheimer's answer to the question of whether China can rise peacefully is no: the result will be intense security competition, with a considerable possibility of war.

	Defensive realism (Waltz)	Offensive realism (Mearsheimer)
How much power do states want?	enough for security	as much as possible, relative to others
What restrains them?	balancing by others; expansion is self-defeating	only the limits of their capabilities
Expected US–China outcome	managed competition	intense security competition

A caution belongs here. Realism predicts **competition**, not war. In Waltz's version it often predicts restraint, since overexpansion provokes balancing. Even Mearsheimer's version predicts security competition, of which war is one possible outcome among several. A risk analyst who reads realism as a forecast of war has misread it; one who reads it as a forecast that the economic relationship will be subordinated to the strategic one has read it correctly.

3.4 Power transition and the Thucydides's Trap

Balance-of-power theory treats an equal distribution of capabilities as the condition of stability. A different tradition reverses the claim.

Power transition theory — the theory, due to A. F. K. Organski, that peace is most secure when a dominant power holds a clear preponderance and that war is most likely when a dissatisfied challenger approaches parity with it (Organski 1958; Organski and Kugler 1980).

Organski's *World Politics* (1958) described the international system as a hierarchy rather than a balance. At the top sits a dominant power that built the existing international order and benefits from it; below are great powers, some satisfied with the order and some not. Industrialisation drives the rise of new powers, and their growth is uneven. Danger arises when a challenger that is *dissatisfied* with the order — that did not write its rules and does not benefit from them as much as it believes it should — draws close to the dominant power in capabilities. At that point the challenger has both the motive and, increasingly, the means to revise the order, and the dominant power has reason to act before its lead disappears. In empirical work, parity is often operationalised as the challenger reaching around 80 per cent of the dominant power's capabilities. On the standard capability index discussed below, China crossed that threshold relative to the United States in 1975, two decades before it passed it — which says as much about the index as about the transition.

Organski and Kugler (1980) took the theory to data in *The War Ledger*, and a substantial literature followed, including the essays collected in Kugler and Lemke (1996). The tradition connects to Robert Gilpin's account of hegemonic war, in which the order established by a dominant power becomes misaligned with the underlying distribution of power as others grow, until a war redistributes both (Gilpin 1981). It connects too to Paul Kennedy's history of the great powers, which argued that dominant powers tend to overextend their commitments relative to their economic base (Kennedy 1987).

Power transition theory has an obvious affinity with Chapter 2. There, the question was whether the rules of the economic order would survive the shrinking share of their authors. Here, the theory supplies a mechanism: a rising power that did not write the rules is likely to be dissatisfied with them, and as its capabilities approach those of the rule-writer, its ability to revise them grows.

3.4.1 Allison's sixteen cases

Graham Allison's *Destined for War* (2017) brought the argument to a mass audience under a classical name. Thucydides, the historian of the war between Athens and Sparta in the fifth century BCE, wrote, in the translation Allison uses, that "it was the rise of Athens and the fear that this instilled in Sparta that made war inevitable" (Allison 2017).

From the archive · Thucydides, fifth century BCE

The Peloponnesian War began in 431 BCE and ended, after a long truce and its collapse, with the defeat of Athens in 404. Its historian, Thucydides, was an Athenian general who had been exiled after failing to prevent the loss of a city to Sparta's forces, and who spent his exile collecting the accounts of both sides.

The sentence Allison made famous comes early in the history, where Thucydides distinguishes the immediate grievances that both sides publicly cited — disputes over Corcyra and Potidaea, over alliances and trade — from what he calls the truest cause, the one least spoken of: the growth of Athenian power and the alarm it caused in Sparta. That distinction between the stated occasion and the structural cause is the analytical move every realist since has borrowed.

It is worth reading what Thucydides does *not* say. He does not say that rising powers always fight ruling ones; he says this war had this cause. He was writing about a particular conflict between two particular cities, one a maritime and commercial empire and the other a land power with a very different political system. The generalisation to a law of history is Allison's, and the critics' first objection is to the generalisation, not to Thucydides.

Allison and the Thucydides's Trap project at Harvard identify **sixteen** cases in the past five hundred years in which a rising power challenged a ruling one. **Twelve** ended in war. The four that did not — among them the United States overtaking Britain in the early twentieth century, and the Cold War rivalry between the United States and the Soviet Union — are presented as evidence that war is not inevitable, but that avoiding it requires deliberate and painful adjustment by both sides.

From the archive · Britain and the United States, 1895–1903

At the end of the nineteenth century the United States overtook Britain as the world's largest economy, and its navy was growing. The two powers had serious quarrels. In 1895 a boundary dispute between Venezuela and British Guiana brought a sharply worded American demand that Britain accept arbitration, and talk of war on both sides of the Atlantic. Britain accepted arbitration, and a tribunal issued its award in 1899. In 1901 the Hay–Pauncefote Treaty released the United States from an earlier agreement to share control of any canal across Central America, clearing the way for an American-controlled canal at Panama. In 1903 a tribunal settled the boundary between Alaska and Canada largely in the American favour, with the British member siding with the Americans against the Canadian members' objections.

In each dispute, the ruling power gave way. Britain, facing a rising Germany closer to home and overstretched across its empire, chose to accommodate the rising power in the Western Hemisphere rather than contest it. The case is the strongest evidence that a transition can be peaceful. It is also a reminder of the price: the accommodation was made, in part, at the expense of a third party — Canada — whose interests the ruling power judged less important than the relationship with the rising one.

For a firm, the analogy is uncomfortable and useful. A peaceful transition between great powers is not costless for everyone else. The terms on which it is reached may be paid by actors who were not at the table.

3.4.2 The critics

The Thucydides's Trap has been criticised on four grounds.

Case selection. The cases were assembled with knowledge of the outcome. What counts as a "rising" power, a "ruling" power, and a "challenge" is elastic enough that peaceful transitions — which leave less of a trace in history than wars — may be under-counted, inflating the proportion that ended in war. A ratio of twelve to sixteen is only as informative as the rule that produced the sixteen.

Nuclear weapons. No pre-1945 case involved two powers each capable of destroying the other. Nuclear deterrence changes the expected cost of great-power war so drastically that it is not clear any pre-nuclear case is informative about the present. The Cold War, the one transition-like rivalry between nuclear powers in the dataset, did not end in war between the principals.

Interdependence. The United States and China are bound by trade, investment and finance far more tightly than Athens and Sparta were. Liberal theory holds that interdependence raises the cost of conflict and so makes it less likely. The realist reply is that Britain and Germany were also major trading partners in 1914, and fought anyway; Copeland (2015) argues that what matters is not the level of trade but leaders' expectations of its future, so that interdependence restrains only as long as it is expected to last. Chapter 4 takes up the debate with data.

Measuring "rise". The trap depends on knowing when a challenger is approaching parity, and that depends on how power is measured. Beckley (2018) argues that the conventional gross indicators — GDP, population, composite indices of capability — systematically overstate the power of large, populous countries, because they ignore the costs a state bears in governing, feeding, policing and protecting its population. A country with a large GDP and a large population may have much less surplus available for projecting power than the gross figures imply. Beckley proposes measures that account for such costs, and on them the gap between the United States and China looks larger and the transition further off.

The last criticism can be examined directly, because it turns on the choice of measure.

3.5 Measuring power

Every theory in this chapter turns on a quantity — the distribution of capabilities, parity, relative power — that none of them can observe directly. Power is a latent concept. What can be observed are its supposed ingredients: people, money, industry, arms. The standard way of combining them in quantitative international relations was devised in the 1960s at the University of Michigan by J. David Singer and his collaborators on the Correlates of War project.

Composite Index of National Capability (CINC) — a yearly index of a state's material capabilities, equal to the unweighted average of its shares of the state-system total on six indicators: total population, urban population, iron and steel production, primary energy consumption, military personnel and military expenditure (Singer, Bremer, and Stuckey 1972).

The construction has three steps. For every state in the international system in a given year, take six indicators, grouped into three dimensions: demographic (total population, urban population), industrial (iron and steel production, primary energy consumption) and military (military personnel, military expenditure). For each indicator, compute the state's share of the system total that year. Average the six shares. The result is a number between 0 and 1; summed over all states, it comes to about 1 — not exactly, because a state with a missing component is averaged over the components it has. In the current release, the National Material Capabilities dataset version 7.0, the coverage runs from 1816 to 2022, one row per state-year.

The design has real virtues. It is transparent. It is long: two centuries of comparable data, which is what allows Organski's and Mearsheimer's arguments to be confronted with evidence at all. And by using shares of the system total, it sidesteps the problem of comparing military spending in nineteenth-century pounds with twenty-first-century dollars — each year is compared only with itself. Mearsheimer's own measures of latent power in the *Tragedy* rely on similar industrial ingredients for the nineteenth and early twentieth centuries.

3.5.1 What it rewards

The design also embodies a theory of power, and it is a theory of the 1960s. Two of the six components are population and two are heavy industry. CINC therefore rewards **size and smokestacks**. A large, populous, industrialising country — one that produces a great deal of steel and burns a great deal of energy — will score highly even if its technology is behind, its productivity low and its military poorly equipped.

The consequence is visible in the data. In version 7.0 of the National Material Capabilities dataset, China's CINC first exceeds that of the United States in 1995; the order reverses for a single year in 2002, and China has led in every year since 2003. Other measures tell a different story. At purchasing-power parity, China's GDP passed that of the United States in 2014, according to the World Bank's data; at market exchange rates it never has. On military expenditure the gap is wider still: about a quarter of the American figure in 2022 in the Correlates of War series, about a third in 2025 in the Stockholm International Peace Research Institute's (SIPRI) database, whose Chinese figures are its own estimates. One underlying reality, four answers: 1995, 2014, not yet, and not close.

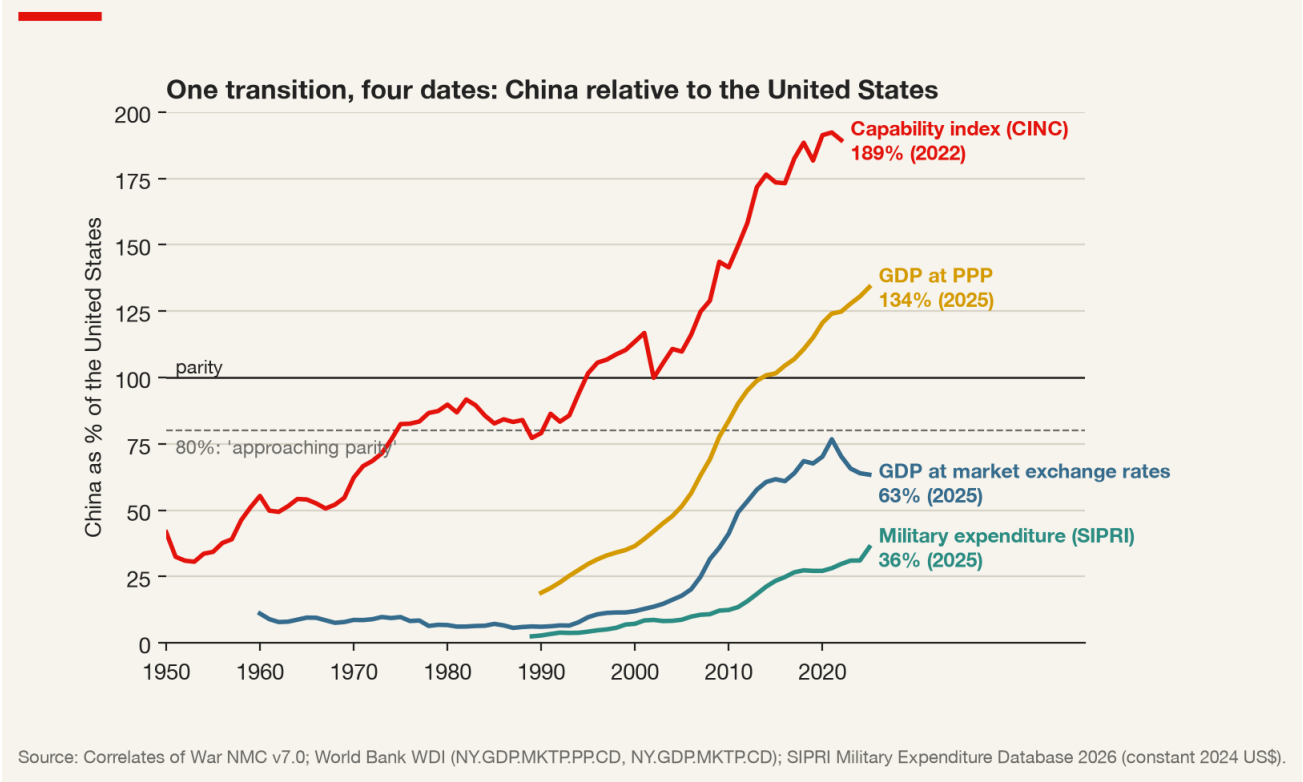


Figure 3.1: China relative to the United States: past parity on the capability index since 1995 and at purchasing-power parity since 2014, but at 63 per cent of US GDP at market exchange rates and 36 per cent of US military spending in 2025. Source: COW NMC v7.0; World Bank WDI; SIPRI (constant 2024 US dollars).

Figure 3.1 puts the four series on one scale, with lines at parity and at the 80 per cent threshold, and Table 3.2 dates the crossings. The measures disagree not only about the level but about the *shape* of the transition. On the capability index China was already at 79 per cent of the United States in 1990, before its take-off showed in any output statistic, and entered the danger zone in 1975, in the last years of the Cultural Revolution. At purchasing-power parity it stood at 19 per cent in 1990 and reached parity in 2014. At market exchange rates it rose from 6 per cent to 77 and fell back as the renminbi

weakened and American nominal growth outpaced China's. On military money it went from under 3 per cent to just over a third. Four histories: a transition long completed, one recently completed, one stalled, one not begun.

Table 3.2: When did China approach parity with the United States? Source: Correlates of War, National Material Capabilities v7.0; World Bank, World Development Indicators (NY.GDP.MKTP.PP.CD, NY.GDP.MKTP.CD); SIPRI Military Expenditure Database, 2026 release.

Measure	Source	China as % of US, first year ≥ 80%	First year ≥ 100%	Latest value
Composite Index of National Capability	COW NMC v7.0	1975	1995	189% (2022)
GDP at purchasing-power parity	World Bank WDI	2010	2014	134% (2025)
GDP at market exchange rates	World Bank WDI	not reached (peak 77%, 2021)	not reached	63% (2025)
Military expenditure, constant 2024 US$	SIPRI	not reached	not reached	36% (2025)

The measures sort themselves by what they weigh. The two that put China ahead count physical volumes — tonnes of steel, the domestic price of a haircut; the two that put the United States ahead count things priced in world markets. Purchasing-power parity is the better guide to what a government can buy at home; market exchange rates to what it can buy abroad, and to the market a foreign firm earns revenue in. A multinational that reports its Chinese sales in dollars lives in the market-exchange-rate world, whatever the capability index says.

The size bias is visible even earlier in the series. In the 1860s, 1870s and most of the 1880s — decades in which China was losing wars to European powers and ceding territory and treaty ports — its CINC exceeded that of the United States, because its population was vastly larger and American industry had not yet grown to its later scale. No historian of the period would call Qing China the more powerful of the two. The index does, for a quarter of a century, and for the same reason that it dates the modern transition early.

Because the index is a simple average, the size bias can be taken apart component by component (Figure 3.2). In 2022 China produced 54 per cent of the system's recorded iron and steel and consumed 30 per cent of its primary energy, against 4 and 14 per cent for the United States; it held 18 per cent of world and of urban population, against about 4 and 3. Only on military expenditure does the United States lead — 42 per cent against 11.

Steel alone contributes nine of China's 23 points. Without it, China still leads on the other five components, by 17 points to 14; on the two military components alone the order reverses, 10 against 24. In 1995, the crossover year, the two countries each produced about 13 per cent of the system's steel, and nearly all of China's subsequent gain came from the steel and energy columns — the construction boom of the 2000s and 2010s and the coal-fired power that drove it. That capacity could in principle be turned to shipbuilding and munitions. But a measure of national power whose largest driver is a sector Chinese policy has spent a decade trying to shrink needs careful reading.

3.5.2 What it misses

CINC is equally instructive for what it cannot see. It has no component for **technology** — the quality of weapons, the sophistication of industry, the capacity to innovate. It has no component for **alliances**, although a state with many capable allies commands far more power than its own figures suggest. It has nothing on **financial power**: the role of a state's currency in international payments, the depth of its capital markets, its control of the institutions through which money moves. And it has nothing on **control of networks** — the payment systems, cloud platforms and communications infrastructure that Farrell and Newman (2019) identify as instruments of coercion (Chapter 2). These are precisely the forms of power that reach firms most directly, and Chapters 8, 10 and 11 return to them.

The choice of measure is not a technicality, as Beckley's critique shows: on gross resources China caught up long ago; on net resources, quality-weighted military capability or control of networks, the picture changes. Each measure is defensible. None is neutral.

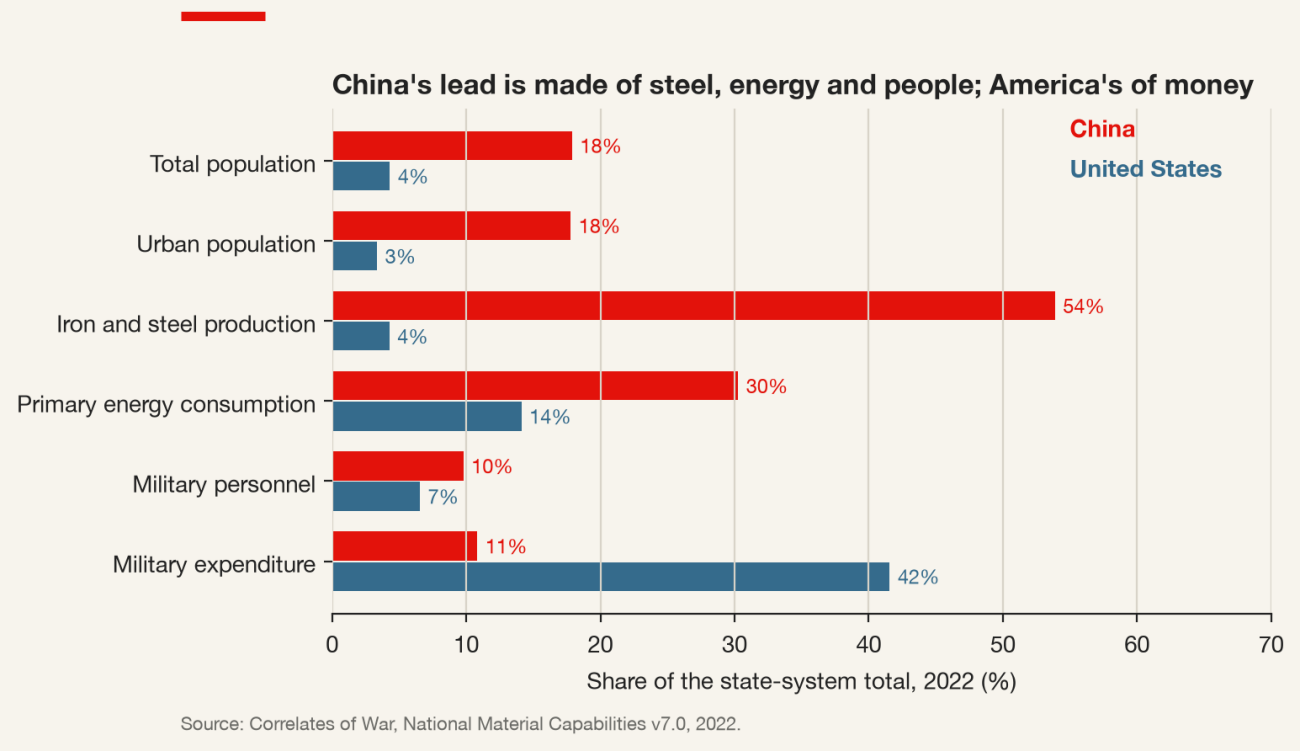


Figure 3.2: China's capability lead in 2022 rests on iron and steel (54 per cent of the system total against 4 per cent for the United States), energy and population; the United States leads only on military expenditure (42 per cent against 11). Source: Correlates of War, National Material Capabilities v7.0.

3.6 Reading two centuries in one chart

The long-run chart of capability shares is a compressed history of the modern state system, and reading it well requires knowing where its seams are, so that the anomalies are recognisable as anomalies.

The first seam is that the chart is not a chart of a fixed set of countries. The Correlates of War data describe the members of the state system in each year — some 17,000 state-years in the current release — and membership changes. China and Japan enter the data in 1860. Germany, coded `GMY`, disappears in 1945 and returns in 1991; in between, the Federal Republic and the Democratic Republic appear as separate states, `GFR` and `GDR`. Japan is absent from 1946 to 1951, under occupation. The Soviet Union, by contrast, is coded as `RUS` throughout, so that Russia, the USSR and Russia again form one continuous series. Each produces a break or a continuity that has nothing to do with power and everything to do with coding. The codes themselves look like ISO codes and are not — the United Kingdom is `UKG`, France `FRN` — so that any combination with the World Bank data of Chapter 2 has to be made through a concordance rather than by matching letters.

The second seam is arithmetic. Missing values of the six components are coded as `-9`, not left blank, and the index is averaged over whatever components a state has. Summed over all states, CINC should equal one; in practice the yearly total runs from 1.000 to about 1.075, highest in 1860 and generally above one in the nineteenth century, when the components are most often missing.

The third seam is the joining of sources over time. Between 2016 and 2017 the urban population recorded for the United States falls from about 198 million to about 110 million, while its total population continues to rise; the American share of the world's urban population, as the dataset counts it, drops from about 8 per cent to about 3 per cent in a single year. No migration explains that. It is a change in source or definition, and because urban population is one of six components, it removes nearly a tenth of the United States' CINC score — from 0.133 to 0.122 — overnight. China's urban population jumps in the same year in the opposite direction. A reader who saw the gap between the two powers widen in 2017 would be reading a revision, not a shift in power.

The fourth thing is the substantive story, and read with these cautions it is clear (Figure 3.3). Britain holds the largest share from 1816 until the 1890s — a third of the system's capabilities at the start of the series, and still more than a quarter in 1860; the United States passes it briefly in 1892 and for good in 1897, in the years when, as the archive above recounts, Britain chose to accommodate it. The two World Wars violently rearrange the distribution: the American share spikes to 38 per cent in 1919, as wartime spending is counted against a devastated Europe, and peaks in 1945 at 38.4 per cent of the system's capabilities, the highest any state reaches in two centuries. The Cold War appears as Waltz's bipolarity made visible — two powers far above the rest — and on this index the Soviet Union actually leads from 1971 to 1988, a reminder of how much weight the measure gives to steel, energy and soldiers and how little to technology. From the 1990s, China's share rises past both, to about 23 per cent in 2022 against about 12 per cent for the United States.

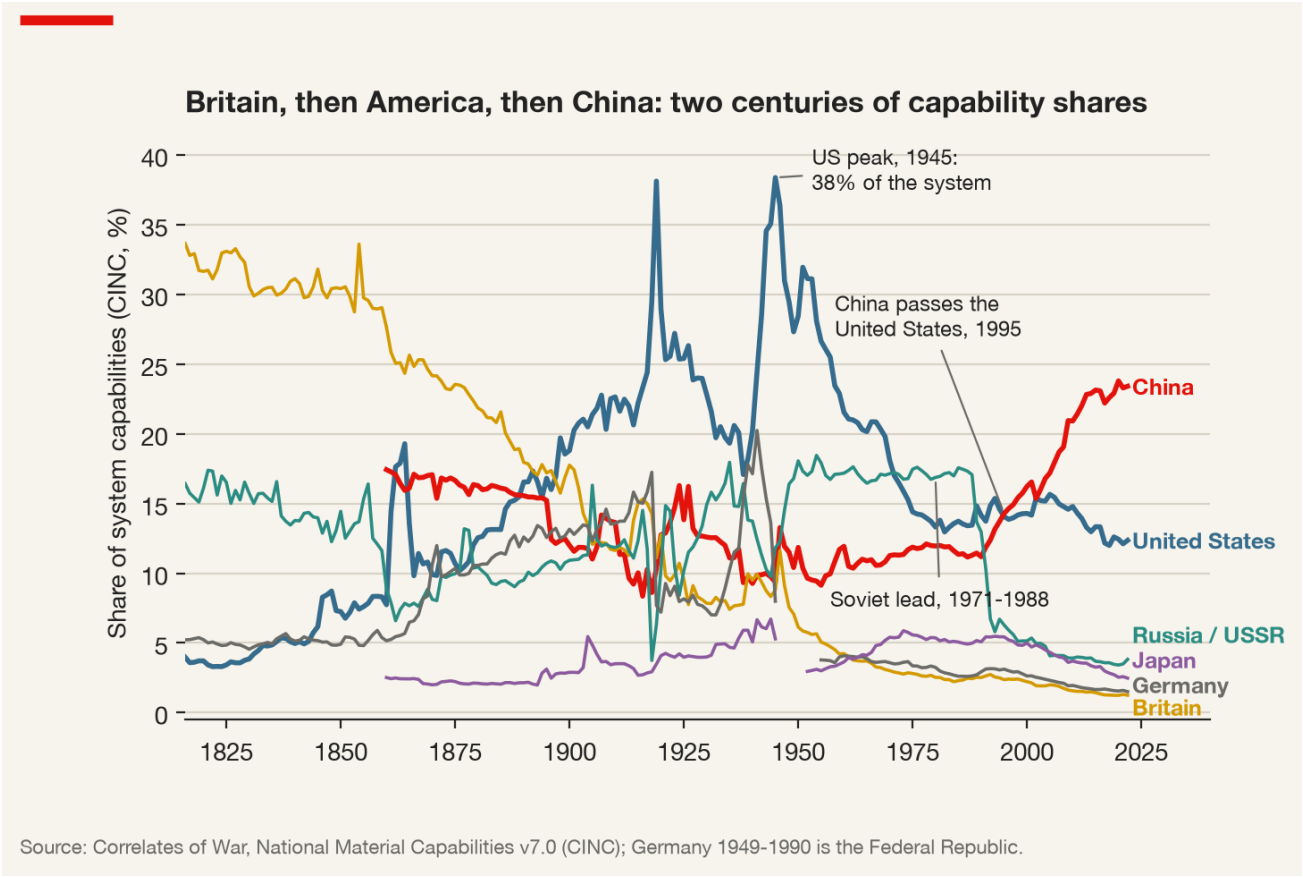


Figure 3.3: Britain leads until the 1890s, the United States from 1897 (peaking at 38 per cent in 1945), the Soviet Union on this index from 1971 to 1988, and China since 1995. Source: Correlates of War, National Material Capabilities v7.0.

Two features of the chart deserve more attention. The first is the shape of American dominance: the 1945 peak reflects the destruction of every other industrial power, and it decays fast — 28 per cent in 1950, 14 per cent by 1990. A present-day share of 12 per cent is not a collapse; what changed after 1990 was which line sat above the American one. The second is that for the period in which most Western managers formed their picture of the world, the index shows a Soviet Union at or above the United States — wrong for the reason the Chinese figures need caution today: the index counts inputs, not how efficiently they are used. The Soviet line falls from about 18 per cent in 1985 to under 7 in 1992, partly because post-1991 Russia is a smaller state than the Soviet Union it continues in the data.

3.6.1 Bipolar, multipolar, or something else?

The chart also bears on a question that has occupied realists since the Soviet Union dissolved in 1991: what is the *polarity* of the system now? The 1990s were widely described as a unipolar moment, with the United States unmatched. Waltz expected that unipolarity would not last, because other states would balance against a power with no counterweight. Whether the

present system is returning to bipolarity — the United States and China — or moving to a multipolar one with several significant powers is contested, and the answer depends, once again, on the measure.

Polarity has its own measurement tradition: Ray and Singer (1973) proposed an index of the *concentration* of capabilities among the major powers. A simpler reading takes the shares of the three largest states each year. A system is bipolar when the gap between the second and the third is wide, multipolar when it is narrow.

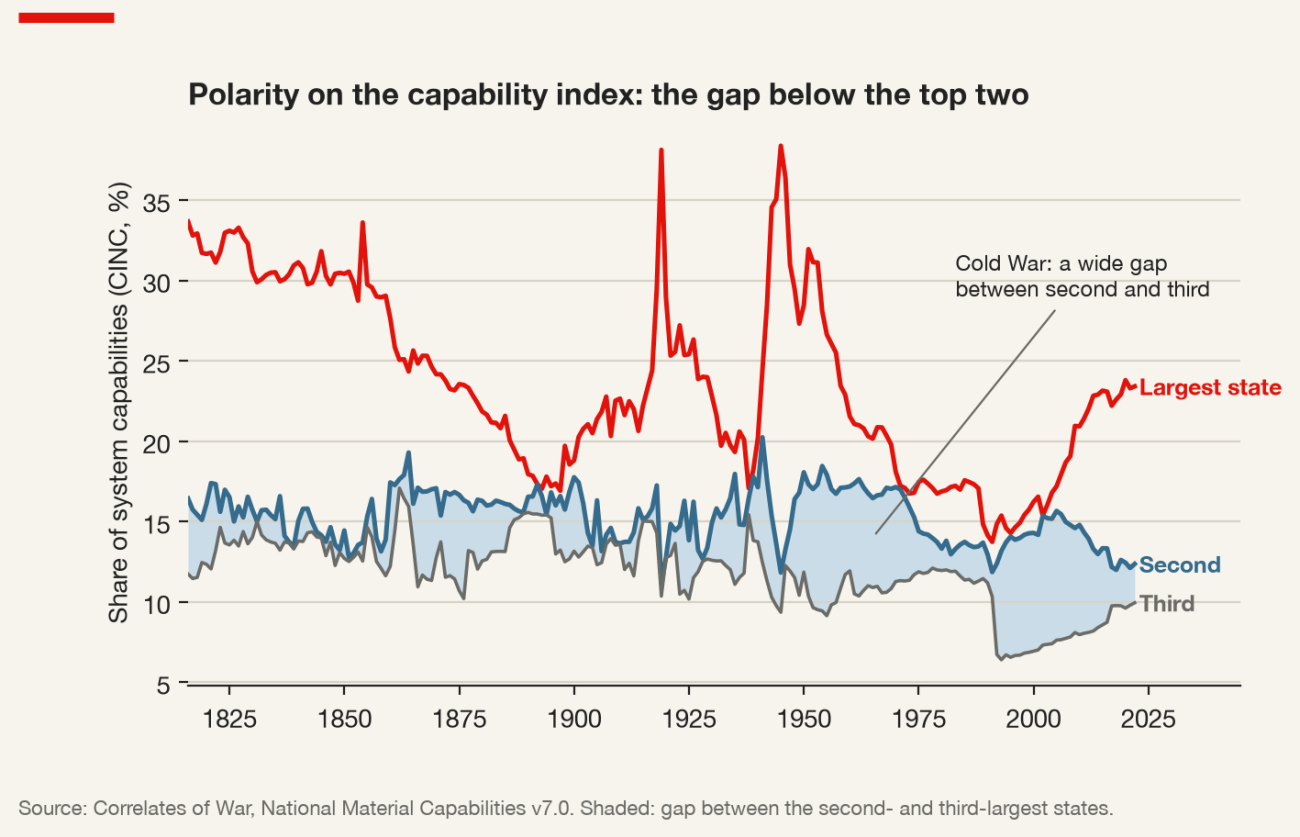


Figure 3.4: On the capability index, the gap between the second- and third-largest states, wide in the Cold War and after 1991, had narrowed to about two and a half points by 2022. Source: Correlates of War, National Material Capabilities v7.0.

In 1950 the largest state held 28 per cent of the system's capabilities, the second 18 and the third 12 (Figure 3.4). The gap was wider still in the mid-1990s, but the index names the top two in 1995 as the United States and China, at 14.3 and 14.1 per cent. On CINC there was no "unipolar moment" — the opposite of the reading of Wohlforth (1999), who argued that the post-Cold War system was unipolar and, because unipolar, durable, on measures that weight quality: defence spending, research and development, output at market prices.

By 2022 the band has narrowed to about two and a half points: China 23 per cent, the United States 12, India nearly 10. India's share reached 80 per cent of the American one in 2017 — the year of the American urban-population revision — and stood there in 2022. A mechanical reader would say the United States is "approaching parity" from below with India, whose military spending in 2025 was about a tenth of the American figure. Yet India's population, steel and energy are large and growing; if latent power is what they measure, the data point to a three-cornered distribution.

The debate over the present configuration is a debate about weights. Brooks and Wohlforth (2016) argued that technology-intensive military power makes it harder for a rising state to convert economic size into superpower capabilities, and treated China as, at most, an emerging potential superpower. Others see a return to bipolarity, or a world of several centres. Each answer is a choice of yardstick.

Nor is balancing always military. Pape (2005) argued that states facing an American lead too large to balance by arms would turn to *soft balancing* — denying territory, entangling diplomacy in institutions, economic strengthening — to raise

the costs of American action. Its instruments are commercial, which makes it the form that reaches firms: the refused port call becomes a refused licence, and the coalition becomes a payment system built to work around the dominant power's.

The question matters to firms because the two configurations imply different environments. A bipolar system tends to produce two blocs with clear membership, so that the firm's problem is being made to choose a side. A multipolar system produces more hedging states — governments that trade with both sides and bargain between them — and therefore more room for firms to locate operations in third countries open to both. The geography of "connector" economies, which Chapter 4 examines with trade data, is a direct consequence of which of the two worlds firms are in.

3.6.2 The military money

If the capability index flatters size, military expenditure flatters quality, and it is the measure on which the unipolar reading rests. Figure 3.5 divides each country's spending in the SIPRI database, in constant 2024 dollars, by the world total, which SIPRI reports from 1988.

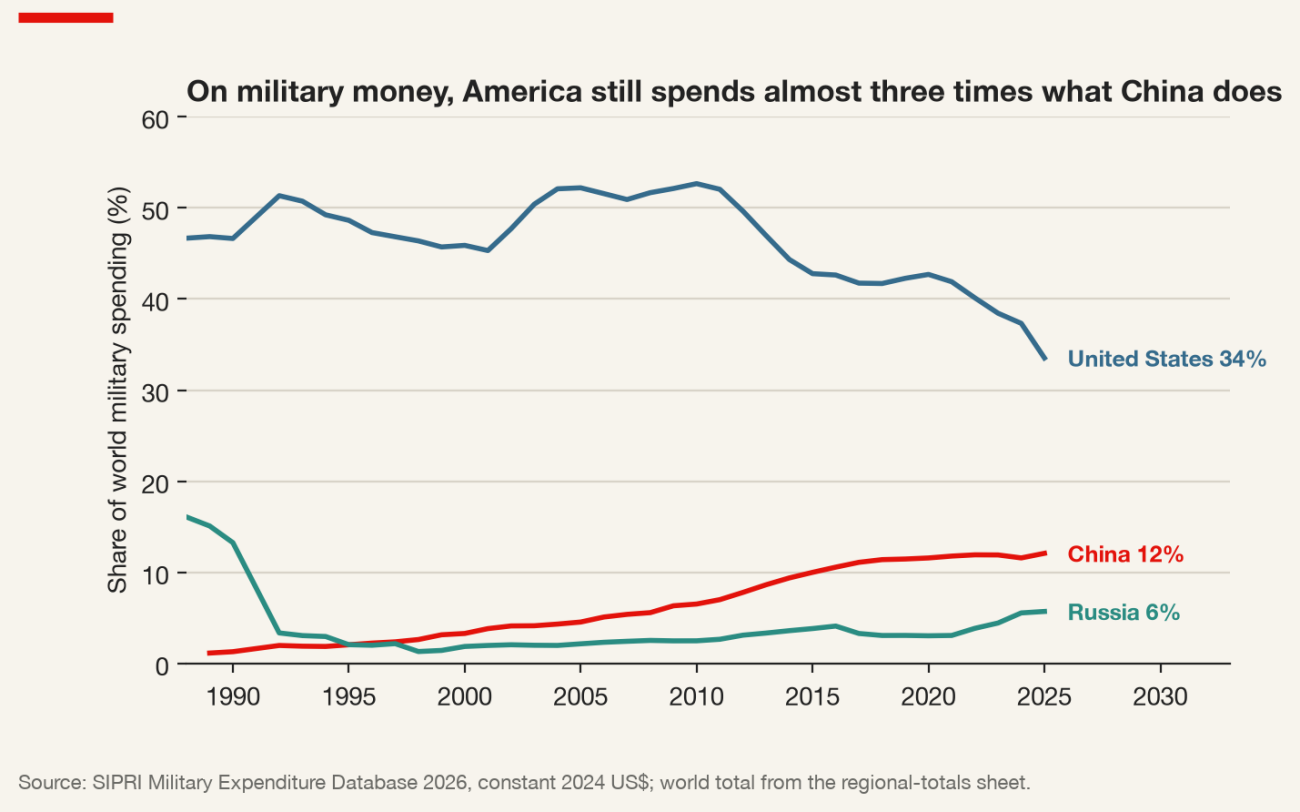


Figure 3.5: The American share of world military spending fell from 53 per cent in 2010 to 34 per cent in 2025; China's rose to 12 per cent and Russia's recovered to 6. Source: SIPRI Military Expenditure Database, 2026 release, constant 2024 US dollars.

The United States spent 47 per cent of the world's military budget in 1988 and 53 per cent at the peak in 2010, during the wars in Iraq and Afghanistan. Its share fell to 34 per cent in 2025, as much through others' growth as through restraint: world spending rose from about 1,961 billion dollars in 2015 to 2,771 billion in 2025, in constant 2024 prices, while American spending fell in 2025 from 1,005 to 929 billion. China's rise has been steady rather than spectacular — from 196 to 335 billion dollars over the decade, 12 per cent of the world total. The United States still spent 2.8 times as much.

Table 3.3: The ten largest military spenders in 2025, which together account for 72 per cent of the world total. Figures for China, Russia and Saudi Arabia are, in whole or part, SIPRI estimates. Source: SIPRI Military Expenditure Database, 2026 release (constant 2024 US dollars).

Rank	Country	Spending, 2025 (US$ bn, constant 2024)	Share of world (%)	Change since 2015 (%)
1	United States	929.2	33.5	+11
2	China	335.0	12.1	+71
3	Russia	158.2	5.7	+111
4	Germany	106.7	3.9	+128
5	India	93.3	3.4	+53
6	United Kingdom	88.0	3.2	+32
7	Saudi Arabia	81.5	2.9	−20
8	Ukraine	77.6	2.8	+1,507
9	France	64.5	2.3	+21
10	Japan	59.5	2.1	+60

Table 3.3 is a portrait of a system rearming. Russia's spending more than doubled in a decade and Ukraine's rose sixteen-fold; Germany's more than doubled; Japan's rose by 60 per cent. Each increase is defensive in the mind of the state that makes it, and read by others as a reason for their own — the security dilemma in a table.

Spending has its own measurement problems. Market exchange rates understate what a lower-cost economy buys, since Chinese soldiers and shipyard workers are paid less than American ones; purchasing-power parity raises China's figure but overstates what it can buy of weapons priced in world markets. China's official budget excludes items SIPRI adds back, which is why SIPRI's figure is higher and marked as an estimate. And much of the American budget goes on personnel, health care and global bases that a power concentrating on its own region does not need.

3.7 How a firm would know the balance is shifting

The second half of the chapter's question — how would a multinational know that the balance is shifting? — has two answers, a structural one and a behavioural one.

The structural answer is the one the data above supply: track the distribution of capabilities on several measures, and watch the ratios. A challenger's share approaching the dominant power's on one measure, while still far behind on others, is a transition in progress and a transition contested; it is the situation in which the security dilemma is most acute, because each side can read the evidence as it wishes. Structural indicators of this kind, in the typology of Chapter 1, explain *why* tension is building. They do not say *when* it will matter.

The behavioural answer is to watch what states do with the rules. A shifting balance shows up first not in war but in the instruments of economic statecraft: export controls, investment screening, sanctions and counter-sanctions, restrictions on data, rules about where a firm may source a component. These are the measures by which states convert relative-gains concerns into policy, and they are observable — in official gazettes, in company disclosures, in the event datasets of later chapters — long before any military confrontation. For a firm, the most reliable signal that the balance is shifting is a change in which rules it is being asked to follow.

3.8 What it means for the multinational

The realist frame leads to three implications for a firm, each of which recurs later in the book.

The balance of power sets the rules of trade. Who can impose sanctions, export controls and market-access conditions depends on who holds capabilities others need: the largest market, the dominant currency, the critical technology. A firm's exposure to a state's economic statecraft is a function of how much of its business depends on capabilities that state controls.

A transition changes which rules a firm follows. A firm operating in both the United States and China increasingly faces two compliance regimes — and, since China adopted its own instruments of counter-sanction in 2020 and 2021, the possibility of penalties in one jurisdiction for complying with the other's rules. The question "is this transaction legal?" has acquired a second clause: "legal where, and under whose law?".

Relative gains turn commerce into security. Products that would make a rival stronger — advanced semiconductors, the tools that make them, artificial intelligence, batteries, aerospace — move from the trade ministry's jurisdiction to the security establishment's. For firms in these sectors, market access is no longer negotiated mainly on economic grounds, and commercial arguments about mutual benefit carry less weight than they did.

Mapped onto the five channels of this book, the balance of power runs through every one, but most directly through **market access** (which markets will remain open to firms from which bloc) and **assets and people** (where conflict, if it comes, will fall). It reaches **inputs** through export controls on critical goods, **finance and payments** through the dominant currency, and **data and operations** through the control of networks.

3.9 A short review of the literature

Realism's modern structural form begins with Waltz (1979), who explained state behaviour by anarchy and the distribution of capabilities and judged bipolarity the more stable configuration. The security dilemma that underlies much of it was named by Herz (1950) and given its analytical form by Jervis (1978), who related its severity to the offence–defence balance. Grieco (1988) made relative gains the realist objection to liberal institutionalism, and Powell (1991) showed that the weight states put on relative gains depends on whether force is a live option between them. Mearsheimer (2001) pushed the structural argument to its offensive conclusion — power maximisation and regional hegemony — and applied it to China.

A parallel tradition locates danger in hierarchy rather than balance. Organski (1958) introduced power transition theory; Organski and Kugler (1980) and the contributions in Kugler and Lemke (1996) tested it on data. Gilpin (1981) gave a systemic account of hegemonic war, and Kennedy (1987) a historical account of how great powers rise and decline relative to their economic base. Allison (2017) reframed the transition debate for the US–China relationship through sixteen historical cases.

Measurement runs through all of it. Singer, Bremer, and Stuckey (1972) introduced the Composite Index of National Capability and the Correlates of War data on which much of the quantitative literature rests. Beckley (2018) argued that gross indicators overstate the power of populous states and proposed net measures, which change the dating of the US–China transition. Farrell and Newman (2019) identified a form of power — control over network hubs — that no capability index was designed to capture.

The through-line is that every theory of the balance of power is also a claim about how to measure it, and that the measurement is where theories can be checked against the world — and where they most often disagree.

3.10 Which kind of power the firm depends on

Realism begins from an absence. There is no government above states, and so the rules on which firms rely — the open market, the convertible currency, the procedure for settling a dispute — are kept only as long as the states that hold the relevant capabilities find it in their interest to keep them. That is not a prediction of war. Defensive realists expect restraint; offensive realists expect security competition; power transition theorists expect danger only where a dissatisfied challenger nears parity. What all three share is the claim that the economic relationship between great powers is, in the end, subordinate to the strategic one. Export controls, investment screens and counter-sanctions are what that subordination looks like before anyone fires a shot.

The evidence of this chapter adds a second claim, about measurement. Every theory of the balance of power turns on a quantity — capabilities, parity, relative power — that cannot be observed directly, and the answer to the question "has the transition happened?" depends on how the quantity is built. On an index that rewards population and heavy industry, China overtook the United States in 1995, and Qing China was ahead of it in the 1870s. On output at purchasing-power parity it did so in 2014; at market exchange rates it has not; on military spending it is far behind. Each number is correctly computed;

none is *the* balance of power. A board slide that says "China has overtaken the United States" without saying on what has said nothing a decision can use.

For a multinational, the useful question is therefore not which measure is right in general, but which kind of power its own exposures depend on. A firm whose risk lies in market access needs to know who holds the largest market and the willingness to close it. A firm whose risk lies in inputs needs to know who controls the technology, the mineral or the equipment that cannot be replaced quickly. A firm whose risk lies in finance needs to know whose currency clears its payments, and a firm whose risk lies in data needs to know whose networks its operations run through. On some of these dimensions the transition is far advanced; on others it has barely begun; on the forms of power that reach firms most directly — currency, networks, critical technology — the standard capability index does not look at all.

A crossover on a capability chart, then, does not tell a multinational that war is coming — power transition theory makes war likelier near parity only if the challenger is dissatisfied, and dissatisfaction is not on the chart. Nor does it say that the firm's Chinese revenue is at risk or its American suppliers safe. It says that, on one measure built in the 1960s around steel, energy and soldiers, and ending in 2022, the distribution of material capabilities has shifted. Whether that shift reaches the firm depends on whether the measure can see the kind of power the firm actually depends on — and, as the chapters on trade, sanctions, cyber operations and space will show, the kinds that matter most are often the ones it cannot.

Chapter 4

Trade, Conflict and Geoeconomic Fragmentation: From Commercial Peace to Rival Blocs

4.1 An old promise, and an old embarrassment

The idea that commerce keeps the peace is older than the discipline that now tests it. Montesquieu put it in a sentence in 1748 — peace is the natural effect of trade — and the sentence did a great deal of work for the next century and a half. Richard Cobden built a political career on it: the repeal of the Corn Laws in 1846 and the Anglo-French commercial treaty of 1860 that carries his name were sold, in part, as instruments of peace between nations that had spent much of the previous century fighting each other. The argument was not that merchants are nicer than generals. It was that exchange creates gains, that war destroys them, and that a state which stands to lose a great deal from fighting will think harder before it fights.

The argument's most famous restatement came at the worst possible moment.

From the archive · Angell, 1910

In 1910 Norman Angell, a British journalist, published *The Great Illusion* (Angell 1910). Its claim is usually remembered as "war has become impossible", which is not what it said. Angell's claim was narrower and more interesting: that in an economy of credit, contracts and cross-border investment, conquest no longer pays. A victor cannot seize the wealth of an industrial country, because that wealth consists of claims, trading relationships and productive organisation that collapse when they are seized. The economic case for war, he argued, rested on an illusion — the belief that national wealth is a stock of things that can be carried off.

The book sold in enormous numbers and was translated widely. Four years later Britain and Germany — two of each other's most important trading partners, bound by finance, shipping and insurance as well as by goods — were at war. The war went on for more than four years and destroyed exactly the wealth Angell said it would destroy.

The episode is usually told as a refutation. It is better read as a lesson about which claim was refuted. Angell was largely right that the war would be ruinous for the victors as well as the vanquished. What failed was the inference from *war does not pay* to *states will therefore not make war*. Leaders in 1914 were not computing the net present value of conflict, or were computing it with expectations about the war's length and cost that turned out to be badly wrong. Angell received the Nobel Peace Prize in 1933, as the next European war was being prepared. The commercial peace has lived with the embarrassment of 1914 ever since, and every serious version of the argument is a version that tries to explain it.

This chapter takes the argument seriously, sets out the realist reply, asks what a century of evidence has settled, and then turns to the question that has replaced it. The old question was whether trade pacifies. The question that matters to a multinational in the 2020s is different: if states have begun to trade more with friends than with rivals, how does a firm find out which side each of its markets is on, and how fast the lines are hardening? That question has an empirical answer, assembled from public data, and the second half of the chapter shows how the answer is built and how far it can be trusted.

4.2 Four reasons trade might pacify

The modern version of the argument is less a single claim than a family of mechanisms, and they are worth keeping apart because they fail in different ways.

Commercial peace — the proposition that economic interdependence between two states lowers the probability that they will come into militarised conflict with each other. It is a claim about *dyads* — pairs of states — and it is probabilistic: it predicts fewer disputes, not none.

The first mechanism is **opportunity cost**. Conflict disrupts trade, and the more a dyad trades, the more it stands to lose. This is Angell's mechanism with the conquest argument stripped out, and it was formalised by Polachek (1980), who modelled conflict as an activity whose price rises with the gains from trade that it puts at risk. Its weakness is the weakness of 1914: it assumes leaders weigh those costs, and weigh them accurately.

The second is **signalling**. Interdependent states have a menu of costly actions short of force — threatening, restricting or suspending economic ties — that allow them to reveal how much they care about an issue. A state that is willing to lose trade to make a point is telling its rival something credible about its resolve, and credible information reduces the risk of miscalculation that rationalist accounts of war identify as its main cause. On this account trade matters not because it makes war expensive but because it makes communication possible.

The third is **domestic coalitions**. Trade creates exporters, importers, port operators, banks and their employees, all of whom have something to lose from a rupture with the partner. They lobby. A government contemplating confrontation faces an organised constituency on the other side, and the larger the trade, the larger the constituency. This mechanism is the one most familiar to a multinational, because the multinational is frequently the constituency.

The fourth is **institutions**. Trade agreements create committees, forums and dispute-settlement procedures through which grievances can be processed before they escalate, and they create repeated interaction among officials who come to know each other. Mansfield and Pevehouse (2000) found that pairs of states belonging to the same preferential trading arrangement are less prone to militarised disputes, and that the effect strengthens as trade between them grows — a result that has made institutions the most robust strand of the liberal argument.

Statistical work on dyads from the 1990s onwards gave these mechanisms empirical weight. Oneal and Russett (1997), using dyads from 1950 to 1985, titled their findings with some confidence — the classical liberals were right — and the programme culminated in Russett and Oneal (2001), which set interdependence alongside democracy and shared membership of international organisations as the three legs of a "Kantian" peace. Dyads that trade more relative to their GDP experience fewer militarised disputes. That correlation is real. What it means is a different question.

4.3 The realist reply: interdependence as exposure

The realist critique does not deny that trade creates gains. It denies that gains are what states care about most. In an anarchic system — the world described in Chapter 3, with no authority above states to enforce agreements — a state's first concern is survival, and every economic relationship is evaluated for what it does to security.

Seen that way, interdependence is a polite word for **vulnerability**. If a state depends on a rival for food, fuel or critical components, the rival holds a lever over it. Kenneth Waltz drew the conclusion that great powers will try to limit such dependence, and that where they cannot, they may be tempted to secure supplies by other means (Waltz 1979). The relationship that liberals see as a bond, realists see as a hostage.

The argument has a precise early statement in economics, and it was written by someone watching it happen.

Albert Hirschman, a refugee economist writing during the Second World War, analysed how Germany in the 1930s had organised its foreign trade with the smaller states of south-eastern Europe (Hirschman 1945). The terms were often generous. Germany bought their agricultural exports in quantities no one else would, through bilateral clearing arrangements that tied payment to purchases of German goods. The result was that a large share of each small country's trade came to depend on a single partner, while each of them was a trivial share of Germany's. Hirschman separated two effects. The *supply effect* is the gain in resources that trade brings a state. The *influence effect* is the power a state acquires over partners for whom the relationship matters much more than it does to the state itself — because breaking it would cost them dearly and cost it little. The influence effect depends not on the volume of trade but on its *asymmetry* and its *concentration*, and to measure concentration Hirschman proposed an index that, after a later rediscovery by Herfindahl, became the standard measure of market concentration in competition policy. The intellectual ancestor of every modern dependency dashboard was designed to measure geopolitical leverage.

The index is the square root of the sum of the squared shares of each partner in a country's exports, times a hundred: 100 if everything goes to one partner, close to zero if exports are spread evenly across many. Table 4.1 computes it from the exporter-reported goods exports in the IMF's partner-country trade statistics (International Monetary Fund 2026b).

Table 4.1: Hirschman's index of export-partner concentration (100 = all exports to one partner) and the largest partner's share of goods exports. Source: IMF, International Trade in Goods by partner country (exporter-reported); author's calculations.

Exporter	Index 2021	Index 2024	Largest partner, 2024	Its share, 2024 (%)
Mongolia	83.2	92.0	China	91.9
Mexico	80.9	83.3	United States	83.2
Canada	76.0	76.7	United States	76.5
Vietnam	35.8	36.3	United States	30.4
Russia	21.9	33.5	China	25.6
South Korea	32.4	30.4	China	19.5
United States	27.4	26.8	Canada	16.9
China	23.6	21.1	United States	14.7

The table sorts itself into Hirschman's two kinds of country. Mongolia sent 92 per cent of its recorded exports to China in 2024; Mexico and Canada sell three-quarters or more of theirs to the United States. For them the largest partner is a matter of national survival, while for China and the United States no single partner takes more than 17 per cent of exports. That asymmetry, not the volume of trade, is what the influence effect is made of. The striking movement is Russia's: China's share of its exports rose from 13.9 per cent in 2021 to 25.6 per cent in 2024, and the index from 21.9 to 33.5. Russia stopped publishing detailed customs data in 2022, so its later figures rest partly on estimates, but the direction is clear — the turn away from Europe handed its new principal customer the leverage Hirschman described. South Korea moved the other way, China's share of its exports falling from 25.3 to 19.5 per cent.

Two further realist arguments sharpen the point. The first concerns **relative gains**. Chapter 3 showed why a state in an anarchic system may reject a deal that makes it better off if the deal makes its rival better off by more. Trade usually benefits both partners, but rarely equally, and a state that expects to face its partner as an adversary must worry that today's gains will fund tomorrow's threat.

The second is Joanne Gowa's argument about **security externalities** (Gowa 1994).

Security externality — the effect of trade on the military potential of the partner. Because the gains from trade raise a partner's economic capacity, and economic capacity can be converted into military power, trade with an ally strengthens one's own side while trade with an adversary strengthens the other. States should therefore trade more with allies than with adversaries.

Gowa and Mansfield (1993) tested the prediction and found that alliances do increase trade between their members, and that the effect is stronger in bipolar than in multipolar systems — exactly where the security stakes of strengthening the wrong partner are clearest. The prediction matters for this chapter because it is, word for word, the hypothesis that the fragmentation literature now tests: that trade will reorganise itself along the lines of geopolitical alignment.

The most important refinement of the realist position came from Dale Copeland, and it rescues something from both sides. Copeland's claim is that what matters is not the level of trade today but **expectations of future trade** (Copeland 1996, 2015). A dependent state that expects trade to continue has every reason to keep the peace, because the liberal gains are real and are expected to keep flowing. The same state, at the same level of dependence, becomes dangerous when it comes to expect that it will be cut off — because then dependence is a vulnerability about to be exploited, and acting before the window closes can look rational. Interdependence is pacifying or provocative depending on what leaders believe about the future.

From the archive · Washington and Tokyo, 1941

Copeland's most cited illustration is the road to Pearl Harbor. Through the 1930s Japan depended on the United States for a large part of its oil and of other strategic materials, while it was waging war in China. As Japan advanced into Indochina, Washington tightened export controls step by step, and in the summer of 1941 the United States froze Japanese assets, which

in practice cut Japan off from American oil; Britain and the Netherlands, which controlled the other accessible supplies, followed.

Japan's leaders now faced a dependence that had turned, in their reading, from an asset into a countdown: stocks of oil would run down, the navy would lose the capacity to act, and the terms on offer from Washington — withdrawal from China — were ones they would not accept. In December 1941 they attacked the American fleet at Pearl Harbor and moved south against the oil-producing Dutch East Indies.

The episode does not show that trade causes war; Japan's expansion was the cause of the embargo before the embargo became a cause of the attack. What it shows is the mechanism Copeland identifies. The same trading relationship that had given Japan a reason for restraint gave it, once expectations of continued supply collapsed, a reason for haste. For a firm the lesson transfers directly: the moment of maximum danger in a dependent relationship is not when trade is high, but when one side comes to believe it is about to end.

4.4 What the dyads say, and why they disagree

By the end of the 1990s the two camps had produced findings that pointed in opposite directions. Oneal and Russett (1997) found interdependence pacifying. Barbieri (1996), using dyads from 1870 to 1938 and measures that distinguished the salience of a relationship from its symmetry, found that extensive interdependence was associated with *more* militarised disputes, not fewer, and later extended the case against the liberal view at book length (Barbieri 2002). Much of the disagreement turned out to be about measurement — trade relative to whose GDP, counted from whose customs records, over which years, against which definition of a dispute — which is a warning that applies to every measure in this book.

Edward Mansfield's review in the *Oxford Handbook of International Political Economy* is the best short guide to where the debate now stands (Mansfield 2021). Four of its points organise everything that follows.

The first is that **the relationship runs both ways**. Conflict reduces trade: states that expect to fight trade less, and states that are fighting trade hardly at all. A negative correlation between trade and conflict is therefore what we would observe whether trade causes peace, peace causes trade, or both are driven by something else — a shared alliance, a shared regime type, geographical distance. This is reverse causality in its classic form, and it is fatal to any naive reading of a dyadic correlation. It returns in the fragmentation literature in a new form: when trade between two blocs falls, we must ask whether the trade fell because the relationship deteriorated, or whether the deterioration is what the falling trade is measuring.

The second is that **institutions matter**. The finding of Mansfield and Pevehouse (2000) on preferential trading arrangements has proved more durable than the bare correlation between trade and peace, perhaps because institutions supply the information and the repeated interaction that the signalling and coalition mechanisms require.

The third is that **context matters**. The pacifying effect of trade differs with regime type, with the level of development, and with the distribution of power between the partners. A finding pooled across all dyads in all decades is an average over very different relationships, and an average can describe none of them well.

The fourth is the one that matters most for a business school: **the question has changed**. As great-power competition returned, research turned from asking whether trade pacifies to asking how states use trade as an instrument of conflict. That is the question of Chapters 8, 9 and 10 — cyber, economic diplomacy and sanctions — and it is the reason this chapter ends not with dyadic dispute data but with the geography of trade itself.

4.5 Bilateral ties, multilateral exits

One contribution reshaped the debate by asking which trade matters. Martin, Mayer, and Thoenig (2008) — economists rather than political scientists — separated two quantities that earlier work had often lumped together.

Bilateral versus multilateral openness — *bilateral* openness is how much two states trade with each other; *multilateral* openness is how much each trades with the rest of the world. The first raises what the pair loses from fighting each other. The second lowers it, because a state with many alternative partners can replace the trade a conflict destroys.

The logic is that of any outside option. If country A trades heavily with B and with almost no one else, a war with B is ruinous. If A trades heavily with B but also with fifty other countries, losing B is costly but survivable: the fifty others

become substitutes. Their model predicts, and their empirical work supports, both effects at once: bilateral trade lowers the probability of a bilateral conflict, and multilateral openness raises it.

	Effect on the risk of conflict between A and B
Bilateral trade between A and B	**lowers** it: more to lose from fighting each other
Multilateral openness of A and B with everyone else	**raises** it: substitutes make losing the partner cheaper

The implication is uncomfortable for the liberal view and was meant to be. **Globalisation lowers the cost of bilateral conflict.** The very process that multiplied trading relationships after 1990 made each individual relationship less essential, and therefore made breaking any one of them cheaper. A world of many alternative suppliers is a world in which a state can afford to fall out with one of them.

There is a corollary for the multinational that is easy to miss. Martin, Mayer and Thoenig describe the cost of switching *for states*, aggregated across an economy. A government that decides to reduce its dependence on a rival can point to a world of substitutes. The firm whose particular supplier sits in that rival, whose tooling is calibrated to that supplier's components, whose certifications name that plant, does not get the aggregate substitution elasticity. It gets its own contracts, which do not become cheaper to break because the national economy has alternatives. Geopolitical decisions are taken at the level where switching is cheapest and paid for at the level where it is dearest.

The rise of multilateral openness is a single series. Figure 4.1 plots world exports plus imports as a share of world GDP (World Bank 2026): 25.8 per cent in 1970, 38.1 per cent in 1990, a peak of 60.3 per cent in 2008. Then the rise stopped. The 2022 value of 62.1 per cent is the arithmetic of energy and food prices rather than of integration, and by 2024 the ratio was back at 56.7 per cent.

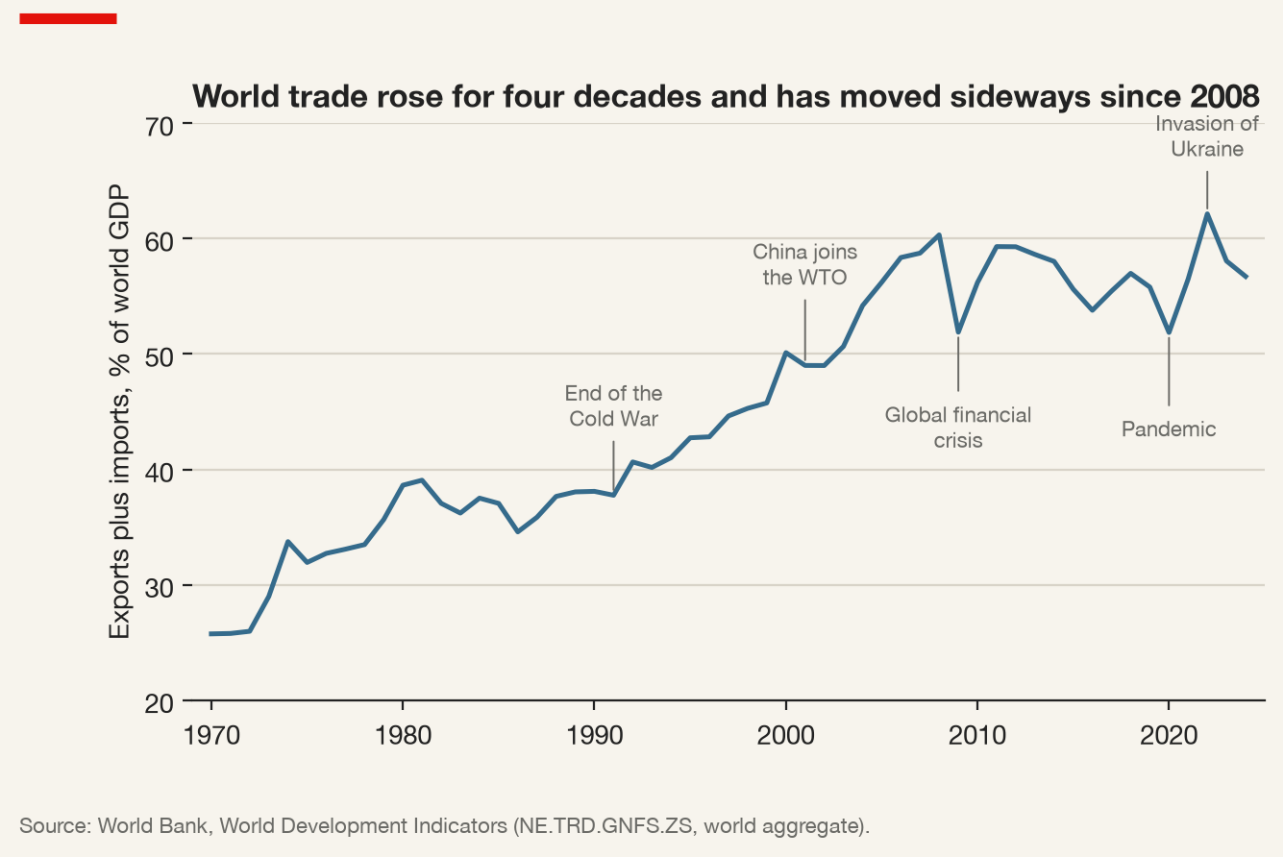


Figure 4.1: World trade rose from about a quarter of world GDP in 1970 to 60 per cent in 2008 and has moved sideways since; the 2022 high reflects commodity prices. Source: World Bank, World Development Indicators, NE.TRD.GNFS.ZS (world); the preliminary 2025 aggregate is omitted.

On the argument of Martin, Mayer, and Thoenig (2008), four decades of rising openness made any single bilateral rupture cheaper. On a fragmentation reading, the plateau since 2008 shows the expansion reaching its political limits a decade before the tariffs of 2018. No aggregate ratio can choose between the two: flat openness is compatible with trade reorganising along political lines beneath an unchanged total, which is what the second half of this chapter looks for.

4.6 From "does trade pacify?" to "how is trade used?"

The fourth of Mansfield's points has a theoretical counterpart that has become central to the study of geoeconomics. Farrell and Newman (2019) observe that globalisation did not produce a flat network of equal partners. It produced hubs: the dollar clearing system, the messaging network banks use for cross-border payments, a handful of firms that control critical technologies, a small number of undersea cables and cloud providers. States with jurisdiction over a hub acquire two kinds of power over everyone who must pass through it. The *panopticon* effect is the ability to see — to gather information from the flows that cross the hub. The *chokepoint* effect is the ability to deny — to cut a target off from the network.

This is Hirschman's influence effect, rebuilt for a networked economy. It also turns the liberal argument on its head. In the liberal story, dense economic ties reduce conflict by raising its cost. In the account of weaponised interdependence, dense ties are the instruments through which conflict is conducted. Interdependence does not stop states competing; it changes the terrain on which they compete, and the multinational, which builds and uses those networks, finds itself standing on that terrain.

The policy response has been to reorganise the networks themselves — to reduce exposure to a rival's chokepoints and to build one's own. That reorganisation has a name.

4.7 Fragmentation: three words, three policies

The vocabulary of the early 2020s distinguishes three policies that are often run together in public debate, and a firm that cannot tell them apart will misread its own exposure.

Friend-shoring — a term popularised in US policy debate in 2022, most prominently by the US Treasury Secretary — proposes to move supply chains towards trusted partners: to keep trading, but with friends. **De-risking** — the European Commission's term, adopted in 2023 to describe its approach to China — proposes to reduce dependence on a rival in critical goods and technologies while preserving trade overall. **De-coupling** is the end point that both of the others say they are trying to avoid: separate economic blocs with little exchange between them.

Geoeconomic fragmentation — a policy-driven reversal of global economic integration, in which states redirect trade, investment, technology and payments according to geopolitical alignment rather than cost (Aiyar et al. 2023). It is distinguished from ordinary changes in trade patterns by its cause: it is driven by security and alignment, not by prices or technology.

The definition, from an IMF Staff Discussion Note that set out the research agenda, contains a claim that is hard to test: that the cause is geopolitical. A fall in trade between two countries may be the result of a tariff, a sanction, a shift in comparative advantage, a change in exchange rates, or a firm's decision to build a factory closer to its customers. Fragmentation is a hypothesis about *why* trade patterns change. What the data can show directly is only *whether* they change along geopolitical lines — whether the lines that matter for trade start to coincide with the lines that matter for foreign policy.

That narrower question is the one Gopinath and her co-authors set out to answer.

4.8 How Gopinath and co-authors draw the lines

Gopinath et al. (2025), first circulated as an IMF working paper (Gopinath et al. 2024), ask whether the global economy has begun to split into blocs since Russia's full-scale invasion of Ukraine in February 2022. Their design has three steps, and each embeds a choice that a reader of the result should know about.

First, define the blocs. Countries are grouped into a US-leaning bloc, a China-leaning bloc and a non-aligned group on the basis of their foreign-policy alignment as revealed by voting in the UN General Assembly. The choice is deliberate. Alliances are too sparse to classify most countries; trade itself cannot be used, because trade is the outcome; and General Assembly votes are cast by nearly every state on hundreds of resolutions a year, recorded publicly and comparable across countries. Section 2 of the paper states exactly which measure, which reference period and which cut-offs the authors use. Any simpler classification — including the illustrative rule described later in this chapter — departs from theirs, and the size of the departure is part of what any comparison with their result has to report.

Second, compare trade within and between blocs before and after the invasion. The comparison is made in a gravity framework — the workhorse model of international trade, in which bilateral trade is explained by the economic size of the partners, the distance between them and a set of other frictions — so that the change attributed to bloc membership is the change left over once the usual determinants of bilateral trade have been accounted for. The paper also examines foreign direct investment, where the same question is asked of announced investment projects.

Third, look for connectors. Non-aligned countries whose trade with *both* blocs grew after 2022 are candidates for rerouted trade: goods that once moved directly between the blocs may now move through a third country, sometimes after some processing, sometimes after very little. Countries such as Mexico and Vietnam are the obvious candidates.

Independent evidence on the United States points the same way and sharpens what "connector" means. Freund et al. (2024), working with US imports at the ten-digit tariff line between 2017 and 2022, find that China's share of US imports fell from 22 to 16 per cent and attribute the decline to the tariffs imposed from 2018. The imports were replaced by large developing countries with a revealed comparative advantage in the products concerned — and those countries were deeply integrated into China's own supply chains and saw their imports from China grow faster, especially in strategic industries: to displace China as an exporter to the United States, a country had to embrace China's supply chains. Within products the pattern looked like "China plus one" rather than broad diversification, and there was no consistent evidence of reshoring. China remained the largest supplier of goods to the United States in 2022. Alfaro and Chor (2023) reach a compatible conclusion from a longer view of US sourcing: direct imports from China have fallen while Vietnam and Mexico have gained share, and it is unclear whether the reallocation reduces dependence on supply chains linked to China at all. The bilateral line has moved; the network beneath it has moved much less.

Figure 4.2 shows the same movement in US-reported imports. China's share rose from 8.6 per cent in 2000 to 21.6 per cent in 2017, fell to 16.5 per cent in 2022 — the fall Freund et al. (2024) measure at the tariff line — and to 13.4 per cent in 2024; the provisional figure for 2025, a year of much higher tariffs, is 9.0 per cent. Mexico has been the largest source since 2023, and Vietnam rose from 2.0 to 5.7 per cent. The cost was not borne by Chinese exporters alone: the 2018 tariffs were passed almost entirely into the prices paid by American buyers (Amiti, Redding, and Weinstein 2019; Fajgelbaum et al. 2020).

The red line records where goods crossed the border from, not where their value was added.

The main finding of Gopinath et al. (2025) is that after the invasion trade *between* the two blocs fell relative to trade *within* them, by a margin the paper estimates with care and which no summary, including this one, should be trusted to round; foreign direct investment shows a similar pattern. The authors are careful in two directions. The fragmentation they measure is real and is concentrated in strategic sectors. It is also small compared with the Cold War, when trade between the Western and Soviet blocs was a small fraction of what it is now. Both halves of that conclusion matter to a firm. The first says the lines are hardening. The second says that the world is a long way from two separate economies, and that a strategy built on the assumption of complete de-coupling is a strategy built for a world that does not yet exist.

Bloc membership is fixed at a reference date, so countries that drift are classified by where they stood; the non-aligned group mixes countries that lean slightly and countries that genuinely balance. None of this undermines the paper; it defines what the paper licenses.

4.9 Turning alignment into a number

The whole design rests on a single variable that few readers outside political science have met, and it is worth understanding how it is made, because every choice in its construction travels into the result.

A General Assembly roll call records, for each resolution, whether each state voted yes, no or abstained. The simplest way to measure the alignment of two states is to count how often they vote the same way. Agreement scores of this kind have

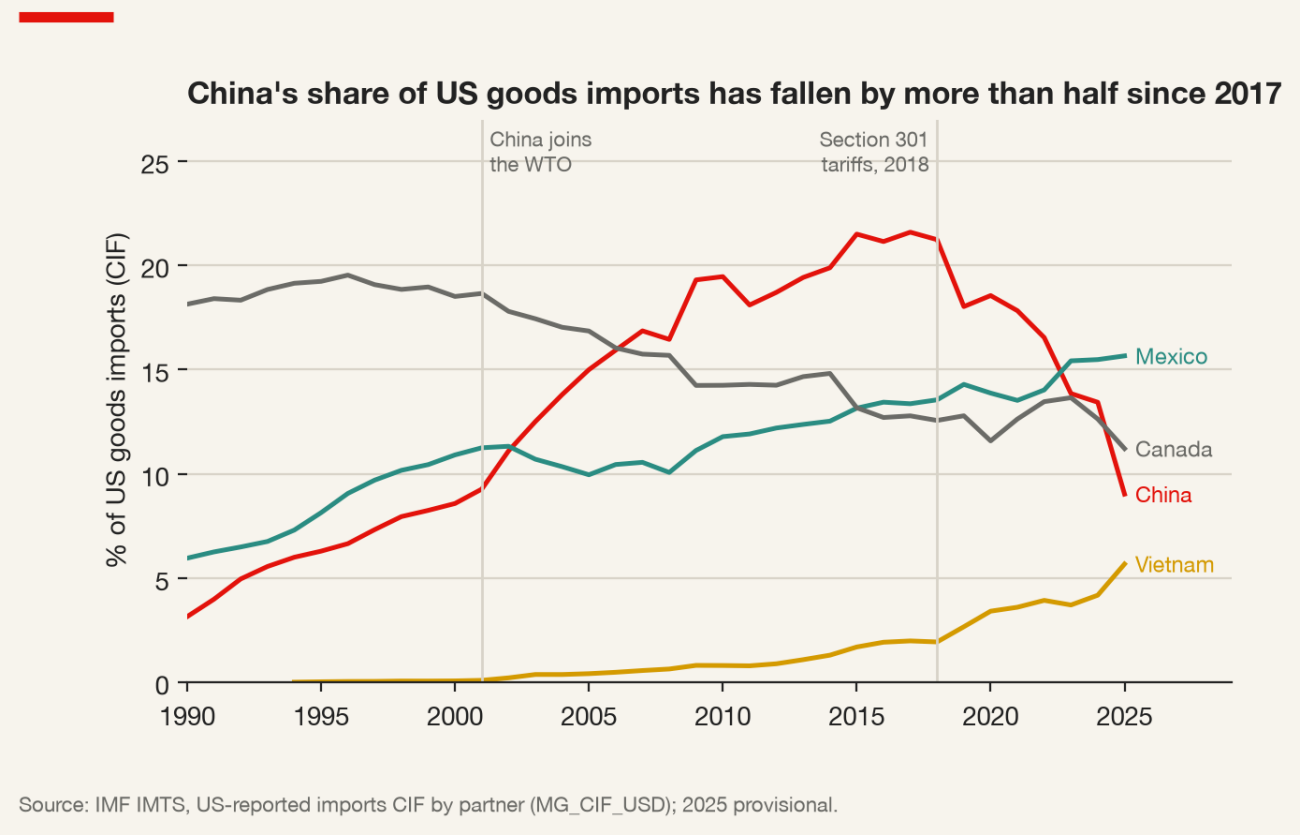


Figure 4.2: China's share of US goods imports fell from 21.6 per cent in 2017 to 9.0 per cent in 2025 (provisional), while Mexico and Vietnam gained. Source: IMF, International Trade in Goods by partner country, US-reported imports (CIF).

been used since the 1960s, and their limits were documented by Signorino and Ritter (1999), who showed that the common measures confound genuine similarity with the base rate of agreement on resolutions everyone supports. A deeper problem is comparability over time. The agenda changes from one year to the next. If the Assembly votes on more resolutions about the Middle East in one year and more about decolonisation in another, a change in a state's agreement with the United States may reflect the agenda rather than the state.

Bailey, Strezhnev, and Voeten (2017) solve the problem with a spatial model. Each state is assumed to have a position — an *ideal point* — on an underlying dimension, and each resolution divides the dimension into the positions that favour it and those that oppose it. The model estimates both states' positions and resolutions' cut-points from the pattern of votes. Comparability over time comes from resolutions that are repeated in essentially identical form across years: because the same text is voted on again, it anchors the scale, and movement by a state relative to those anchors is movement by the state. Voeten (2000) had earlier shown that after the Cold War the main dimension of Assembly voting separates states by their stance towards the US-led liberal order — which is why the dimension can be read as alignment with, or against, the United States.

Ideal point — a state's estimated position on a single underlying dimension of UN General Assembly voting, derived from all its roll-call votes and designed to be comparable over time. Higher values indicate positions closer to those of the United States. It measures revealed foreign-policy alignment in the Assembly; it does not measure trade, alliance membership, military co-operation or friendship.

The data come from the Dataverse record maintained by Erik Voeten and co-authors (Voeten 2026). Version 39, released in July 2026, estimates ideal points by calendar year, where older releases used Assembly sessions; the file still carries a session column, and the calendar year is the session number plus 1945. That change of meaning is not a pedantic detail: a figure built on the older file and a figure built on the newer one can disagree for reasons that have nothing to do with geopolitics. In 2021, the last full year before the invasion, the file covers 193 countries, with the United States at 2.41 and China at −0.41 on the scale.

Figure 4.3 traces five states on the scale. The Soviet Union sits near the bottom for most of the Cold War, swings to 1.30 in 1995 and has hovered around zero since, with a dip to −0.59 in 2022–2024. Germany has drifted from 1.86 in 1985 to 0.97 in 2025. The United States is alone at the top: in 2021 only Israel, the United Kingdom and Micronesia were within a point of it.

That isolation means the midpoint between the United States and China lies at 1.00 — above Japan (0.67), South Korea (0.83) and Italy (0.99).

An ideal point is a continuous variable. A bloc is a category. Getting from one to the other requires a rule, and the rule is a modelling choice. The simplest serviceable rule runs: a country is US-leaning if it is closer to the United States than to China by more than a chosen margin, China-leaning if the reverse holds, and non-aligned otherwise. The margin is a parameter, which means it can be moved — and when it moves, countries change bloc and the result changes with them. A result that survives a range of reasonable margins is evidence of fragmentation. A result that appears at one margin and vanishes at another is evidence of the margin. On this scale the rule puts, at a margin of zero, 170 of 193 states on China's side and 23 on America's; fixed cut-offs on the scale serve better, as the section after next shows.

The trade side of the comparison has its own decisions. The source is the IMF's International Trade in Goods by partner country, the dataset formerly known as the Direction of Trade Statistics (International Monetary Fund 2026b). It reports, for each reporting country and each partner, the value of goods exports and imports in US dollars. Two things about it shape any analysis built on it. The first is the **mirror problem**: A's reported exports to B are not equal to B's reported imports from A, because of timing, of valuation (exports are recorded free on board, imports including cost, insurance and freight), of re-exports and of confidentiality. An analysis must pick one side and say which, because the two sides can diverge most where trade is routed through third countries — precisely where the interesting movement is. The second is that the values are in **current US dollars**. When commodity prices rise, the dollar value of trade rises without a single extra barrel or tonne crossing a border, and 2022 was a year of very high energy and food prices. Trade "within" a bloc that includes large energy exporters can rise for reasons that have nothing to do with alignment.

The final step is the index. Each series — trade within the US-leaning bloc, within the China-leaning bloc, between the two, and involving the non-aligned — is divided by its own value in a base year and multiplied by a hundred. Indexing removes the enormous differences in level between the series and lets the eye compare their *trajectories*, which is what fragmentation is about. It also hands enormous weight to the base year. Index to 2021 and the post-COVID rebound of 2021 is built into the denominator; index to 2019 and it is not. Two honest analyses of the same data can therefore disagree

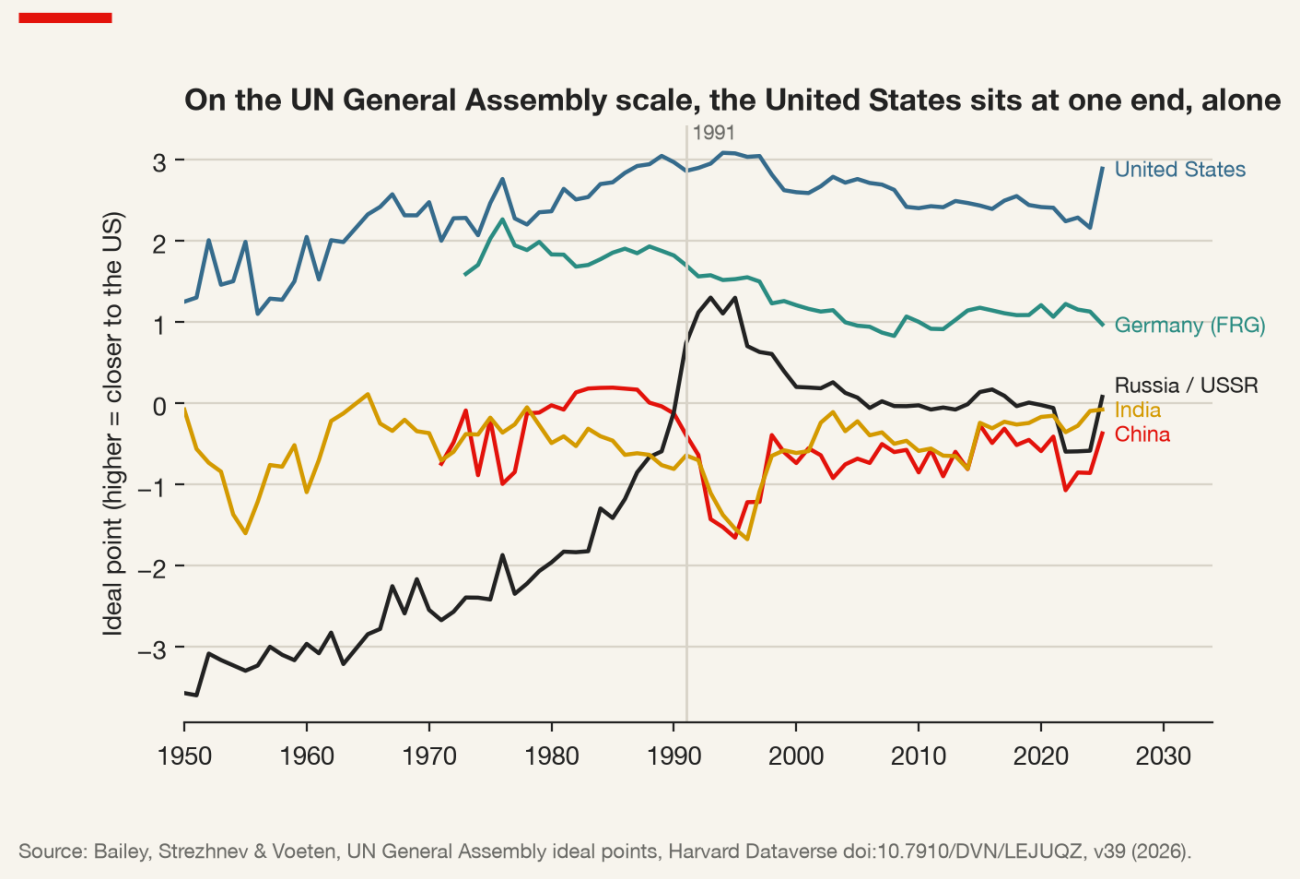


Figure 4.3: On the General Assembly scale the United States sits at one end, alone; Russia swung towards it after 1991 and back to the middle. Source: Bailey, Strezhnev and Voeten, UN General Assembly ideal points, Harvard Dataverse, version 39 (2026).

about the size of the break simply because they chose different base years, and the base year belongs in the caption of any such figure, not in a footnote.

4.10 What the public data show

The exercise below uses 2021 ideal points and exporter-reported goods exports for 2015–2024, the last year with near-complete reporting; the 193 countries with an ideal point account for 94.6 per cent of world exports (Taiwan and Hong Kong drop out). The main rule uses fixed cut-offs: US-leaning at 0.5 or above, in a natural gap below the Western cluster; China-leaning at −0.2 or below; non-aligned in between. It yields 52 US-leaning states (North America, Europe, Japan, South Korea, Australia, Turkey, Ukraine), 109 China-leaning (including Vietnam, Indonesia, Saudi Arabia and South Africa) and 32 non-aligned, among them India, Brazil, Mexico — and Russia, at −0.06 in 2021. Drawn on 2022 votes, Russia would have been China-leaning. The Assembly measures voting, not alliances, and the reference year is a choice.

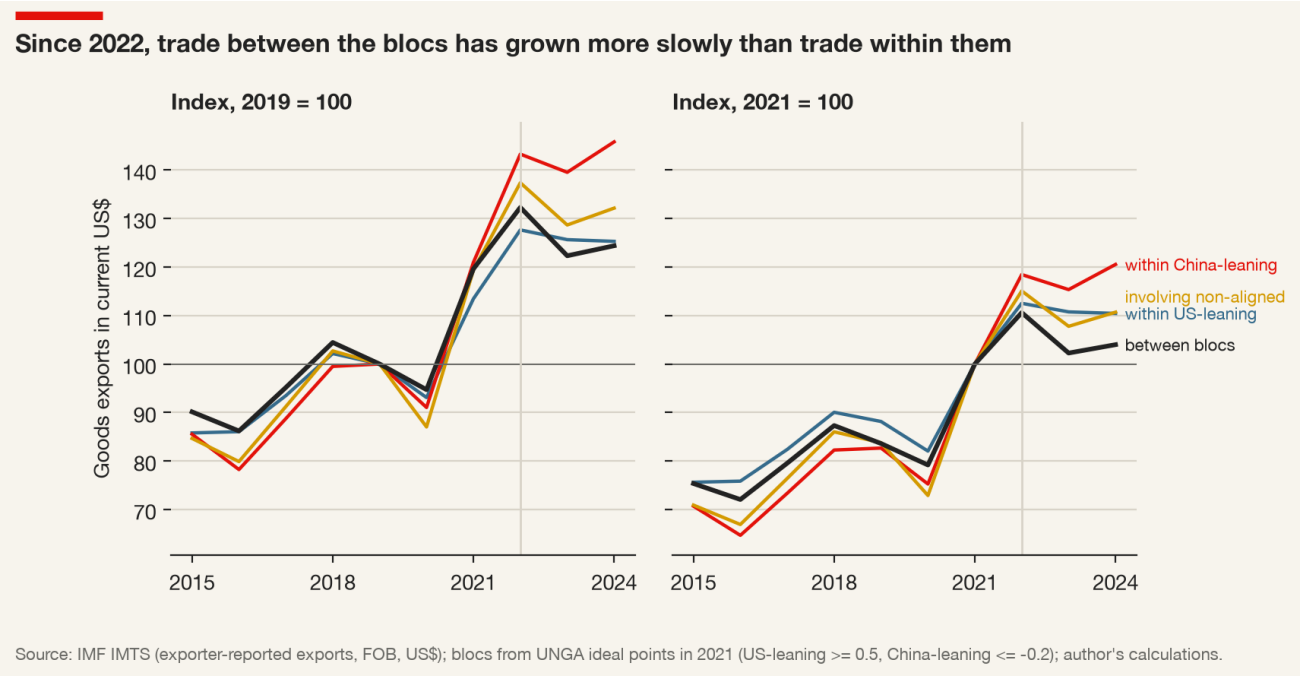


Figure 4.4: Between 2021 and 2024 goods trade between the blocs grew by 4 per cent in current dollars, against 10 per cent within the US-leaning bloc and 21 per cent within the China-leaning bloc; with 2019 as the base the gap is smaller. Source: IMF, International Trade in Goods by partner country (exporter-reported exports, FOB); blocs from 2021 UN General Assembly ideal points; author's calculations.

Figure 4.4 shows the result. Indexed to 2021, trade between the blocs was 4 per cent higher in 2024; pooled trade within them was 12.5 per cent higher, a gap of 8.5 percentage points. Indexed to 2019 the gap narrows to 4.9 points, and the panel shows where most of it comes from: not a separation of the US-leaning bloc from the rest, but fast growth of trade among China-leaning countries, many of them commodity exporters whose earnings rose with prices. Values are not volumes, and part of what looks like alignment is the price of oil.

Table 4.3: How the bloc rule changes the result: states per bloc (2021 ideal points) and growth of goods trade in current dollars, 2021–2024. Source: IMF, International Trade in Goods by partner country; UN General Assembly ideal points, version 39; author's calculations.

Rule	US-leaning	China-leaning	Non-aligned	Between-bloc share of trade, 2021 (%)	Between-bloc growth (%)	Within-bloc growth (%)	Gap (pp)
Closer to Washington or Beijing, margin 0	23	170	0	44.3	6.2	12.9	−6.7
Closer to Washington or Beijing, margin 0.5	6	155	32	20.1	9.0	17.6	−8.6
Cut-offs +0.3 / −0.3	54	94	45	25.0	2.8	12.3	−9.5
Cut-offs +0.5 / −0.2 (main)	52	109	32	26.3	4.0	12.5	−8.5
Cut-offs +0.5 / −0.4	52	77	64	21.1	1.6	12.1	−10.5
Cut-offs +0.7 / −0.1	40	119	34	23.5	5.0	12.5	−7.5
Main, Russia moved to China-leaning	52	110	31	28.6	−1.7	13.3	−15.0

Table 4.3 and Figure 4.5 move the lines. Under every rule, between-bloc trade grew more slowly than within-bloc trade; when the scale alone decides, the gap runs from 6.7 to 10.5 points. Its size is not a fixed property of the world: it nearly doubles, to 15.0 points, when Russia alone moves to the China-leaning bloc, because between-bloc trade then includes the collapse of Russia's trade with Europe. These are descriptive ratios without gravity controls, not estimates of the effect of geopolitics; they show only that the separation reported by Gopinath et al. (2025) is visible in raw public data and that its size depends on where the lines are drawn.

Table 4.4: The largest non-aligned traders: goods exports plus imports with each bloc, as reported by the exporting country, current US dollars; blocs from 2021 ideal points under the main rule. Source: IMF, International Trade in Goods by partner country; author's calculations.

Non-aligned economy	Trade with both blocs, 2024 (US$ bn)	With US-leaning bloc, change 2019–24 (%)	With China-leaning bloc, change 2019–24 (%)
Mexico	1,142	+34	+80
India	950	+40	+35
Russia	540	−63	+85

Non-aligned economy	Trade with both blocs, 2024 (US$ bn)	With US-leaning bloc, change 2019–24 (%)	With China-leaning bloc, change 2019–24 (%)
Brazil	509	+33	+70
Chile	149	+22	+47
Pakistan	89	+13	+32
Argentina	88	+19	+15
Colombia	84	+21	+21

Table 4.4 looks for connectors. All but Russia traded more with both blocs in 2024 than in 2019, and trade with the China-leaning bloc generally grew faster — Mexico's by 80 per cent — which is the pattern Freund et al. (2024) found at the product level. Russia is not a connector but a country that changed sides in trade while staying in the middle of the Assembly. Country totals cannot show the link at the heart of the connector story, inputs from one bloc becoming exports to the other; that needs product-level and input–output data. What the table shows a firm is where to look.

4.11 What fragmentation does to a multinational, channel by channel

This book follows five channels through which geopolitical risk reaches a firm. Fragmentation touches all of them, but not equally, and a firm that can say which channel matters most for its own business has already done most of the analysis.

Market access is the most direct. A firm whose markets sit in different blocs faces the prospect that the rules governing its sales — tariffs, export controls, procurement preferences, product standards — will diverge between them. The first practical step is unglamorous: classify the firm's most important markets by alignment, and look not only at where they sit but at whether they are drifting.

Inputs are where friend-shoring and de-risking bite. Fragmentation in the data is concentrated in strategic sectors — semiconductors, critical minerals, advanced machinery, energy — and so is the policy pressure. A maker of consumer goods may see little beyond rising transport and compliance costs; a maker of semiconductor equipment may find its suppliers, customers and licences reorganised by law. A supplier in a connector country may be the cheapest route around a bloc boundary — and the next target of rules-of-origin enforcement, because governments that wish to make fragmentation stick have every reason to close the routes around it.

Finance follows the same lines. The between-bloc decline Gopinath and co-authors find in foreign direct investment reaches firms through investment screening, restrictions on outbound investment in sensitive technologies, and the cost of capital for projects that straddle the boundary. Payments, too, run through hubs of the kind Farrell and Newman (2019) describe, and access to them is a matter of jurisdiction.

Assets and people are exposed in the connector countries. A plant built to serve both blocs from a non-aligned location is a bet that the location will stay non-aligned and that both blocs will keep accepting its goods. Staff who move between blocs face visa regimes and export-control rules on the technical knowledge they carry.

Data and operations fragment as well: rules on data localisation, on cloud providers and on cross-border transfers increasingly follow the same alignment, and a firm whose systems are integrated across blocs may find that integration itself becomes a compliance problem.

Across all five channels the same distinction from Martin, Mayer and Thoenig applies. The aggregate economy has substitutes; the firm has contracts. When a government decides that de-risking is affordable because the country's imports can be sourced elsewhere, it is making a claim about averages; the firm's own substitutes may be few, slow and expensive. If the data show de-risking — selective reductions in strategic sectors with overall trade preserved — a blanket exit from the other bloc overreacts; if they show de-coupling, waiting overreacts the other way. The evidence so far is closer to the former, and the right response is sectoral.

A tempting but wrong conclusion is that non-aligned markets are safe because they are on neither side. They are where between-bloc trade is being rerouted, and they are therefore where enforcement, scrutiny and secondary sanctions will concentrate. Non-alignment is a position in a contest, not an exemption from it.

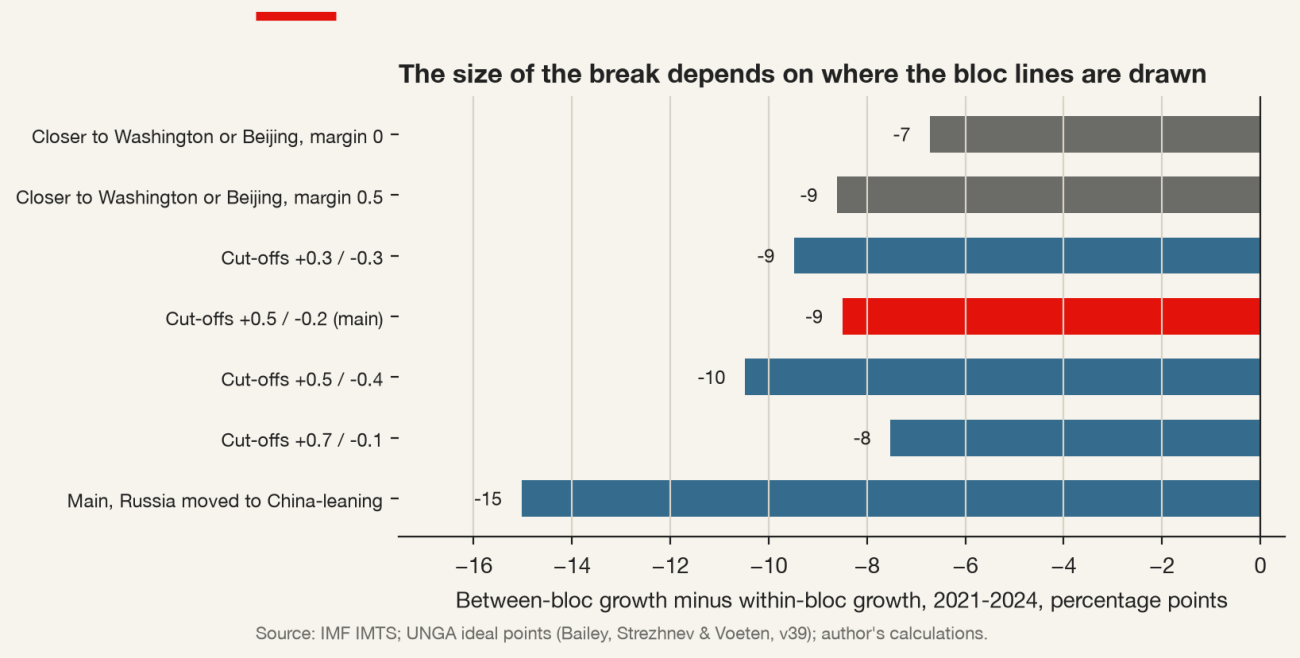


Figure 4.5: Between-bloc trade lagged within-bloc trade under every rule tried, by 7 to 15 percentage points over 2021–2024; moving Russia alone nearly doubles the gap. Source: IMF, International Trade in Goods by partner country; UN General Assembly ideal points; author's calculations.

4.12 A short review of the literature

The commercial peace has a long pre-history in political philosophy and a short history as a testable proposition. Angell (1910) gave it its most influential popular form and its most notorious timing. The modern empirical literature begins with Polachek (1980), who modelled conflict as an activity priced by the gains from trade it endangers. Oneal and Russett (1997) and Russett and Oneal (2001) established the dyadic correlation between interdependence and peace, alongside democracy and international organisations; Barbieri (1996) and Barbieri (2002) challenged it with longer series and different measures, and much of the subsequent dispute concerned measurement as much as theory. Mansfield and Pevehouse (2000) showed that preferential trading arrangements dampen conflict, and Mansfield (2021) is the most recent and compact review of the whole field.

The realist tradition starts from Hirschman (1945), whose distinction between the supply and influence effects of trade, and whose concentration index, remain the foundation of the study of economic dependence as power. Waltz (1979) placed dependence within structural realism as a vulnerability states will seek to limit. Gowa and Mansfield (1993) and Gowa (1994) identified the security externalities of trade and showed that allies trade more with each other. Copeland (1996), developed at length in Copeland (2015), reconciled the two camps by making expectations of future trade the variable that decides whether interdependence restrains or provokes. Martin, Mayer, and Thoenig (2008) separated bilateral from multilateral openness and showed that globalisation can lower the cost of bilateral conflict.

The contemporary turn is from trade as a restraint to trade as an instrument. Farrell and Newman (2019) recast interdependence as a network with hubs that confer panopticon and chokepoint power. Aiyar et al. (2023) defined geoeconomic fragmentation and set out its channels; Gopinath et al. (2024), published as Gopinath et al. (2025), provided the first systematic evidence that trade and investment have begun to reorganise along lines of foreign-policy alignment; Freund et al. (2024) and Alfaro and Chor (2023) show, for US imports, that the reallocation away from China runs largely through countries tied into China's own supply chains. The measurement of that alignment rests on a parallel literature: Signorino and Ritter (1999) on the pitfalls of agreement scores, Voeten (2000) on the post-Cold War structure of General Assembly voting, and Bailey, Strezhnev, and Voeten (2017) on dynamic ideal points comparable over time. Le Billon and Shykora (2020) extends the analysis from manufactured goods to commodities, showing how supply chains themselves carry conflict.

The through-line is a shift in what is being explained. The older literature asked whether economic ties prevent war. The

newer one asks how the expectation of conflict reorganises economic ties — and therefore treats trade data not as a cause of peace but as a record of how states expect each other to behave.

4.13 What the evidence does not license

Suppose the lines separate after 2022 exactly as Gopinath et al. (2025) report. The temptation is to read the separation as a finding about geopolitics. It is a finding about a classification, a currency and a date, and each carries part of the result.

It is a **correlation with a date**: sanctions on Russia, the post-pandemic rebound and dearer commodities all moved trade in 2022, and a gravity model separates only some of them. **Bloc membership is a modelling choice**: in Table 4.3 one country's placement nearly doubles the gap. **Values are not volumes**, as the commodity exporters' 2022 figures show. And the result is **aggregate**: a firm's products may sit in a sector where fragmentation is far stronger than the average, or in one where it is absent.

What the evidence does establish is narrower and still worth having: the broad pattern is visible in public data under transparent choices, a firm's markets can be placed in blocs under a stated rule whose fragility can be shown, and the connectors can be identified — where, on the evidence of Freund et al. (2024), dependence on the rival has often been relabelled rather than removed.

Angell was right about the costs of war and wrong about what leaders would do with that knowledge; Copeland located the danger in expectations rather than in levels of trade. The fragmentation data are best read in the same spirit: not as proof that geopolitics has caused trade to split, but as a record of what states and firms now expect of one another. The evidence does not license the conclusion that alignment caused the separation, that a firm's own sector follows the aggregate, that its non-aligned markets are safe, or that blocs drawn at one margin in one year will be the blocs that matter in five. It licenses something more modest and more useful: the knowledge that the expectation of a divided world is already reorganising the network the firm depends on, in places the averages do not show.

Part II

Who governs

Chapter 5

International Institutions: Who Finances, Insures and Referees the Firm

5.1 A hotel in New Hampshire, and what it still decides

Ask a manager which international institutions matter to her firm and the first answer is usually the United Nations. Ask instead who lent the money for the port her goods leave from, who insured the plant her company built in a country whose currency may not be convertible next year, who will hear the case if that country's government takes the plant, and who decided which firms were allowed to bid for the contract to build the port in the first place — and the list becomes long, specific and almost entirely economic. The World Bank, its private-sector arm, its insurer, its arbitration centre; a regional development bank in Manila, Abidjan or London; and, increasingly, a bank in Beijing or Shanghai that did not exist a little over a decade ago.

This chapter is about those institutions. It treats them in two ways at once, because a multinational has to. They are **business infrastructure**: sources of long-term finance, guarantees, political risk insurance, contracts and dispute settlement in exactly the markets where private finance is scarcest and political risk highest. And they are **arenas of power**: bodies governed by states, whose votes are unequally distributed, whose rules are bargained, and whose decisions — as the evidence later in this chapter shows — can be bought. A firm that relies on such an institution to manage its geopolitical risk is relying on a rule; it is also relying on the coalition of states that keeps the rule in force. The question that runs through the chapter is which of the two it is really relying on, institution by institution and issue by issue.

The story starts, as the order it describes did, in July 1944.

From the archive · Bretton Woods, 1944

In July 1944 delegations from forty-four countries met at a resort hotel in Bretton Woods, New Hampshire, to design the post-war monetary order. The war in Europe was not over. The two plans on the table were British and American: John Maynard Keynes proposed an International Clearing Union with its own unit of account and pressure on creditor as well as debtor countries to adjust; Harry Dexter White, for the US Treasury, proposed a more modest stabilisation fund to which members would contribute and from which they could draw. The American plan largely prevailed, for the reason a realist would give: the United States was the creditor, and the creditor wrote the rules.

The Articles of Agreement of the International Monetary Fund were adopted on 22 July 1944, alongside those of the International Bank for Reconstruction and Development, and entered into force in December 1945. The design choice that matters most for this chapter was made then and has never been reversed: each member's **quota** — its subscription to the Fund — determines both how much it can borrow and how many votes it has. Money and voice were tied together from the first day. Eighty years later the argument about whether the new institutions of Asia and the BRICS complement or contest the Bretton Woods order is, at bottom, an argument about that tie.

5.2 Why would states under anarchy build institutions at all?

Chapter 3 described a system that has no government above states, and Chapter 4 a world in which trade is reorganising itself along geopolitical lines. Yet the same system is thick with institutions. Two traditions explain why, and they make opposite predictions for a firm.

The liberal institutionalist answer, given its canonical form by Keohane (1984), is functional. States that would gain from cooperation often fail to achieve it because they cannot observe each other's behaviour, cannot make credible promises, and fear being exploited. Institutions lower the **transaction costs** of agreement, supply **information** about compliance, and lengthen the **shadow of the future** by turning one-off bargains into repeated games. Keohane's point was that this logic survives the decline of the hegemon that created the institutions: once built, they are cheaper to maintain than to replace, and states keep obeying them because reneging would be costly.

The realist answer, set out most sharply by Mearsheimer (1994), is that institutions reflect the distribution of power. The strong build them to serve their interests, and when an institution's rules collide with a powerful state's interests, the rules give way. On this view an institution has no independent effect on behaviour; it is a stage on which power is exercised, and its apparent successes are cases in which states would have behaved the same way without it.

International institution — a persistent set of rules, usually with an organisation to administer them, that states have agreed to and that prescribes behaviour, constrains activity and shapes expectations. The economic institutions in this chapter add a balance sheet: they lend, guarantee, insure or invest, and their votes are attached to capital.

Neither account is a strawman, and a firm does not need to choose between them in general. It needs to know **which institution, on which issue, behaves like which**. A MIGA guarantee, an IFC loan covenant and an ICSID award are rules that have, on the whole, been honoured; the WTO's Appellate Body is a rule whose enforcement collapsed when its most powerful member stopped appointing judges. That is an empirical question, and the rest of the chapter treats it as one.

A third idea is needed to read the last decade. Morse and Keohane (2014) observed that dissatisfied states do not only lobby inside an institution; they also go around it. They may move an issue to a different existing body — *regime shifting* — or build a new one — *competitive regime creation*. They called the result **contested multilateralism**.

Contested multilateralism — the use of multilateral institutions to challenge the rules, practices or missions of other multilateral institutions, either by shifting an issue to an existing alternative forum or by creating a new institution to compete with the old one (Morse and Keohane 2014).

The Asian Infrastructure Investment Bank was their leading example of the second strategy, and it is where this chapter ends up. To see why it was built, one has to start with the votes inside the institutions it was built beside.

5.3 The votes behind the money

The International Monetary Fund lends to governments, never to firms. It matters to a multinational because its programmes set the conditions in which the firm operates: whether the host currency floats, whether capital may leave, whether subsidies and public-sector wages are cut, whether the government can pay its suppliers. A country in an IMF programme has its macroeconomic policy partly written in Washington.

The Fund now has 191 members. Its governance follows the 1944 design: quotas determine votes. As of 30 April 2026 the United States held 16.49% of total voting power, Japan 6.14% and China 6.08%; the other founding BRICS each held less than 3% (International Monetary Fund 2026a). Article III of the Articles requires an 85% majority of total voting power for any change in quotas, so a member holding more than 15% can block a quota reform alone — and only one member does.

Voting power and capital — in a shareholder-governed institution, a member's votes are a function of its capital subscription (plus, usually, a small equal allocation of basic votes). Voting power is therefore a lagging indicator of economic weight: it moves only when members agree to reallocate capital, and those whose share would fall can often block the reallocation.

The lag is the hinge of this chapter. China's economy is far larger relative to the American economy than a vote of 6.08% against 16.49% suggests, and reform has been slow. The Sixteenth General Review of Quotas, approved in December 2023, raises total quotas by half but does so **equiproportionally**: every member's share stays where it was. Even that increase had not taken effect by October 2026 — the IMF had extended the deadline for members' consents to 15 November 2026 — and the realignment of shares was left to a Seventeenth Review for which, at the time of writing, guiding principles had been endorsed but no formula agreed. The same pattern holds at the World Bank: in the IBRD the United States held 15.98% of votes, Japan 6.88% and China 5.75% as of 30 June 2026 (World Bank Group 2026).

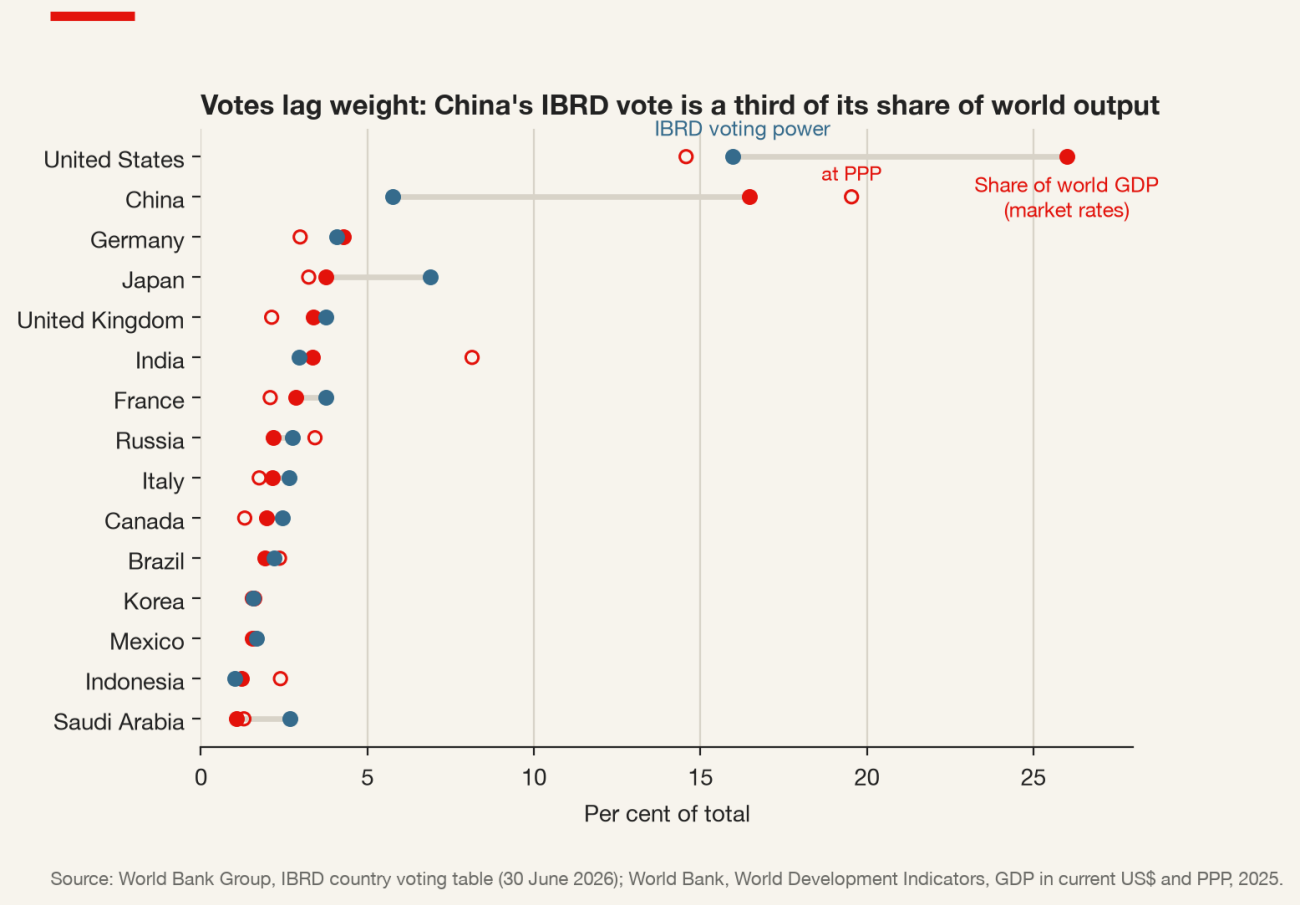


Figure 5.1: Votes lag weight: IBRD voting power on 30 June 2026 (blue) against share of world GDP in 2025 at market rates (red) and at PPP (hollow). Source: World Bank Group, IBRD country voting table; World Bank, World Development Indicators.

Figure 5.1 puts the voting table beside the economic weight it is supposed to lag. Two members stand far from where votes and output would coincide, and they are the two that matter most. China produced 16.5% of world output at market rates in 2025 and 19.5% at purchasing-power parity, yet holds 5.75% of the votes — about a third of its weight on the first measure. The United States holds 15.98% of the votes against 26.0% of output at market rates, which looks like under-representation too, until one notices that at purchasing-power parity its share of output is 14.6%, below its vote. The measure of weight is not neutral: market exchange rates favour rich economies with strong currencies, purchasing-power parity favours large emerging ones, and the IMF's quota formula blends the two with openness, variability and reserves because each member prefers the measure that flatters it. Choosing the yardstick is the first round of the bargain.

Japan holds 6.88% of IBRD votes against 3.75% of world output, Saudi Arabia 2.67% against 1.08%, France and the United Kingdom 3.75% each, above their output shares; India's 2.95% is far below its 8.1% of output at purchasing-power parity. A realignment requires the over-represented — Saudi Arabia apart, the United States' closest allies — to give up votes.

Table 5.1: Voting power in the four World Bank Group institutions with capital-based votes, per cent of the total, 30 June 2026. Source: World Bank Group country voting tables, parsed by the book's figure script.

Institution	United States	Japan	China	India	Russia
IBRD (lends to governments)	15.98	6.88	5.75	2.95	2.74
IDA (concessional lending)	9.52	8.43	2.55	2.95	0.29
IFC (private sector)	16.91	7.03	2.90	3.95	3.08
MIGA (political risk insurance)	14.81	4.16	2.61	2.53	2.61

Table 5.1 reads the same tables across the Group, and the pattern runs against intuition: China's voice is weakest in the institutions that deal with firms. It holds 5.75% of the votes in the IBRD, which lends to governments, but 2.90% in the IFC, which invests in companies — less than India — and 2.61% in MIGA, which insures them. The United States holds about a sixth of both. The institutions most directly in the business of a firm's political risk are those in which the challenger's weight is smallest. (IDA's votes follow contributions to a fund replenished every three years; there Japan nearly matches the United States.)

Two other institutions shape the conditions of business without lending a cent. The World Trade Organization, founded in 1995 with 166 members today, holds the tariff bindings and non-discrimination rules on which a firm's market access rests. Its Appellate Body has had no members since 30 November 2020, because appointments require consensus and the United States has withheld it; a subset of members now appeal through an interim arbitration arrangement instead. The rule still exists; whether it binds depends on which members have opted in. The OECD's relevance is quieter. Its Arrangement on Officially Supported Export Credits, in force since 1978, limits the terms that participating governments' export-credit agencies may offer, so that exporters compete on price and quality rather than on subsidised finance — and China is not a participant.

5.4 The World Bank Group as the firm's toolkit

The World Bank is usually described as a development institution, which it is. For a multinational it is also five distinct services, run by five legally distinct institutions with overlapping but different memberships.

The **IBRD** (1944) lends to middle-income and creditworthy governments; the **International Development Association** (1960) gives grants and concessional credits to the poorest. Neither lends to firms. Both matter to firms because the money they lend is spent buying goods, works and services, and the World Bank's Procurement Framework, in force since July 2016, requires that much of that buying be done through open international competition, with notices published on United Nations Development Business and the Bank's own site. Procurement is the least glamorous business channel of the multilateral system; for an engineering, construction or equipment firm it is often the largest.

Procurement has a disciplinary side. Firms found to have engaged in fraud, corruption, collusion or coercion in Bank-financed contracts are **debarred**, and their names published. Since the Agreement for Mutual Enforcement of Debarment Decisions of 9 April 2010, a debarment of more than one year by the World Bank Group, the Asian Development Bank, the African Development Bank Group, the European Bank for Reconstruction and Development or the Inter-American

Development Bank Group is enforced by the other four. A sanction at one bank becomes a sanction at five — which makes integrity compliance in one country a condition of market access in dozens.

The **International Finance Corporation** (1956) invests in the private sector directly: loans, many in local currency, equity stakes, guarantees, and the mobilisation of other investors. Its Managed Co-Lending Portfolio Program lets institutional investors — insurers, pension funds, sovereign funds — lend alongside IFC in a portfolio that mirrors its own. For a firm, IFC is a co-investor whose presence lowers the cost of other finance and signals to the host government that the project has a powerful friend.

The **Multilateral Investment Guarantee Agency** (1988) is the institution in this chapter most directly in the business of geopolitical risk. It insures cross-border investors and lenders against five political perils, and in fiscal year 2026 it issued a record US$10.6 billion of guarantees for sixty projects, bringing its gross exposure to US$41.4 billion (Multilateral Investment Guarantee Agency 2026). Since July 2024 the Group's guarantee products have been brought together in a single platform housed at MIGA.

Political risk insurance — cover for a cross-border investor or lender against losses caused by government action or political violence: currency inconvertibility and transfer restriction, expropriation, war and civil disturbance, breach of contract by the host government, and non-honouring of a sovereign financial obligation. Multilateral cover adds something private cover cannot: the insurer's shareholders include the host state.

That last clause is the deterrence argument for multilateral insurance. A host government that expropriates a MIGA-insured investor is not only in dispute with a foreign firm; it is in dispute with an agency of which it is itself a member, which will pay the claim and then seek recovery from it, and whose other members include the governments that decide its access to World Bank lending. The insurance works, in part, because the coalition behind it is large. It also means that the insurer is governed by states: in MIGA the United States held 14.81% of votes as of 30 June 2026 (World Bank Group 2026), and the cover a firm can buy reflects what that coalition is willing to underwrite.

The **International Centre for Settlement of Investment Disputes** (Convention in force 1966) provides the forum. A firm whose host state has consented — usually through a bilateral investment treaty — can bring a claim against the state before an arbitral tribunal rather than the state's own courts. ICSID had registered 1,118 cases by mid-2026, and of the cases registered in fiscal year 2026 a quarter concerned oil and gas and almost a fifth mining; more than three in five were brought under bilateral investment treaties (International Centre for Settlement of Investment Disputes 2026).

The case database ICSID publishes online shows how recent this is (Figure 5.2). Counting cases under the Convention and the Additional Facility, it lists 1,115 by mid-October 2026 (the Centre's own statistics are compiled slightly differently). For its first quarter-century the forum was nearly idle: 35 cases between 1972 and 1996, fewer than one and a half a year. The caseload then rose in steps — 18 a year on average in 1997–2006, 37 a year in 2007–2016, 54 a year in 2017–2025, with a peak of 66 in 2021 — and two-thirds of all the cases registered by the end of 2025 were registered after 2011. The timing is the signature of the treaty network: most of the investment treaties that give investors standing were signed in the 1990s.

The spikes have names. In 2003, 17 of the 30 cases registered were against Argentina, brought by investors in utilities and energy whose tariffs had been frozen and converted into devalued pesos during the crisis of 2001–2002. In 2011 and 2012, 19 of 85 were against Venezuela, as its expropriations reached the forum before its denunciation took effect. In 2015, 15 of the 52 cases were against Spain, brought by investors in renewable energy after the government cut the subsidies that had attracted them — claims made under the Energy Charter Treaty, by European investors against a European state. The last episode is the corrective: the forum protects whoever holds a treaty against whichever state signed one, rich states included. Several European Union members, and then the Union itself, have since left the Energy Charter Treaty — states leaving a rule when it began to bind them.

Table 5.2: The ten states most often named as respondents in ICSID cases (Convention and Additional Facility), by mid-October 2026. Source: ICSID online case database, counted by the book's figure script.

Respondent state	Cases	First registered	Latest registered	Pending
Argentina	61	1997	2026	8
Venezuela	56	2000	2025	7
Mexico	51	1997	2026	18
Peru	46	1998	2026	21
Spain	44	1997	2024	10

Respondent state	Cases	First registered	Latest registered	Pending
Egypt	38	1984	2026	3
Colombia	29	2016	2026	19
Ukraine	26	1998	2026	9
Romania	25	2001	2026	9
Panama	24	2006	2026	10

Table 5.2 lists the states most often sued. Ten states account for 36% of all cases. Six are in Latin America, which combines dense treaty networks with large mining and energy sectors. The "pending" column is the one a manager should read: Peru has 21 cases open, Colombia — whose first case came only in 2016 — 19, and Mexico 18.

Two findings temper what a firm should expect of the forum. Investors lose more often than the publicity suggests: Pelc (2017) shows that their success rate is low and that much of the litigation challenges general, non-discriminatory regulation, which tribunals are reluctant to condemn. And the forum disciplines states through reputation as well as law: Allee and Peinhardt (2011) find that countries named as respondents at ICSID receive less foreign direct investment afterwards, and less still when they lose.

Investor–state arbitration — a procedure in which a foreign investor brings a claim directly against a host state before an international tribunal, under consent given in a treaty, a contract or a national investment law. The award is binding; whether it is paid depends on the state.

The forum is a rule a state can leave. Bolivia denounced the ICSID Convention in 2007, Ecuador in 2009 and Venezuela in 2012; Ecuador rejoined in 2021, and Honduras, which gave notice in 2024, rejoined in 2026. The investor protection a firm buys when it structures an investment through a treaty jurisdiction is real, but it is protection under a regime whose membership moves with the politics of host states.

5.5 Does influence inside an institution have a price?

Everything so far assumes that the institutions apply their rules to all comers. The realist suspicion is that they do not — that the votes in the tables above are used, and that influence is traded. The sharpest test of that suspicion comes from the institution whose decisions bind every firm in the world.

The UN Security Council has fifteen members: five permanent, with a veto, and ten elected by the General Assembly for two-year terms by regional group, five each year, without immediate re-election. Before 1966 there were six elected seats. Substantive decisions need nine votes and no permanent-member veto, and measures adopted under Chapter VII of the Charter — sanctions, arms embargoes, authorisations of force — bind every member state, and therefore every firm operating in one. The arithmetic gives the elected members leverage: the permanent five cannot pass anything without at least four of them.

Kuziemko and Werker (2006) asked whether that leverage is paid for. Their identification idea is what makes the paper so persuasive. Rotation onto the Council follows regional queues and informal norms, not a country's need for aid in a particular year. So instead of comparing members with non-members — who differ in a hundred ways — they compare a country **with itself**, in the years before, during and after its term. If aid rises during the term and falls back afterwards, the seat is the likeliest explanation: poverty, alliances and politics do not rotate on a two-year cycle.

Event time — time measured relative to an event rather than by the calendar: year 0 is the first year of a Council term, −1 the year of the election, +2 the first year after the term. Re-indexing every country's series to its own event aligns cases that happened decades apart and lets one ask what typically happens around the event.

They found that US aid to a developing country rises substantially when it rotates onto the Council — by about 59% in the paper's headline estimate — and that aid through the UN rises too, more modestly. The increase is concentrated in the term years and disappears afterwards. And it is larger in years when the Council matters more, measured by how much the press covered it. That last result turns a pattern into a mechanism: if aid followed membership regardless of what the Council was doing, rotation might proxy for something else; that it tracks the **salience** of Council business is what makes vote buying the most plausible reading.

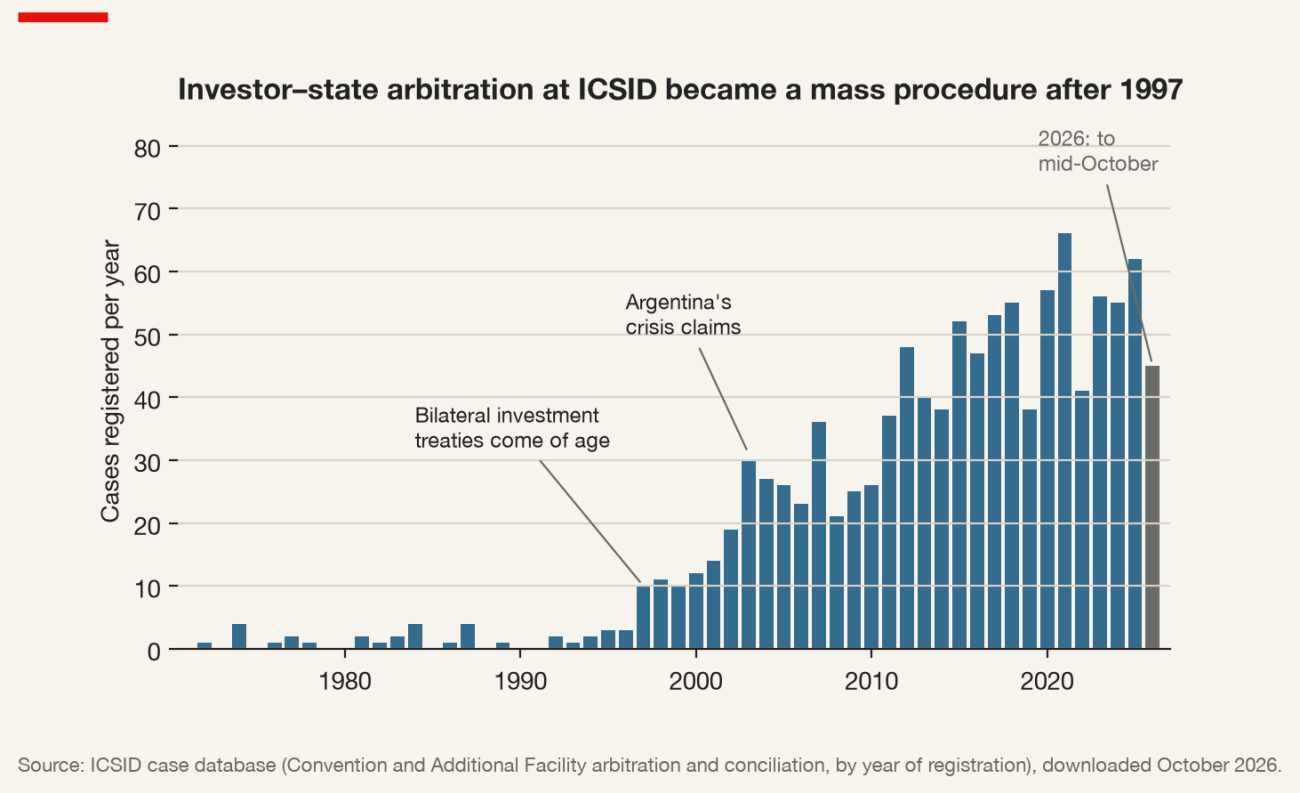


Figure 5.2: Investor–state arbitration became a mass procedure after 1997: ICSID cases registered per year, 1972 to mid-October 2026 (last bar partial). Source: ICSID online case database.

The distinction between finding and interpretation matters: the paper shows that aid moves with membership and with salience; *bribery* is the authors' word for the most plausible mechanism, not something the data observe. No transfer in the data is labelled as payment for a vote, and the paper does not claim to have seen one.

For this chapter the decisive step is what came next. The same research programme looked inside the economic institutions. Temporary Council members receive more World Bank projects (Dreher, Sturm, and Vreeland 2009a) and IMF programmes with fewer conditions attached (Dreher, Sturm, and Vreeland 2009b); Vreeland and Dreher (2014) assemble the evidence across channels. Kilby (2009) showed that World Bank disbursements respond less to a borrower's compliance with conditions when the borrower is aligned with the United States. Stone (2011) built a general account of how the dominant shareholder exercises **informal** influence over the IMF when its interests are at stake, while leaving the formal rules to operate otherwise. The voting tables of the previous section are not trivia. Who holds the votes shapes who gets the money — and the money decides where the contracts, the co-investment and the insurance are.

5.6 Regions with banks of their own

The World Bank model was copied, region by region, with a different balance of shareholders each time. The **Inter-American Development Bank** (1959) was the first; its private-sector arm, now IDB Invest, has operated since 1986. The **African Development Bank** followed in 1964, the **Asian Development Bank** in 1966, the **European Bank for Reconstruction and Development** in 1991, to finance the transition of the former Soviet bloc. The **European Investment Bank** (1958) is the EU's own bank, owned by its member states, and the **Islamic Development Bank** (operations from 1975) serves the members of the Organisation of Islamic Cooperation. Sub-regional banks fill in the map — CAF in Latin America, owned mainly by its borrowers; the Eurasian Development Bank in the post-Soviet space, founded by Russia and Kazakhstan.

Each has a private-sector window, a trade-finance programme that guarantees the letters of credit of local banks so that international banks will confirm them, and a procurement market. And each tells a geopolitical story through its shareholder

register. Japan and the United States each held 15.6% of the Asian Development Bank's shares at the end of 2025; the United States holds about 30% of the votes in the Inter-American Development Bank; Nigeria is the largest single voter in the African Development Bank; Saudi Arabia holds the largest share of the Islamic Development Bank.

Three features of the regional banks matter to a firm in ways the World Bank's rules can obscure. The first is that **procurement eligibility is foreign policy**. The Asian and Inter-American banks restrict contracts to firms and goods from member countries; the EBRD's are open to firms from any country; the African Development Bank opens contracts financed by its concessional fund to all while restricting those financed by the bank itself to members. Whether a Canadian, Chinese or Turkish firm may bid on a road in Bangladesh or Bolivia depends on whether its home state paid in.

The second is that a development bank can become an **instrument of sanctions**. On 4 April 2022 the EBRD's Board of Governors suspended Russia's and Belarus's access to the bank's resources and decided to close its offices in Moscow and Minsk (European Bank for Reconstruction and Development 2022). Both remain shareholders. A firm that had counted on EBRD co-finance in either country lost it overnight, through a decision taken by votes in which the host state took part and lost.

The third is that **shareholding moves**, slowly. The EBRD has expanded into sub-Saharan Africa, with Benin, Côte d'Ivoire and Nigeria becoming countries of operation in 2025; the Asian Development Bank amended its charter for the first time, with effect from March 2026, to lend more against the same capital; the Eurasian Development Bank cut Russia's share below half at the end of 2022. Each move redraws the map of where a firm finds support, and on whose terms.

Multilateral development bank — a bank owned by several states that borrows on capital markets against its members' paid-in and callable capital and lends, invests and guarantees for development purposes. Its rating, and therefore its cost of funds, depends on the creditworthiness of its shareholders and on its standing in the dollar and euro bond markets.

5.7 Building beside the old house: the new institutions

The new institutions of the last decade are a response to the lag between votes and weight. They come in three kinds: development banks, reserve pools, and the national policy banks of China.

From the archive · London and Beijing, March 2015

On 12 March 2015 the British government announced that it intended to join China's proposed Asian Infrastructure Investment Bank as a founding member — the first major Western economy to do so. Within weeks other European governments had followed, among them France, Germany and Italy. The United States and Japan stayed out, and are still out. On 29 June 2015 the bank's Articles of Agreement were signed in Beijing by fifty countries; they entered into force on 25 December 2015, and the bank opened on 16 January 2016 with fifty-seven founding members.

For this chapter the episode is a demonstration of Morse and Keohane's mechanism. China had sought a larger voice in the IMF and the World Bank for a decade, without a realignment of shares. It then built an institution in which it held the largest share by design — and invited the Europeans in. Once they were in, the new bank could not easily be dismissed as a Chinese instrument, and the old ones lost their monopoly on legitimacy in infrastructure finance. Competitive regime creation does not need to replace the incumbent to change the bargaining inside it.

The **Asian Infrastructure Investment Bank** has an authorised capital of US$100 billion. Its membership had reached 111 approved members by October 2026, 106 of them full members (Asian Infrastructure Investment Bank 2026). China holds 30.5% of subscriptions and 26.37% of votes; India, the second-largest member, holds 7.53% and Russia 5.92%. The Articles require a "super majority" — two-thirds of governors representing at least three-quarters of total voting power — for the most important decisions, among them capital increases and the election of the president. China's quarter of the votes is therefore a veto, as the United States' sixth of the IMF's is. The board is non-resident, a deliberate contrast with the full-time resident boards of the Bretton Woods institutions. The bank signed co-financing frameworks with the World Bank and the Asian Development Bank in its first months and co-financed many of its early projects with them; it lends to private as well as sovereign borrowers, and offers non-sovereign loans in a range of local currencies.

The **New Development Bank** was created by the five original BRICS under an agreement signed at Fortaleza on 15 July 2014 and is headquartered in Shanghai. Its initial subscribed capital of US$50 billion was divided **equally** among the founders, and voting power equals subscribed shares; the founders' combined votes may not fall below 55%, and non-borrowing members' combined votes may not exceed 20% (BRICS Governments 2014). Bangladesh, the United Arab

Emirates, Egypt, Algeria and, in 2026, Uzbekistan have since joined; several other countries have been admitted but have not yet completed accession. Its 2022–2026 strategy set a target of 30% of financing in local currencies. The equal-shares rule is the most interesting design choice in the new architecture: it prevents China from dominating the bank by capital, at the price of limiting how much capital the bank can raise.

The reserve pools are less visible to firms but matter to the country risk they face. The **BRICS Contingent Reserve Arrangement**, signed on the same day in Fortaleza, commits US$100 billion — China 41, Brazil, India and Russia 18 each, South Africa 5 — to members facing balance-of-payments pressure; but only 30% of a member's maximum access is available without an IMF programme in place. The **Chiang Mai Initiative Multilateralisation** among the ASEAN members, China, Japan and Korea, in force since 2010, is a US$240 billion network of currency-swap commitments, 40% of which has been available without an IMF link since 2021; its surveillance unit, AMRO, became an international organisation in 2016. Both are regional insurance that defers, for most of its value, to the Fund they were meant to supplement.

IMF-delinked portion — the share of a regional financing arrangement that a member can draw without having an IMF programme in place. It measures how far the arrangement is an independent lender of last resort rather than a co-financier of the Fund: 30% for the BRICS Contingent Reserve Arrangement, 40% for the Chiang Mai Initiative Multilateralisation.

The third kind of new institution is not multilateral at all. China's two policy banks, the China Development Bank and the Export-Import Bank of China, both created in 1994, and the Silk Road Fund, established in December 2014, are the financial core of the Belt and Road Initiative announced in 2013. AidData's tracking of Chinese official finance counts US$1.34 trillion of loans and grants to 165 low- and middle-income countries between 2000 and 2021 (AidData 2023). Its terms differ from those of the multilateral banks in ways that matter: Horn, Reinhart, and Trebesch (2021) show that a large part of China's overseas lending has gone unreported to the international debt statistics, much of it on commercial rather than concessional terms and often collateralised; Dreher et al. (2022) trace its aims and effects project by project. Gelpern et al. (2023), analysing a hundred contracts between Chinese state-owned lenders and governments in twenty-four developing countries, found three features that set them apart from other creditors' contracts: unusual confidentiality clauses that bar the borrower from revealing the terms or even the existence of the debt; arrangements that seek advantage over other creditors, such as lender-controlled revenue accounts and promises to keep the debt out of collective restructuring; and cancellation, acceleration and stabilisation clauses that could allow the lender to influence the borrower's domestic and foreign policies. For a firm, a host government that has borrowed on such terms has a debt position — and a creditor to placate in a crisis — that no public statistic shows. For a non-Chinese firm the opportunities are case by case. Contracts on Chinese-financed projects have often gone to Chinese contractors, and the standards that govern them are set bilaterally.

BRICS itself, finally, is not an institution: it has no charter and no secretariat, and its two treaty bodies are the NDB and the CRA. Its expansion matters to a firm as a map of alignment. Egypt, Ethiopia, Iran and the United Arab Emirates became members on 1 January 2024 and Indonesia in January 2025; Saudi Arabia is listed by the BRICS chairs but has never formally confirmed its membership, and its status is best described as ambiguous. A category of partner countries, created at the Kazan summit in October 2024, numbered ten by mid-2025. The New Delhi summit of September 2026 added no members or partners and asked for continued work on cross-border payments in local currencies — work, not a launched system.

How much alignment does membership actually signal? Figure 5.3 places the members of the club, its partner countries and the G7 on the scale estimated by Bailey, Strezhnev, and Voeten (2017) from votes in the UN General Assembly, for the 2025 session. The G7 sit to the right, from Japan at 0.66 to the United States, alone at 2.89. The BRICS founders have a median of −0.37, the five members admitted in 2024 and 2025 a median of −0.70, and the ten partner countries −0.56 — all close to the median of the whole Assembly, −0.31. Between them the members and partners span the range from Cuba (−1.47) to Russia (0.08), and 68% of the Assembly's 191 members fall inside that range. Saudi Arabia, whose membership is ambiguous, sits at −0.51, in the thick of it.

BRICS, then, is not a bloc in the sense an alliance is: its members vote as the Assembly's large majority votes, and the expansion of 2024–2025 widened rather than sharpened the club. What sets it apart is distance from the G7 — every member and partner sits to the left of every G7 member — not internal cohesion. Membership signals a preference for an order with more than one centre, not a commitment to a rival one. (The scale is re-estimated each session, and a few salient resolutions can move a country: Russia's place near the middle in 2025 is not a settled alignment.)

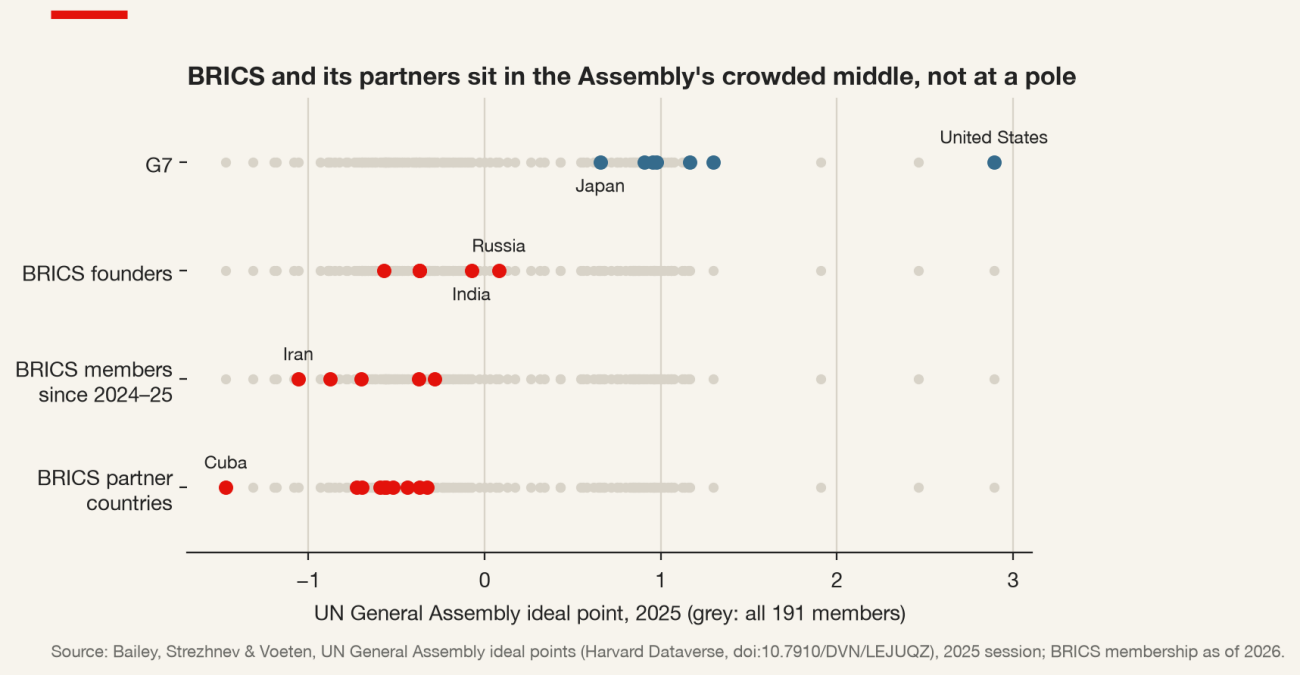


Figure 5.3: BRICS and its partners sit in the Assembly's crowded middle, not at a pole: UN General Assembly ideal points, 2025 (grey: all 191 members). Source: Bailey, Strezhnev and Voeten, UN General Assembly ideal points (Harvard Dataverse).

5.8 Complement or counter-hegemony?

The question every analyst asks of the new institutions is whether they complement the Bretton Woods order or contest it. Put that way, it has no answer; split into four tests, it does.

On **votes**, the new banks give emerging economies the voice that quota reform at the Fund has not — and reproduce its structure. China's veto in the AIIB mirrors the American veto in the IMF; the NDB avoids a dominant shareholder only by fixing equal shares among the founders.

On **standards**, the evidence leans towards complement. The AIIB adopted its own environmental and social framework and co-financed many of its first projects with the established banks; Wang (2017) and Humphrey (2022) both stress how much of the older banks' operating model the new ones borrowed. The contest is sharper outside the multilateral banks, in bilateral Chinese lending that sits outside the OECD export-credit rules and the cross-debarment agreement, on terms that are often confidential.

On **currency**, local-currency lending by the AIIB and the NDB reduces the currency mismatch that has ruined many infrastructure borrowers, which any development economist would call a complement. The BRICS work on local-currency settlement is something else: an effort to reduce members' dependence on the dollar and the payment chokepoints through which sanctions are enforced (Chapter 10).

On **sanctions**, the test came in March 2022. Within a week of Russia's full-scale invasion of Ukraine, the AIIB announced that all activities relating to Russia and Belarus were on hold and under review, and the NDB put new transactions in Russia on hold — Russia being one of its five founders. Both banks depend on dollar funding markets. When the choice was between a founding member and access to the market that funds them, they chose the market. That is the strongest single piece of evidence that the new multilateral banks are, for now, inside the system they were thought to contest.

The verdict, then, is **both**, and the reason is structural. The new institutions do not need to replace the old ones to change the bargaining inside them; they need only to exist as an **outside option**. Ikenberry and Lim (2017) frame the AIIB as institutional statecraft that pursues influence partly through the existing order rather than against it. Humphrey and Michaelowa (2013) had already shown, for Latin America, that borrowers with access to a bank controlled by borrowers prefer it to the

banks controlled by creditors. A host state with several lenders can shop among them, and the conditions any one of them attaches bind less.

5.9 What the firm should do with all this

Throughout this book, geopolitical risk reaches a firm through five channels: market access, inputs, finance, assets and people, and data and operations. The economic institutions of this chapter touch all five, but they open four specific **business channels** — procurement, finance, insurance and dispute settlement — and each comes with its own geopolitical risk.

Procurement is a market-access channel. The projects financed by the World Bank, the regional banks, the AIIB and the NDB buy through published procedures; a firm that tracks project pipelines and procurement notices sees demand years before it reaches the market. The risk is eligibility: whether the firm may bid at all depends on its home state's shareholding, and a cross-debarment for misconduct in one country closes the channel everywhere.

Finance and co-investment belong to the finance channel. IFC, IDB Invest, the EBRD and the private-sector windows of the Asian and African banks lend and invest alongside firms, increasingly in local currency and increasingly by mobilising institutional investors. A development bank in the capital structure lowers the cost of other finance and raises the political cost to a host government of interfering. The risk is conditionality, and decisions that respond to the largest shareholders.

Insurance and guarantees protect assets and people. MIGA, the World Bank Group's guarantee platform, the Islamic Development Bank's insurer and the trade-finance programmes of the regional banks turn political risk into a premium. The risk is that the cover is itself political: what a multilateral insurer will underwrite, and in which countries, reflects its shareholders' view of those countries.

Dispute settlement — ICSID and the treaty network behind it — protects assets against the host state. The new institutions have built nothing comparable. The risk is exit: states denounce the Convention or terminate treaties.

Three further risks cut across the channels. **Sanctions exposure** runs through the institutions as well as around them: suspension at the EBRD, freezes at the AIIB and the NDB. **Alignment** is signalled by membership — BRICS status, AIIB or EBRD shareholding, a Belt and Road memorandum. **Competing standards** mean that a project bankable by one lender may not be bankable by another.

An institution is a risk **mitigator** when the coalition that sustains it is stable, and a risk **source** when it is not. A firm's dashboard should track the coalition, not only the rule: who holds the votes, who sits on the Council, who has been suspended, who has joined which club.

A guarantee, a co-financing agreement or a treaty is a promise by a coalition of states. Read the coalition before the contract.

5.10 How influence becomes evidence

Institutions leave good data: they publish their rules, members and votes. The IMF publishes each member's quota and votes, the World Bank Group publishes voting tables for each of its institutions, and the AIIB publishes each member's subscription and voting power (International Monetary Fund 2026a; World Bank Group 2026; Asian Infrastructure Investment Bank 2026). Set side by side, China's and the United States' shares across the three summarise this chapter: a creditor's veto in the old house, a challenger's veto in the new one. Activity — MIGA's issuance, ICSID's caseload, AidData's record of Chinese official finance — is a second series; membership and its changes, a third.

The subtler evidence is an institution's own rules turned into a natural experiment. The UN publishes the list of elected Council members by year; the United States publishes its aid by country and year. Neither source contains the quantity Kuziemko and Werker study. It exists only once the two are joined, every country-year is re-indexed to the start of a term, and aid is expressed relative to the country's own pre-term level. That last step carries the design: each country serves as its own control, and large and small recipients are put on the same footing.

This is also why re-estimates do not reproduce the published numbers exactly. Aid statistics are revised and re-expressed in different constant dollars; country codes change; some countries serve several terms, so that the years after one term are

the years before the next. A re-estimate on current data that finds the same shape corroborates the finding; one that finds a different magnitude may simply be recording a disagreement between data vintages.

5.10.1 A re-estimate on current data

That warning is not hypothetical. Figure 5.4 applies the simplest version of the design to the data available in 2026. The Council's page lists 204 elected terms beginning between 1970 and 2010. Dropping countries that were OECD members by 1973 and keeping the terms in which the United States obligated aid to the country in each of the seven years around the term leaves 129. US aid is taken in logs, net of the average across all recipients that year (removing swings in the whole aid budget), and relative to the country's own average over the three pre-term years. If the seat were bought, the line would jump in the shaded years.

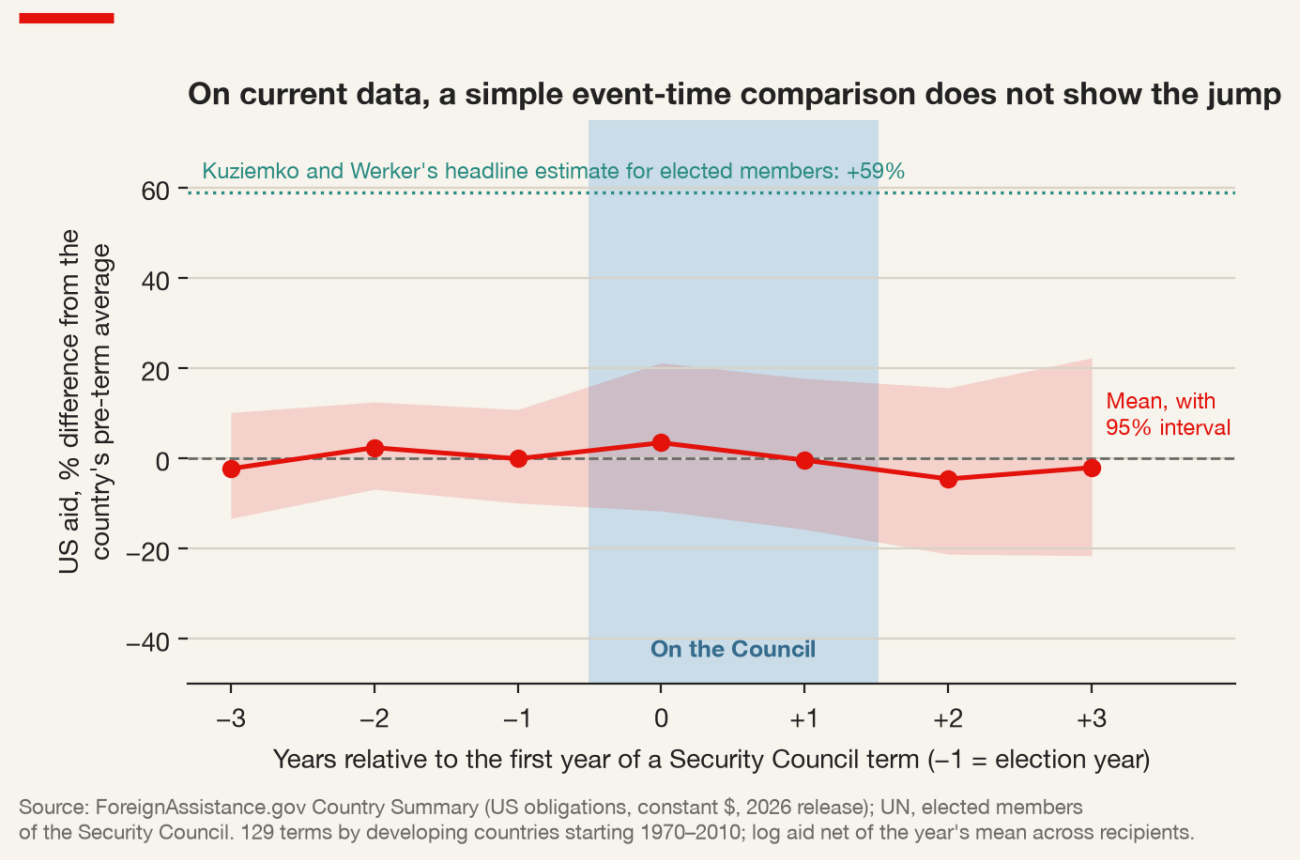


Figure 5.4: On current data, a simple event-time comparison does not reproduce the jump: US aid around 129 Council terms starting 1970–2010, relative to each country's pre-term average, net of year effects, with a 95% interval. Source: ForeignAssistance.gov Country Summary (obligations, constant dollars); United Nations, elected members of the Security Council.

It does not. In the two term years aid is on average 1.5% above the pre-term level (95% interval: −13% to +18%), and in 57% of terms it is no higher. Splitting the period or dropping the year adjustment changes little (Table 5.3); the published 59% lies outside every interval.

Table 5.3: US aid during a Security Council term relative to the country's own three pre-term years, under four specifications. "Term years" averages event years 0 and +1. Source: ForeignAssistance.gov Country Summary (US obligations, constant dollars, 2026 release); United Nations; computed by the book's figure script.

Specification	Terms	Term years vs pre-term average	95% interval	Terms above baseline
1970–2010, net of year effects	129	+1.5%	−13% to +18%	43%
1970–2010, raw logs	129	+2.2%	−12% to +19%	49%
1970–1989, net of year effects	55	+4.3%	−17% to +31%	45%
1990–2010, net of year effects	74	−0.5%	−19% to +22%	41%

This is a disagreement, not a refutation. The published estimate comes from regressions with country and year fixed effects on 1946–2001 and the aid data of the early 2000s; the series here is the 2026 release, in fiscal years and re-based constant dollars. The balanced window drops every term in which a country received no US aid in some year — and a country that goes from nothing to something on joining the Council is exactly what a vote buyer would produce. Averaging all terms also dilutes an effect the paper finds concentrated in salient years. What the exercise establishes is narrower: the effect is not visible in the simplest picture on current data, so whoever relies on it relies on the paper's design, not on a pattern anyone can see by joining two public tables — which is why the follow-up studies, with other institutions and other designs, matter as much as the original.

5.11 A short review of the literature

The theoretical frame is the exchange between Keohane (1984), for whom institutions persist because they reduce the costs of cooperation, and Mearsheimer (1994), for whom they reflect the distribution of power and constrain it little. Ikenberry (2011) reads the post-war order as an American project that bound its creator as well as others, which is why its institutions outlived the moment of American dominance. Morse and Keohane (2014) supply the concept that organises the last decade — contested multilateralism, through regime shifting and competitive regime creation.

On the politics inside the Bretton Woods institutions, Woods (2006) examines how the IMF and the World Bank's relations with their borrowers are shaped by their largest shareholders, and Stone (2011) models the informal influence of the dominant shareholder alongside the formal rules. The empirical programme on the price of influence starts with Kuziemko and Werker (2006) on US and UN aid to elected Security Council members, and extends to World Bank projects (Dreher, Sturm, and Vreeland 2009a), IMF conditionality (Dreher, Sturm, and Vreeland 2009b) and World Bank disbursements (Kilby 2009); Vreeland and Dreher (2014) synthesise it. Humphrey and Michaelowa (2013) show that shareholder composition shapes borrowers' choices among development banks.

On investment protection, Büthe and Milner (2008) find that international trade agreements raise FDI into developing countries by making policy commitments more credible, and Simmons (2014) shows that developing countries signed bilateral investment treaties, and tied their hands most tightly, when their bargaining position was weak — after which investors sued governments with surprising regularity. Pelc (2017) explains why investors nonetheless lose so often, and Allee and Peinhardt (2011) show that being sued, and losing, costs a host state investment. On the new institutions, Wang (2017) gives an early assessment of the AIIB and the NDB as complements to and potential competitors of the established banks; Ikenberry and Lim (2017) read the AIIB as Chinese institutional statecraft; Humphrey (2022) places both in a broader account of multilateral development banks under geopolitical change. On Chinese official finance, Dreher et al. (2021) build the project-level dataset and estimate its growth effects, Dreher et al. (2022) extend the analysis of its aims and impacts, Gelpern et al. (2023) open its contracts, and Horn, Reinhart, and Trebesch (2021) document the scale of lending absent from official debt statistics.

5.12 What the evidence does not license

Suppose the pattern the published research reports is right, whatever the simple re-estimate above fails to show: aid flat before a term, higher during it, back down afterwards. That shows, about as cleanly as observational data can, influence being priced inside an international institution. It shows less than it seems to.

It is evidence about **one** institution, **one** donor, and the decades the data cover. It does not show that the World Bank, MIGA, the AIIB or the NDB sell their decisions; the follow-up literature makes a case for some of them, with its own designs and its own limits. It does not tell a firm how a particular host government on the Council will vote, or whether a sanctions resolution will survive the next change of government in a permanent member. Nor does it show that the rules a firm relies on are hollow — only that they rest on a coalition.

What the evidence licenses is a change in where the analyst looks. This chapter opened by asking whether a firm that relies on an international institution is relying on a rule or on the coalition of states that keeps the rule in force. The evidence says it always relies on both, and that the coalition is the part that moves: quotas lag, seats rotate, members join new clubs and leave old forums, and the largest shareholders exercise informal influence when their interests are engaged. A firm cannot know in advance whether it will meet Keohane's institution or Mearsheimer's. It can know who is in the coalition behind each guarantee, loan and treaty it holds, what that coalition is being offered elsewhere, and how long the offer is likely to last.

Chapter 6

Private Authority, Standards and Global Regulation: Who Writes the Rules Firms Obey

6.1 Count the rules in a telephone

Pick up a mobile phone and ask how many rules it obeys. Its radio speaks a language specified by 3GPP, a partnership of standards organisations in which the engineers of competing telecommunications firms draft the technical specifications for each generation of mobile networks. Its wireless connection follows a family of specifications developed under the IEEE, a professional association of engineers. Its charging port follows a specification written by an industry forum. The factory that assembled it is probably certified to ISO 9001 for quality management and ISO 14001 for environmental management; its data centre partners to ISO/IEC 27001 for information security. The company that sold it reports its emissions under the Greenhouse Gas Protocol, a standard developed by a research institute and a business council, and its suppliers of tin, tantalum, tungsten and gold carry out due diligence on conflict minerals under frameworks developed by an intergovernmental organisation and audited by private firms.

Now ask how many of those rules were written by a government. Very few, and in most cases none. Some have since been made compulsory or referenced in procurement and trade law, but they were drafted and revised by engineers, auditors, industry associations and non-governmental organisations. The multinational that sells the phone spends far more of its compliance effort on these rules than on any treaty.

The quantity of such rules can be counted, at least for the largest body that writes them. ISO publishes its catalogue as open data: every document, with its type, edition and date (International Organization for Standardization 2026). From 1972, the first year the catalogue records International Standards as such, to 1979, ISO published about 436 a year; between 2020 and 2025 it published about 1,190 — nearly three times as many (Figure 6.1). Some 21,500 are in force, before counting ISO's other documents or the output of the IEC, the ITU and the industry consortia.

The composition has changed as much as the volume. In the 1990s a third of the standards ISO published (32 per cent) were new editions of existing ones; since 2020, 53 per cent. Both management-system standards discussed below were revised again in 2026. Compliance is a subscription, not a state, and whoever sits on the committee doing the rewriting holds the rules continuously — which is why the question of who sits there matters.

Chapter 5 looked at intergovernmental institutions — the organisations states build and in which states vote. This chapter is about the other half of global governance, the rules that come from somewhere other than states, and about why that half is increasingly geopolitical.

Private authority — the capacity of non-state actors (firms, industry associations, standards bodies, non-governmental organisations) to make rules that others treat as binding, even though no state has enacted them and no court may enforce them directly.

For a multinational, private authority is two things at once. It is a **risk**: a standard written elsewhere, by actors the firm does not control, can become a condition of access to a market. And it is a **lever**: a firm that shapes a standard shapes the market the standard governs, and may write its own technology, its own processes and its own costs into the rules its competitors must follow. The question this chapter asks is therefore twofold: who writes the rules a multinational actually follows, and with whose power — and does trade spread those rules up or down?

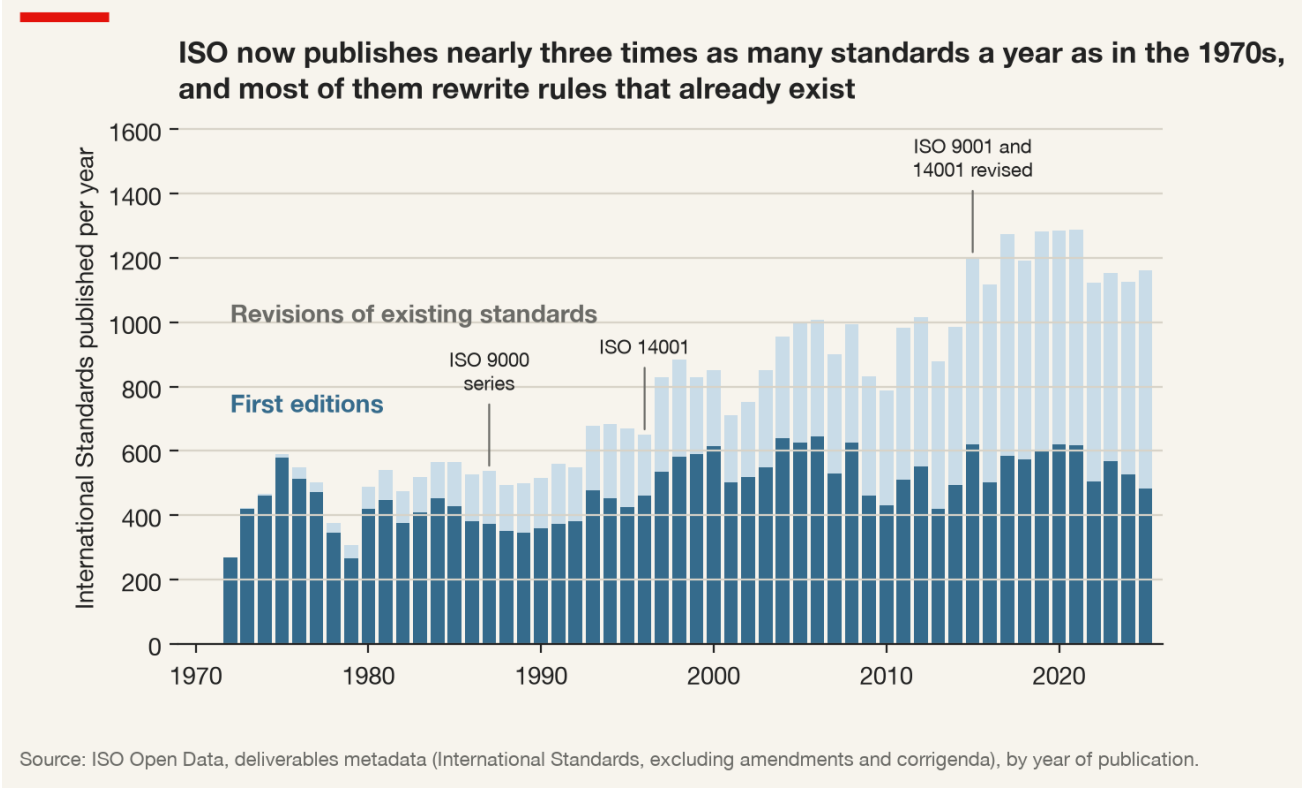


Figure 6.1: ISO publishes nearly three times as many International Standards a year as in the 1970s, and since 2020 more than half have been revisions. International Standards by year of publication, 1972–2025; amendments and corrigenda excluded. Source: ISO Open Data, deliverables metadata.

6.2 Where private authority comes from

For most of the twentieth century, private authority in international affairs was close to a contradiction in terms: international relations studied states, and rule-making was what states did. The concept entered the field in the 1990s, as globalisation multiplied the transactions that crossed borders faster than states could agree on rules to govern them.

Susan Strange made the argument most sharply. In *The Retreat of the State* she contended that markets, firms and a variety of non-state actors had come to exercise authority over outcomes — prices, investment, technology, credit — that had once been the preserve of governments, and that the study of international political economy had to follow power wherever it actually lay (Strange 1996). Cutler, Haufler, and Porter (1999) gave the concept its name in an edited volume on private authority in international affairs, focusing on the inter-firm arrangements — industry norms, business associations, private regimes — through which firms govern their own transactions. Hall and Biersteker (2002) extended it to a broader set of actors, from credit-rating agencies to transnational NGOs and even illicit networks, and asked a question that has never been fully settled: what makes private authority *authority*, rather than simply market power or persuasion?

The usual answer is **legitimacy combined with consequences**. A rule has authority when those it governs accept that it ought to be followed, and when failing to follow it has costs. Private rules acquire those costs through markets rather than courts. A supplier that is not certified loses the contract; a forest company whose timber is not certified loses access to the retailers whose customers demand certification; a firm whose emissions report does not follow the recognised protocol loses credibility with investors.

"Voluntary" does not mean "optional". A standard required by a firm's buyers, insurers or investors is, for that firm, as compulsory as a law — and sometimes more so, because a law can be challenged in court and a lost customer cannot.

6.3 The governance triangle

If rules can come from states, firms and NGOs, it helps to have a map. Kenneth Abbott and Duncan Snidal provide one (Abbott and Snidal 2009). They study what they call regulatory standards-setting schemes — arrangements that write standards for business conduct, monitor compliance, and sometimes certify it — and place each scheme in a triangle whose three corners are the three types of actor that can write rules.

Regulatory standards-setting (RSS) scheme — in Abbott and Snidal's terms, an arrangement through which states, firms, NGOs or combinations of them set standards for the conduct of firms, and often monitor or certify compliance. Its position in the governance triangle is given by the share each of the three actor types has in its governance.

Schemes near a **corner** are run by one type of actor. An industry code of conduct, drafted by firms for firms, sits near the firm corner. A certification scheme created by an activist organisation sits near the NGO corner. A traditional intergovernmental rule sits near the state corner. Schemes in the **interior** are hybrids. The Forest Stewardship Council, founded in 1993, brings together environmental organisations, social organisations and firms in a multi-stakeholder structure, and sits on the edge between NGOs and firms. The UN Global Compact, launched in 2000 by the UN Secretary-General as a set of principles to which firms commit, brings states, firms and civil society together and is often placed near the centre. ISO itself is harder to place: its members are national standards bodies, some of them government agencies and many of them private associations whose committees are staffed largely by industry experts — which is why the placement of ISO standards is a matter of argument rather than of fact.

The triangle's most important feature is not on the diagram. Abbott and Snidal stress the **shadow of the state**: even schemes run entirely by private actors are shaped by the possibility that states will regulate if private actors do not, and by states' decisions to endorse, require, reference or ignore private schemes. Firms write codes of conduct partly to pre-empt legislation; NGOs create certification schemes partly because they have despaired of getting legislation; and states frequently pick up private standards and give them legal force, at which point the scheme moves across the triangle without a word of its text changing.

From the archive · Brussels, 2022

For years, mobile phones in Europe were sold with a variety of incompatible chargers. The European Commission encouraged manufacturers to converge voluntarily, and a memorandum of understanding among major producers in 2009 led much

of the market towards a common connector — but not all of it, and not permanently, as connectors evolved and one large manufacturer kept its own.

Meanwhile an industry body, the USB Implementers Forum, had written the specification for a new reversible connector, USB Type-C. It was a private standard in the purest sense: drafted by firms, adopted voluntarily, and spreading because device makers found it convenient.

In 2022 the European Union adopted a directive amending its Radio Equipment Directive to require a common charging port (European Parliament and Council of the European Union 2022a). From the end of 2024, new mobile phones and a range of other portable devices sold in the EU had to charge through USB Type-C, with laptops following in 2026. A specification written by an industry forum became, by a single legislative act, a legal requirement for access to a market of several hundred million consumers. Because few manufacturers wish to design one version of a phone for Europe and another for everywhere else, the EU rule had consequences far beyond the EU.

In the governance triangle the specification did not change at all. What changed was its position: it moved from the firm corner towards the state corner, carried by the shadow of the state that had been hanging over it for more than a decade. And it illustrates in one case both of the chapter's main arguments — that private standards can become conditions of market access, and that the jurisdiction which turns them into law extends its rules well beyond its borders.

For a firm, the triangle is useful because a scheme's position predicts its behaviour. A scheme near the firm corner is easier for firms to change, and correspondingly less trusted by NGOs and regulators. A scheme with strong NGO participation is more legitimate in the eyes of activist consumers and harder for any single firm to steer. A scheme that a state has begun to reference in law is a scheme that may become mandatory, and a firm that ignores it is betting against the shadow.

6.4 Who writes an international standard?

The triangle tells us what kinds of actor write rules. It does not tell us whose rules win when actors from different countries meet in the same room. That is the question Walter Mattli and Tim Büthe ask of the two principal international standards bodies for products, ISO and the International Electrotechnical Commission (Mattli and Büthe 2003).

Some institutional background makes the question concrete. ISO was founded in 1947, and its members are national standards bodies — one per country. Standards are drafted in technical committees and their subcommittees and working groups, where experts nominated by the national bodies — overwhelmingly engineers employed by firms — negotiate the text. Each national body takes positions on drafts and votes on them. The process looks consensual and technical, and in one sense it is. But the experts who sit at the table, the national positions they carry, and the timing with which they bring proposals all determine whose existing practice becomes the international standard. A firm whose product already conforms to the standard that emerges has a head start; a firm whose product does not must redesign.

Mattli and Büthe compare three explanations of who wins. On the first, **technological rationality**, the technically best solution wins whoever proposes it — the engineers' view of themselves. On the second, the **primacy of power**, the standard favours the states with the largest markets — the realist's view of everything. Mattli and Büthe argue for a third.

Institutional complementarity — in Mattli and Büthe's account, the fit between a country's domestic standard-setting institutions and those of the international body. A hierarchical, coordinated national system, which speaks with one voice and passes information efficiently between firms and the national delegation, matches the structure of ISO and IEC and wins more influence than a fragmented, decentralised one, regardless of market size.

Their evidence pointed to the advantage of the European national standards bodies, which are hierarchical and coordinated, over the fragmented, decentralised system of the United States, in which many private standards developers compete and no single body aggregates national preferences. The result was striking because it cut against the realist expectation: the country with the largest market was not the most effective at shaping international product standards. Influence over "technical" rules depends on how a country and its firms are organised to take part, not only on how big they are. Büthe and Mattli (2011) later generalised the argument to accounting standards as well as product standards, showing that the same logic of institutional fit shapes who wins in the private bodies that write rules for global financial reporting.

The lesson for a firm is practical and somewhat uncomfortable. Participation is costly — it takes the time of senior engineers over years — and its benefits are diffuse and delayed. It is easy to conclude that someone else will look after the standard. But if a competitor's national standards body is better organised than one's own, and that competitor's engineers are in the

room, the standard will reflect their technology, and the firm that stayed away will pay to comply with it. In standards, absence is a decision.

6.5 Counting the rule-writers

Influence in a standards committee is hard to observe, but one part of it is visible. Every ISO technical committee and subcommittee is administered by a **secretariat**: a national member body that volunteers to run it, supplies the committee manager, circulates drafts and organises ballots. It gains no extra vote — ISO's rule is one member, one vote — but it holds the pen, the calendar and often the first draft. And it pays: a body that holds many secretariats has shown that it, or the industry behind it, will fund staff and meetings year after year, which is exactly the capacity the institutional-complementarity argument says wins influence.

ISO's open data list, for each of its 787 active committees and subcommittees, the secretariat, the date of creation and the **participating members** — those committed to vote on every draft, as distinct from observers (International Organization for Standardization 2026). Secretariats are strikingly concentrated (Figure 6.2). Six bodies — Germany's DIN, China's SAC, the American ANSI, France's AFNOR, Japan's JISC and the British BSI — hold 71 per cent. Germany alone holds 127. European members, counting the EU, EFTA and the United Kingdom, hold 51 per cent, from about 22 per cent of world GDP: the institutional-complementarity argument in a single comparison.

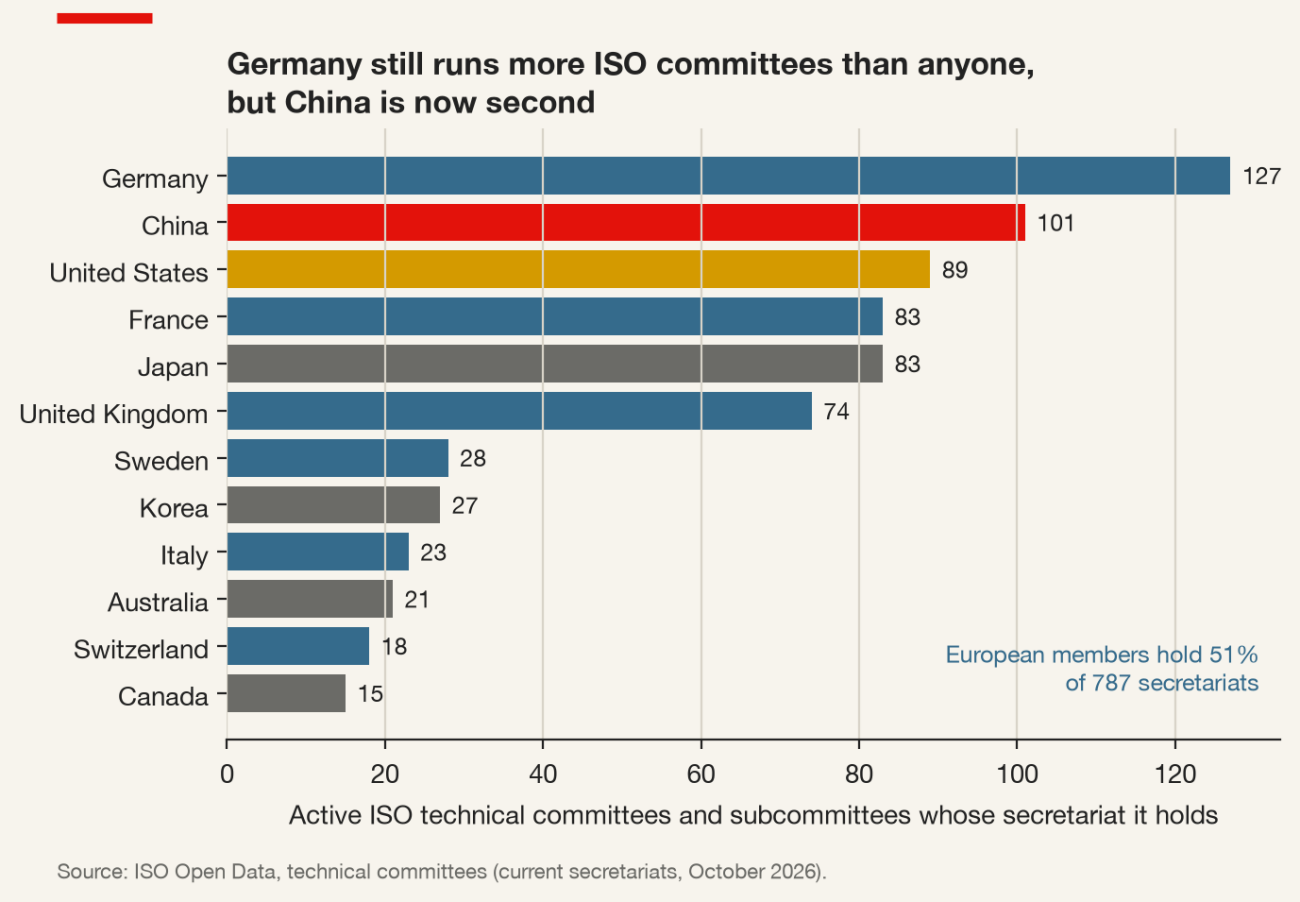


Figure 6.2: Germany still administers more ISO committees than any other country, but China is now second; European member bodies hold half of all secretariats. Active ISO technical committees and subcommittees by secretariat, October 2026. Source: ISO Open Data, technical committees.

Table 6.1 separates participation, which is cheap, from administration, which is not. Korea, India and Russia take part in almost as many committees as Germany but hold 7 to 27 secretariats. The United States, with a quarter of world GDP,

holds fewer than Germany, whose economy is a sixth of its size — the pattern Mattli and Büthe described, still intact: America's many competing standards developers convert economic weight into ISO leadership less efficiently than Europe's centralised bodies.

Table 6.1: Who runs ISO's 787 active committees and subcommittees, who takes part, and how large their economies are, October 2026; GDP share of the economies reporting to the World Bank. Source: ISO Open Data; World Bank, World Development Indicators.

Member body	Country	Secretariats	Share of secretariats (%)	Participating memberships	Share of world GDP, 2024 (%)
DIN	Germany	127	16.1	714	4.2
SAC	China	101	12.8	777	16.9
ANSI	United States	89	11.3	564	26.5
AFNOR	France	83	10.5	652	2.9
JISC	Japan	83	10.5	666	3.8
BSI	United Kingdom	74	9.4	692	3.3
KATS	Korea	27	3.4	612	1.7
BIS	India	14	1.8	593	3.4
GOST R	Russia	7	0.9	643	2.0
ABNT	Brazil	3	0.4	261	2.0

The member whose position has changed is China. SAC is a participating member of 777 committees, more than any other body, and holds 101 secretariats, more than the United States. The data record only today's holder, not the history of transfers, but they record when each committee was created, and holders by vintage tell the story almost as well (Figure 6.3). Of the committees created between 1970 and 2009, China holds 7 per cent; of the 195 created since 2010, 26 per cent — more than the United States and Japan together.

The committees themselves are more revealing. China holds the new committees on rare earths (created in 2015) and lithium (2020), the raw materials at the centre of the geopolitics of magnets and batteries, as well as biogas, coalbed methane and e-commerce. It also holds 17 of ISO's oldest committees, created between 1947 and 1969 — screw threads, ships, textiles, plastics, cranes — taken over as earlier holders withdrew. One pattern stakes out fields where Chinese industry is strong and the rules are unwritten; the other takes over the administration of mature industrial standards. Both are what institutional capacity looks like when a government funds it.

Three cautions apply. ISO is one body among several: secretariats there say nothing about the IEC, the ITU or 3GPP, where the strategic stakes in telecommunications are highest. A subcommittee on chains and sprockets counts the same as one on information security. And administering a committee is not winning its arguments: Rühlig (2023) argues that counts of secretariats and proposals overstate Chinese influence, which depends on the quality of contributions and the bodies' rules, and that the visible hand of the state in Chinese delegations breeds suspicion as well as strength.

6.6 Standards as geopolitics

For most of the post-war period, the politics of standards was the politics of firms and of national standards bodies, and it rarely reached foreign ministries. That has changed. The competition over telecommunications standards, over battery and charging standards, over the rules for artificial intelligence and for the flow of data, has become a competition between blocs, each trying to make its domestic rules the world's.

Standards confer advantages that outlast the meeting at which they were agreed. A standard that incorporates patented technology generates licence revenue for as long as it is used; one that matches a country's industrial practice lowers its firms' costs and raises its competitors'; one that embeds choices about data access or encryption shapes what governments can see. And standards are sticky, so early influence compounds. When governments came to see technical standards as a component of technological leadership, and technological leadership as a component of national power, standard-setting acquired the stakes of geopolitics — and the bodies in which standards are written, which had been designed for engineers, found themselves hosting a contest between states.

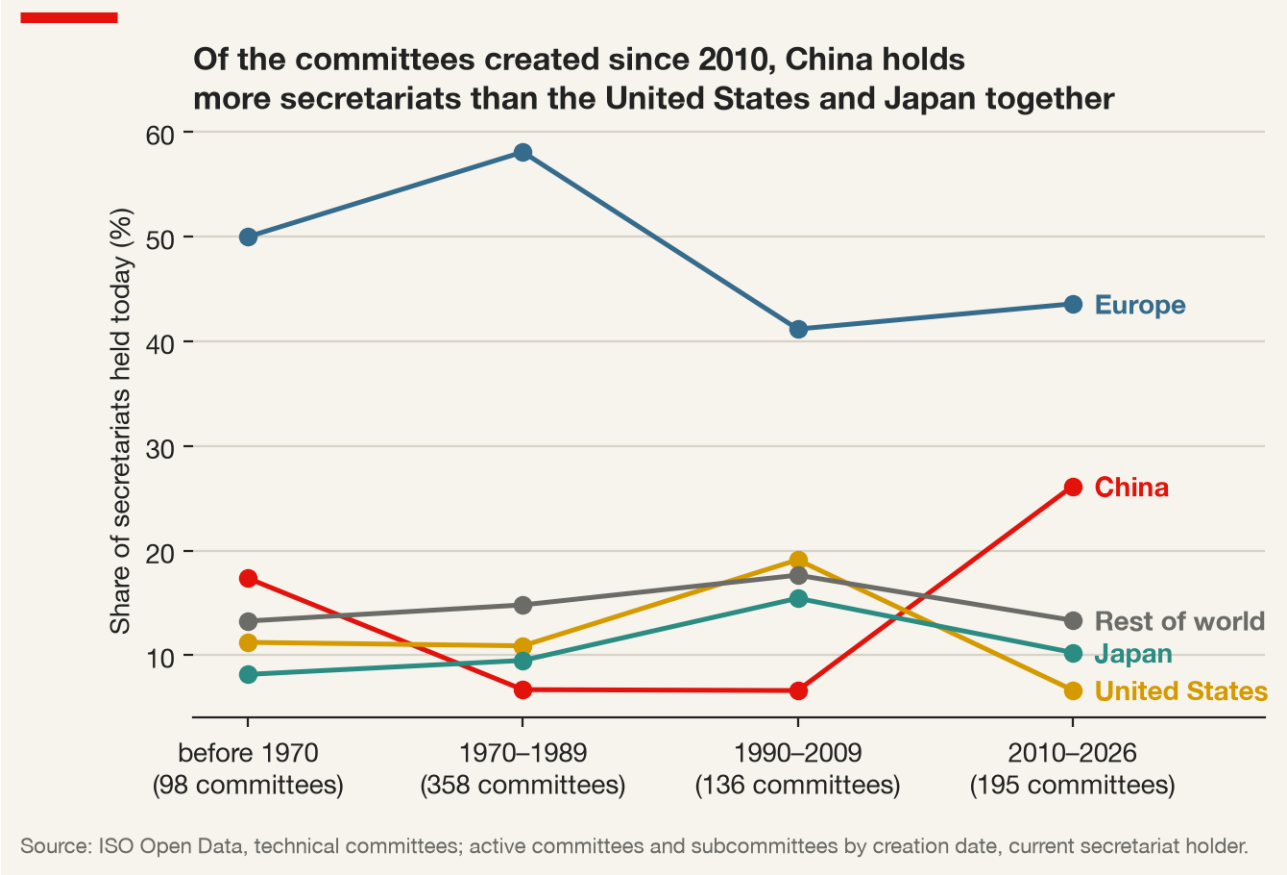


Figure 6.3: Of the ISO committees created since 2010, China holds more secretariats than the United States and Japan together. Share of active committees, by period of creation, whose secretariat each holds today. Source: ISO Open Data, technical committees.

The European Union's version of this contest has been described most fully by Anu Bradford as the **Brussels effect** (Bradford 2020). The EU rarely sets out to impose its rules on the world. It regulates its own market. But when a large market sets a strict rule, firms that sell in it face a choice: produce two versions of their product, one for the strict market and one for everyone else, or produce one version that meets the strictest rule and sell it everywhere. Where production cannot easily be divided — because of economies of scale, because the product is a single global platform, or because maintaining two compliance systems costs more than over-complying in lax markets — firms adopt the strict rule globally, and then often lobby their home governments to adopt it too, so that their less global competitors bear the same costs. Bradford identifies the conditions under which the effect operates: a large market, the regulatory capacity to write and enforce rules, a preference for strict rules, regulation of targets that cannot easily leave the market (consumers rather than mobile capital), and non-divisibility of production. Data protection, chemical safety, competition law and product standards are her main cases; the common charger is a small, clean example.

The Brussels effect is a mechanism of *unilateral* regulatory globalisation, but it shares a logic with private standards. In both cases, rules travel along the channels of trade, carried by firms. That shared logic is precisely what the evidence later in this chapter tests.

The contest has since become explicit. In *Digital Empires* Bradford extends the argument from a single regulator to a rivalry among three (Bradford 2023): an American model that leaves the governance of technology largely to markets and firms, a Chinese model in which the state directs technology to serve its own goals, and a European model that regulates in the name of individual rights. Each jurisdiction seeks to export its model, through its firms and its market, and the battles run vertically — between governments and the technology companies they regulate — as well as horizontally, between the jurisdictions themselves. The European Commission's 2022 strategy on standardisation states the stakes in the language of geopolitics: the geopolitical landscape, it observes, "has significantly shifted in recent years", and other actors "follow a much more assertive approach to international standardisation than the EU and have gained influence in international standardisation committees" (European Commission 2022). For a firm, the change is that the technical committee where its engineers sit has become a place where its home state's interests are being pursued, whether or not the firm shares them.

The three jurisdictions have each written the contest into policy. The Chinese Communist Party's Central Committee and the State Council issued a National Standardization Development Outline in October 2021 (Central Committee of the Communist Party of China and State Council 2021), with targets for Chinese participation in the leadership of international bodies. The European Union amended its standardisation regulation in 2022 so that decisions on European standards supporting EU law are taken only by the national standards bodies of the EU and the European Economic Area, removing the votes that non-European firms held in ETSI, the telecommunications standards institute (European Parliament and Council of the European Union 2022b). The United States published its first national standards strategy for critical and emerging technology in 2023 (The White House 2023). None of these writes a technical standard. All are about who will be in the room — the variable Mattli and Büthe identified, now pursued deliberately by states.

6.7 Of, by, as: what standards do to organisations

Standards are usually associated with stability and sameness: a standard exists so that things will be the same everywhere. Nils Brunsson, Andreas Rasche and David Seidl argue that this view misses most of what is interesting about them (Brunsson, Rasche, and Seidl 2012). They organise research on standards around three perspectives.

The first is **standardisation of organisations**: standards that tell firms how to be organised. ISO 9001 and ISO 14001 are the archetypes. They do not specify what a product must be like or how much pollution a plant may emit. They specify that a firm must have a management *system* — documented procedures, objectives, responsibilities, internal audits, management reviews and a commitment to continual improvement. The second is **standardisation by organisations**: the bodies that write standards, how they work, and how they compete with each other for adopters. The third is **standardisation as organisation**: standards as a way of organising markets and fields without formal hierarchy, coordinating the behaviour of thousands of firms that have no contract with each other and no common owner.

Their central point is that standards are **dynamic**. They are revised, contested, interpreted differently in different places, and — most important for anyone using certification as evidence — **decoupled** from practice.

Decoupling — the separation between the formal structures an organisation adopts to signal conformity to external expectations and the practices it actually follows. A firm certified to a management standard may change its documents and its audits while leaving its operations largely as they were.

The idea is older than the study of standards. Meyer and Rowan (1977) argued that organisations adopt formal structures because those structures are regarded as legitimate in their environment, not necessarily because they make the organisation more efficient, and that organisations protect themselves by decoupling those structures from the technical core of their work. A certificate can be a ceremony.

Whether ISO 14001 certificates are mostly ceremony is an empirical question, and the evidence is mixed in an informative way. Potoski and Prakash (2005) found that US facilities certified to ISO 14001 reduced their pollutant emissions somewhat more than comparable uncertified facilities — the standard's "weak sword", in their phrase, could still cut, because third-party audits and the reputational stakes of certification gave it some force. Christmann and Taylor (2006), studying firms in China, found that whether certification was implemented substantively or symbolically depended on the firm's circumstances, and in particular on whether its customers monitored what it actually did. Certification backed by demanding buyers was more likely to change practice; certification sought only for its signal was more likely to be decoupled. The implication for a multinational is direct. If it relies on its suppliers' certificates as evidence of environmental performance, the certificate's value depends on whether anyone — the multinational included — looks behind it.

6.8 Agents and entrepreneurs

Private authority does not all arise in the same way, and its origin matters for how durable it is. Jessica Green draws the key distinction (Green 2013).

Delegated versus entrepreneurial private authority — *delegated* authority exists when states hand rule-making or implementation to private actors, who act as their agents; *entrepreneurial* authority exists when private actors create rules on their own initiative and others choose to adopt them.

Green's field is carbon governance, and it supplies clean illustrations of both. Under the Clean Development Mechanism of the Kyoto Protocol, private bodies were accredited to validate and verify emission-reduction projects: they exercised authority that states had delegated to them, within rules states had written. The Greenhouse Gas Protocol, by contrast, was created by a research institute and a business council without any state mandate, and became the dominant framework for corporate carbon accounting because firms, investors and eventually regulators adopted it.

The distinction tells a firm what to watch. Delegated authority is as durable as the delegation: the state that handed it over can take it back, rewrite it or abolish the programme, and when the Kyoto mechanisms gave way to new arrangements under the Paris Agreement, the authority of the bodies that served them changed with them. Entrepreneurial authority is as strong as the number of its users. It grows through adoption and network effects, and it can be displaced by a competing standard that attracts more adopters — or captured by a state that writes it into law, at which point it becomes something closer to delegated authority after the fact.

Benjamin Cashore's analysis of forest certification shows how entrepreneurial schemes gain the authority to make rules (Cashore 2002). He calls them non-state market-driven governance systems: they have no state backing, and their authority derives from the market — from retailers and consumers who demand certified products, and from the firms in the supply chain who must therefore decide whether to comply. The Forest Stewardship Council, created by environmental groups with stricter rules, was soon met by industry-backed schemes with laxer ones, and the competition between them became a contest over which would be accepted as legitimate by the buyers whose purchases conferred authority. Private authority, on this view, is not given; it is won, one supply chain at a time, and it can be lost the same way.

6.9 Race to the bottom, or trading up?

With the concepts in place, the chapter's central question can be stated precisely. Trade intensifies competition among producers in different countries. Does that competition push standards down or up?

The worry is the **race to the bottom**. If firms can produce anywhere and sell everywhere, governments compete for investment and exports, and one way to compete is to lower the costs that regulation imposes. Environmental standards are an obvious candidate: lax rules make production cheaper, and countries that compete hard for exports may tolerate pollution that richer or less trade-dependent countries would not. The argument predicts that greater openness to trade will be associated with weaker environmental governance.

The alternative was named by David Vogel (Vogel 1995).

California effect — the tendency of trade to spread *stricter* rules from strict to less strict jurisdictions. When a large, rich market imposes demanding standards on the products sold in it, producers elsewhere who wish to sell there adopt those standards, and may then press their own governments to adopt them too. Trade becomes a mechanism of "trading up".

The name comes from American environmental history. California, with severe air-quality problems and a large car market, set vehicle emission standards stricter than federal ones; manufacturers that wanted to sell in California built cars to meet them, and other states followed. Vogel contrasted it with the **Delaware effect** — the race to the bottom in corporate chartering law, in which American states competed to attract company registrations by offering laxer rules, and Delaware won. The two effects are not contradictory. They describe different regulatory targets. Where the rule governs a *product* that must cross a border to be sold, the importing market can impose it, and trade tends to spread strict rules upward. Where the rule governs a *process* or a *legal form* that can be relocated, jurisdictions compete for the mobile target, and trade can push rules down.

Environmental management standards sit in an interesting middle position. ISO 14001 governs a process — how a plant manages its environmental impacts — and not a product. No importing government can require it at the border. But buyers can, and that is where the question becomes empirical.

From the archive · Rio and Geneva, 1992–1996

ISO 14001 was born of a particular moment. In the run-up to the UN Conference on Environment and Development in Rio de Janeiro in 1992, business leaders organised to present a constructive business response to global environmental concerns, and international standardisation was one of the avenues they explored. ISO and the IEC set up an advisory group on the environment to consider whether international standards for environmental management were needed.

They were not starting from nothing. The British Standards Institution had published BS 7750, a national standard for environmental management systems, in 1992, building on the management-system model that the ISO 9000 quality standards, first published in 1987, had made familiar to firms worldwide. In 1993 the European Union adopted its own Eco-Management and Audit Scheme. ISO created a new technical committee for environmental management, TC 207, in 1993, and in 1996 it published ISO 14001.

The standard reflected its parentage. Like ISO 9001, it specified a management system rather than a performance level: a certified firm commits to comply with applicable law and to improve continually, but the standard sets no emission limit of its own. That design made it adoptable anywhere, by firms in any sector and any regulatory environment — and it made it, from the first, vulnerable to the charge that certification and performance could diverge. It was revised in 2004 and again in 2015, and it became the most widely adopted environmental management standard in the world.

The episode illustrates Mattli and Büthe's argument in miniature. The international standard grew out of a European national standard and a European regulatory scheme, drafted in a committee where coordinated national bodies with a working model to propose were well placed to shape it. Firms in countries with that model already in place adopted the international standard more easily — a pattern documented by Delmas (2002), who traced how differences in institutional environments made adoption faster in Europe than in the United States.

Aseem Prakash and Matthew Potoski put the race-to-the-bottom hypothesis to a direct test (Prakash and Potoski 2006a). ISO 14001 is a useful instrument because it goes beyond what many domestic laws require, is adopted voluntarily, and is counted, country by country, every year. Using a panel of 108 countries over seven years, they asked whether a country's adoption of ISO 14001 depends on adoption in its **export markets** — the countries to which it sells.

Their finding is that trade linkages **encourage** ISO 14001 adoption **when a country's major export markets have themselves adopted it**. Exporters to markets where certification is widespread certify more; exporters to markets where it is rare do not. Trade, on this evidence, does not drive environmental governance down. It transmits the standards of customers to their suppliers — a California effect operating through private rather than public regulation, with buyers rather than governments doing the imposing. The argument was developed at book length in Prakash and Potoski (2006b), which treats voluntary programmes like ISO 14001 as "green clubs" whose value to members depends on the credibility of the club's standards and the visibility of membership to outsiders.

The mechanism is consistent with earlier work on the diffusion of quality certification. Guler, Guillén, and Macpherson (2002) showed that ISO 9000 certificates spread across countries through trade and investment ties with countries where certification was already common, alongside the influence of states and of imitation among firms. Christmann and Taylor

(2001) found, among firms in China, that environmental self-regulation was more likely where firms were linked to multinational owners or exported to developed-country customers. Across these studies the picture is consistent: private standards travel along supply chains, from buyer to supplier, and the supply chains themselves determine where they go.

Gereffi, Humphrey, and Sturgeon (2005) explain why: where buyer and supplier exchange complex information that is hard to codify, value chains become relational or captive, and lead firms specify not only the product but how it is made — exactly the chains through which process standards such as ISO 14001 travel.

For the firm, the conclusion is one of the most practical in this book. A private standard can become a de facto condition of access to a market without a single law changing. Monitoring "regulatory" risk therefore means monitoring not only what governments legislate but what one's customers' home markets adopt.

6.10 How a certificate becomes evidence

The classic source on certification is **the ISO Survey of Management System Standard Certifications**, which counts valid certificates by country and year. It counts *certificates*, not firms or sites; its coverage of certification bodies has changed across editions, so that a jump in one country in one year may reflect reporting rather than adoption; and from 2025 it is compiled from the IAF CertSearch database, a change of method that breaks the series where a reader would most like to extend it. A count is also not an adoption rate: a large economy has more certificates because it has more firms. Adoption must be **normalised** — by GDP, by the number of firms or by exports — and a finding that holds under only one denominator is a finding about the denominator.

The trade side requires a measure of **export-market adoption**: a weighted average of partners' adoption rates, each partner weighted by its share of the country's exports, built from bilateral trade data such as the IMF's International Trade in Goods by partner country (International Monetary Fund 2026b). The weighting is the substance of the argument. The California effect predicts that what matters is adoption among the markets a country actually sells to, in proportion to how much it sells to them; an unweighted average would give a country's smallest customers as much influence as its largest.

The most instructive check is a **placebo**. ISO 9001 is demanded by buyers everywhere and has no environmental content. If its adoption tracks export-market adoption just as closely as ISO 14001 does, the pull may have little to do with environmental governance and much to do with supply chains demanding certificates in general. The finding of Guler, Guillén, and Macpherson (2002), that ISO 9000 certificates spread along trade ties, suggests the placebo would not come back empty. That would change what the finding means — from "trade spreads environmental governance" to "trade spreads certification".

6.11 A newer club: science-based climate targets

The trading-up argument can be checked with a younger private standard whose register is public firm by firm, and which is the kind of rule multinationals now ask their suppliers to meet. The Science Based Targets initiative was launched in 2015 by CDP, the UN Global Compact, the World Resources Institute and WWF to assess whether companies' emission-reduction targets are consistent with the goals of the Paris Agreement. A company submits a target and the initiative validates it, or does not, against criteria it writes itself. In Green's terms its authority is purely entrepreneurial; like ISO 14001 it governs a process rather than a product; unlike the ISO Survey, its register lists every company, its country and the date of each target (Science Based Targets initiative 2026).

The register shows a private standard's diffusion with unusual clarity (Figure 6.4). By October 2026, 12,162 companies had validated near-term targets; only 14 of them have a current target published before 2019. The take-off came in 2021–2023, and Europe led it, with 47 per cent of validated companies. But the largest national group is Japanese — 2,582 companies, more than the United Kingdom (1,569) or the United States (1,170) — and East Asia accounts for 31 per cent against North America's 11. More than half the validated companies (55 per cent) are small and medium-sized enterprises, many of them suppliers whose targets were requested by their customers: the California effect in its private form.

The test is Prakash and Potoski's. A country's own adoption is its number of validated companies per US$100 billion of GDP; its export-market adoption is the same quantity for its trading partners, weighted by their shares of its 2024 merchandise

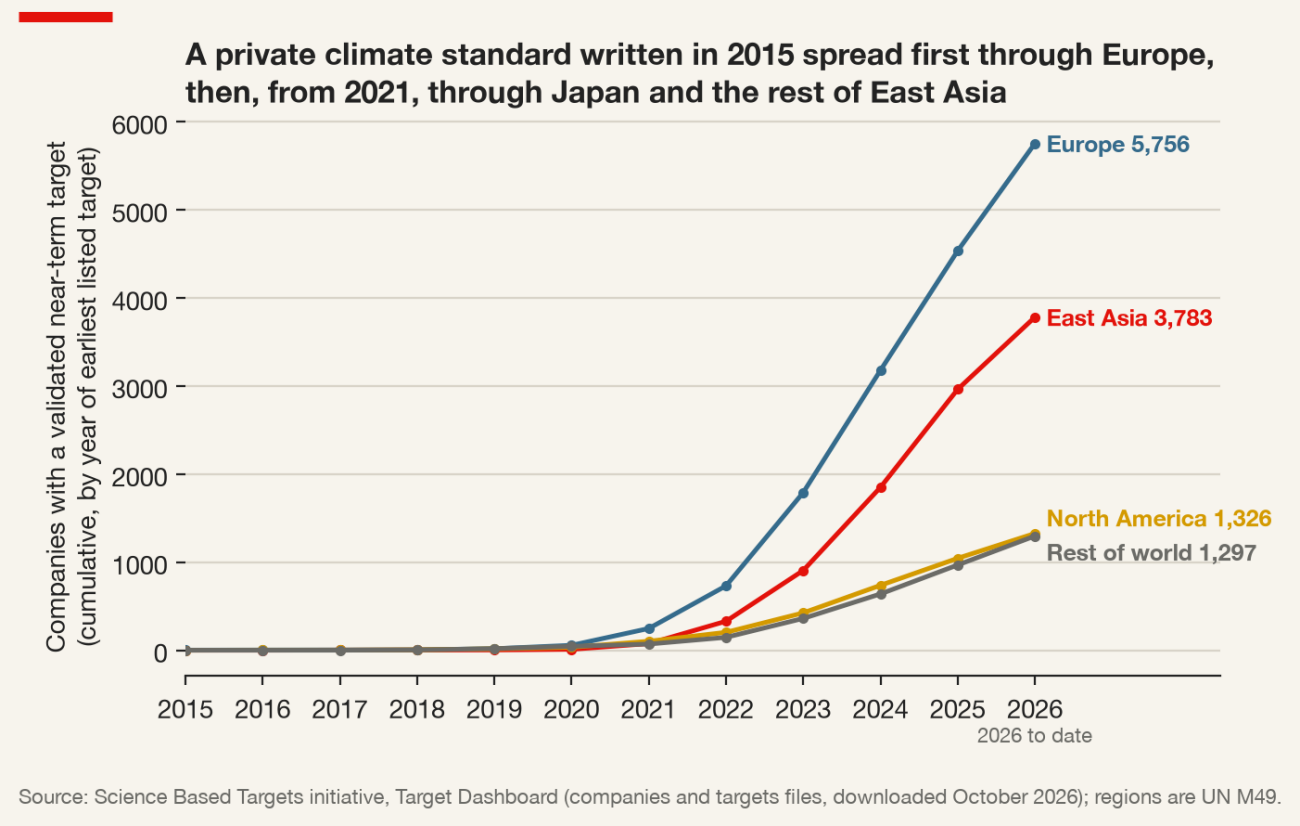


Figure 6.4: A private climate standard written in 2015 spread first through Europe, then, from 2021, through Japan and the rest of East Asia. Companies with a validated near-term emissions target, cumulative by the year of the earliest near-term target currently listed, by UN region; 2026 is to October. Source: Science Based Targets initiative, Target Dashboard.

exports (International Monetary Fund 2026b; World Bank 2026). The sample is the 91 economies with GDP of at least US$50 billion.

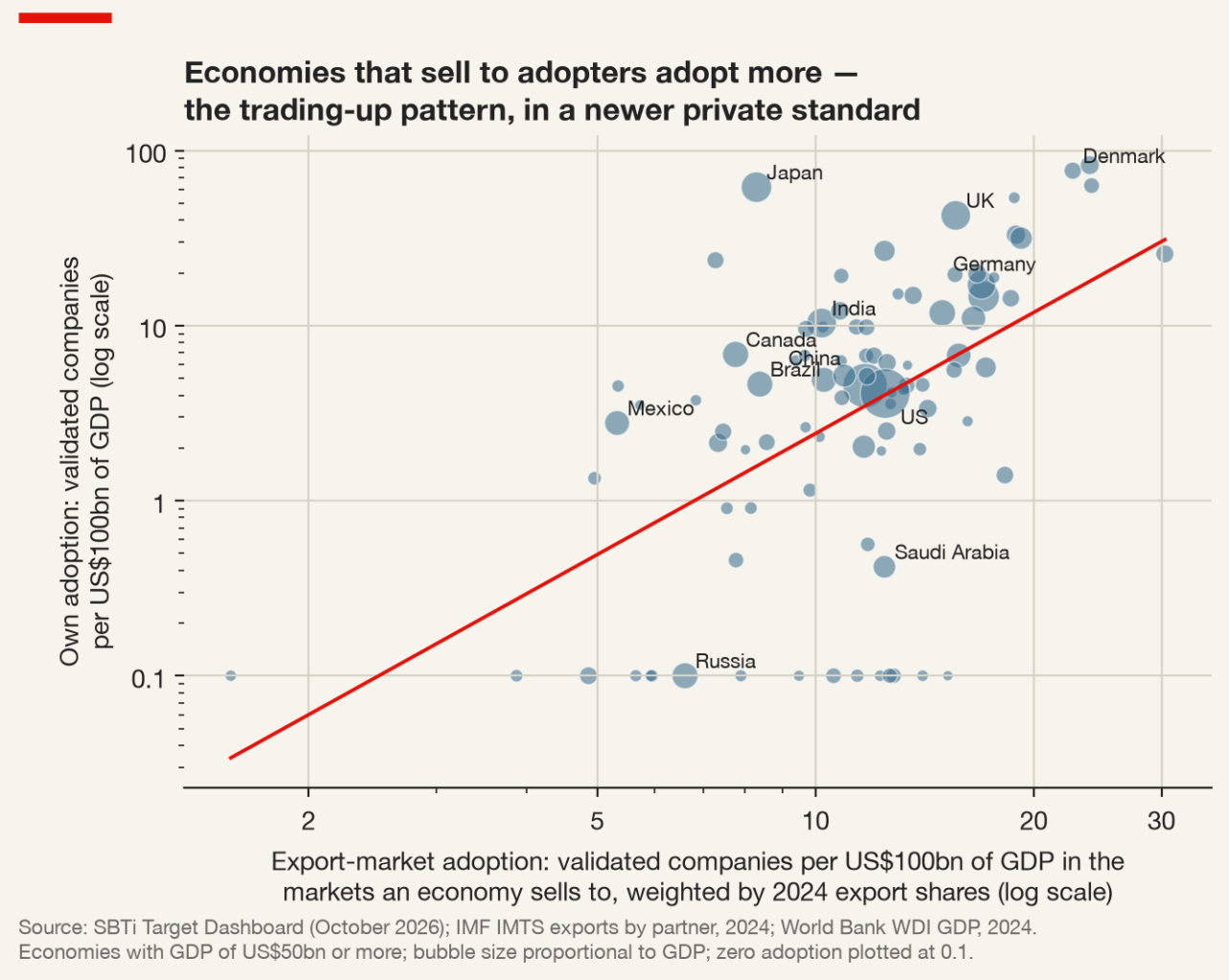


Figure 6.5: Economies whose export markets have adopted science-based targets adopt more themselves. Own and export-market adoption as validated companies per US$100 billion of GDP, log scales; economies with GDP of US$50 billion or more, bubble size proportional to GDP. Source: SBTi Target Dashboard; IMF, International Trade in Goods; World Bank, World Development Indicators.

The pattern is the one the trading-up argument predicts (Figure 6.5); the correlation in logarithms is 0.53. Denmark, with 77 validated companies per US$100 billion of GDP, sells to markets averaging 23; Russia, whose exports go mainly to markets with below-average adoption, has none. Table 6.2 confronts the relationship with its most obvious rival, income. Taken alone, a doubling of export-market adoption is associated with own adoption about five times higher (an elasticity of 2.3), and income per head alone explains slightly more of the variation. With both in the regression the export-market coefficient falls to 1.6 but remains more than four standard errors from zero.

Table 6.2: Export-market adoption and own adoption of science-based targets, OLS. Dependent variable: log of (validated companies per US$100 billion of GDP + 0.1); standard errors in parentheses. Source: SBTi Target Dashboard; IMF; World Bank; author's calculations.

	(1) Export markets	(2) Income	(3) Both
Log export-market adoption	2.30 (0.39)		1.64 (0.36)
Log GDP per head, 2024		0.83 (0.13)	0.64 (0.12)

	(1) Export markets	(2) Income	(3) Both
R^2	0.28	0.33	0.45
Economies	91	91	91

The departures from the line are as instructive as its slope. Japan has 62 validated companies per US$100 billion of GDP while its export markets average 8: large manufacturers asked their supplier networks to set targets, and government programmes helped smaller firms do so. A coordinated domestic system carried the standard further than trade would have — Mattli and Büthe's institutional argument appearing, unbidden, in a diffusion study. The United States and China sit together at about 4, with export markets averaging about 12: the two largest economies lag their customers, for reasons the data cannot separate. Saudi Arabia, whose customers adopt about as much as America's, has almost none, and sixteen economies in the sample have no validated company at all, most of them exporters of oil, gas, metals or farm commodities. A barrel of oil carries no supplier code of conduct; the California effect needs a buyer who cares how a good was made.

The limits are those set out above, and some are sharper. A cross-section shows association, not cause. The measure counts companies, not emissions. And a validated target is a promise, not a reduction: Bjørn et al. (2023) argue that validation is not transparent enough for outsiders to judge how demanding targets are — decoupling again, at the level of the standard rather than the firm. The initiative's authority has itself been contested: in 2024 a board statement that it would consider letting carbon credits count towards reductions (Science Based Targets initiative 2024) drew public objections, including from within the initiative, a reminder that an entrepreneurial standard's legitimacy rests on its adopters' trust. What the pattern shows is that buyers transmit the requirements they have accepted. A supplier's regulatory future is being written in its customers' markets.

6.12 What private authority means for the multinational, channel by channel

The five channels through which this book traces geopolitical risk all run through private authority, and in each the firm is both a rule-taker and a potential rule-maker.

Market access is the most direct channel. A certificate required by buyers is a condition of entry as real as a tariff, and it is set by the buyers and, behind them, by their home markets. When a large market writes a private standard into law, as the EU did with the common charger, the condition hardens and its reach extends through the Brussels effect. A firm that monitors only government regulation will see these changes late; its future requirements will come from what its customers' home markets adopt, privately and in law.

Inputs are where the multinational becomes the imposer rather than the imposed upon. A firm that requires its suppliers to certify is exercising private authority over them, and the evidence on decoupling says that the value of that requirement depends on whether the firm monitors behind the certificate. A supplier's ISO 14001 certificate is evidence that a management system exists, not that the supplier pollutes less.

Finance increasingly runs on private standards too — carbon accounting, sustainability reporting frameworks, credit ratings — each written by a private or hybrid body, each conditioning access to capital, and several being absorbed into law, which moves them across the triangle and changes who can change them.

Assets and people are exposed through the standards that govern labour and human rights in supply chains. A plant certified under a scheme that later loses legitimacy is an asset whose value in the eyes of customers has fallen overnight.

Data and operations are where standards have become most explicitly geopolitical. Telecommunications standards, information-security standards, rules for artificial intelligence and for cross-border data flows are now contested between blocs. A firm whose products embed one bloc's standards may find them excluded from the other's procurement, and a firm whose engineers sit on the committees writing the next generation of a standard is taking part, whether it likes it or not, in a contest between states.

Across the five channels, the strategic choice has three options. The firm can **adopt** a standard written elsewhere, accepting the cost of compliance as the price of access. It can **shape** the standard, investing in participation in the bodies that write it — the lesson of Mattli and Büthe is that this investment pays only if the firm and its national body are organised to use it. Or it can **resist**, backing a competing standard or lobbying against legal adoption — a bet that the shadow of the state will

not fall. The choice depends on where the scheme sits in the governance triangle, on whether its authority is delegated or entrepreneurial, and on whether the firm's own customers are among those who confer it.

6.13 A short review of the literature

The study of private authority in international political economy begins with Strange (1996), who argued that authority over economic outcomes had shifted from states to markets and firms, and with Cutler, Haufler, and Porter (1999) and Hall and Biersteker (2002), who gave the concept its name and extended it from inter-firm regimes to NGOs, rating agencies and other non-state actors. Abbott and Snidal (2009) organised the resulting landscape into the governance triangle and introduced the shadow of the state; Green (2013) distinguished delegated from entrepreneurial private authority and traced both in carbon governance; Cashore (2002) showed how non-state, market-driven schemes such as forest certification gain the authority to make rules through the purchasing decisions of firms and consumers.

The politics of who writes international standards is the subject of Mattli and Büthe (2003), whose argument for institutional complementarity over technological rationality and raw power was generalised in Büthe and Mattli (2011) to the private regulation of products and financial reporting. Bradford (2020) documented the unilateral mechanism by which strict EU rules become global through firms' preference for a single production standard, and Bradford (2023) extended it to a three-way contest over the regulation of technology; the EU's common-charger directive (European Parliament and Council of the European Union 2022a) is a recent and transparent instance of a private specification hardened into law.

On standards as organisation, Brunsson, Rasche, and Seidl (2012) set out the three perspectives — of, by and as organisation — and the dynamism of standards in practice, drawing on the institutionalist insight of Meyer and Rowan (1977) that formal structure may be decoupled from activity. Whether ISO 14001 changes behaviour is addressed by Potoski and Prakash (2005), who find modest emission reductions among certified US facilities, and by Christmann and Taylor (2006), who show that substantive rather than symbolic implementation depends on monitoring by customers.

On the race to the bottom, Vogel (1995) named the California effect and set it against the Delaware effect. Prakash and Potoski (2006a) provided the cross-national test at the centre of this chapter, finding that trade spreads ISO 14001 from export markets to exporters, and Prakash and Potoski (2006b) developed the theory of voluntary programmes as clubs. The diffusion mechanism is corroborated for quality standards by Guler, Guillén, and Macpherson (2002), for Chinese firms by Christmann and Taylor (2001), and for the differing paths of Europe and the United States by Delmas (2002), and explained by the value-chain governance of Gereffi, Humphrey, and Sturgeon (2005). Rühlig (2023) weighs China's rise in standardisation; Bjørn et al. (2023) raise decoupling for climate targets. The through-line is that private rules travel along supply chains, and that the power to write them lies with those organised to take part.

6.14 What the evidence does not license

Suppose the evidence comes out as Prakash and Potoski's argument predicts: countries whose export markets certify heavily certify more themselves. That is not yet proof that trade spreads environmental governance.

The cross-country pattern is a **correlation** before it is a test. Countries whose export markets certify heavily are richer, more industrialised and more embedded in multinational supply chains than others, and any of those differences could produce the pattern on its own; a panel model controls for some of them, a scatter for none. Certificates are **not performance**: the evidence on decoupling says that a certified plant may or may not pollute less than an uncertified one, and the counts say nothing about which. The data carry the scars of their **collection**, and the measure depends on **choices** of denominator and weighting.

The evidence therefore does not license the conclusion that a certified supplier is environmentally better than an uncertified one, that certification caused a country's export success, that a standard voluntary today will stay voluntary, or that the rules a firm's markets follow were written by anyone the firm can vote for. What it does license is the chapter's central claim about authority. Rules that no legislature enacted travel along supply chains, from buyer to supplier, with the force of market access behind them; the bodies that write them reward those organised to take part; and states, increasingly conscious of the stakes, are moving to capture both the bodies and the rules. A multinational that treats standards as a technical matter for its engineers and its quality department has misread where authority lies. Every time it decides whether to adopt, shape or resist a rule it did not write, it is making a judgement about power.

Part III

From judgement to number

Chapter 7

Measuring Risk: How a Political Judgement Becomes a Number

7.1 The thermometer in the wrong room

Every geopolitical risk score that reaches a board has a pedigree, and the pedigree is usually hidden. A consultancy reports that a country's risk has "risen to 71"; a bank's dashboard turns a cell from amber to red; an index published by a central bank spikes in the week an invasion begins. Each of these numbers is the last link of a chain that started with a concept nobody can observe directly — the probability that politics will damage a firm's business — and passed through a sequence of choices about texts, words, coders, units, weights and scales. The choices were made by someone, for reasons, and most of them could have been made differently.

This chapter is about seeing that chain, so that a score one is sold can be audited and a score one builds can be defended. Two words carry most of the weight, and they are worth fixing at the start.

Validity — the degree to which an indicator measures the concept it claims to measure. Is the share of newspaper articles mentioning war threats a measure of geopolitical *risk*, or of editors' attention to geopolitics, or of the number of journalists posted to a war zone?

Reliability — the degree to which the same procedure, applied by someone else to the same material, yields the same number. Would a second coder, following the same rules, put the same article in the same category? Would the same dictionary, run next year on the same archive, return the same count?

The two are independent in one direction only. A measure can be perfectly reliable and entirely invalid: a thermometer that everyone reads identically, hanging in the wrong room. It cannot be valid without being reasonably reliable, because a measure that returns different numbers for the same object is measuring, in part, the person who applied it. The history of measurement in the social sciences is largely a history of trading one against the other — cheap, mechanical procedures that are perfectly reliable and of uncertain validity, against slow human judgement that captures meaning and disagrees with itself.

7.2 The chain, link by link

Write the chain out once, so that its parts have names. A measure of geopolitical risk passes from a **concept** ("geopolitical risk") to an **indicator** (the share of articles that discuss geopolitical tensions), drawn from a **source** (a fixed set of newspapers), processed by a **coding rule** (a list of words grouped into categories), summarised by an **aggregation** (a monthly share, rescaled so that a base period averages 100) and delivered in a **presentation** (a line with spikes labelled "9/11" and "Iraq invasion"). Every one of the risk measures in this book can be written in that form, and every link is a place where the number can part company with the concept.

At the concept link, the danger is that nobody has said what is being measured. "Political risk" in the business literature has meant expropriation, contract repudiation, currency inconvertibility, civil unrest, regulatory change and, more recently, sanctions and export controls — things with different causes, different probabilities and different remedies. An indicator that pools them may be a reasonable summary of the newspapers and a poor guide to any one of the decisions it is meant to inform.

At the indicator and source links, the danger is selection. Newspapers report what is new and dramatic, in the countries their readers care about, in the language they print in. The ten newspapers behind the Caldara–Iacoviello index are American, British and Canadian; a war in a country with no correspondents may barely register. That is a property of the index, not a flaw, but a user must carry it into every inference.

At the coding link, the danger is the gap between words and meaning: a rule that counts "war" cannot tell a war from a price war or a trade war. At the aggregation link, choices about scaling and weighting, innocuous one at a time, decide the result. And at the presentation link — the one most often forgotten — the labels on the spikes tell the reader what to see, and the absence of labels on the troughs tells them what not to.

From the archive · Inferences from enemy propaganda, 1941–1945

The systematic analysis of text as evidence about political intentions did not begin with computers. During the Second World War, analysts in the United States — among them a group at the Federal Communications Commission's foreign broadcast monitoring service — read German radio propaganda not for what it said but for what its saying implied. If the propaganda ministry began preparing the public for a particular kind of news, what did the regime expect to happen? If a theme disappeared from broadcasts, what had changed in the regime's own calculation?

Alexander George later reconstructed this work and asked how well the analysts' inferences had held up once the German archives could be checked (George 1959). His account made two points that every risk analyst should know. The first is that the useful inferences were rarely about the *manifest* content — the frequency of a word — and usually about what the content revealed of the speaker's situation and purposes, which required a model of the speaker. The second is that the analysts were most often wrong when they forgot that the text had been produced *for* an audience, by an actor with reasons to mislead it. A newspaper is not a propaganda ministry, and an earnings call is not a broadcast to a wartime public. But both are texts written by people with purposes, and an index built from them inherits those purposes.

7.3 Content analysis: from reading to inference

The discipline that formalised these practices is content analysis. Its early textbook definition called it a technique "for the objective, systematic, and quantitative description of the manifest content of communication" (Berelson 1952). The modern definition, Krippendorff's, is broader and more demanding.

Content analysis — a research technique for making replicable and valid inferences from texts (or other meaningful matter) to the contexts of their use (Krippendorff 2018). Replicable: a second analyst following the same procedure should reach the same result. Valid: the inference must hold about the context, not merely about the texts.

The shift from "description" to "inference" matters. Nobody builds a geopolitical risk index to learn how often "missile" appears in newspapers; the texts are evidence, and the context is the object.

Krippendorff breaks the procedure into steps, and they map cleanly onto both the published indices and the small hand-coded indices that analysts inside firms build for themselves. **Unitising** decides what a single unit is: an article in the GPR and EPU indices, a transcript in Hassan and colleagues' measure, a headline in a hand-coded index of the news about one country. **Sampling** decides which units enter: every article in ten newspapers, every earnings call of listed firms that hold one, a few dozen headlines drawn at random from a news query. **Recording** assigns each unit to a category, by dictionary or by coders following a protocol. **Reducing** summarises the codes. **Inferring** draws the conclusion about the context: geopolitical risk is high this month; this firm faces unusual political risk this quarter.

7.3.1 Manifest and latent content

The distinction that does most of the work in practice is between what a text *says* and what it is *about*. Manifest content is on the surface: the words, their frequency, their co-occurrence. Latent content requires interpretation: whether a headline reports a threat or an act, whether a statement is ironic, whether a reference to war is literal. Dictionaries read manifest content; human coders can read latent content. The trade-off between them is the trade-off between reliability and validity seen from another angle. A dictionary is perfectly reliable — run it twice and it returns the same count — and blind. A human coder sees meaning, and two human coders see slightly different meanings.

The best measures use both. They encode the analyst's judgement in a dictionary that a machine can apply to millions of documents, and then audit the machine with humans who read a sample and check whether the dictionary's classifications match what the texts are about. That design — a dictionary audited by people — is common to all three of the indices this chapter examines, and the quality of the audit is the best single guide to the quality of the index.

7.4 Three dictionaries, three theories of risk

The three measures compared in Table 7.1 share a family resemblance and differ in ways that matter.

Table 7.1: Three text-based measures of risk, link by link. Source: Caldara and Iacoviello (2022); Baker, Bloom, and Davis (2016); Hassan et al. (2019).

	GPR	**EPU**	**PRisk**
Text	newspaper articles	newspaper articles	earnings-call transcripts
Unit counted	articles containing geopolitical terms	articles containing economy *and* policy *and* uncertainty terms	political bigrams near synonyms of risk or uncertainty
Dictionary	authors' eight categories	authors' term sets, audited by hand	learned from political vs non-political texts
Scaled by	all articles	all articles	transcript length
Level	world; 44 countries	country; global aggregate	firm × quarter

7.4.1 Geopolitical risk as newspaper attention

Caldara and Iacoviello define geopolitical risk as the threat, realisation and escalation of adverse events associated with wars, terrorism and tensions between states that affect the normal and peaceful course of international relations (Caldara and Iacoviello 2022). Their index counts, each month, the share of articles in a fixed set of newspapers that contain words from eight categories: war threats, peace threats, military build-ups, nuclear threats, terror threats, beginning of war, escalation of war and terror acts. The first five form a sub-index of *threats*, the last three a sub-index of *acts*. The series is rescaled so that its average over a base period equals 100, and national indices are built by requiring that the article also mention the country.

The theory of risk embedded in this procedure is explicit: geopolitical risk is what informed newspaper readers in a few large democracies are being told to worry about. The split between threats and acts is not a technical detail. It distinguishes the risk that something will happen from the fact that it has, and the two behave differently — threats can rise for months before an act, and an act can resolve uncertainty as well as create it. Any hand-coding protocol that borrows the eight categories needs a rule for the headline that reports both, and the natural rule is that acts beat threats.

The authors validate the index in two ways. Historically, the archive lets them extend the series back to 1900, and it spikes where a historian would want it to: around the two world wars, at the beginning of the Korean War, during the Cuban Missile Crisis and after 11 September 2001. Economically, higher readings foreshadow lower investment and employment and are associated with a higher probability of disaster and larger downside risks, and the adverse consequences come from threats as well as from acts. They also build industry- and firm-level versions, and find that investment falls more in industries exposed to aggregate geopolitical risk (Caldara and Iacoviello 2022). The first test is of face validity; the second, of predictive validity, is the more demanding.

The published series makes the split concrete (Figure 7.1). The two sub-indices are correlated — 0.59, month by month, since 1985 — but the episodes that dominate them are different. The attacks of 11 September 2001 were an act before they were anything else: in September 2001 the acts sub-index stood at 841, more than eight times its 1985–2019 average, while the threats sub-index stood at 260. The run-up to the invasion of Iraq was the opposite kind of episode. From December 2002 to February 2003, months of ultimatums and inspections, threats climbed from 183 to 297 while acts stayed below about 215; only in March 2003, the month of the invasion, did acts overtake threats, at 429 against 362. The war in Ukraine, too, began in the newspapers as a threat. From October 2021 to January 2022, while Russian forces massed on the border,

the threats sub-index averaged 143 and the acts sub-index 53. In February 2022, the month of the invasion, threats stood at 321 and acts at 105; acts reached 251 only in March.

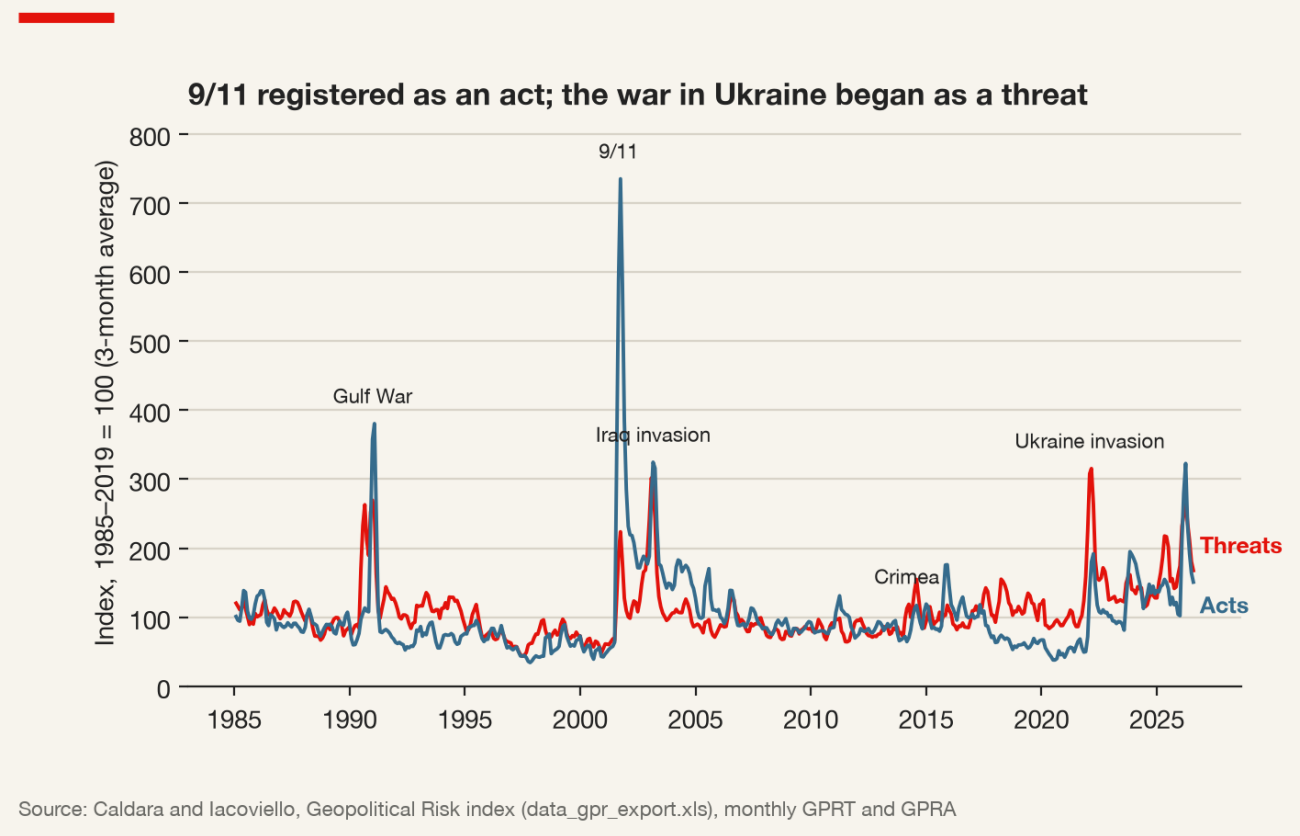


Figure 7.1: Threats and acts are different series: 2001 registered as acts; before the invasions of 2003 and 2022, threats rose first. Monthly sub-indices, 1985–2019 average = 100, three-month centred average. Source: Caldara and Iacoviello, Geopolitical Risk index.

Two lessons follow. The headline GPR, which combines the two, can rise for very different reasons, so a reader who sees a spike should ask which sub-index produced it. And the threats sub-index is the one with a claim to foresight: it rose in the autumn of 2021, when the decisions open to a firm — to build inventory, to move staff, to hedge a currency — could still be taken. The scale deserves a word too. "100" is the average of 1985–2019, so a reading of 146, the value for September 2026, says that the share of articles about geopolitical tension was nearly half as high again as its long-run norm. It does not say that the world was 46 per cent more dangerous.

7.4.2 Policy uncertainty as a three-way conjunction

Baker, Bloom and Davis built the Economic Policy Uncertainty index on a different logic (Baker, Bloom, and Davis 2016). An article counts only if it contains a term from each of three sets: one about the economy, one about policy and one about uncertainty. The conjunction is the clever part. A filter on "uncertainty" alone would capture weather and sport; a filter on "policy" alone would capture every legislative report. Requiring all three narrows the count to articles that discuss uncertainty about economic policy, at the cost of missing articles that discuss it without using any of the listed words.

What distinguishes the EPU work methodologically is the audit. Human readers coded a large sample of articles, and the policy term set was tuned to reduce errors in both directions against their judgements. The audit does not make the index perfect. It makes its error rates *known*, which is the more useful property. The global index is a GDP-weighted average of national indices, published in two versions — weighted by current-price GDP and by purchasing-power-parity GDP — and the choice between them is a small instance of the aggregation decisions discussed below: the second gives large emerging economies more weight, and therefore gives their politics more voice in what is called "global" uncertainty.

Set beside each other, the two newspaper indices tell different histories (Figure 7.2). Over the 355 months from January 1997 to July 2026 that both cover, their correlation is 0.26 in levels and 0.19 in monthly changes — positive, but low enough to say that they measure different things. Their largest episodes barely overlap. Taking the five highest months of each, at least six months apart, the geopolitical risk index peaks in October 2001, March 2003, March 2026, March 2022 and June 2025; the global policy uncertainty index in April 2025, March 2026, May 2020, November 2020 and June 2019. Only March 2026 — the war of the United States and Israel with Iran (Chapter 1) — appears in both lists. The attacks of 2001 and the invasion of Iraq barely moved economic policy uncertainty; the pandemic and the American tariff announcements of April 2025 barely moved geopolitical risk.

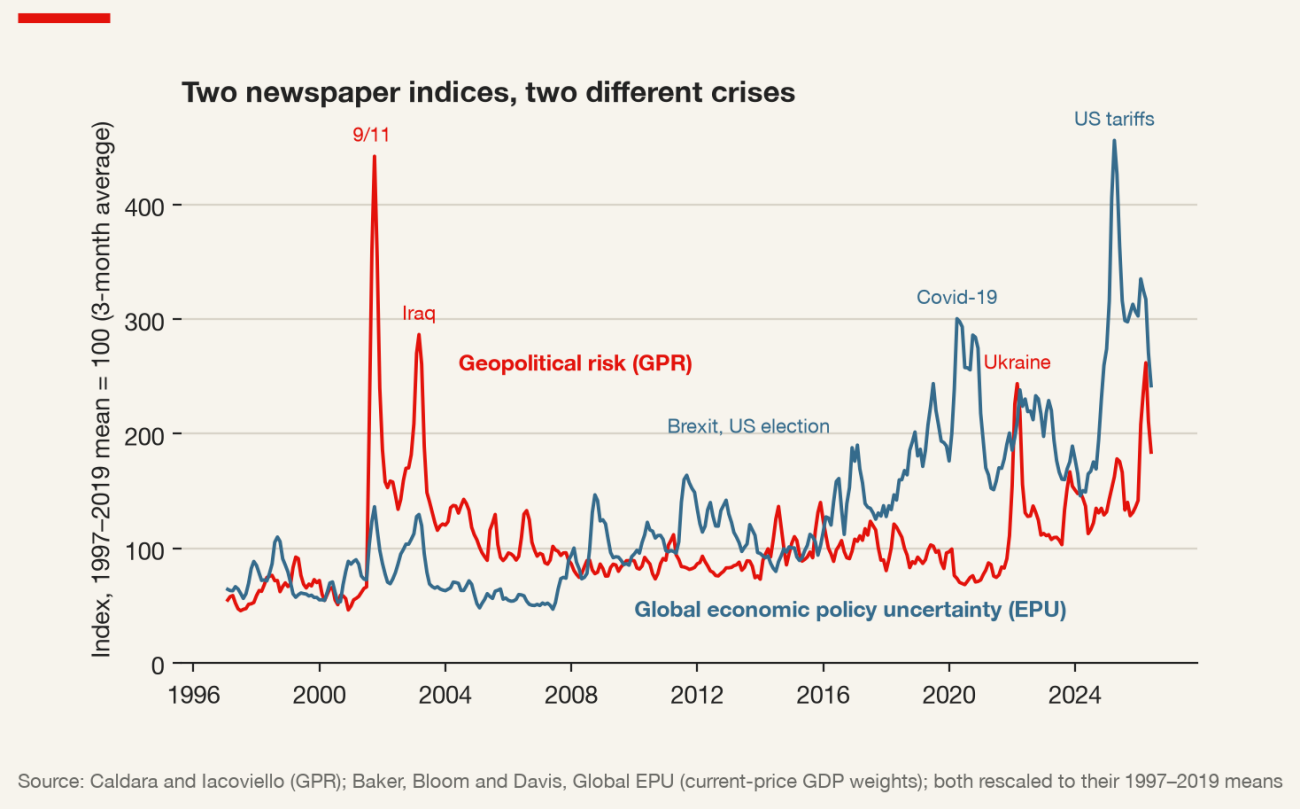


Figure 7.2: The two best-known newspaper indices record different crises: wars and terrorism for one; the pandemic, elections and tariffs for the other. Both rescaled to their 1997–2019 means, three-month centred average. Source: Caldara and Iacoviello; Baker, Bloom and Davis, global EPU weighted by current-price GDP.

This is what the definitions predict. A firm exposed through tariffs and export controls will find its risk in the EPU series and little of it in the GPR series, whose eight categories exclude trade conflict; a firm with plants near a contested border will find the reverse. The weighting of the global EPU index, by contrast, matters much less than the definitions: the versions weighted by current-price and by purchasing-power-parity GDP correlate at 0.997 over the same months. The choice of weights changes whose newspapers count for how much; the choice of concept changes which crises exist.

7.4.3 Political risk as what managers discuss

Hassan, Hollander, van Lent and Tahoun asked a question the newspaper indices cannot answer: how much political risk does an *individual firm* face, and does it matter for what the firm does (Hassan et al. 2019)? Their text is not newspapers but the transcripts of quarterly earnings conference calls, in which senior managers present results and answer questions from analysts. An earnings call is where a firm's leadership, questioned by people paid to find weaknesses, talks about what worries it.

Their dictionary is not written by hand. It is *learned*. The authors took a body of political texts and a body of non-political texts and identified two-word combinations — bigrams — that are frequent in the first and rare in the second. They then

counted, in each transcript, the political bigrams that appear close to a synonym of "risk" or "uncertainty", weighted them by how distinctively political they are, and divided by the length of the transcript. The result, PRisk, is a firm-quarter measure of the share of the conversation devoted to political risk.

Dictionary method — a measurement procedure that assigns texts to categories by counting words or phrases from a pre-specified list. The list may be written by the analyst (GPR, EPU) or learned from labelled texts (PRisk). Either way it is a theory of what the concept looks like in language, and its validity depends on whether that theory holds in the texts to which it is applied.

The proximity requirement deserves a moment. Counting political bigrams alone would measure how much a firm talks about politics — which may reflect opportunity as much as risk, or simply a chief executive's interests. Requiring that the political language appear near the language of risk and uncertainty turns a measure of political *attention* into a measure of political *risk*. It is the same logic as EPU's three-way conjunction, applied at the level of a sentence rather than an article.

The dependence of a dictionary on its domain is not a hypothetical worry. Loughran and McDonald showed that the general-purpose word lists then in common use classified much of the ordinary vocabulary of corporate filings as negative — words such as "liability", "tax" or "cost" that in a financial report are neutral descriptions — and built a finance-specific list in response (Loughran and McDonald 2011). Tetlock's earlier work on newspaper pessimism and stock prices had shown that a dictionary measure of tone could carry real information about markets (P. C. Tetlock 2007). Together they make the point that runs through this chapter: a dictionary works only for the texts it was built for.

7.5 What the firm-level measure found

Hassan and colleagues validated PRisk in three ways. First, its aggregate across firms moves over time with the EPU index, as it should if both capture the same variation in political uncertainty. Second, it varies intuitively across sectors and over time. Third, and most persuasively, firms with high PRisk *behave* as theory predicts firms facing political risk should behave: they cut investment and hiring, and they become more active politically, lobbying more and donating more to politicians. A measure that predicted nothing about behaviour would be a curiosity. A measure that predicts the behaviour the concept implies has passed a test of validity that no amount of reliability could substitute for.

The public release of the data lets anyone see the first of these tests again. The 2022 vintage of the file contains 225,954 firm-quarters for firms headquartered in the United States — 7,073 firms between the first quarter of 2002 and the first quarter of 2022, a median of 2,932 in each quarter. Averaged across firms each quarter, and expressed in standard deviations of the firm-level measure, PRisk tracks the American EPU index closely: the correlation between the two quarterly series is 0.80 (Figure 7.3). Newspapers and earnings calls agree about when political uncertainty was high for listed American firms: at the end of 2008, in the dispute over the "fiscal cliff" at the end of 2012, and above all in the second quarter of 2020, when average PRisk reached 1.02 standard deviations, its highest in the series. They also disagree in instructive places. The confrontation over the federal debt ceiling in the summer of 2011 doubled the quarterly EPU index, from 116 to 235, but moved average PRisk only from 0.56 to 0.64 standard deviations: newspapers wrote a great deal about a crisis that most managers, on their calls, did not treat as a risk to their own business.

The finding that matters most for a multinational concerns where the variation lies. Most of the variation in political risk is not common to all firms at a given time, nor common to all firms in a sector; it is *specific to the firm*. Two companies in the same industry, in the same quarter, can face very different political risk, because they sell to different governments, depend on different regulations, source from different countries or hold different licences. Two further results sharpen the point. The dispersion of political risk across firms widens when aggregate political risk is high, so that the moments at which a national index rises are precisely the moments at which it says least about any one firm. And when the measure is broken down by topic, firms that talk more about the risks of a given political topic increase their lobbying on that topic, and not on others, in the following quarter (Hassan et al. 2019). Managers do not merely worry in general; they worry about something, and act on it.

The band in Figure 7.3 shows the second of those results in the public data. Its edges are the 10th and 90th percentiles of firms in each quarter, and it widens precisely when the average rises: the correlation between the quarterly average and the distance between the two percentiles is 0.95. A quarter of high political risk for the average firm is a quarter in which firms differ from each other more than usual.

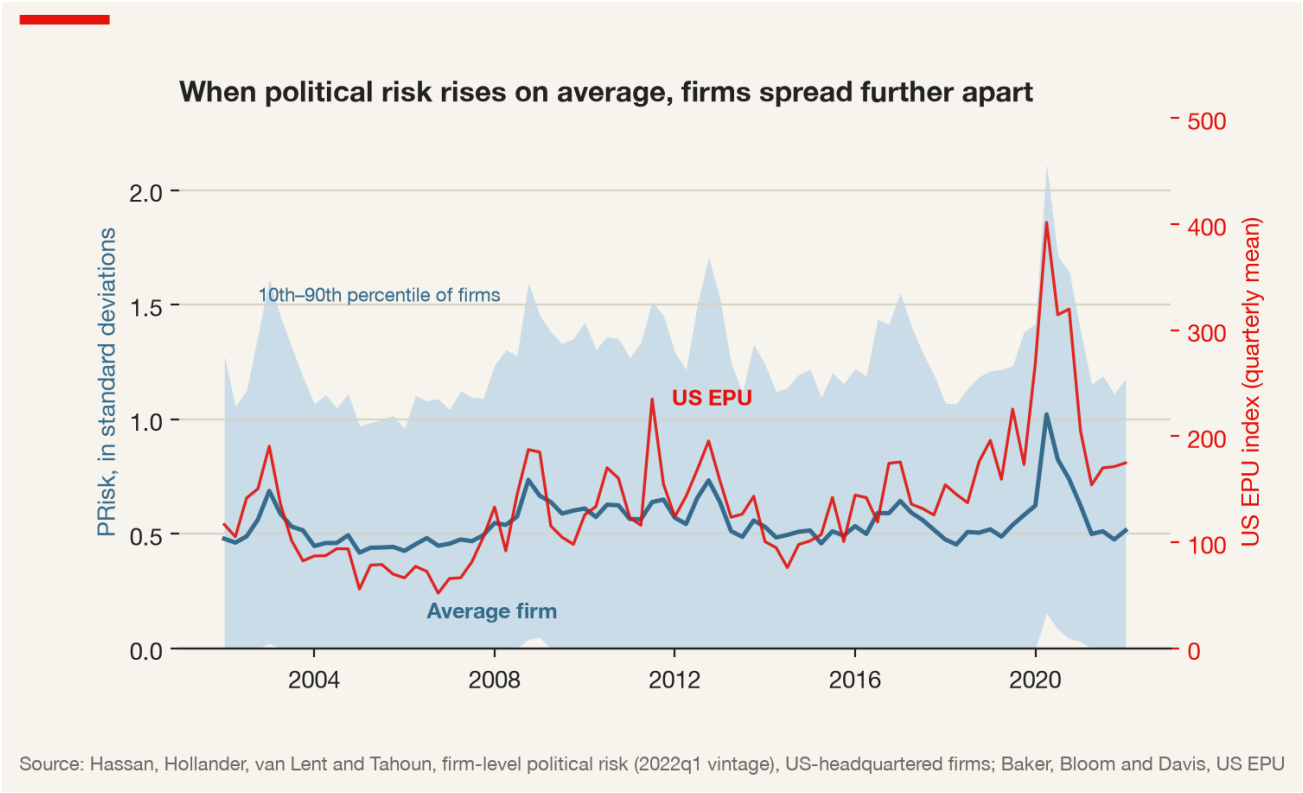


Figure 7.3: Political risk in earnings calls moves with newspaper-based policy uncertainty, and firms spread apart when the average rises. Quarterly mean PRisk of US-headquartered listed firms, in standard deviations, with the 10th–90th percentile range; US EPU on the right axis. Source: Hassan, Hollander, van Lent and Tahoun, firm-level political risk (2022q1 vintage); Baker, Bloom and Davis.

How much of the variation is common and how much belongs to the firm can be measured directly, by asking how much of the variance of PRisk the quarter explains on its own, and how much more the identity of the firm adds (Figure 7.4). The quarter — everything that all American listed firms shared in a given three months, from elections to wars to pandemics — explains about 1 per cent. Permanent differences between firms, the fact that a defence contractor always talks more about politics than a maker of software, explain another 21 per cent. The remaining 78 per cent is variation within the firm from one quarter to the next that other firms did not share. The authors' own decomposition, which uses sectors rather than individual firms, reaches the same conclusion: the common component is a small fraction of the whole (Hassan et al. 2019).

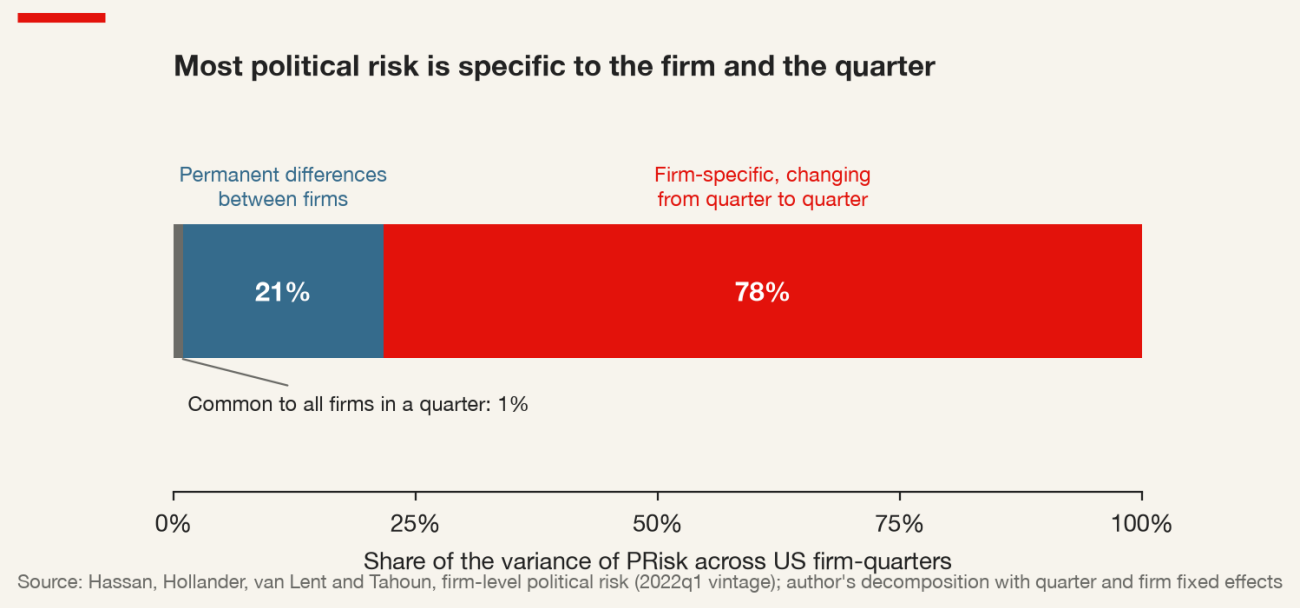


Figure 7.4: Most variation in firm-level political risk is neither common to all firms nor a permanent trait of the firm. Shares of the variance of PRisk across US firm-quarters, 2002–2022. Source: Hassan, Hollander, van Lent and Tahoun, firm-level political risk (2022q1 vintage); decomposition by the author.

Part of the residual is noise — a chief executive's words on one call are partly accidental — so the 78 per cent is an upper bound on the firm-specific risk that matters, not an estimate of it. But the behavioural tests in the paper show that much of it is real, because noise does not make firms lobby on the very topic they talk about. Most of the political risk a firm faces is its own; the part it shares with every other firm is the smallest part.

The implication is uncomfortable for anyone who buys a country-risk score. An aggregate index measures what is common; the political risk that reaches a particular firm is mostly what is not. A board that tracks a national index and believes it is tracking its own exposure is watching the weather report for a region while the risk is in the plumbing of its own building.

This is where the five channels used throughout this book earn their place. Political risk reaches a multinational through **market access** (who may sell what to whom), **inputs** (what can be bought, and from where), **finance** (who will lend, insure, settle), **assets and people** (what can be seized, damaged or detained) and **data and operations** (what can be stored, transferred or shut down). A newspaper index measures, at best, the climate in which these channels operate; PRisk measures how worried a firm's own managers are, which integrates the channels from the inside. Neither tells the firm which channel is exposed. That is the job of a composite exposure index built for a named firm, channel by channel, and the arithmetic of such indices is the subject of the second half of this chapter.

7.6 Two coders and a protocol

The obvious measure of reliability is **percent agreement**: the share of items on which two coders chose the same category. It is intuitive and misleading. Two coders who assigned codes at random, in proportion to how often each category occurs,

would agree often. In a sample where four headlines in five are not about geopolitics at all, two coders who both write "not geopolitical" for every item will agree on most of them, while having measured nothing.

Cohen's coefficient corrects for this by comparing observed agreement with the agreement that chance alone would produce, given each coder's own distribution of codes (Cohen 1960).

Cohen's kappa — a measure of agreement between two coders on a nominal scale, corrected for chance: $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is the observed share of agreements and p_e the agreement expected if the two coders assigned categories independently at their own marginal rates. A kappa of 1 is perfect agreement; 0 is agreement no better than chance; negative values are worse than chance.

If coder A puts $n_{A,k}$ of N items in category k and coder B puts $n_{B,k}$ there, the expected agreement is

$$p_e = \sum_k \frac{n_{A,k}}{N} \cdot \frac{n_{B,k}}{N}.$$

Take forty headlines, collapsed into three categories — not geopolitical, threat, act — with coder A's codes in the rows and coder B's in the columns, as in Table 7.2:

Table 7.2: Two coders, forty headlines: an illustrative agreement table (hypothetical counts).

	B: 0	**B: threat**	**B: act**	**A total**
A: 0	**20**	3	1	24
A: threat	2	**8**	0	10
A: act	0	1	**5**	6
B total	22	12	6	40

The coders agree on the diagonal, $20 + 8 + 5 = 33$ items, so $p_o = 33/40 = 0.825$. Chance agreement is $(24 \cdot 22 + 10 \cdot 12 + 6 \cdot 6)/40^2 = 684/1600 = 0.4275$. Kappa is therefore $(0.825 - 0.4275)/(1 - 0.4275) \approx 0.69$. An agreement rate of better than four in five shrinks by almost a fifth once chance is removed, because both coders put more than half the headlines in the dominant category, where much of their agreement was to be expected.

What counts as good enough is a convention, and the conventions differ. Landis and Koch proposed a set of labels that have become widely used, under which values between 0.61 and 0.80 are "substantial" (Landis and Koch 1977). Krippendorff, for his alpha — a generalisation that handles more than two coders, missing data and ordinal or interval scales — asks for at least 0.800 before data are relied on, and accepts values from 0.667 only for tentative conclusions (Krippendorff 2018; Hayes and Krippendorff 2007). By the first convention the worked example is respectable; by the second it would support only a tentative claim.

7.6.1 What disagreement is for

The worst thing to do with disagreement is to average it away. Disagreements are where the concept is fuzzy, and they are information. Some will be errors: a coder misread a headline, or forgot a decision rule. Others will reveal that the protocol did not anticipate a case — a headline about sanctions, a cyber operation, a military exercise that might be a build-up or might be routine. Those are the valuable ones. The right response is to **revise the protocol**, write down the new rule, and then adjudicate.

Any protocol modelled on the GPR categories contains a gap of exactly this kind. Sanctions, export controls and trade wars are not among Caldara and Iacoviello's eight categories, so a protocol that follows them faithfully must assign such headlines to "not geopolitical" — at best with a flag. Yet a headline about export controls on semiconductors may matter more to a firm's inputs than one about a distant military exercise. The gap is not a mistake in the GPR index, which measures what it says it measures; it is a reminder that an index built to track war and terrorism may not be the index a firm needs.

7.6.2 Machines as coders

The newest temptation is to hand the coding to a machine: a classifier trained on headlines people have already coded, or a general-purpose language model asked in plain words to apply the protocol. Such systems read latent content better than any dictionary and cost little per headline, but they relocate the problem rather than escape it. A machine coder is a third coder, and its agreement with the humans can be measured with the same coefficient. Automated methods, in Grimmer and Stewart's words, are no substitute for careful thought and close reading, and require extensive and problem-specific validation (Grimmer and Stewart 2013). A classifier that agrees with people on headlines about elections may disagree on headlines about export controls; only a fresh human-coded sample will say.

7.7 Big data: detection, not measurement

Hand-coded samples are often drawn from automated news monitors, of which the GDELT Project, which turns news media worldwide in many languages into searchable records, is the best known. It is the most visible example of big data in political risk: a source that promises *coverage* — every country, every day — and *speed*, without a research assistant reading anything.

The promise is real, and the traps are four. The first is **selection**. GDELT records what is reported, by the sources it monitors, in the languages it processes. Its coverage of sources has changed over time, so a rising count of articles about a country may mean more sources rather than more events. The second is **duplication**: one wire-service story republished by hundreds of outlets is counted hundreds of times. The third is **machine coding**, which misreads irony, quotation and historical reference, and errs in ways that are not random. The fourth is the most seductive: "N = all" is never all, and a very large sample drawn by a biased process is still biased — only more confidently so.

The fourth trap has been formalised. Meng showed that the error of an estimate from a large non-random sample depends on the correlation between being in the sample and the quantity being measured, and that even a tiny such correlation can make a sample of millions worth less than a small random sample (Meng 2018). The parable everyone cites is the story of a search-based model of influenza that, after an initial success, substantially overestimated flu prevalence for long stretches, because the behaviour generating the data changed while the model did not (Lazer et al. 2014). Neither is about geopolitics; both describe what can happen to an event count built from online news.

More data is not better measurement. Millions of machine-coded news records are a sample drawn by a process nobody controls. Forty headlines drawn at random from a well-defined query and coded by two people with a written protocol can be more *valid* for a specific question — not because forty is more than millions, but because the analyst knows what the forty are.

The working rule for a risk analyst follows. Use big data for **detection**: something is happening, here, now, and it is worth a closer look. Use small, carefully coded data for **measurement**: how much is happening, compared with what, and whether it has changed. A monitoring system that flags a spike in news about a firm's supplier country is doing detection well. A board paper that reports that "risk in the supplier country has doubled" on the strength of the same spike is doing measurement badly.

7.8 Composite indicators and the arithmetic of judgement

The third measurement task is the one that a multinational is most likely to perform itself: combining several indicators into one number. Country-risk ratings, governance indices, supplier-risk scores and a firm's own exposure index are all composite indicators. They are attractive for the same reason they are dangerous: they compress several dimensions into one figure that a board can read at a glance, and they hide, in the compression, a series of decisions that determine the ranking.

The standard guide is the handbook produced by the OECD and the European Commission's Joint Research Centre (OECD and European Commission Joint Research Centre 2008). It sets out ten steps, from the theoretical framework to the presentation of results. Five of them decide most of the outcome.

The **theoretical framework** says which dimensions of risk the index captures and why. Without one, the index becomes a collection of whatever data were available — which is to say, an index whose content was chosen by the data suppliers rather than by the analyst. For a firm's exposure index, the natural framework is the five channels: each indicator must be defended as a measure of exposure through a named channel, or it does not belong. An index of a firm's exposure to sanctions risk, for example, might draw on the share of revenue earned in sanctioned or sanctioning jurisdictions (market access), on the share of critical inputs sourced from them (inputs) and on the banks through which the firm settles in them (finance) — each defensible, and each measuring a different route by which the same political event would reach the accounts.

Missing data is where indices quietly lie. The country GPR indices cover 44 countries; the country EPU indices about twenty. A composite that drops every country missing an indicator will cover only large, well-documented economies — and the countries it silently drops are, very often, the risky ones, about which fewer people write. Imputing a value is a claim about a country nobody measured; flagging the gap is honest and less tidy.

Normalisation puts indicators measured in different units on a common scale. The simplest method, min–max, maps each indicator to the interval from 0 to 1:

$$x^{*} = \frac{x - \min x}{\max x - \min x} \qquad \text{or, where higher is safer,} \qquad x^{*} = \frac{\max x - x}{\max x - \min x}.$$

The second form matters more than it looks. The Worldwide Governance Indicators' measure of political stability is scored so that higher means *more* stable; in a risk index it must be inverted, or a stable country will score as risky. The indicators also changed scale in their 2025 edition, moving to a 0–100 score and recalculating their history, so that values from different editions cannot be mixed; the methodology behind the indicators is set out in Kaufmann, Kraay, and Mastruzzi (2011). They are themselves composites, pooling many sources in an unobserved-components model that reports a margin of error for every estimate — a discipline most commercial risk scores do not impose on themselves. Other normalisations — z-scores, ranks, distance to a reference country — produce different rankings from the same data, because they treat outliers and gaps differently. Min–max is sensitive to the extremes: a single country with an extraordinary value compresses everyone else into a narrow band.

Weighting and aggregation combine the normalised indicators. The usual form is a weighted average,

$$CI = \frac{\sum_{j} w_{j}\, x_{j}^{*}}{\sum_{j} w_{j}},$$

and the weights can be equal, derived statistically (from principal components, say), or set by experts. Each choice is defensible and none is neutral. Equal weights are a judgement that every dimension matters equally, which is rarely what anyone believes. Statistical weights reward the indicators that move together, which may simply be the redundant ones. Expert weights encode the expert.

The aggregation rule carries a further assumption that is easy to miss.

Compensability — the property of an aggregation rule under which a high score on one dimension can offset a low score on another. Additive aggregation is fully compensatory: a country with excellent governance and an active war can average to "medium". Geometric aggregation is partially compensatory, penalising unbalanced profiles; non-compensatory rules rank countries without allowing any trade-off at all.

For a risk index, full compensability is often the wrong assumption. A firm does not experience the average of a country's risks; it experiences the worst of them that touches its operations. Excellent contract enforcement does not compensate for a plant inside a war zone. Additive aggregation remains the default because it is transparent, but whoever uses it should show the profile behind each score.

From the archive · The Human Development Index, 1990 and 2010

The most influential composite indicator of the past half-century was designed to argue with a single number. The first *Human Development Report*, published by the United Nations Development Programme in 1990 under the direction of Mahbub ul Haq, proposed the Human Development Index as an alternative to ranking countries by income per head (United Nations Development Programme 1990). It combined three dimensions — a long and healthy life, knowledge and a decent standard of living — each normalised against minimum and maximum values, and averaged them with equal weights. It was a simplification built to change a conversation, and it did.

Twenty years later the 2010 report changed the aggregation from an arithmetic to a geometric mean of the three dimension indices and revised the indicators of education and income (United Nations Development Programme 2010). The reason given was compensability: under the arithmetic mean, a country could make up for poor health with high income, one for one; under the geometric mean, poor performance in one dimension weighs more heavily. The change re-ranked countries. Ravallion then showed that the new formula implied trade-offs between longevity and income that few would accept if they were stated openly — in effect, valuations of an extra year of life that varied enormously with a country's income, and were very low in the poorest countries (Ravallion 2012).

The weights and the aggregation rule were value judgements; changing them changed the ranking; and the implicit prices they carried became visible only when someone did the algebra. A firm's exposure index embeds trade-offs of the same kind, and the board deserves to see them stated.

7.9 Sensitivity: the ranking that one weight overturns

The last of the decisive steps is the one most often skipped. **Sensitivity analysis** asks how much the results move when the choices change: a different normalisation, different weights, a different treatment of missing data, a different aggregation rule. Saisana, Saltelli and Tarantola set out how to do this systematically, by treating each methodological choice as an uncertain input and examining the distribution of ranks it produces (Saisana, Saltelli, and Tarantola 2005). The simplest version needs no special software. Record the ranking under equal weights; then under each indicator alone; then count how many countries move by more than a few places. A country that stays near the top under every reasonable choice is robustly high-risk. A country whose rank swings from fifth to twenty-fifth depending on one weight is not "fifth"; it is a country about which the index has no firm opinion.

This is why a composite worth showing to a board should be presented with weights the reader can move, or at least with the rankings that alternative weights would produce. The point is not interactivity for its own sake. It is that a composite which hides its weights is an opinion dressed as a measurement, and a composite whose weights are visible and adjustable is an argument the reader can test. A board that sees a supplier country fall from first to ninth when one weight is halved has learnt that the decision rests on a judgement about that indicator, not on the index.

7.9.1 Two indicators, forty-four countries

A small composite shows how quickly these choices bite. Take the 44 countries for which Caldara and Iacoviello publish a country index, and combine two indicators: the average country GPR for 2025, which measures the share of newspaper articles that discuss geopolitical tension and mention the country, and the political stability score of the Worldwide Governance Indicators for 2024, from the 2025 edition, inverted so that a higher value means less stability. Normalise each by min–max, and average them with a weight w on GPR and $1 - w$ on instability. Nothing could be simpler, and almost nothing about the result is stable (Figure 7.5 and Table 7.3).

The two indicators barely agree to begin with: across the 44 countries, the correlation between the GPR ranking and the instability ranking is 0.09. Under GPR alone the United States is the riskiest country in the set; under instability alone it is twentieth. Venezuela is first on instability and nineteenth on GPR; Tunisia sixth on instability and last on GPR. With equal weights, the United States comes first, Israel second and Russia third. Move the weight on GPR from a quarter to three quarters and 13 of the 44 countries move by more than five places; the United Kingdom moves from twentieth to fourth, China from nineteenth to sixth, Germany from twenty-third to twelfth.

Table 7.3: Rank (1 = riskiest of 44) as the weight on country GPR changes; the twelve highest under equal weights. Source: author's calculation from country GPR (2025 average) and WGI political stability (2025 edition, 2024).

Country	Instability only	25% GPR	Equal weights	75% GPR	GPR only
United States	20	6	1	1	1
Israel	2	2	2	3	4
Russia	7	3	3	2	3
Venezuela	1	1	4	7	19
Ukraine	16	10	5	5	5

Country	Instability only	25% GPR	Equal weights	75% GPR	GPR only
Türkiye	5	4	6	9	10
Egypt	3	5	7	10	13
United Kingdom	27	20	8	4	2
India	8	9	9	11	12
Colombia	4	7	10	14	30
China	19	19	11	6	6
Tunisia	6	8	12	15	44

The cause of the movements is the lesson. The country GPR measures how often the newspapers write about a country in a geopolitical context, and the newspapers write about the United States, the United Kingdom, China, France and Germany as *actors* — the states that send forces, impose sanctions and convene summits — as well as about the places where conflict happens. A validity problem at the indicator link has become a ranking problem at the aggregation link. Min–max normalisation then amplifies it. In 2025 the United States' country index averaged 3.61 and the median of the 44 was 0.16, so that 38 of the 44 countries sit in the bottom quarter of the GPR range once it is scaled from 0 to 1. The GPR component separates a few heavily reported countries from everyone else and says almost nothing about the differences among the rest; a z-score or a rank would produce another ranking again.

The top is robust: Israel and Russia are in the first three under every weight from a quarter to three quarters. The middle is not. Whether China ranks above Colombia, or the United Kingdom above India, is decided by a weight nobody could defend to the first decimal. A board shown only the equal-weight column has been shown an opinion about the weight.

7.10 A short review of the literature

Content analysis as a social-science method was codified in the mid-twentieth century. Berelson (1952) gave the classic definition centred on manifest content, and George (1959) reconstructed the wartime analysis of enemy propaganda that showed both the power and the dangers of inference from text. Krippendorff (2018) is the standard modern treatment, recasting the method around replicable and valid inference and setting out the reliability apparatus; Hayes and Krippendorff (2007) make the case for Krippendorff's alpha as a general standard. The chance-corrected agreement coefficient is Cohen (1960), and the labels most often attached to its values are Landis and Koch (1977).

Grimmer and Stewart (2013) is the essential guide to automated analysis of political texts, and its insistence on problem-specific validation is the spine of this chapter. Gentzkow, Kelly, and Taddy (2019) survey text as data in economics. In finance, P. C. Tetlock (2007) showed that dictionary measures of media tone carry information about markets, and Loughran and McDonald (2011) showed how badly general-purpose dictionaries misread specialised documents.

The three indices at the centre of the chapter are Caldara and Iacoviello (2022) for geopolitical risk, Baker, Bloom, and Davis (2016) for economic policy uncertainty and Hassan et al. (2019) for firm-level political risk. On big data, Lazer et al. (2014) is the cautionary tale and Meng (2018) the formal account of why sample size does not rescue a biased selection process.

Composite indicators have their own methodological literature. OECD and European Commission Joint Research Centre (2008) is the practitioner's handbook and Saisana, Saltelli, and Tarantola (2005) the statement of sensitivity and uncertainty analysis as a quality test. Kaufmann, Kraay, and Mastruzzi (2011) document the Worldwide Governance Indicators, a composite of composites. The Human Development Index (United Nations Development Programme 1990, 2010) and the critique of its implicit trade-offs in Ravallion (2012) are the clearest case study of how aggregation rules carry value judgements.

7.11 What the numbers do not license

Each measure in this chapter comes with a sentence it does not license. Where two of them part company, the question is not which is right but which link of which chain explains the gap.

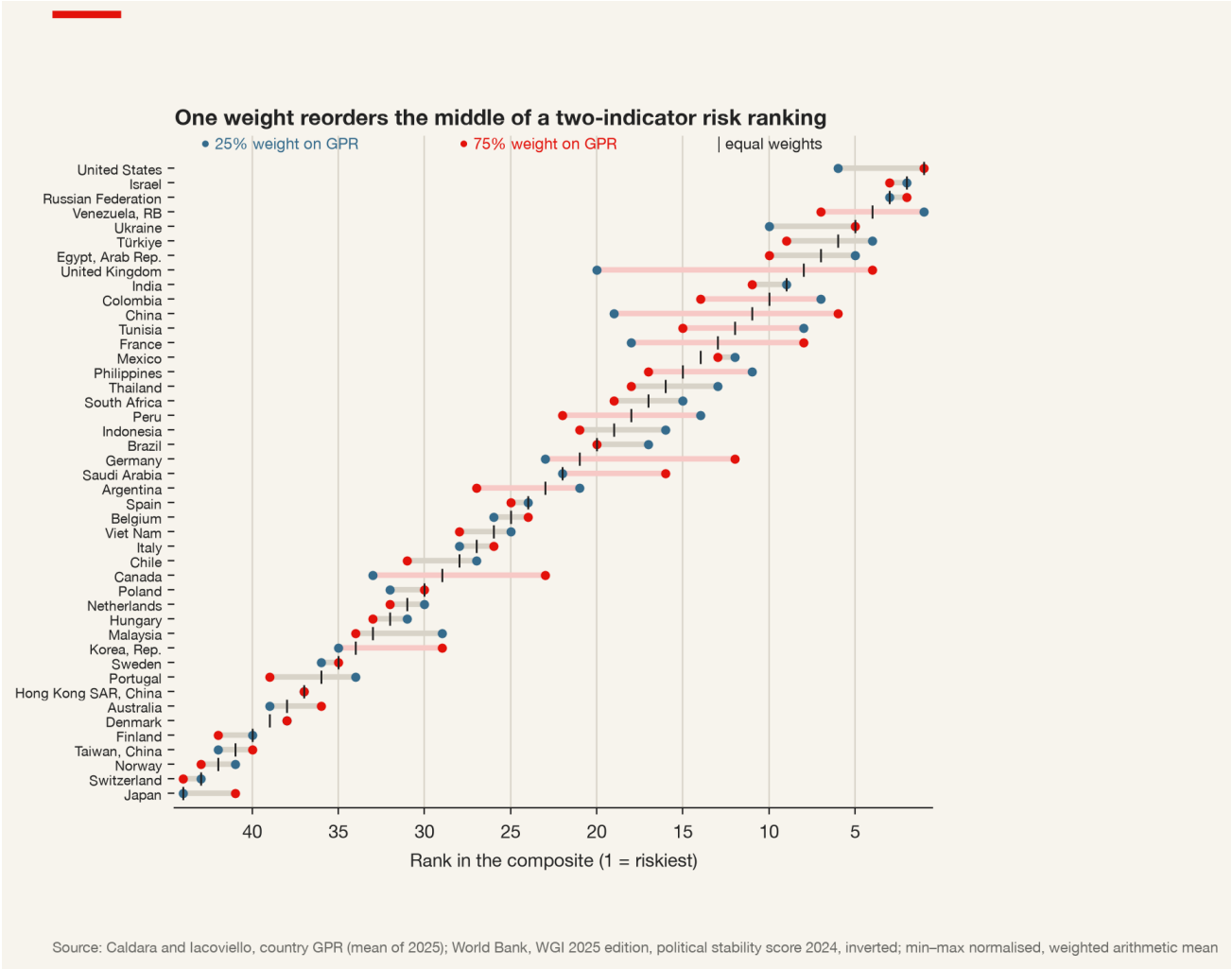


Figure 7.5: In a two-indicator ranking of 44 countries the top is robust and the middle is decided by the weight. Rank with 25 per cent (blue), 50 per cent (tick) and 75 per cent (red) of the weight on country GPR, the rest on inverted political stability. Source: Caldara and Iacoviello, country GPR, 2025 average; World Bank, Worldwide Governance Indicators, 2025 edition, political stability score for 2024; min–max normalisation and weighting by the author.

The line of average firm-level political risk does not tell a particular multinational how much political risk *it* faces. It is an average over American listed firms, and the central finding of the paper that built it is that most political risk is specific to the firm — precisely the variation an average removes. A board can use the line to see when political uncertainty was high for corporate America as a whole; it cannot use it to see its own exposure, which would require scoring its own calls, or at least those of its closest peers.

A hand-coded index of the news about a country, however carefully its coders agree, does not license a claim about how geopolitical that country's situation is. A few dozen headlines from a news monitor that leans towards English-language media, coded against a protocol that places sanctions outside geopolitics, measure what one slice of the press said, filtered through one theory of what counts. What a good kappa licenses is a statement about how reliably two trained readers can apply that protocol — and the list of disagreements licenses something more valuable, which is a map of the places where the protocol must change before anyone relies on it.

A composite exposure index does not license a ranking presented as fact. It licenses a ranking *under stated weights*, together with the weight that would overturn it. That is less than a board usually wants, and more than it usually gets.

The argument of this chapter can be compressed into a five-part audit, to be put to every risk score a multinational builds or buys: what concept, from which sources, coded by whom or by what, normalised and weighted how, and how sensitive to each of those choices. None requires a statistician; each requires only that the score's makers have written down what they did. The scores that survive the audit are not thereby true. They are *measurements* — numbers whose errors can be named and whose disagreements with other numbers can be traced to a decision someone took. A score that cannot answer the five questions is somebody's opinion, with decimals; and the danger of such a score is not that the opinion is wrong, but that the decimals make it impossible to argue with.

Part IV

The channels

Chapter 8

Cyber Geopolitics: Restraint, Rivals and Weaponised Networks

8.1 The war that was supposed to happen

For most of the past three decades, the public conversation about cyberspace and international politics has been conducted in the future tense. A "cyber Pearl Harbor" would switch off a country's power grid; a handful of skilled operators would humble a great power; the next war would be won or lost before a shot was fired. The scenarios were vivid, and they were not absurd: modern economies do run on networked systems, and those systems can be broken. What the scenarios lacked was a record. They were claims about what *could* happen, offered in the absence of systematic evidence about what *had*.

The opposite caricature is just as easy to state. On this view cyber operations are simply crime and espionage conducted with better tools — a nuisance for security departments, of no strategic consequence, and no business of a book on geopolitics. It, too, is a claim about evidence, and it, too, is usually made without any.

This chapter takes the question to the data. It does so through the work of Brandon Valeriano and Ryan Maness, who built a dataset of cyber incidents between rival states and asked what the record actually shows (Valeriano and Maness 2015), and through the argument of Henry Farrell and Abraham Newman that the networks globalisation built are not flat, and that states with jurisdiction over their hubs can turn them into instruments of coercion (Farrell and Newman 2019). The two works answer different questions. The first asks how states use offensive cyber operations against each other. The second asks how states use the architecture of the global economy — including its digital architecture — to watch and to deny. A multinational needs both answers, because it sits inside the architecture and in the path of the operations.

The question for the firm is narrower and more practical than the question for the strategist. It is not whether cyberspace is a new domain of war. It is *through which channels inter-state rivalry in cyberspace reaches a firm's operations, and how its decision architecture should absorb it*. The answer, this chapter argues, is that the firm is rarely the intended target of strategic cyber operations, frequently their collateral, and increasingly a piece of the infrastructure over which states compete.

8.2 Three activities that look alike

Three kinds of activity are routinely run together under the heading "cyber attack", and the first task is to separate them.

Espionage steals information: state secrets, negotiating positions, intellectual property, the personal data of officials. Its typical actors are intelligence services and their contractors, and its typical intrusion — into a ministry, a defence supplier or a law firm advising on a merger — is designed never to be noticed. **Disruption and degradation** interrupt or damage systems: a denial-of-service attack that floods a ministry's website, malware that wipes the disks of a company's computers, an operation that manipulates an industrial control system. Its actors are military and intelligence units and the proxies they sponsor. **Crime** extracts money: ransomware that encrypts a hospital's records, fraud against a bank's payment systems. Its actors are criminal groups.

Cyber operation — the use of computer code or network access by an organised actor to obtain information from, disrupt, degrade or manipulate a target's information systems. Espionage, disruption and crime are all cyber operations; they differ in purpose and actor, not in technique. The same intrusion may serve any of the three.

The boundaries blur, and the blurring is itself a geopolitical fact. States tolerate criminal groups that operate from their territory against foreign targets, hire them, shelter them, and occasionally use their tools. An intrusion that begins as espionage leaves access that can later be used for disruption. A ransomware campaign run by a criminal group may serve a state's purposes by sowing disorder in a rival's economy, without the state having ordered it. When a ransomware attack by a criminal group believed to operate from Russia forced the shutdown of the Colonial Pipeline in the United States in May 2021 (Cybersecurity and Infrastructure Security Agency 2023), it was crime in its motive and geopolitics in its consequences: the question of what the Russian state knew, tolerated or could have prevented became a matter between governments.

For a multinational the distinction matters because the three activities call for different responses. Espionage is a threat to the firm's intellectual property and its negotiating position, and the defence is largely technical and organisational. Disruption is a threat to its operations, and the defence includes resilience — the ability to keep working when systems fail. Crime is a threat to its cash and its reputation, and it is the one where law enforcement, insurers and the firm's own willingness to pay become part of the calculation. A risk register that lists "cyber" as a single line has not yet done the first piece of analysis.

From the archive · Estonia, 2007

In April 2007 the Estonian government moved a Soviet-era war memorial, the Bronze Soldier, from the centre of Tallinn to a military cemetery. The decision provoked protests by members of the country's Russian-speaking minority and an angry response from Moscow. Over the following weeks a wave of distributed denial-of-service attacks hit Estonian government websites, banks, newspapers and political parties. Estonia, among the most digitised societies in Europe, found its online services intermittently unreachable.

Estonian officials pointed to Russia. The Russian government denied involvement, and the public record never established a chain of command; much of the traffic came from botnets and from individuals mobilised on Russian-language forums. The episode thus contained, in compressed form, nearly every feature that later analysts would identify in state cyber conflict: an existing political rivalry, an operation of low technical sophistication and limited physical harm, high political salience, and attribution that remained contested.

Its institutional legacy was substantial. NATO's Cooperative Cyber Defence Centre of Excellence was established in Tallinn the following year, and under its auspices a group of international lawyers produced the *Tallinn Manual*, an attempt to state how existing international law applies to cyber operations (Schmitt 2013). The episode was widely described at the time as the first cyber war. With hindsight it looks more like the first clear case of what Valeriano and Maness would call restraint: a rival state's displeasure expressed through disruption that stopped well short of destruction.

8.3 The restraint thesis

Valeriano and Maness built their dataset around a simple design choice that shapes everything they found. They did not try to catalogue every cyber operation in the world. They took pairs of states with a history of enduring hostility — **rivals**, in the sense the international-conflict literature has used for decades — and recorded the cyber incidents between them. The unit is the incident, nested in campaigns; the population is rival dyads; the evidence is public. The dataset has since been extended as the Dyadic Cyber Incident and Campaign Data (DCID), whose version 2.0 covers incidents beginning between 2000 and 2020 (Maness et al. 2022).

Cyber incident (in DCID) — an individual cyber operation launched by one state against another within a rival dyad, with a start date, a method, a target type, an objective and a severity score. Incidents can be grouped into **campaigns**, sustained series of operations with a common purpose. The unit is the incident between rivals, not the operation in general.

The choice of rivals as the population is a theoretical commitment, not a convenience. The study of international conflict has long found that a small number of pairs of states account for a large share of militarised disputes, and that these enduring rivalries have their own dynamics: hostility persists, crises recur, and each side interprets the other's moves in the light of a long history. If cyber operations are a new instrument of old conflicts, rivalries are where they should appear; if they are something genuinely new — an instrument that lets any state hurt any other cheaply — they should appear outside rivalries too. Restricting the data to rivals makes the first hypothesis testable and the second invisible — a limitation to which this chapter returns when the rival-dyad record is set beside a wider one.

From that record they drew four propositions.

The first is **restraint**. States use cyber operations far less often, and far less destructively, than the "cyber war" narrative predicts. Most incidents in the data are of low severity: defacements of websites, denial-of-service attacks, espionage. Operations that destroy physical property or take lives are rare. The image of cyberspace as a domain of constant, escalating attack does not survive contact with a systematic count.

The public release of the data lets the claim be checked directly (Figure 8.1). Of the 433 incidents in the September 2022 release of DCID, 393 — nine in ten — are coded 2, 3 or 4 on the severity scale: harassment and disruption, the theft of information from one network, or widespread intrusion into several. Eighteen are coded 5, for intrusions that attempted damage, and five are coded 6, the level the codebook reserves for widespread destruction in a network critical to national security. Those five are the canonical cases: the two Stuxnet entries against Iran, starting in 2009 and attributed one to the United States and one to Israel; the Shamoon wiper and the attack on the gas producer RasGas, both attributed to Iran and directed at Saudi Arabia in 2012; and the reported American effort to sabotage North Korean missile launches, coded for 2015. Nothing in two decades is coded 7 or above — the levels at which the codebook places deaths, the disruption of a national economy and the destruction of critical infrastructure.

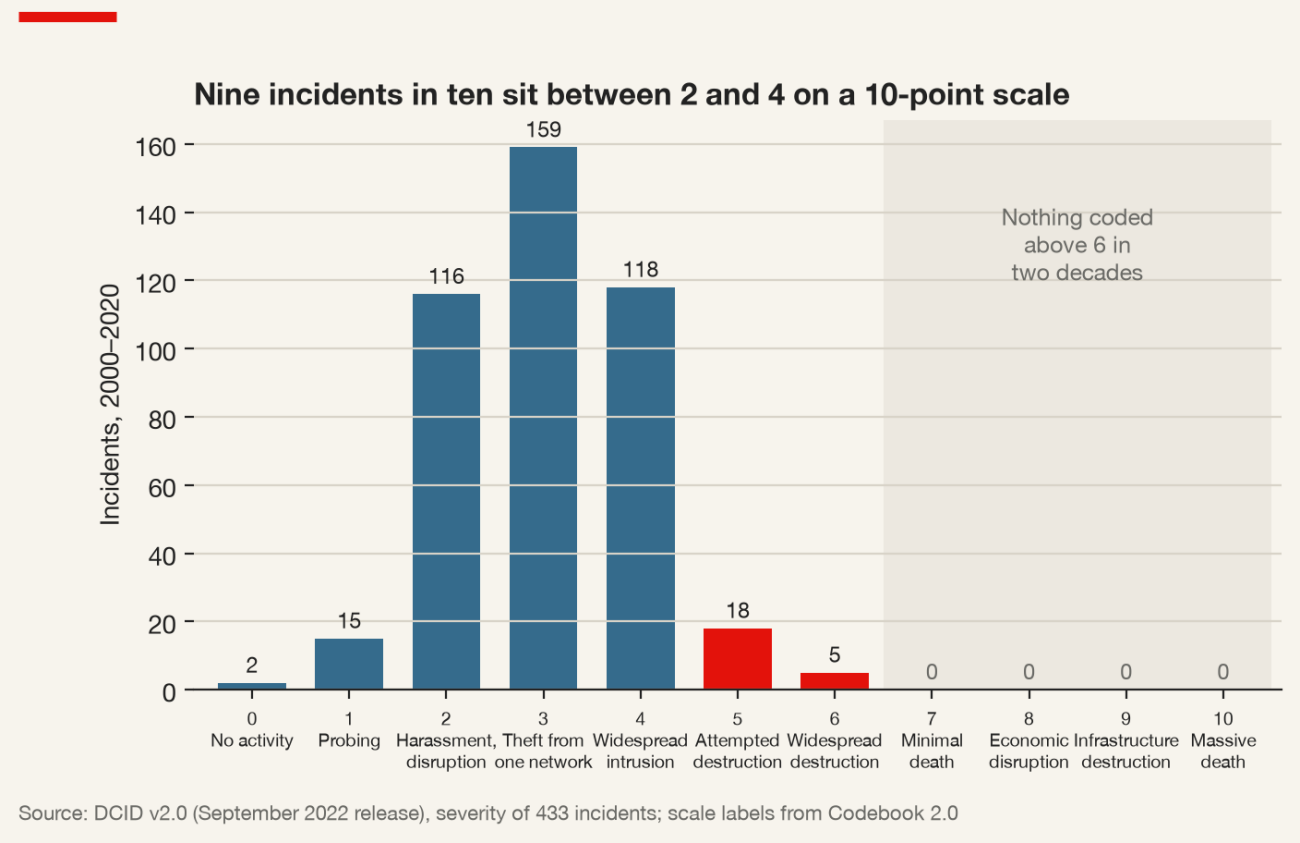


Figure 8.1: Nine in ten publicly attributed cyber incidents between rival states sit at 2–4 on a ten-point severity scale, and none above 6. Incidents beginning 2000–2020 by severity, with the codebook's labels. Source: Dyadic Cyber Incident and Campaign Data, version 2.0 (September 2022 release), and its Codebook 2.0.

The scale is worth reading for what it ranks low. The operation that cost firms more than any other in the record, NotPetya, appears twice: once against Ukraine, coded 5, and once against the United States, coded 4 — below Stuxnet and Shamoon. The WannaCry ransomware of 2017, attributed to North Korea, is coded 4 in both its American and its British entries, although it shut down hospital services in Britain. The scale measures harm as a state experiences it, against networks critical to national security, and that is the right scale for testing a theory of how states fight. It is not a scale of cost to the firms that are in the way, and an analyst who sorted incidents by it to rank commercial exposure would have the order wrong.

Cyber restraint — the empirical pattern, and the theoretical argument, that states with offensive cyber capabilities use them sparingly and at low severity against their rivals, holding back the most destructive options. Restraint is a claim about *strategic* severity, not about frequency of nuisance or about commercial cost.

The second is that cyber conflict follows **regional rivalries**. The dyads that account for most of the incidents are those that were already hostile before anyone had a network: the United States and China, Russia and its neighbours and the West, Israel and Iran, India and Pakistan, the two Koreas. Cyber operations do not create new conflicts; they are a tool in old ones. That finding matters for anyone who expected the low cost of entry to cyber operations to flatten the international hierarchy. The states that are most active in cyberspace are, by and large, the states that were already contending.

The concentration is easy to state in numbers (Figure 8.2). The 433 incidents fall into 52 directed pairs of initiator and target, and the five largest — China against the United States (59 incidents), Russia against the United States (40), North Korea against South Korea (37), Russia against Ukraine (28) and Iran against the United States (21) — account for 43 per cent of the total; the ten largest for 61 per cent. Four initiators, China (115 incidents), Russia (113), Iran (63) and North Korea (59), account for 81 per cent of the record, and the United States is the target of 32 per cent of it. The United States appears as initiator in 24 incidents, and Israel in ten. Those numbers carry the warning set out below about attribution: they are the operations that someone chose to make public and that the coders could attribute, and the states with the most capable intelligence services and the most active public attribution are, unsurprisingly, more often found as victims than as perpetrators.

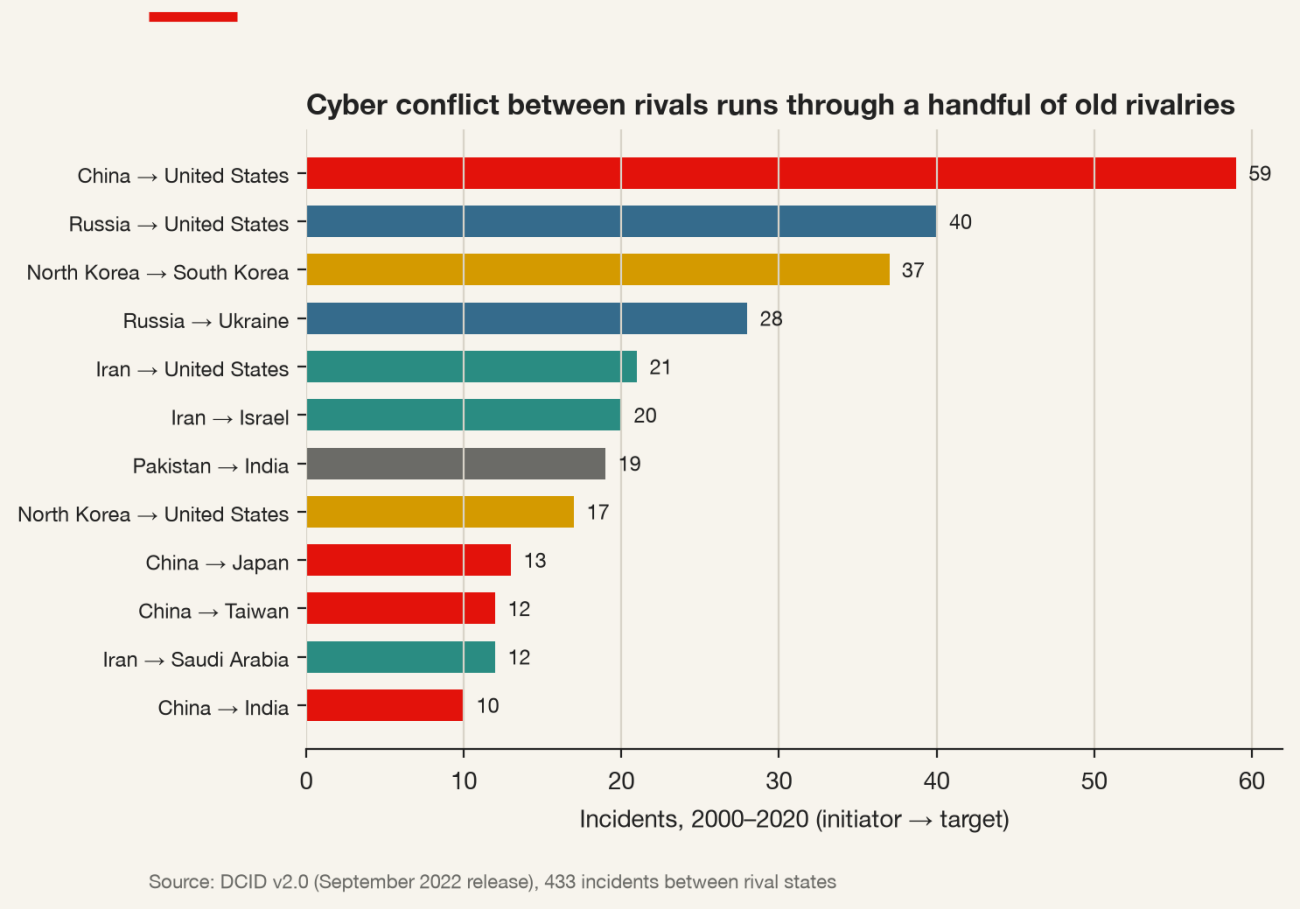


Figure 8.2: The record of cyber conflict between rivals is dominated by a few old rivalries: the five largest directed pairs account for 43 per cent of all incidents. The twelve largest pairs, 2000–2020, by initiator. Source: Dyadic Cyber Incident and Campaign Data, version 2.0.

The third proposition explains the first: **why** states hold back. The fourth is that cyber operations have **limited coercive effect**. They rarely change the target's behaviour. They are better at signalling displeasure and at gathering intelligence than at compelling a rival to do something it would otherwise not do — a conclusion that the same authors developed at greater length in later work on cyber strategy (Valeriano, Jensen, and Maness 2018).

The restraint thesis did not arrive in an empty field. Thomas Rid had argued, provocatively, that cyber war in the sense of an act of force that is violent, instrumental and political had never occurred, and that what was called cyber war was better

understood as sabotage, espionage and subversion — old activities in a new medium (Rid 2012, 2013). Erik Gartzke made a related argument from the logic of force: an attack that cannot be followed up by conventional power changes nothing durable, so cyber operations are likely to be adjuncts to force rather than substitutes for it (Gartzke 2013). Lucas Kello took the opposite position, arguing that the cyber revolution genuinely strained the concepts of international relations because it blurred war and peace into a condition of permanent, sub-threshold "unpeace" (Kello 2013, 2017). Valeriano and Maness's contribution was to bring data to an argument that had been conducted largely through concepts and anecdotes, and the data came down closer to Rid and Gartzke than to the prophets of catastrophe.

8.4 Why states hold back

The reasons for restraint are worth taking one at a time, because each has an analogue in the firm's own calculation.

Escalation. A destructive cyber operation against a rival invites a response, and the response need not stay in cyberspace. A state that switches off a rival's grid has made a move whose consequences it cannot control. In a relationship where both sides have conventional and sometimes nuclear forces, the incentive to stay below the threshold that would provoke them is strong.

Revelation. A cyber capability depends on a vulnerability the target has not fixed. Using it reveals it, and once revealed it can be patched — in the target's systems and everyone else's. The most sophisticated capabilities are therefore wasting assets, worth more held in reserve than spent.

Blowback. Malicious code, once released, can be captured, analysed and reused. A weapon that can be reverse-engineered and turned back on its maker, or on the maker's allies and firms, is a weapon to be used with care.

Access. The value of being inside a rival's networks is often greater than the value of damaging them. A state that can read a rival's communications has every reason not to do anything that would alert the rival to its presence. Espionage, in this sense, restrains disruption.

These incentives are not the whole story. Ben Buchanan has argued that intrusions made for defensive purposes — to understand a rival's capabilities, or to pre-position against a future threat — look identical, from the target's side, to preparation for attack, and that this creates a security dilemma in which each side's prudence alarms the other (Buchanan 2017). Restraint in the use of destructive operations is compatible with constant, mutual intrusion, and the intrusions themselves are a source of instability.

The war in Ukraine has tested the restraint thesis harder than anything in the dataset, which ends in 2020. Russian cyber operations against Ukraine have been extensive since 2014, and they intensified around the full-scale invasion of February 2022. The best systematic study of the earlier period found that cyber operations had little discernible effect on events on the battlefield (Kostyuk and Zhukov 2019), which is consistent with the fourth proposition: cyber operations are weak instruments of compulsion even in war. Whether the restraint thesis holds *as of today* is an open question. A pattern measured to 2020 is evidence about 2000–2020, and the war has supplied episodes that sit awkwardly within it.

One of them began in the first hours of the invasion. On 24 February 2022 an operation against the KA-SAT satellite network operated by the American company Viasat disabled large numbers of the modems its customers used in Ukraine and elsewhere in Europe. The apparent target was Ukrainian military and government communications. Among the collateral casualties was the remote monitoring and control of wind farms in central Europe, whose operators used the same service. In May 2022 the European Union, the United Kingdom and the United States attributed the operation to Russia (UK Government 2022). The episode is consistent with restraint in the narrow sense — it was an adjunct to a conventional invasion, not a substitute for one — and it is a warning about everything the restraint thesis does not cover. A commercial satellite network was a target because the Ukrainian state relied on it, and the firms that suffered had no connection to the war except that they were customers of the same hub.

The most famous exception proves the rule in an instructive way. The operation later known as Stuxnet, discovered in 2010, sabotaged centrifuges at Iran's uranium-enrichment plant at Natanz by manipulating the industrial control systems that ran them. It was widely attributed to the United States and Israel, neither of which has formally acknowledged it. It is the clearest case in the public record of code causing physical destruction to a rival's strategic asset, and it fits the restraint thesis better than the catastrophe narrative: it was narrowly targeted, designed to look like malfunction rather than attack, and aimed at delaying a programme rather than at provoking a war. It also illustrates blowback. The code escaped the plant's isolated network, spread to computers around the world, and was analysed in detail by security researchers, so that

its techniques became available to everyone. For firms, the lesson was that industrial control systems — the machinery of plants, pipelines and utilities — were now a target class, and that the code designed to attack one of them could end up on their own networks.

Low strategic severity is not low commercial cost. An operation that barely registers on a scale of harm between states — a wiper that destroys no infrastructure and kills no one — can halt a company's production for weeks. The restraint thesis is about what states do to each other. The firm's question is what happens to the firms that happen to be in the way.

8.5 Weaponised interdependence

Valeriano and Maness study offensive operations. Farrell and Newman study something quieter and, for a multinational, probably more consequential: the way in which the networks of the global economy can be used by the states that sit at their centre (Farrell and Newman 2019).

Their starting observation is that the networks globalisation built — for finance, for communications, for the internet's physical and logical infrastructure — are not flat meshes in which every node is equal. They are highly asymmetric, with a few hubs through which most traffic passes. Network economics explains why: hubs are efficient, and efficiency attracts more traffic, which makes the hub more central still. The internationalisation of the 1990s was conducted on the assumption that this centralisation was a technical fact with no political content. Farrell and Newman's argument is that the assumption was wrong. A state that has jurisdiction over a hub can use it.

Panopticon effect — the use of jurisdiction over a network hub to *see* what flows through it: to gather information about the activities of other states, firms and individuals whose traffic passes through the hub.

Chokepoint effect — the use of jurisdiction over a network hub to *deny* others access to it: to cut a bank out of a payment system, a firm out of a software supply, a country out of a market for critical components.

The two effects need each other. The panopticon tells a state where to apply the chokepoint, and the chokepoint gives the panopticon its teeth. Farrell and Newman's cases make both concrete. The disclosures by Edward Snowden in 2013 showed the extent to which American intelligence agencies had used the concentration of global internet traffic and data storage in American firms and on American territory to collect information at scale — the panopticon. The disconnection of sanctioned Iranian banks from the SWIFT financial-messaging network in 2012, after the European Union adopted the necessary legal measures, showed how a hub located in one jurisdiction could be used to deny a whole country's banking system its normal connection to the world — the chokepoint.

The implication for a multinational is uncomfortable in two ways. The first is that the firm's exposure to geopolitical risk depends on where the hubs it uses are located and whose law applies to them. A firm whose payments clear through one jurisdiction, whose data are stored with providers subject to another, and whose software updates arrive from a third has three sets of exposure to the coercive choices of three states, most of which have nothing to do with the firm.

The second is more uncomfortable still: **the firm itself can be the hub**. A cloud provider, a payment network, a telecommunications operator, a maker of widely used software — each is infrastructure that its home state can watch through or weaponise, and that rival states therefore have reason to treat as a target, to exclude, or to replace with a domestic alternative. A firm in that position is not a bystander to geopolitical competition. It is part of the terrain. This is the bridge to sanctions in Chapter 10, where the chokepoint becomes a legal instrument, and to space in Chapter 11, where the hubs are in orbit.

8.6 Attribution: evidence about evidence

Every dataset of state cyber operations rests on an act of attribution. Somebody has to say who did it. And attribution in cyberspace has four properties that make it unlike the identification of the attacker in almost any other kind of conflict event.

Attribution — the identification of the actor responsible for a cyber operation, at three levels: the machines used, the individuals or groups who operated them, and the state, if any, that directed or sponsored them. Public attribution is the decision to announce the third.

It is **slow**. Forensic analysis takes weeks or months; public attribution by governments often comes a year or more after the event. It is **probabilistic**. Technical indicators — the code, the infrastructure, the working hours of the operators, the language settings left in a file — can be faked, borrowed or reused, and a careful operator plants false ones. It is **political**. Governments decide when to attribute publicly, to whom, and with how much of their evidence, and they decide in the light of their relations with the accused state and their wish to protect their own sources. And it is **selective**. What is attributed publicly is what someone wanted known.

Rid and Buchanan put the point sharply: attribution is what states make of it (Rid and Buchanan 2015). It is not a technical fact waiting to be discovered but a judgement, assembled from technical, operational and strategic evidence, and communicated for a purpose. Lindsay offered a partial corrective: for the most consequential operations the set of capable actors is small, and the attribution problem is therefore less severe than it is for routine intrusions — which is one reason deterrence of major attacks remains possible (Lindsay 2015). Both views lead to the same conclusion about data.

An event dataset of cyber operations is a record of public attributions, not a record of operations. That sentence is the most important caveat attached to every count in this chapter. It has predictable consequences for what the data show. States that attribute publicly — mostly Western democracies, with active security services and a free press — appear more often as victims than states that do not. Operations by allies against each other are rarely attributed in public and are therefore nearly invisible. Operations that succeeded in remaining undetected are, by definition, absent. A dyad with no incidents in the data may be a dyad with no operations, or a dyad in which neither side chose to say anything.

8.7 When the firm is collateral

The restraint thesis concerns the intentions of states. The cost to firms is driven by something else: the technical reach of what states release.

From the archive · NotPetya, June 2017

On 27 June 2017, the eve of a Ukrainian public holiday, malware spread through the update mechanism of a Ukrainian accounting program widely used by businesses operating in the country. It looked like ransomware, displaying a demand for payment, but its encryption could not be reversed: it was a wiper dressed as a ransom note. Within hours it had spread beyond Ukraine through the internal networks of any company that ran the software at a Ukrainian site. In February 2018 the governments of the United States and the United Kingdom, among others, publicly attributed the operation to the Russian military.

The apparent target was Ukraine. The victims included some of the largest firms in the world. The shipping company Maersk lost the use of systems across its global operations. The pharmaceutical company Merck and the food company Mondelez reported very large losses, as did logistics and manufacturing firms in many countries. None of them had a quarrel with anyone; they had a subsidiary, an office or a supplier in Ukraine, running a program everybody used (Greenberg 2019).

The aftermath carried a second lesson. When the companies claimed under their property insurance, insurers invoked the **war exclusions** that such policies had carried for generations, arguing that an operation by a state's military was an act of war. The litigation lasted years and did not go uniformly the insurers' way; in Merck's case, New Jersey courts held that the traditional exclusion, written with armed conflict in mind, did not cover the attack. The market's response was to write explicit exclusions for state-backed cyber operations into its contracts, which the Lloyd's market required from 2023. The effect is that a firm's cyber cover may now end exactly where geopolitics begins.

Three lessons follow for the firm. The first is that **proximity in cyberspace is technical, not political**. Firms were hit by NotPetya because they ran a piece of software in a country at war, not because anyone chose them. A firm's exposure to a state's cyber operations depends on its network architecture — what it connects to, what it updates from, what its subsidiaries can reach — as much as on its political profile.

The second is that **insurance is not a backstop**. A risk transfer that excludes the scenario the board is most worried about does not transfer that risk. The question to ask of a cyber policy is not whether it covers "cyber attacks" but what it says about operations attributed to states — and who decides the attribution.

The third is that **the board owns it**. Securities regulators increasingly require listed firms to disclose material cyber incidents and to describe how their boards oversee cyber risk. In the United States, rules adopted in 2023 require listed companies to report material incidents promptly and to describe their risk management and governance of cybersecurity in their annual

reports (U.S. Securities and Exchange Commission 2023). The geopolitical part of cyber risk — the part that arises from the rivalry of states — is thereby made a matter of corporate governance, whatever the firm's own view of it.

8.8 Digital sovereignty and the splintering network

Weaponised interdependence has a defensive mirror. States that fear the panopticon and the chokepoint of others respond by asserting control over the networks within their own jurisdiction: where data must be stored, who may access them, which foreign equipment may be installed, which vendors may operate. The vocabulary varies — digital sovereignty, data localisation, trusted vendors, technological self-reliance — and the effect is the same. The single global network on which the internationalisation of the 1990s assumed firms could operate is splintering into jurisdictions with different rules.

Three instruments matter most for a multinational. **Data-localisation and transfer rules** require that certain data be stored within a country or restrict their movement abroad; the invalidation by the Court of Justice of the European Union in 2020 of the framework that governed transfers of personal data from the European Union to the United States showed how quickly a legal basis for a firm's data architecture can disappear. **Security reviews and vendor bans** exclude foreign suppliers from critical networks; the restrictions imposed on Huawei's telecommunications equipment by the United States and a number of its allies from 2019 onward forced operators to redesign and in some cases to strip out networks they had already built. **Access laws** give governments the power to compel firms under their jurisdiction to hand over data, including data stored abroad; the United States' CLOUD Act of 2018 is the best-known example, and it sits awkwardly beside European data-protection law.

For the firm, these instruments generate three kinds of cost. **Duplication**: separate data centres or cloud regions per jurisdiction, separate supply chains for network equipment, separate compliance regimes. **Conflicting obligations**: one state's law requires a disclosure that another's forbids, and the firm must choose which to break. **Vendor risk**: a supplier banned in one market may be mandatory, or at least dominant, in another, so that a single global architecture becomes impossible. None of these costs arises from a cyber operation. All of them arise from the geopolitics of cyberspace.

8.9 Where cyber risk sits in the firm

The five channels through which geopolitical risk reaches a multinational help locate cyber risk precisely. It runs most obviously through **data and operations**: the operation that halts production, the localisation rule that forces a redesign. It runs through **assets and people**, when intellectual property is stolen or staff in a jurisdiction are compelled to provide access. It runs through **finance**, when insurers exclude state-backed operations and when payment hubs become chokepoints. It runs through **inputs**, when a vendor is banned or a software supply chain is compromised — NotPetya travelled through a software update. And it runs through **market access**, when a firm is excluded from a market as a security risk, or when its home state's use of the firm as a hub makes it unwelcome abroad.

Inside the firm, the risk is distributed across several actors, and the distribution is part of the problem. The **board** sets risk appetite and oversees disclosure; its distinctive question is *which state actors would want our data or our downtime, and why?* The **chief information security officer** turns threat intelligence into controls and owns incident response, but typically reports through a technology function with limited geopolitical analysis. The **chief risk officer** is the natural integrator of cyber risk with country, sanctions and supply-chain risk — the view this book takes. The **insurer** prices the risk and, increasingly, draws the line at state-backed operations. And **regulators** impose disclosure, localisation and vendor rules from outside. A firm in which each of these actors holds a piece of the picture and nobody holds the whole has a decision architecture designed for crime and espionage, not for geopolitics.

What should change for a firm with a plant in a country that sits in one of the densest cells of the rivalry matrix? Not, in the first instance, its technology. The first change is to the map: an inventory of which systems at that site connect to which systems elsewhere, which software is updated from which vendors, and which data cross which borders — because it is along those connections that an operation aimed at a government travels into a firm. The second is to segmentation: a subsidiary in a contested jurisdiction whose network can reach the group's core systems is a bridge for anyone who compromises it, and the cheapest geopolitical control a firm can buy is often a wall between the two. The third is to contracts: what the insurance policy says about operations attributed to states, what the cloud provider's terms say about government access in each jurisdiction, and what suppliers have agreed about notifying the firm when they are compromised. The fourth is to

governance: who in the firm asks, before an investment rather than after an incident, which states would want the firm's data or its downtime.

8.10 What the record shows, and how it was made

The descriptive picture at the heart of Valeriano and Maness's book — incidents by year, by method and by severity, and the matrix of who targets whom — can be rebuilt from the public release of DCID, and the codebook repays reading because it is where the concepts of this chapter become columns.

The unit is one incident between an initiating state and a target state within a rival dyad. The dyad's two states are recorded in the order of their Correlates of War codes, so that the first state of the pair is not necessarily the initiator; the initiator is recorded separately, and the target is whichever state of the dyad did not initiate. The method is coded on a scale that runs from vandalism through denial of service and network intrusion to network infiltration, with sub-codes for kinds of malicious software. Severity is coded on a scale the codebook defines from 0 to 10, of which the released file uses only the values 0 to 6 — a discrepancy that, if the codebook's scale applies, is itself consistent with restraint: in two decades of rivalry, nothing in the record reaches the upper part of the scale. The type of target distinguishes private and non-state targets from governmental ones, civilian and military, and the objective distinguishes disruption, short- and long-term espionage and degradation. The September 2022 release contains 433 incidents with start dates from September 2000 to January 2021.

Two properties of the data follow directly from the arguments above. The first is that the population is rival dyads. A cell of the initiator × target matrix that is empty does not mean "no cyber conflict"; it means "no rivalry in the data", because pairs of states that are not coded as rivals never enter. The second is that the incidents are public attributions. The matrix is therefore a picture of who has been publicly *accused* of operating against whom — itself a geopolitical artefact, shaped by which governments attribute and when.

The record does say something about what the operations were *for* and *whom* they hit, and both answers matter to a firm (Table 8.1). Espionage, short- or long-term, is the objective of 263 incidents, three in five; disruption of 120; and degradation — the deliberate damage of a target's systems — of 50. And 175 of the 433 incidents, 40 per cent, are coded as directed at private or non-state targets rather than at governments or militaries. Even in a dataset built to study conflict between states, the target was a company or another non-state body in two incidents in five.

Table 8.1: Cyber incidents between rival states by type of target and objective, 2000–2020. Source: Dyadic Cyber Incident and Campaign Data, version 2.0.

Target	Disruption	Short-term espionage	Long-term espionage	Degradation	Total
Private or non-state	47	50	57	21	175
Government, non-military	59	64	37	18	178
Government, military	14	18	37	11	80
Total	120	132	131	50	433

The methods have shifted too (Table 8.2). In the first decade of the record, vandalism and denial-of-service attacks — the defacements and floods of the Estonian kind — made up a third of the incidents. From 2015 they made up a fifteenth, and network intrusions, the quiet entry that serves espionage and prepares disruption, rose from 40 of 97 incidents to 157 of 216. The change is consistent with Buchanan's account of a world of constant, mutual intrusion, and it matters to a firm because an intrusion, unlike a defacement, may not be noticed at all.

Table 8.2: Cyber incidents between rival states by method and period of start (the last period includes one incident begun in January 2021). Source: Dyadic Cyber Incident and Campaign Data, version 2.0.

Method	2000–2009	2010–2014	2015–2021	Total
Vandalism	13	14	2	29
Denial of service	20	18	13	51
Network intrusion	40	70	157	267
Network infiltration (malware)	24	18	44	86
Total	97	120	216	433

A second dataset, the European Repository of Cyber Incidents, makes both properties visible (Zettl-Schabath et al. 2025). It records incidents with state and non-state initiators, including criminal, hacktivist and unattributed operations, and in its version 1.3 it contains 3,414 incidents with start dates from 2000 to 2024 — several times the size of DCID (Table 8.3). Restricting it to incidents attributed to state or state-affiliated actors leaves 1,014; restricting it to states alone leaves 329. The gap between those numbers, and between all of them and DCID's 433, is a measure of how much the answer to "how much cyber conflict is there?" depends on the definition of the population and of attribution (Figure 8.3).

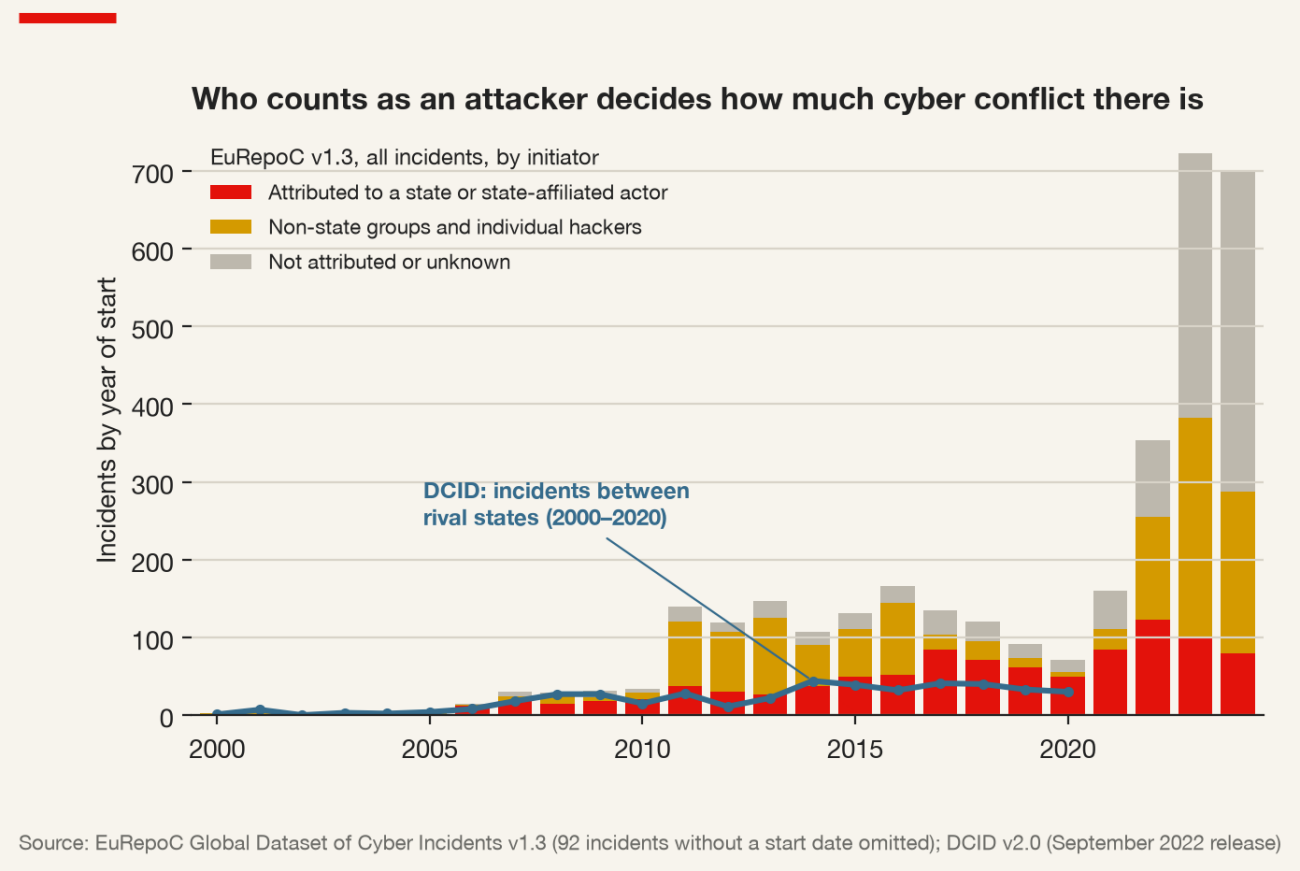


Figure 8.3: How much cyber conflict there is depends on who is counted as an attacker. Incidents by year of start in the European Repository of Cyber Incidents, by type of initiator, against the incidents between rival states in DCID. Source: EuRepoC Global Dataset of Cyber Incidents, version 1.3 (92 incidents without a start date omitted); Dyadic Cyber Incident and Campaign Data, version 2.0.

The wider record shows the restraint thesis in a different light. In DCID's peak year, 2014, it records 44 incidents between rivals. EuRepoC records 71 incidents of all kinds beginning in 2020, 354 in 2022, 723 in 2023 and 700 in 2024. Part of that growth is real — the war in Ukraine and the hacktivist campaigns around it — and part is the repository's own expanding coverage of sources, the selection problem that every event count built from reports carries. What changes most

is the composition. Of the incidents EuRepoC dates to 2000–2010, 58 per cent are attributed to a state or a state-affiliated actor; of those dated 2011–2020, 41 per cent; of those dated 2021–2024, 20 per cent. The growth is overwhelmingly in incidents attributed to non-state groups or to nobody, and half of all incidents in the repository name critical infrastructure or a corporate target among their receivers — a share no lower, 51 per cent, among those attributed to states.

Where the two datasets overlap, they agree about the principal initiators (Figure 8.4). China is named in 27 per cent of DCID's incidents and in 26 per cent of EuRepoC's state-attributed ones; Russia in 26 and 20 per cent; Iran in 15 and 13; North Korea in 14 and 14; the United States in 6 and 4. Nearly three-quarters of EuRepoC's state-attributed incidents name at least one of the four. Two independent projects, with different populations, coders and sources, converge on the same short list — which is either evidence that the list is right or evidence that both rely on the same attributions by the same governments and security firms, and the honest answer is that it is some of each.

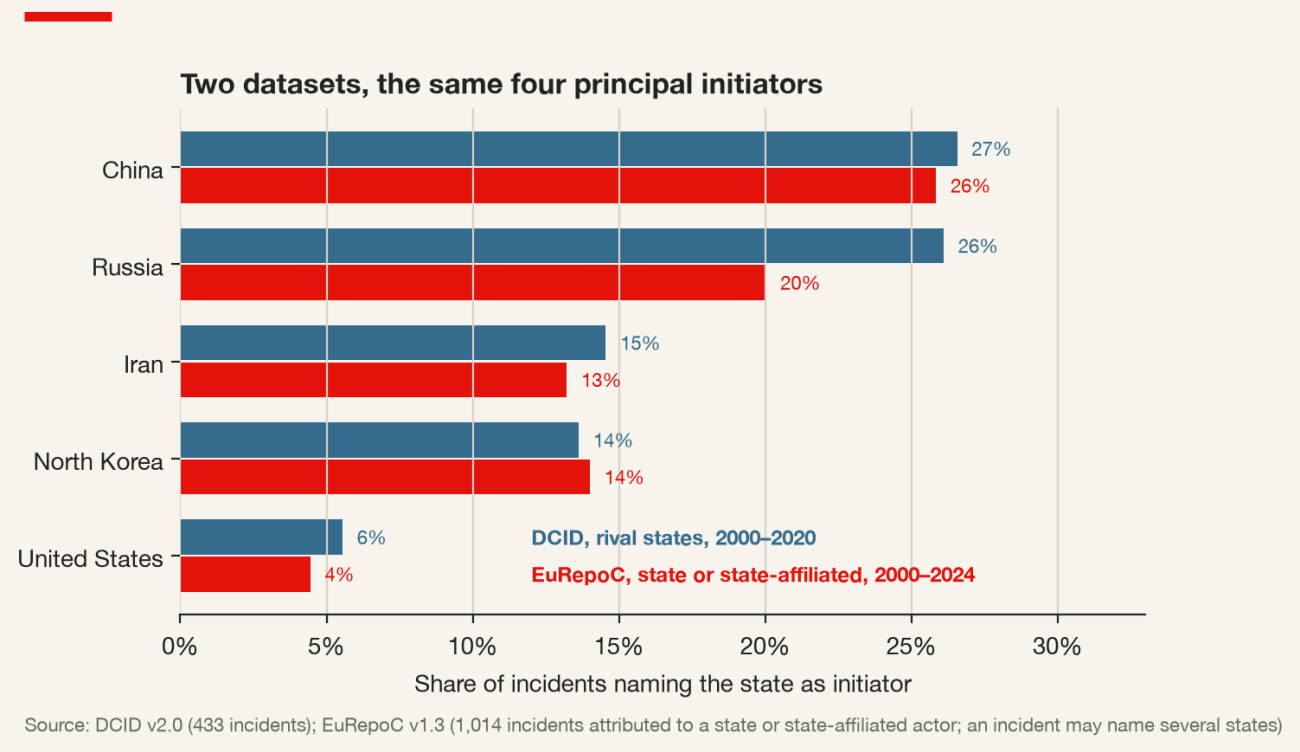


Figure 8.4: Two datasets with different populations converge on the same four principal initiators. Share of incidents naming each state as initiator, in DCID and in the EuRepoC incidents attributed to a state or state-affiliated actor. Source: Dyadic Cyber Incident and Campaign Data, version 2.0; EuRepoC Global Dataset of Cyber Incidents, version 1.3.

Table 8.3: Two records of cyber conflict and the populations behind them. Source: Dyadic Cyber Incident and Campaign Data, version 2.0; EuRepoC Global Dataset of Cyber Incidents, version 1.3.

Record	Who can be an initiator	Start dates	Incidents
DCID v2.0	rival states	2000–2020 (one in January 2021)	433
EuRepoC v1.3, all incidents	anyone, including unknown	2000–2024	3,414
… attributed to a state or state-affiliated actor			1,014
… of which to a state alone			329

Record	Who can be an initiator	Start dates	Incidents
… attributed to non-state groups or individual hackers			1,230
… not attributed, or initiator unknown			1,170

The two datasets do not contradict each other. They answer different questions: how rival states use cyber operations against each other, and how much publicly reported cyber activity of any origin there has been. A firm that reads the first as if it answered the second will underestimate its exposure several times over; one that reads the second as if it measured state strategy will see a war where there is mostly crime.

8.11 A short review of the literature

The debate over whether cyberspace is a domain of war was framed in the early 2010s. Rid (2012) and Rid (2013) argued that cyber war had not taken place and would not, reclassifying cyber operations as sabotage, espionage and subversion; Gartzke (2013) argued from the logic of force that cyber operations would complement rather than replace conventional power; Kello (2013) and Kello (2017) argued, against both, that the cyber revolution unsettled the conceptual foundations of international relations. The legal question of how existing international law applies to cyber operations is set out in Schmitt (2013).

Systematic evidence arrived with Valeriano and Maness (2015), whose dataset of incidents between rival states supported a thesis of restraint and regional rivalry. The dataset has been extended through successive releases, of which Maness et al. (2022) is the current public version; Maness and Valeriano (2016) examine whether cyber incidents shift the broader conflict and cooperation between rivals, and Valeriano, Jensen, and Maness (2018) develop the argument that cyber operations are weak instruments of coercion. Kostyuk and Zhukov (2019) tested cyber operations in the war in eastern Ukraine against battlefield events and found little effect. Buchanan (2017) sets out the cybersecurity dilemma created by intrusions that are defensive in purpose and threatening in appearance.

On attribution, Rid and Buchanan (2015) treat it as a political judgement assembled from technical and strategic evidence, and Lindsay (2015) argues that its difficulty is overstated for consequential operations. On the economic architecture of coercion, Farrell and Newman (2019) is the foundational statement of weaponised interdependence, with its panopticon and chokepoint effects. Wider datasets that include non-state actors, such as Zettl-Schabath et al. (2025), now allow the rival-dyad picture to be set against the full record of attributed incidents, and the firm's experience of state operations is documented most vividly in Greenberg (2019).

8.12 What the evidence does not license

The record of cyber conflict between rivals is informative. It shows that incidents concentrate in a few dyads, that most are of low severity, and that the methods have shifted over two decades. It is tempting to read the matrix of initiators and targets as a map of where a multinational's cyber risk lies. It is not, and the reasons are the arguments of this chapter.

The matrix does not license a firm to conclude that it is safe in a country whose cells are empty. An empty cell means no publicly attributed incident between *rivals*; it says nothing about operations that were never attributed, about operations between states that are not coded as rivals, or about criminal groups, which account for much of the cyber harm firms actually suffer. Nor does a dense cell tell a firm that it is a target. The incidents in that cell were aimed mostly at governments, and the firm's exposure to them depends on whether it is technically in the path — whether it runs software, holds data or provides infrastructure that an operation against the government might traverse. NotPetya appears in the data as two incidents in two dyads, Russia against Ukraine and Russia against the United States; its cost fell on firms in dozens of countries. The attack on the satellite network in 2022 would appear, if at all, as an incident between Russia and Ukraine; its casualties included wind farms in Germany.

Nor does the evidence for restraint license complacency. It is a pattern in state behaviour to 2020, measured on a scale of harm between states. It does not bound the commercial cost of a low-severity operation, and it does not tell a firm what has happened since a large-scale war began in Europe. Restraint describes what states have chosen not to do to each other. It is silent about what they have done to the networks that happen to connect them, and about the coercion that weaponised interdependence allows without a line of malicious code being written.

What the evidence does support is a reordering of the firm's questions. The first is not "are we a target?" — for most firms the honest answer is "probably not, deliberately" — but "what are we connected to?": which of the firm's networks, suppliers, software vendors and hubs lie in the path of the rivalries the record shows to be dense. The second is "whose jurisdiction are we in?": which states can see through, or cut off, the services on which the firm depends. The third is "who pays?": what the firm has agreed with its insurer about the operations that a government will one day attribute to a state. A firm that can answer those three questions has turned cyber risk from a line in the information-technology budget into what the evidence says it is — a geopolitical exposure, carried mostly by firms that nobody meant to hit.

Chapter 9

Economic Diplomacy Redefined: Two-Level Games, Embassies and the Firm

9.1 From serving firms to screening them

For most of the period covered by Chapter 2 — the decades in which the liberal economic order was built out and then globalised — economic diplomacy had a settled meaning and a settled direction. Diplomats negotiated trade agreements, opened markets, promoted exports and courted investors. Trade commissioners in embassies found distributors for small exporters; ministers led delegations of chief executives on state visits; export credit agencies insured contracts that private insurers would not touch. The state served the firm, and the firm's success abroad was counted as the state's success.

That meaning has not disappeared, but it has been joined by others, and the direction of travel has partly reversed. The same governments that promote their firms abroad now screen foreign firms at home, restrict what their own firms may sell and to whom, and draw companies into public quarrels conducted on social media. A multinational today is still the beneficiary of economic diplomacy. It is also, increasingly, its object and its instrument — and sometimes a diplomat in its own right.

Economic diplomacy — the use of the instruments of foreign policy for economic ends, and of economic instruments for foreign-policy ends. The first covers trade negotiation, export and investment promotion; the second covers sanctions, export controls, aid and investment screening used to pursue security or political goals.

Commercial diplomacy — the part of economic diplomacy that serves particular firms and sectors: trade commissioners, market intelligence, introductions, advocacy in disputes, trade missions, export credit and investment promotion. Its unit of account is the firm, not the economy.

Three shifts have redefined the field. The first is **securitisation**: trade, investment and technology are now treated as matters of national security, so that a firm faces screening, export controls and "trusted vendor" rules not only from rival states but from its own side. The second is the move **from promotion to coercion**: the networks that were built to promote exports — financial messaging, payment systems, technology supply chains — are the same networks that states now use to deny access, as the chokepoints of Chapter 8 showed and the sanctions of Chapter 10 will show in detail. Home-state support can become home-state constraint overnight. The third is **digitalisation**: diplomacy is increasingly conducted in public, on social media, addressed to domestic and foreign audiences at once, and firms are pulled into it — through boycotts, hashtags and official denunciations — without having chosen to be.

This chapter takes the redefinition in three steps. It begins with the theory that best captures how international bargaining is shaped by domestic politics, Robert Putnam's two-level game, and with the argument of Corneliu Bjola and Ilan Manor that digital diplomacy has changed how that game is played. It then turns to the empirical question at the heart of the evidence: whether the most traditional instrument of commercial diplomacy, the embassy, actually promotes exports, as Andrew Rose found. Finally it turns to the relationship between states and firms directly — bargaining, screening and the choice a firm faces about when to lean on its home state.

9.2 Two boards at once

Putnam's article begins from a frustration with theories of international relations that treated the state as a unitary actor, and with theories of domestic politics that ignored the international environment (Putnam 1988). His alternative is to model an international negotiation as two games played simultaneously.

At **Level I**, negotiators representing their governments bargain with each other to reach a tentative agreement. At **Level II**, each negotiator's domestic constituencies — legislatures, parties, interest groups, bureaucracies, public opinion — decide whether to ratify it. Ratification need not be formal; any domestic process that can kill an agreement counts. An agreement survives only if it is acceptable at both levels, on both sides.

Win-set — for a given Level II constituency, the set of possible Level I agreements that it would ratify. An agreement is possible only where the two sides' win-sets overlap. The size of each win-set depends on the distribution of power and preferences at home, on the domestic institutions that govern ratification, and on the strategies of the negotiators themselves.

Three lessons follow, and each has a direct application to firms.

The first is that **larger win-sets make agreement more likely**. If many possible deals would be ratified at home, the negotiators have more room to find one that the other side can also accept. But a large win-set also leaves a negotiator open to being pushed: the other side knows that concessions can be extracted without endangering ratification.

The second, and the most counter-intuitive, is that **a small win-set can be a bargaining asset**. A negotiator who can say, credibly, "my parliament will never accept that" has a reason to refuse that the other side cannot argue away. Thomas Schelling had made a version of this point about the power of binding oneself; Putnam located the binding in domestic politics. The credibility matters. A constraint that is invented for bargaining purposes is a bluff, and bluffs can be called.

The third is that **negotiators play both boards at once**. A skilled negotiator does not simply report the constraints of Level II; he or she works to shape them — building support at home for a deal that is coming, and reaching across to the other side's Level II to make its ratification easier. Putnam called the latter "reverberation": a move at Level I that changes the balance of opinion in the other country's domestic politics.

Putnam's framework also explains two features of international agreements that otherwise look irrational. The first is **involuntary defection**: an agreement that a negotiator sincerely accepted fails at Level II because the domestic coalition behind it collapses. The second is the role of **side-payments and issue linkage**: a negotiator who cannot win ratification on the merits of a deal can enlarge the win-set by compensating the losers at home, or by linking the deal to a concession on another issue that brings a new constituency into the coalition. Both have a direct bearing on firms. The firms and sectors that stand to lose from an agreement are the ones a government must compensate or overrule; the firms that stand to gain are the ones it mobilises. And the domestic institutions that govern ratification — a legislature's voting rules, the role of sub-national governments, the timing of elections — determine how large the win-set can be, which is why a firm's analysis of a negotiation that affects it should begin with the other side's ratification process rather than with the negotiators' public statements.

9.2.1 Where the multinational sits

Draw the two-level game on a board — two negotiators facing each other at Level I, each standing on a circle of domestic constituencies at Level II — and ask where a multinational belongs. The answer is that it belongs in *several* circles at once. At home, an export-oriented firm or industry is one of the constituencies whose support enlarges the win-set for a trade agreement; firms lobby, testify and fund campaigns, and governments count their support before signing. Abroad, the same firm's subsidiaries, plants and employees are part of the *other* country's Level II, where they may be assets (a large employer can argue for an agreement locally) or hostages (a visible foreign firm is a convenient target for retaliation if negotiations go badly).

The double position changes what a firm should expect from negotiations that concern it. A firm whose home government is negotiating with a host government on its behalf is not a spectator. It is part of the domestic politics that determine what its own negotiator can accept, and part of the foreign domestic politics that determine what the other side can accept. The question it must ask is which of its positions — lobbyist at home, employer abroad, symbol of its home country — the negotiators will use, and whether it can control that use.

The renegotiation of the North American Free Trade Agreement between 2017 and 2018 shows the game at work, with firms on every board. The United States imposed tariffs on Canadian and Mexican steel and aluminium in June 2018, on national-security grounds, while the negotiation was under way — a Level I move aimed squarely at the other sides' Level II, where steel and aluminium producers and their workers would press their governments to settle. Canada's system of supply management in dairy was a constraint at its own Level II, defended by a well-organised domestic constituency, and the eventual agreement opened only a limited part of the dairy market. Automobile manufacturers, whose supply chains crossed all three borders many times, were Level II actors in all three countries at once, and the new rules of origin for

vehicles were written with their production networks in view. For each firm involved, the question was not only what the agreement would say but which domestic coalition in which country would decide whether there was an agreement at all.

9.3 The two-level game goes digital

Putnam wrote in 1988, about a world in which Level II was reached mainly through legislatures, interest groups and the press. Bjola and Manor revisit his model three decades later through the negotiations over Iran's nuclear programme, which culminated in the Joint Comprehensive Plan of Action of July 2015 (Bjola and Manor 2018). Their argument is that **digital diplomacy** — ministries, embassies and leaders communicating directly with publics on social media — gives negotiators a new instrument for acting on Level II, both their own and the other side's.

The mechanics are straightforward. Before social media, a negotiator who wanted to build domestic support for a deal worked through intermediaries: parliamentary allies, friendly editors, interest groups. A negotiator who wanted to reach the other side's public had few channels at all. Digital platforms remove the intermediaries. Foreign ministers and their staff can address their own public directly, framing the negotiation as it happens, and can address the other side's public in its own language, bypassing its government. In the Iran negotiations, Bjola and Manor show, governments used these channels to shape how the negotiation was perceived at home and abroad, attempting to widen the win-sets on which ratification depended while the bargaining was still going on.

Two implications matter for multinationals. The first is **speed**. Domestic opinion now moves within the negotiation rather than after it, so that the outcome a firm plans around can change faster, and for reasons visible to anyone who follows the right accounts. The second is **exposure**. A firm's brand can be pulled into a digital campaign by either government — praised as a model of cooperation, denounced as a symbol of the other side, or made the subject of a boycott that a ministry amplifies. Neither role is chosen by the firm.

The Iran agreement also illustrates the fragility of win-sets over time, which is the deeper lesson for planning. The agreement was concluded and ratified in the sense Putnam meant; it then lost its domestic support in one of the countries that had made it. The United States withdrew in 2018 and reimposed sanctions, including sanctions aimed at foreign firms that continued to do business with Iran. European companies that had re-entered the Iranian market after 2015 found that the domestic politics of a third country — the United States — determined whether they could stay. The two-level game, for a multinational, is played on more boards than two.

9.4 Do embassies sell?

The most traditional instrument of commercial diplomacy is the diplomatic post itself: the embassy, high commission or consulate, with its commercial section, its trade commissioners and its network of contacts. Governments spend substantial sums maintaining these networks and justify part of the cost by their contribution to trade. Rose asked whether the justification holds (Rose 2007).

His design combined, in a single recent cross-section, data on the embassies and consulates that 22 large exporting countries maintain in some 200 destinations with their bilateral exports, using the framework that economists have used to explain bilateral trade since Jan Tinbergen first applied it in the 1960s (Tinbergen 1962).

Gravity model — the empirical relationship in which bilateral trade between two countries rises with their economic size and falls with the distance between them, by analogy with Newton's law of gravitation. Modern versions add measures of trade costs — common language, shared borders, colonial ties, trade agreements — and controls for each country's overall openness to trade with all partners.

The model's empirical success is remarkable — few relationships in economics fit cross-country data as well — and its theoretical foundations were put on a firm footing by Anderson and van Wincoop, who showed that what matters is not only the trade cost between two partners but each partner's trade costs with everyone else, which they called multilateral resistance (Anderson and Wincoop 2003). Head and Mayer's survey is the practical guide to estimating it (Head and Mayer 2014). For Rose's purposes the gravity model is a way of asking a precise question: *holding constant the size of the partners and the distance between them, and the other standard determinants of trade, do countries export more to places where they keep more diplomatic posts?*

His answer was yes. Exports are higher where a country keeps more posts, and the association survives the standard gravity controls. Bilateral exports rise by approximately six to ten per cent for each additional consulate abroad, and the effect varies by exporter and is non-linear: consulates have smaller effects than the creation of an embassy (Rose 2007). The size should be read off his tables, by exporter and type of post, rather than remembered from a summary.

9.5 Three objections

Rose's result faces three objections, and a fourth problem that lies in the data. Each is a general problem in inferring the effect of a policy from a cross-section, and each will recur in the evaluation of sanctions in Chapter 10.

9.5.1 Reverse causality

Countries open posts where they already trade. A foreign ministry deciding where to open a new consulate looks, among other things, at where its exporters are active and where its nationals travel and invest. If posts follow trade, then a cross-section in which posts and trade go together tells us that ministries are sensible, not that embassies generate exports. The two explanations predict the same snapshot.

They predict different *dynamics*. If posts promote exports, then pairs of countries in which a post is opened should see faster subsequent growth in exports than comparable pairs in which it is not. If posts follow exports, then export growth should come first and the post should follow. Telling the two apart requires data over time, and even then requires that the timing of openings not be driven by anticipated growth — a ministry that opens a consulate because it expects a market to boom will produce a post that "precedes" growth without causing it. Rose took the objection seriously and argued that his evidence was consistent with posts having an effect, but a cross-section cannot settle the question. The networks themselves do change, and the changes offer a first, rough test: between 2017 and 2023 the major diplomatic services opened some posts and closed others, and a relationship that holds in the cross-section should leave a trace in the trade of the pairs whose posts changed (Figure 9.3). A trace, however, is not proof, for the reason just given — a post opened in anticipation of growth will appear to cause it.

9.5.2 Omitted variables

Large, rich and nearby partners receive more posts *and* more trade, for reasons that have nothing to do with each other. The gravity controls are designed to absorb exactly this, but a raw scatter of exports against posts does not include them. The simplest correction — dividing exports by the partner's GDP — removes the most obvious confounder and shows how much of the raw relationship was simply size. Other confounders are harder: a shared language, a large diaspora, a history of colonial ties or a security alliance may raise both the number of posts and the volume of trade.

9.5.3 Measurement

A post is not a budget. A small consulate with two staff and a large embassy with a commercial section of fifty count the same in a tally of posts, and an honorary consulate run by a local businessman may count too. The type of post matters, which is why embassies and consulates must be counted separately. The direction of the measurement problem is not obvious. Consulates are often placed in commercial cities precisely to serve trade; yet Rose found the creation of an embassy to matter more than an additional consulate, which may say as much about what an embassy signals as about what it does.

9.5.4 The zeros

There is a fourth problem, which the data make unavoidable. The most comprehensive public record of diplomatic networks, the Lowy Institute's Global Diplomacy Index, lists posts that exist (Lowy Institute 2024). A country with no post in a partner does not appear — the zero is missing, not zero — and must be constructed. Once constructed, the zeros create a second difficulty: the logarithm of zero is undefined, so on a chart with logarithmic axes the partners with no posts fall off the plot. Dropping them silently changes the sample in a way that is correlated with the outcome, because the partners with no posts are disproportionately the small and distant ones with little trade. The same problem in the trade variable — pairs of countries that do not trade at all — led Santos Silva and Tenreyro to argue that the standard practice of estimating gravity equations in logarithms is biased, and to propose an estimator that keeps the zeros in (Santos Silva and Tenreyro 2006). The lesson for a chart is simpler: say how many observations a log scale has removed, and show what the relationship looks like with and without them.

A trend line with a high R^2 and a tiny p-value does not show that embassies cause exports. It shows a strong association in a cross-section. The R^2 measures how closely the points cluster around the line; the p-value measures how surprising the slope would be if there were no association at all. Neither says anything about which way the causation runs, or whether a third variable drives both.

9.6 The rest of the toolbox, and what the evidence says

Embassies are one instrument among several. The literature on the others is useful both for what it finds and for how it finds it, because each study faces the same objections as Rose.

State visits and trade missions. Nitsch examined the travel of the heads of state of France, Germany and the United States from 1948 to 2003 and found that state and official visits are associated with exports higher by about 8 to 10 per cent; when he turned to a difference-in-differences design to address reverse causality, the effect on export growth was strong but short-lived (Nitsch 2007). Head and Ries studied Canada's trade missions, which during the 1990s took the prime minister abroad with large delegations of provincial premiers and business leaders. Their conclusion was sobering. Canada does trade more than normal with the countries to which it sent missions, but the missions do not seem to have caused the increase: in their preferred specification, with country-pair fixed effects, the effects are small, negative and mainly insignificant (Head and Ries 2010).

From the archive · Team Canada, 1994

In November 1994 the Canadian prime minister, Jean Chrétien, led a delegation to China that included most of the provincial premiers and hundreds of business representatives. The mission was branded "Team Canada", and it became the model for a series of similar missions over the following years, to destinations in Asia, Latin America, Europe and elsewhere. Each mission announced a large total of contracts and agreements signed during the visit; according to the Canadian government, the missions generated tens of billions of dollars in new business deals (Head and Ries 2010).

The announced totals were the wrong evidence. Critics pointed out that many of the agreements were memoranda of understanding, that some were deals that had been negotiated before the trip and timed to be signed during it, and the missions were sent to markets that were already growing. Head and Ries's later evaluation used the missions as a natural test of whether high-profile commercial diplomacy changes trade flows, and compared Canada's trade with mission destinations before and after the missions against the trade it would otherwise have been expected to have. They found that the missions had small, negative and mainly insignificant effects.

The episode is a clean illustration of the reverse-causality objection to Rose. Governments direct their most visible commercial diplomacy at the markets they expect to matter — which are, very often, the markets that would have grown anyway. A ministry that evaluates its missions by the deals announced during them, or by the subsequent growth of trade with the destinations, will always conclude that the missions succeeded.

Export promotion agencies. Lederman, Olarreaga and Payton surveyed national export promotion agencies and found that, on average, their budgets are associated with higher exports, with diminishing returns as budgets grow and with considerable differences in effectiveness depending on how the agencies are organised (Lederman, Olarreaga, and Payton 2010). **The body of evidence as a whole.** Moons and van Bergeijk assembled 32 empirical studies published between 1986 and 2011 into a meta-analysis and found a positive and significant effect of economic diplomacy on trade and investment — with the

exception of state visits — and reported effects that are sensitive to the design of the individual studies (Moons and Bergeijk 2017). The fair summary is that commercial diplomacy probably helps at the margin, that the size of its effect is hard to pin down, and that the most visible instruments are the most vulnerable to the selection problems that inflate naive estimates.

9.7 Triangular diplomacy

The two-level game puts firms inside the domestic politics of states. A second tradition puts them across the table. John Stopford and Susan Strange argued that the changing structure of the world economy had turned diplomacy into a triangle (Stopford, Strange, and Henley 1991; Strange 1992). Alongside the familiar **state–state** diplomacy of governments, there is **state–firm** diplomacy, in which governments bargain with multinationals over the terms of investment — where a plant will be built, what share of inputs will be sourced locally, what will be exported — and **firm–firm** diplomacy, in which companies form alliances, joint ventures and consortia that shape markets as surely as treaties do. Governments, on this account, compete for world market shares through the firms they attract, and firms compete partly by bargaining with governments.

The bargaining has a dynamic, and Raymond Vernon described it before Stopford and Strange formalised the triangle (Vernon 1971). Before an investment is made, the firm holds the cards: it can choose among countries, and each host must offer terms good enough to win. Once the investment is made — once the mine is dug, the plant built, the pipeline laid — the firm's capital is sunk and cannot be moved. The host's bargaining power rises and the firm's falls. Terms that were agreed when the firm was mobile can be renegotiated when it is not.

Obsolescing bargain — the tendency for a multinational's bargaining power over a host government to decline after its investment is sunk, so that the terms agreed at entry become vulnerable to renegotiation, higher taxation or, in the limit, expropriation. The bargain "obsolesces" because the conditions that made it acceptable to the host no longer hold.

From the archive · The nationalisation of Chilean copper, 1971

In the 1960s the large copper mines that dominated Chile's exports were owned by two American companies, Anaconda and Kennecott. The terms on which they operated had been renegotiated more than once during the decade, in each case with the Chilean state taking a larger share — first through taxation, then through partial state ownership under the "Chileanisation" policy of President Eduardo Frei. In July 1971, under President Salvador Allende, the Chilean Congress unanimously approved a constitutional amendment nationalising the large copper mines. The compensation formula allowed the government to deduct "excess profits" earned in previous years, and the deductions largely eliminated what the companies would otherwise have been owed.

The episode is the textbook case of the obsolescing bargain, and it was a geopolitical event as much as a commercial one. The companies' home government treated the nationalisation as a matter of foreign policy, and the dispute became entangled with the wider hostility of the United States to Allende's government in the years before the coup of 1973. Kobrin's study of expropriations across developing countries from 1960 to 1979 placed Chile's in a wider wave, in which natural resources — where capital is most sunk — were prominent (Kobrin 1984). For a firm, the lesson has two halves: sunk capital weakens bargaining power over time, and a dispute over that capital can quickly stop being a dispute between a firm and a host and become one between two states.

9.8 Screening: diplomacy aimed at firms

The newest element of economic diplomacy reverses the direction of the oldest. Commercial diplomacy helps a state's firms abroad; **investment screening** examines foreign firms at home, and blocks or conditions their acquisitions on grounds of national security.

The United States has had a screening body, the Committee on Foreign Investment in the United States (CFIUS), since the 1970s, but its reach and visibility have grown sharply. In 2006 the acquisition by Dubai Ports World of a British company that operated terminals at several American ports, approved under the existing procedures, provoked a political storm in Congress; the company agreed to divest the American operations, and the following year Congress tightened the law. A further reform in 2018 widened the committee's jurisdiction to non-controlling investments in businesses involving critical technologies, critical infrastructure and sensitive personal data. The European Union adopted a framework regulation for

the screening of foreign direct investment in 2019 (Regulation 2019/452), which created a mechanism for member states and the Commission to share information and comment on each other's cases. Canada added a national-security review to the *Investment Canada Act* in 2009, and has since used it in a widening range of sectors.

The widening is the important part. Screening began as a defence matter — who may own a supplier of military equipment. It now reaches data, critical minerals, infrastructure, semiconductors and artificial intelligence, which is to say much of the modern economy. For a multinational this is geopolitical risk through the **market access** and **assets** channels at once: an acquisition can be blocked, a completed one unwound, and a firm's ownership structure — who its shareholders are, where its parent is incorporated — becomes a determinant of what it may buy. It is also, in Putnam's terms, a Level II phenomenon: screening decisions are made by bodies that answer to domestic politics, and they are subject to the same reverberation as any other part of a negotiation between states.

9.9 When to lean on the home state

Commercial diplomacy is not only promotion. When a firm's investment is threatened by a host government — an expropriation, a cancelled licence, a discriminatory regulation — its home state's embassy is one of the places it turns. Gertz, studying the commercial work of American embassies, showed that diplomats routinely intervene on behalf of their firms in disputes with host governments, and argued that commercial diplomacy is therefore in part an instrument of political-risk management: a way of raising the cost to a host of mistreating a foreign investor (Gertz 2018).

That protection has a price, and the price has risen with the redefinition this chapter describes. In a dispute that remains commercial, home-state backing helps: the host weighs the diplomatic cost of mistreatment against the gain. In a confrontation between states, being seen as an instrument of one of them can make the firm a target of the other. A firm that accepts its ambassador's offer to raise its mining concession with a host president has gained an advocate and has also, in the host's eyes, become part of its home state's foreign policy. If relations between the two states deteriorate, that association will not be forgotten. Blackwill and Harris describe a world in which states use economic instruments systematically for geopolitical ends (Blackwill and Harris 2016); in such a world, the line between being protected by one's home state and being used by it is thin.

Canada's experience after December 2018 shows how quickly a firm can move from beneficiary to hostage. When Canadian authorities arrested a senior executive of Huawei in Vancouver at the request of the United States, China detained two Canadian citizens and, in March 2019, suspended the export permits of major Canadian canola exporters, citing quality concerns. The exporters had done nothing in the dispute; they were simply the most visible Canadian commercial interest in China that could be squeezed without large cost to China itself. Their home state could protest, and did, but it could not restore their market access without conceding on an issue that was not theirs. In Putnam's terms, they had become instruments for acting on Canada's Level II — a constituency whose losses were meant to raise the domestic cost of the government's position.

The decision can be framed with the five channels used throughout this book. Leaning on the home state may protect **market access** and **assets** in a dispute, but it may endanger **market access** in a rival's market, **inputs** sourced from a rival, **finance** from banks that fear sanctions, and the safety of **people** posted in a country where the firm has become a symbol. And it may draw the firm into the **data** demands that come with being seen as part of a state's infrastructure. The question is not whether a firm should ever accept help. It is whether the firm has thought about what the help will mean if relations between the two states change — and whether it is prepared to be, in the eyes of its host, its government's instrument.

9.10 What the diplomatic networks show

The Lowy Institute's index records the networks of 66 countries and territories — G20, OECD and Asian states, with data collected up to November 2023 — at the level of the individual post. Each of its roughly 7,700 entries for 2023 is one post, with the sending country, the host country, the city and the type of post, which distinguishes embassies and high commissions from consulates, permanent missions and other offices. Earlier releases allow the 2023 networks to be compared with those of 2017, which cover 60 senders.

The headline the index made famous is a change at the top. In 2017 the United States kept the largest diplomatic network in the record, with 274 posts against China's 271. By 2023 the order had reversed: China kept 274 and the United States

271, with Türkiye, Japan and France following at around 250 each (Lowy Institute 2024). The difference at the top is small, and a tally of posts says nothing about the size of the staff in them or the quality of their work. But the composition is revealing. China's lead rests on its consulates — 91 against the United States' 83 in 2023 — which are the posts most directly concerned with trade, investment and nationals abroad, while the United States keeps more permanent missions to international organisations. A state's diplomatic network is a revealed preference: it shows where a foreign ministry has chosen to spend scarce money, and the shift at the top is a shift in where two great powers expect their commercial and political interests to lie.

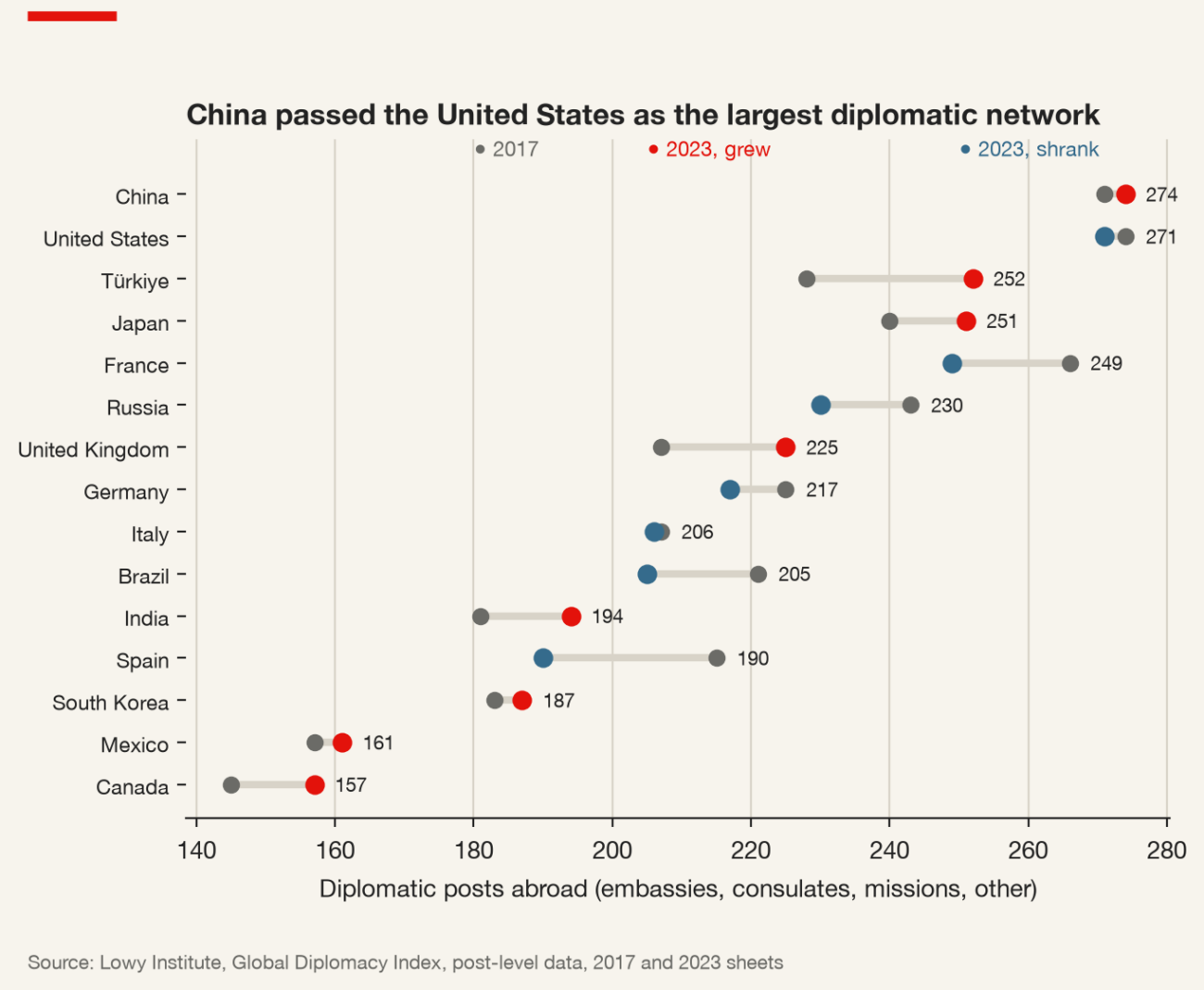


Figure 9.1: China passed the United States as the largest diplomatic network between 2017 and 2023, while France, Russia, Brazil and Spain shrank theirs. Posts abroad of the fifteen largest networks in 2023, against the same senders in 2017. Source: Lowy Institute, Global Diplomacy Index, post-level data.

The post-level data show how the reversal happened, and the story is less dramatic than the headline (Figure 9.1 and Table 9.1). China gained three posts in six years and the United States lost three. China added nine embassies, from 164 to 173, and closed four consulates; the United States kept its embassies almost unchanged, at 168, and closed seven consulates, from 90 to 83. The more striking movements are further down. Türkiye added 24 posts and now has the most consulates of any country in the index, 93; the United Kingdom added 18. France lost 17 posts, mostly consulates, and Russia 13 — twelve of them consulates, over a period that includes the expulsions and closures that followed the invasion of Ukraine.

Table 9.1: The ten largest diplomatic networks in 2023, by type of post, and their size in 2017. Source: Lowy Institute, Global Diplomacy Index, post-level data (2023 and 2017 sheets).

Sender	Embassies and high commissions	Consulates	Permanent missions	Other	Total 2023	Total 2017	Change
China	173	91	8	2	274	271	+3
United States	168	83	12	8	271	274	−3
Türkiye	145	93	12	2	252	228	+24
Japan	152	66	10	23	251	240	+11
France	158	72	18	1	249	266	−17
Russia	143	74	10	3	230	243	−13
United Kingdom	156	51	11	7	225	207	+18
Germany	148	56	11	2	217	225	−8
Italy	124	74	8	0	206	207	−1
Brazil	133	58	12	2	205	221	−16

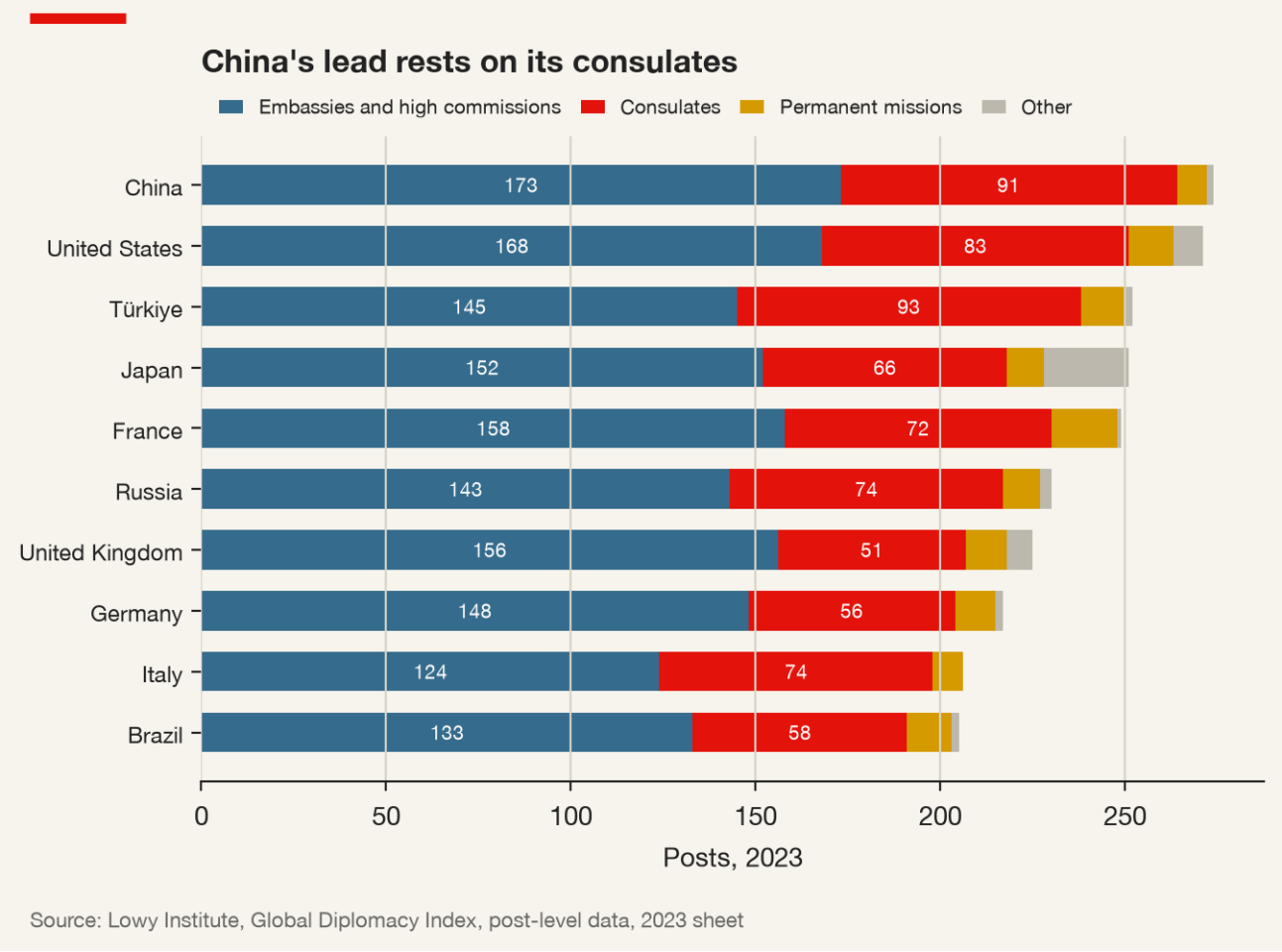


Figure 9.2: China's lead over the United States rests on embassies and consulates; the United States keeps more permanent missions and other offices. Posts of the ten largest networks by type, 2023. Source: Lowy Institute, Global Diplomacy Index, post-level data.

Across all the senders the index covers in both years, the networks grew rather than shrank: the 60 foreign services present in 2017 kept 85 more posts in 2023, and 33 of them grew while 21 shrank (Figure 9.3). That pattern sits awkwardly with any simple story of diplomatic retrenchment, and it is the raw material for the dynamic test of Rose's result: if posts promote exports, the pairs in which Türkiye, Ireland or the United Kingdom opened posts after 2017 should show faster export growth than comparable pairs in which they did not, and those in which Spain or France closed posts slower growth. Six years and a pandemic are a short and noisy window, and the selection problem applies to closures as much as to openings: a ministry closes a consulate where it expects little.

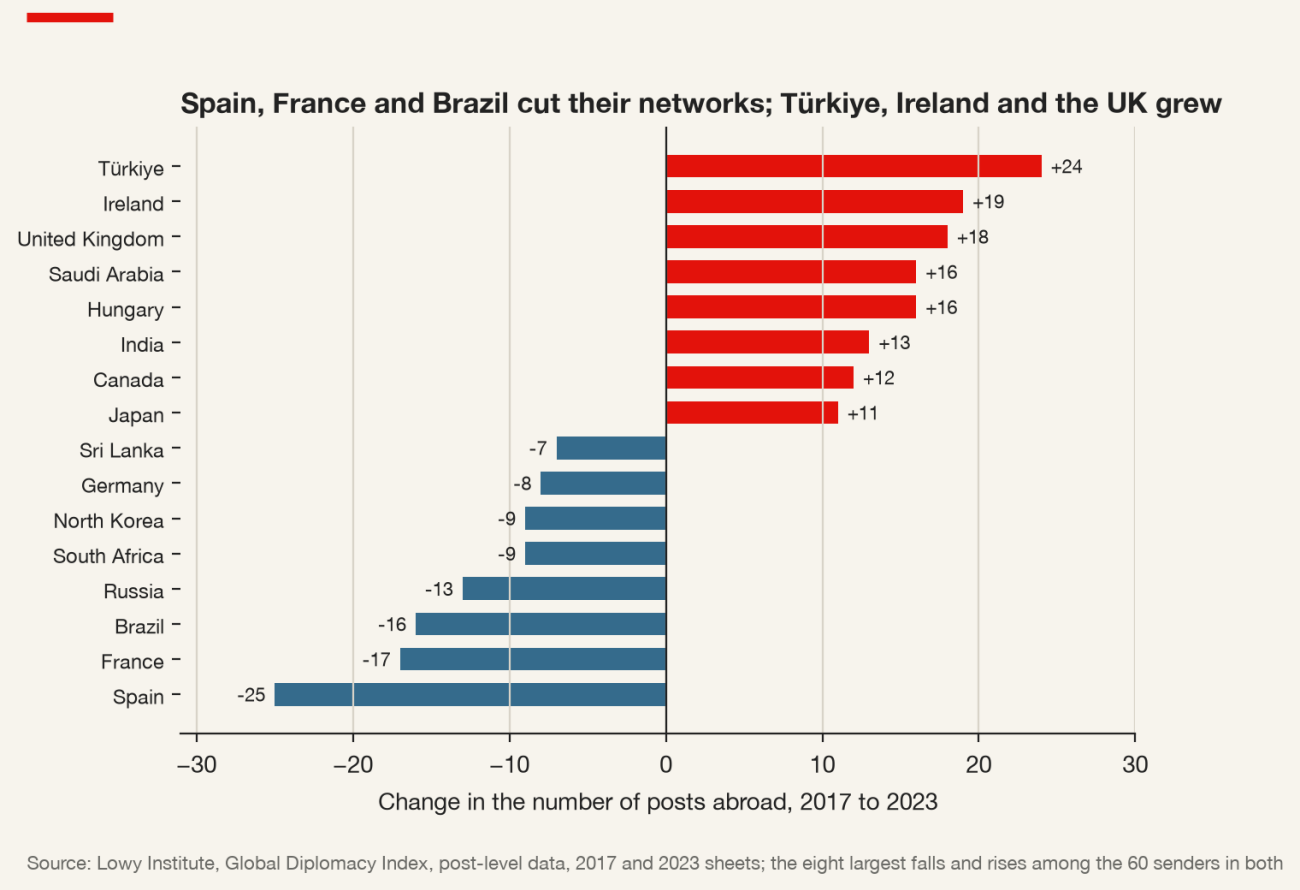


Figure 9.3: Most foreign services grew their networks between 2017 and 2023; a few cut them sharply. The eight largest falls and rises in the number of posts among the 60 senders in the index in both years. Source: Lowy Institute, Global Diplomacy Index, post-level data.

Three features of the data carry the arguments of this chapter into any test of Rose's result. The first is that the record is at the wrong level: the gravity question is about pairs of countries, and the record is about posts. Turning posts into a count per pair is a reduction in Krippendorff's sense, and like every reduction it discards information — the city, the type of post — that the analysis may later need. The second is the missing zeros: the pairs in which a country keeps no post must be built from the list of partners in the trade data, and without them the analysis can only compare partners with one post against partners with several, which is not Rose's question. The third is the exports themselves. The IMF's bilateral trade statistics record each pair's trade twice, as one country's exports and the other's imports, and the two often disagree; using the exporter's own report is a choice, and scaling it by the partner's GDP is the simplest available correction for the most obvious omitted variable.

The type of post deserves a moment of its own. Embassies and high commissions are the standard bilateral posts; consulates and consulates-general are usually in commercial or populous cities other than the capital, and are the posts most directly concerned with trade and with citizens abroad. Permanent missions are a different thing altogether: a mission to the United Nations in New York or Geneva, or to an international organisation elsewhere, is located in the host country but is accredited to the organisation, not to the host. Counting it as a bilateral post would credit Switzerland or the United States with

diplomatic attention that is really directed at multilateral bodies. Deciding which types to count is a measurement decision of exactly the kind Chapter 7 asked to be made explicitly.

A humbler feature matters as much: the index identifies countries by name only, with inconsistent spellings, while trade statistics use codes, so every link between the two depends on a concordance built by hand — and an error there changes the slope more than any substantive objection.

9.10.1 Six diplomatic services, one test

The choices can be made concrete for one exporter and then repeated for others. Take Canada's embassies and consulates in 2023, leaving aside permanent missions and other offices; count them for every partner; and set them against Canada's exports of goods to that partner in 2023, as reported by Canada to the IMF, and the partner's GDP. Of the 188 partners with both trade and GDP data, 93 have no Canadian embassy or consulate at all — half the sample that a logarithmic axis would drop without comment. Together they take 1.2 per cent of Canada's exports to the 188.

The raw relationship is as steep as anyone could want (Figure 9.4). The median partner with no Canadian post buys about $11 million of Canadian goods a year; with one post, $187 million; with two, $906 million; with three, $5.2 billion; and the three partners with four or more, $22.6 billion — the United States alone hosts sixteen Canadian posts. Scale exports by the partner's GDP and the staircase almost disappears. The median share of the partner's GDP that it spends on Canadian goods is 0.085 per cent where Canada has no post, and between 0.12 and 0.14 per cent wherever it has one, two, three, or four and more. Most of the slope in the left panel is the size of the partner.

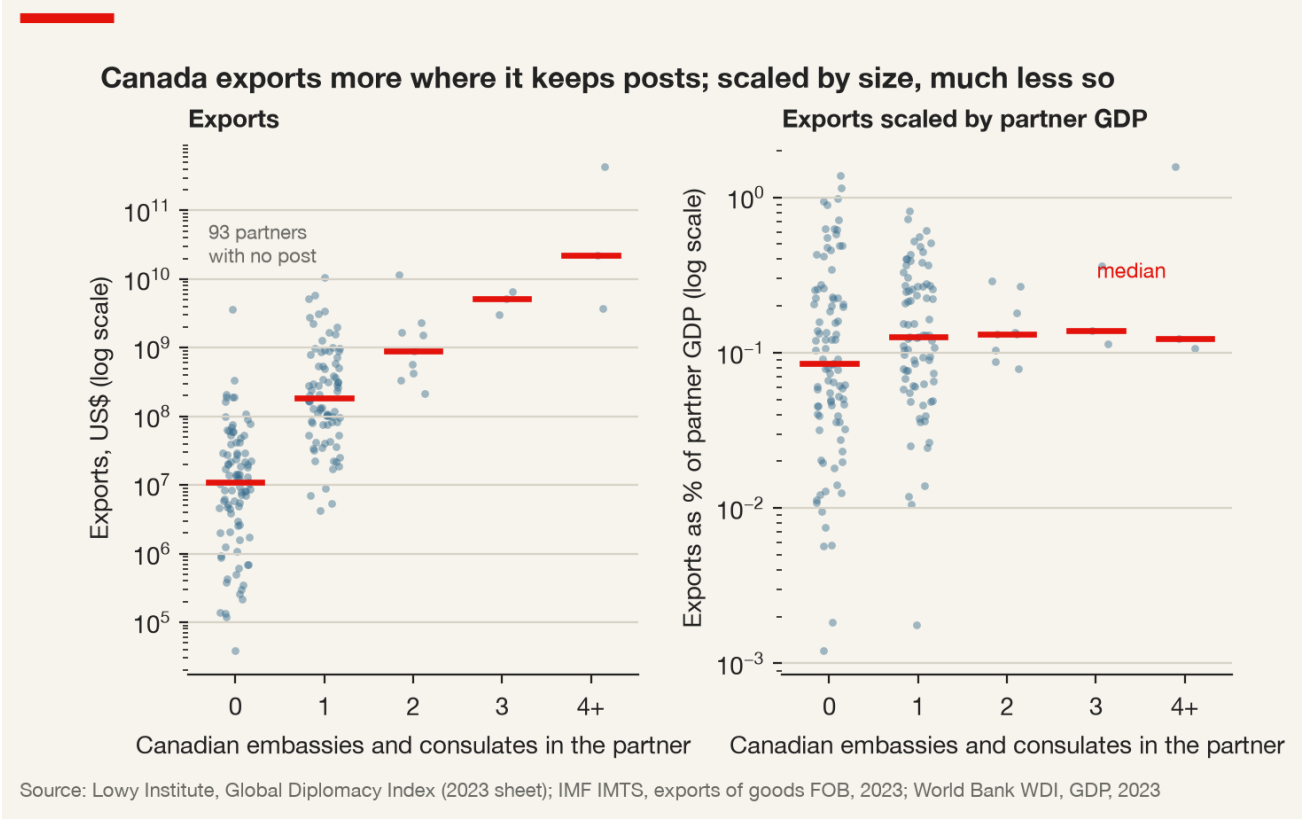


Figure 9.4: Canada's exports rise steeply with the number of its posts in a partner; scaled by the partner's GDP, they barely rise after the first post. Each dot is a partner, 2023; red bars are medians. Source: Lowy Institute, Global Diplomacy Index; IMF, International Merchandise Trade Statistics, exports of goods FOB; World Bank, World Development Indicators.

The same exercise for six foreign services gives Table 9.2. Each column is a choice that an analyst might make without comment. The first is the naive regression of log exports on log posts, among partners with at least one post: the elasticity

is between 2.6 and 3.6 for all six, with an R^2 between 0.29 and 0.44 — the respectable fit this chapter warned about. Taken literally, Canada's 2.62 would mean that doubling the posts in a market multiplies exports by six. Adding the partner's GDP cuts the coefficient to between a fifth and a third of its naive value for four of the six senders; for Germany it disappears, and only Australia's keeps two-thirds of its size. Scaling exports by GDP instead leaves an R^2 of 0.04 or less everywhere. The last two columns keep every partner, including those with no post, using the logarithm of one plus the number of posts. Ordinary least squares gives positive coefficients for all six; Santos Silva and Tenreyro's Poisson estimator (Santos Silva and Tenreyro 2006), which keeps the scale of trade and therefore gives large partners great weight, gives large positive coefficients for Canada and the United States and coefficients near zero or negative for China, Japan and Germany. Canada's is driven by a single partner that takes 78 per cent of its exports.

Table 9.2: Coefficient on posts (embassies and consulates) in regressions of a sender's exports of goods on its posts in each partner, 2023, under five specifications. The first three use partners with at least one post and log posts; the last two use all partners and the log of one plus posts, controlling for log GDP. No distance or other gravity controls: an illustration of fragility, not an estimate of an effect. Source: calculation by the author from the Lowy Institute's Global Diplomacy Index, IMF IMTS and World Bank WDI data.

Sender	Partners	Without a post	Naive elasticity (R^2)	Controlling for GDP	Exports ÷ GDP (R^2)	All partners, OLS	All partners, Poisson
Canada	188	93	2.62 (0.35)	0.68	0.45 (0.04)	0.86	1.79
United States	194	30	3.07 (0.39)	0.71	0.31 (0.01)	0.82	1.70
China	193	24	2.79 (0.44)	0.55	−0.04 (0.00)	0.85	−0.23
Japan	190	41	3.21 (0.40)	1.10	0.48 (0.02)	1.38	−0.21
Germany	190	48	2.85 (0.29)	−0.06	0.23 (0.01)	0.59	−0.07
Australia	184	100	3.58 (0.34)	2.28	0.54 (0.01)	3.54	0.60

None of these regressions includes distance, a common language, a shared border or the multilateral-resistance terms that Anderson and Wincoop (2003) showed to be necessary, and none addresses reverse causality. They are not estimates of the effect of a post, and they are not meant to be. What they show is that the sign and the size of the association depend on choices about the sample, the scaling and the estimator that a chart on logarithmic axes does not display — and that across six of the world's largest diplomatic services, the only robust feature of the numbers is how much they move.

9.11 A short review of the literature

The theory of how domestic politics shapes international bargaining is Putnam (1988), whose two-level game and win-set remain the standard vocabulary; Bjola and Manor (2018) extend it to digital diplomacy through the Iran nuclear negotiations. On the relationship between states and firms, Vernon (1971) introduced the dynamic later called the obsolescing bargain, Kobrin (1984) documented the wave of expropriations that gave it empirical content, and Stopford, Strange, and Henley (1991) and Strange (1992) recast diplomacy as a triangle of state–state, state–firm and firm–firm relations. Blackwill and Harris (2016) describe the return of geoeconomics — the systematic use of economic instruments for geopolitical ends — that has redefined the field.

The empirical evaluation of commercial diplomacy rests on the gravity model, introduced by Tinbergen (1962) and given its modern theoretical foundation by Anderson and Wincoop (2003); Head and Mayer (2014) is the practical guide and Santos Silva and Tenreyro (2006) the warning about logarithms and zeros. Rose (2007) is the central study of diplomatic posts and exports. Nitsch (2007) examines state visits, Head and Ries (2010) Canada's trade missions, Lederman, Olarreaga, and Payton (2010) export promotion agencies, and Moons and Bergeijk (2017) synthesises the econometric evidence in a meta-analysis, and Lowy Institute (2024) is the most comprehensive public record of diplomatic networks. Gertz (2018) shows commercial diplomacy at work in investment disputes, connecting the promotion of firms abroad to the management of political risk.

9.12 What the evidence does not license

Set a country's exports to each partner against the number of posts it keeps there, on logarithmic axes, and the cloud of points slopes upwards with a respectable fit — for each of the six services in Table 9.2. The temptation is to read it as Rose's finding confirmed: posts promote exports.

The association does not license that conclusion, for a multinational or for a foreign ministry. It does not show that opening a post in a market will raise exports there, because posts are opened where exports are expected, and the snapshot cannot tell the two directions apart. It does not show that the relationship is about diplomacy at all until the size of the partner has been removed, and it may show something quite different once embassies and consulates are separated. And if the zeros have been dropped — silently, by a logarithmic axis — it does not show the relationship for the partners where the question is most interesting: the markets in which a country has no presence yet. The studies reviewed above point the same way. The more carefully they deal with selection, the smaller and shorter-lived the effect of commercial diplomacy tends to become, and the most visible instruments — the state visit, the trade mission — fare worst.

For a firm, the more useful reading is the reverse of the naive one. The density of a home state's diplomatic network in a market is a measure of how much that state already cares about the market — a measure of the political attention a firm can expect when it needs help, and of the political attention it will attract when relations sour. The canola exporters of 2019 operated in a market where their government kept an embassy and several consulates; the network did not protect them, and their visibility as Canada's commercial presence in China is what made them useful to squeeze.

The redefinition of economic diplomacy with which this chapter began changes what a firm should ask of its home state. The old question was how much help was available. The new one is what the help will cost if the two states fall out — and the answer depends on the two-level game, the screening regime and the rivalry in which the firm's markets sit. In each market that matters, a firm is some combination of its home state's client, instrument and hostage. Knowing the proportions, and how fast they could change, is the part of geopolitical risk that no gravity equation will estimate for it.

Chapter 10

Economic Sanctions: Do They Work, and Whom Do They Hit?

10.1 The instrument of first resort

When a government wants to do something about a foreign state's behaviour and is not prepared to go to war over it, the menu is short. It can protest, which costs nothing and changes nothing. It can withdraw an ambassador, which is visible and mostly symbolic. Or it can impose economic sanctions: restrict what its firms and banks may sell to, buy from, lend to or invest in the offending state, its government, its companies or its leading citizens. Over the last three decades the third option has moved from the end of the menu to the top. Sanctions are now what governments reach for first, in response to invasions and coups, nuclear programmes and election interference, human-rights abuses and cyberattacks. The Global Sanctions Data Base documents both the rise in the number of active cases since the end of the Cold War and the change in what those cases are made of (Felbermayr et al. 2020).

This chapter asks whether the instrument is an efficient mechanism for *regulating* geopolitical risk — a way of raising the cost of aggression without the cost of war — or a *source* of it. The two are not mutually exclusive, and the argument here is that, for a multinational, the second matters more than the first. A firm does not need to know whether the sanctions on a given state will achieve their political goal. It needs to know whether they will cut it off from a customer, a supplier, a bank or an asset; how far the effective perimeter of the measures extends beyond their legal text; and what the target can do to it in return.

Economic sanctions — restrictions on economic relations imposed by one or more *senders* on a *target* — a state, a government, a firm or an individual — in order to change its behaviour, weaken it, constrain what it can do, or signal disapproval. The instrument is the restriction; the objective is political.

The definition's last clause lists four objectives, and the answer to "do sanctions work?" depends on which of them the instrument is held to. A measure that fails to change a regime's policy may still succeed in making its weapons programme slower and more expensive, or in telling allies and voters at home that something has been done. Much of the disagreement in the literature comes from asking the same question of cases that were never trying to do the same thing.

10.2 A very old weapon

Economic coercion is older than the states that now practise it. Thucydides records the Megarian decree, by which Athens barred the merchants of Megara from the harbours of its empire, among the grievances that preceded the Peloponnesian War; Napoleon's Continental System tried to defeat Britain by closing the continent to British trade, and was undone by smuggling and by the costs it imposed on France's own allies. Both contain, in miniature, the problems this chapter returns to: sanctions hurt the sender's own merchants, invite evasion through third parties, and have uncertain political effects.

The modern idea — sanctions as an organised, collective alternative to war — dates from the League of Nations. Article 16 of its Covenant obliged members to sever trade and financial relations with any member that resorted to war in breach of its commitments. The idea was tested once, and the test is the founding case of the literature.

From the archive · The League of Nations and Italy, 1935

Italy invaded Ethiopia in October 1935. Within days the League's Council found that Italy had resorted to war in breach of the Covenant, and the members agreed a package of measures that came into force in November: an arms embargo, a ban on loans and credits to the Italian government, a ban on imports of Italian goods, and an embargo on the export to Italy of a list of raw materials.

What was left off the list mattered more than what was on it. Oil was not embargoed; nor were coal and steel, and the Suez Canal, through which Italy supplied its army in East Africa, stayed open. The great powers feared driving Italy into Germany's arms, and the United States was outside the League. Italian forces took Addis Ababa in May 1936, and the League lifted the sanctions in July.

The lesson drawn at the time was that collective sanctions had failed. The lesson worth drawing now is subtler. The senders wanted to be seen to act while protecting their own interests, and the gap between the announced objective and the instrument deployed was visible to everyone, including the target. Every later debate about whether sanctions "work" is, in part, a debate about whether the measures studied were ever designed to.

The United Nations Charter, drafted a decade later, put the idea on a firmer legal footing: Article 41 allows the Security Council to decide on "measures not involving the use of armed force", including the "complete or partial interruption of economic relations", and makes them binding on all members. For most of the Cold War the veto kept the provision almost idle. The Council imposed mandatory sanctions on Southern Rhodesia after its unilateral declaration of independence in 1965 and an arms embargo on South Africa in 1977, and little else. Sanctions in this period were overwhelmingly unilateral and American. The end of the Cold War unblocked the Council, and the 1990s became the "sanctions decade".

10.3 Four ways to describe a sanction

Before anything can be counted it must be classified, and classification embeds a theory of what sanctions are for. Four dimensions are useful, and the GSDB codes three of them directly.

The **sender** may be one state acting alone, a group of states acting together — the European Union is the most important plurilateral sender, its measures adopted by unanimity in the Council — or the United Nations Security Council, whose decisions bind every member. The distinction tells a firm how many jurisdictions it must satisfy and how much room a competitor from a non-sending state has to take its place.

The **type** describes the instrument: restrictions on trade (exports, imports, or both, partial or complete), on finance (loans, investment, payments, asset freezes), on travel, on arms, on military assistance, and a residual category. The GSDB records each as a separate indicator, so a single case can be of several types at once — which is why any stacked count of cases by type exceeds the number of cases.

The **scope** distinguishes measures aimed at a whole economy from measures aimed at named persons, entities or sectors.

Comprehensive versus targeted sanctions — *comprehensive* sanctions restrict all, or nearly all, economic relations with a target country; *targeted* or "smart" sanctions restrict relations only with named individuals and entities (asset freezes, travel bans), with specific sectors (energy, finance, defence), or in specific goods (arms, dual-use items, luxury goods). The distinction is about who is meant to bear the cost.

The **objective** is what the sender says it wants. The GSDB distinguishes changing a policy, destabilising a regime, resolving a territorial conflict, preventing a war, ending one, fighting terrorism, protecting human rights and restoring or defending democracy. Many cases have more than one, and the stated objective is not always the real one. A sender that announces "policy change" may be content with "punishment", and one that announces "democracy" may care more about signalling to its own electorate. Coders work from the public record, so the objective variable is a record of what senders said, not of what they wanted.

10.4 Do they work? Coding success

The modern empirical literature begins with one book in three editions. Gary Clyde Hufbauer and Jeffrey Schott, later joined by Kimberly Ann Elliott and Barbara Oegg, assembled every sanctions episode of the twentieth century they could document and asked of each a question that sounds simple: did it succeed (Hufbauer et al. 2007)?

Their answer to the problem of defining success has become the field's reference point, so it is worth stating precisely. Each episode is scored on two components. The first is the **policy result**: how far the outcome the sender sought was actually achieved, on a scale from failure to success. The second is the **contribution of sanctions**: how far the sanctions, as opposed to other factors — military force, a change of government for unrelated reasons, a negotiation that would have happened anyway — contributed to that outcome. Each component is scored from 1 to 4, and the two are multiplied into a success score. A case counts as a success when the product reaches 9 or more: that is, when the outcome was at least partly achieved *and* sanctions made at least a modest contribution to it.

Sanctions success (Hufbauer et al.) — an episode in which the sender's policy goal was at least partly achieved and sanctions made at least a modest contribution to that achievement, operationalised as a product of the two scores, each from 1 to 4, of 9 or more. Success is a joint judgement about an outcome and a cause.

On this definition, roughly a third of the episodes in the study count as at least partial successes. The rate varies systematically with the circumstances. Sanctions succeed more often when the goal is modest — the release of a prisoner rather than the overthrow of a regime — when the target is economically small and weak relative to the sender, when target and sender were previously on friendly terms and trade heavily, and when the measures are imposed quickly and decisively rather than ratcheted up slowly, giving the target time to adapt. They succeed less often when the goal is major, when the target is an adversary, and when the target can turn to other suppliers and buyers.

That last condition was formalised by Bryan Early as *sanctions busting*: third countries whose firms step into the trade the sanctions forbid, for profit or for strategic reasons, and so supply the target with the alternatives that make resistance affordable (Early 2015). It is the reason multilateral sanctions are not straightforwardly more effective than unilateral ones. The more senders there are, the fewer alternatives remain for the target, but the harder it is to agree on anything strong, and the more likely the package resembles the League's of 1935.

The "about a third" figure appears in policy debates as a fact about the world. It is nothing of the kind. It is a fact about one coding of one set of cases, produced by one team's judgements on two four-point scales, and changing any of those three elements changes the number.

10.5 Pape's audit, and why the debate never closed

The most consequential demonstration of this came from Robert Pape, who in 1997 went back through the cases Hufbauer and his colleagues had counted as successes in their second edition and asked whether each one survived scrutiny (Pape 1997). His conclusion was that most did not. In many, he argued, force or its threat had done the work; in others the target had made no real concession, or none attributable to sanctions. By his reckoning only five of the forty claimed successes held up, and he concluded that economic sanctions rarely achieve major political goals and should not be expected to.

Elliott replied that the glass was half full rather than empty (Elliott 1998). Sanctions, she argued, had never been claimed to be a substitute for force in pursuit of major goals, and judged against the modest goals for which they are typically used they did better than Pape allowed; moreover, their record had improved as they were used more often in concert, after the Cold War. Pape answered that sanctions still did not work (Pape 1998). The exchange is usually remembered as a debate about effectiveness. It is better understood as a debate about coding: about what counts as a concession, how much a contribution must be, and what the counterfactual is.

The counterfactual problem is the deep one, and David Baldwin had set it out a decade earlier (D. A. Baldwin 1985). To say that sanctions failed is to say that some other instrument would have done better. If the alternative was war, a sanction that achieved nothing at little cost may still have been the better choice; if the alternative was diplomacy, a sanction that achieved a little at great cost may have been the worse one. Effectiveness, on Baldwin's account, is always comparative and always relative to cost, and a success rate computed without reference to the alternatives answers a question no policymaker faces.

A second line of argument asks *how* sanctions are supposed to work, and finds the naive answer wanting. Johan Galtung, studying the sanctions on Rhodesia, described what he called the naive theory: economic pain causes political disintegration, which forces the government to comply (Galtung 1967). The Rhodesian evidence suggested the opposite. Pressure produced a rally around the government, a turn towards self-sufficiency, and domestic winners — smugglers, import substitutes, holders of scarce foreign exchange — with an interest in the sanctions continuing. Jonathan Kirshner later generalised the

point: sanctions work, when they work, by changing the relative power of groups within the target, and their effect therefore depends on who inside the target bears the cost and who profits from it (Kirshner 1997).

The lesson for an analyst. "Do sanctions work?" has no answer independent of how a case, an objective and a success are coded, and of what the comparison is. Which objective counts when a case has several, whether cases under way enter the denominator, how large a contribution must be: each coding decision moves the share judged successful.

10.6 The cases we never see

There is a deeper problem with any success rate computed from imposed sanctions, and it was put most clearly by Daniel Drezner. His first book began from a puzzle: states impose sanctions most often on their adversaries, yet adversaries are the targets least likely to concede, because they expect future conflict and fear that any concession will be exploited; allies concede more readily, but are sanctioned less often (Drezner 1999). The second step of the argument concerns the threat (Drezner 2003). If a target that would concede does so as soon as sanctions are threatened, the episode ends before any measure is imposed. The cases that reach imposition are disproportionately those in which the target had decided to resist.

Selection effect of threats — the bias that arises when sanctions most likely to succeed do so at the threat stage and are therefore never imposed. Samples of imposed sanctions over-represent hard cases, so imposed sanctions look less effective than the instrument is. The most significant effects of economic coercion may be the ones that never appear in a list of sanctions.

The implication for measurement is direct. A dataset that records only impositions cannot estimate the effectiveness of the instrument, because it has selected its cases on a variable correlated with the outcome. This is the reason the Threat and Imposition of Economic Sanctions dataset exists. T. Clifton Morgan, Navin Bapat and their colleagues recorded threats as well as impositions, first for 1971–2000 (Morgan, Bapat, and Krustev 2009) and then, in an extended version, for 1945–2005 (Morgan, Bapat, and Kobayashi 2014). In TIES a case begins with a threat or an imposition, whichever comes first, and the dataset records whether sanctions were in fact imposed. A substantial share of its cases never were.

The selection argument also cuts the other way: imposition is a costly signal of the sender's seriousness, and some of its effects may come from the signal rather than the pain. Neither effect can be read off a count of imposed cases. A dataset that records threats can at least show where the cases go.

10.6.1 What the record of threats shows

TIES contains 1,412 cases that began between 1945 and 2005. In 845 of them sanctions were eventually imposed; in the other 567 — two cases in five — the episode ended without any measure taking effect. Figure 10.1 shows when the cases began. The post-Cold War surge is there, as in every count of sanctions: 547 cases began in the 1990s, against 226 in the 1980s, and the single busiest year was 1993, with 87. The share that never got past a threat moves more than the total: a third in the 1950s and 1960s, more than half in the 1980s, a third again in the 1990s and almost half between 2000 and 2005. Counted by impositions alone, the 1980s look quiet. They were a decade of threats.

The more telling evidence is in how cases end. TIES codes a final outcome on a single scale that distinguishes the stage at which the episode closed: the target may acquiesce completely or partially, the two sides may negotiate a settlement, the episode may end in stalemate, or the sender may back down — each either at the threat stage or after imposition (Morgan, Bapat, and Kobayashi 2014). Table 10.1 counts the 1,024 cases with a coded outcome; the remaining 388, most of them imposed sanctions still in force when coding stopped, have none.

Table 10.1: How sanctions episodes end, by stage. Stage is read from the outcome code (1–5 at the threat stage, 6–10 after imposition). Source: Morgan, Bapat and Kobayashi, TIES v4.0; computed by the author.

Final outcome	Ended at the threat stage	Ended after imposition
Complete acquiescence by the target	146 (31%)	121 (22%)
Partial acquiescence by the target	56 (12%)	61 (11%)
Negotiated settlement	60 (13%)	132 (24%)
Stalemate	23 (5%)	71 (13%)

Final outcome	Ended at the threat stage	Ended after imposition
Sender backs down	188 (40%)	166 (30%)
Cases with a coded outcome	**473**	**551**

Two findings stand out. The first is Drezner's prediction in its plainest form: complete acquiescence is more common at the threat stage (31 percent of cases that ended there) than after imposition (22 percent). Targets that will give in tend to do so before the measures bite. The second is less often noticed: threats are frequently bluffs. In 40 percent of the cases that ended at the threat stage the sender backed down without imposing anything, and a sender that backs down after imposition — lifting measures the target has not complied with — is almost as common, at 30 percent. A "success rate" that counts only targets that comply hides the fact that senders often quit first.

Figure 10.2 splits the same outcomes by the issue at stake, and the selection effect turns out to be concentrated where Drezner said it would be. Half of the TIES cases concern trade practices or economic reform alone — many of them disputes between the United States and its allies over market access. In those economic disputes, 31 percent of the cases settled at the threat stage ended with the target's complete acquiescence, against 11 percent of those that went on to imposition, where negotiated settlements and stalemates dominate. In political and security disputes, the difference between the stages almost disappears (30 against 27 percent), and what changes instead is the sender's resolve: half of the political threats ended with the sender backing down. Allies concede to threats over trade; adversaries facing threats over security mostly wait to see whether the sender means it.

10.7 From comprehensive to smart

The sanctions decade of the 1990s produced the instrument's most damaging case, and that case changed how sanctions are designed.

From the archive · Iraq, 1990–2003

Four days after Iraq invaded Kuwait in August 1990, the Security Council adopted Resolution 661, imposing comprehensive sanctions: a ban on all trade with Iraq except medical supplies and, in humanitarian circumstances, foodstuffs. The war of 1991 expelled Iraq from Kuwait; the sanctions stayed, tied now to the disarmament of Iraq's weapons programmes.

They stayed for more than a decade, and their humanitarian consequences became the central fact of the debate about them. Malnutrition, the collapse of water and health services and a rise in mortality were documented by UN agencies and researchers, although the size of the excess mortality and its attribution to sanctions rather than to war damage and the regime's choices remain contested. In 1995 the Council created the Oil-for-Food Programme (Resolution 986), allowing Iraq to sell oil under UN supervision to pay for food and medicine. The programme relieved the shortages and was in turn corrupted through surcharges and kickbacks, and the subsequent inquiry named companies from many countries.

The regime survived the sanctions intact. Those who bore their cost were the population and, through the programme's scandals, the firms that had traded with the regime under UN licence.

The Iraqi experience made it difficult for senders to defend comprehensive sanctions on a whole population, and the response was the turn to *targeted* measures. In intergovernmental processes around the turn of the century — at Interlaken on financial sanctions, at Bonn and Berlin on arms embargoes and travel bans, at Stockholm on implementation — governments and experts worked out how to aim sanctions at those responsible: freeze the assets and ban the travel of named leaders and financiers, embargo arms and the goods that sustain a war, restrict specific sectors.

Targeted sanctions created problems of their own. Listing is an executive act with severe consequences, and in the *Kadi* judgment of 2008 the European Court of Justice annulled the EU's implementation of a UN listing for disregarding the listed person's rights of defence; review procedures and a UN Ombudsperson followed. For firms the lesson was that lists change, are litigated, and may be annulled — and that a listing in one jurisdiction may not exist in another.

Whether targeting works better is an empirical question with a mixed answer. The most comprehensive study of UN targeted sanctions distinguishes three purposes — to coerce a change in behaviour, to constrain the target's ability to act, and to signal a norm — and finds that the measures do better at constraining and signalling than at coercing (Biersteker, Eckert, and Tourinho 2016). On humanitarian cost the evidence is clearer for the comprehensive kind: Dursun Peksen finds that

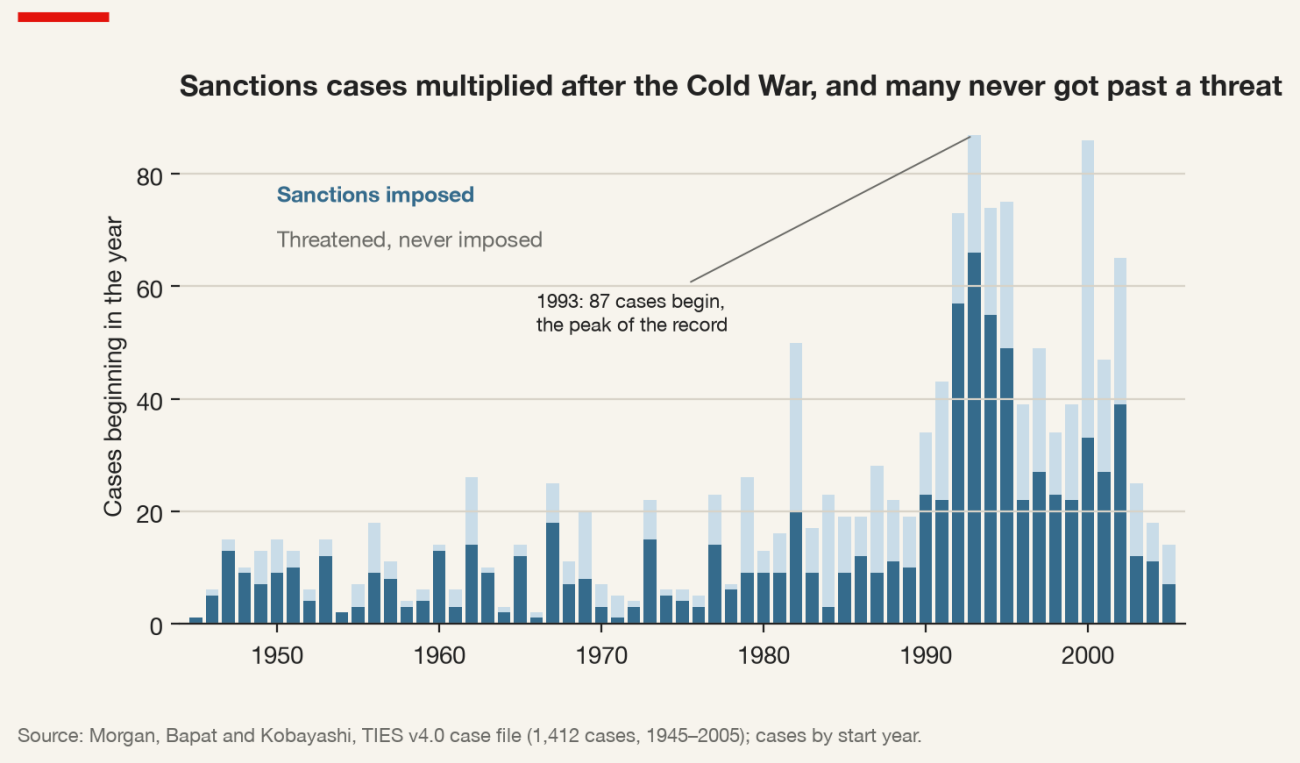


Figure 10.1: Sanctions cases multiplied after the Cold War, and two in five never got past a threat. Cases in TIES by the year they began, divided by whether sanctions were eventually imposed. Source: Morgan, Bapat and Kobayashi, TIES v4.0.

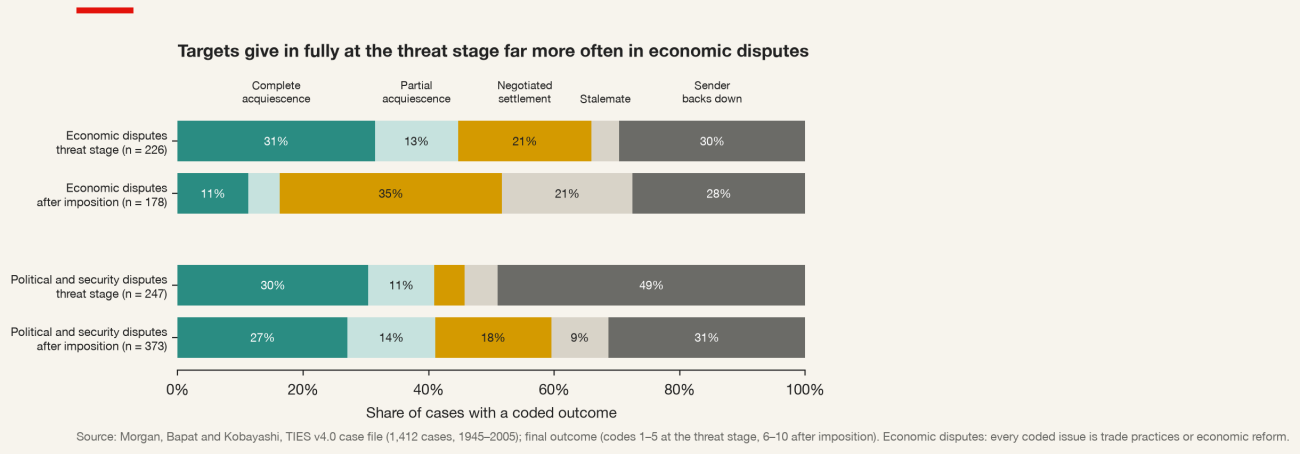


Figure 10.2: Targets give in fully at the threat stage far more often in economic disputes. Final outcomes of TIES cases by stage and by issue; economic disputes are cases whose every coded issue is trade practices or economic reform. Source: Morgan, Bapat and Kobayashi, TIES v4.0.

sanctions are associated with a deterioration of the target government's respect for physical-integrity rights, and that the effect is stronger for comprehensive measures (Peksen 2009). The targets of sanctions, in other words, tend to pass the cost down.

10.8 The financial turn

The second change in the design of sanctions since 2000 has been the rise of financial measures, and it is the change that matters most to a multinational.

Henry Farrell and Abraham Newman explained it in the framework introduced in Chapter 8 (Farrell and Newman 2019). Global economic networks — payment systems, financial messaging, correspondent banking, the internet's infrastructure — are not flat. They have hubs, and the hubs are disproportionately under the jurisdiction of a few states, above all the United States. A state with jurisdiction over a hub can exploit it in two ways: as a *panopticon*, using the information flowing through the hub to see what others are doing, and as a *chokepoint*, denying others access to the hub and therefore to the network. International dollar payments clear, in the end, through banks with access to the US financial system; banks around the world communicate payment instructions through SWIFT, a Belgian cooperative subject to European law. Cutting a bank off from either is a sanction of a different order from an embargo on its country's exports.

This is why targeted does not mean contained. A financial sanction is aimed at named banks, but every firm that pays or is paid through those banks is affected, and the banks of third countries — which need access to the dollar system more than they need any single customer — will cut off a sanctioned client long before the law requires them to. Sanctions on banks, insurers and shippers propagate down every supply chain that depends on them.

The measures of 2022 against Russia showed the reach of the instrument and its new forms: the freezing of a large part of the Russian central bank's foreign reserves, the disconnection of major Russian banks from SWIFT, and, from December 2022, a price cap on seaborne Russian crude, initially set at 60 US dollars a barrel. The cap did not try to keep Russian oil off the market, which would have raised the world price; it allowed Western insurers, shippers and financiers to service cargoes only if the oil had been sold at or below the cap. The chokepoint was the maritime services the trade relied on, and the market's answer — an older, opaquely owned tanker fleet operating outside those services — showed that a sanction built on a chokepoint lasts only as long as the chokepoint does.

10.9 The firm in the middle

Sanctions are aimed at states, governments and the people who run them. They are implemented by firms. Banks, exporters, shippers and insurers decide, transaction by transaction, whether a payment, shipment or contract falls within the rules, and bear severe penalties if they decide wrongly. The firm is therefore not a bystander to sanctions policy. It is its executor, and its executor of last resort.

10.9.1 Lists, ownership and the perimeter

Each sending jurisdiction maintains its own legal architecture. In the United States, sanctions programmes rest mostly on the International Emergency Economic Powers Act of 1977 and are administered by the Treasury's Office of Foreign Assets Control, whose Specially Designated Nationals list is the most consulted list in global commerce. In the European Union, restrictive measures are adopted by Council decision and given effect by regulations binding every person and firm in the Union. In Canada, they rest on the *Special Economic Measures Act*, the *United Nations Act* and the *Justice for Victims of Corrupt Foreign Officials Act*. The lists overlap but do not coincide, and a firm operating across jurisdictions must satisfy all of them.

The list is only the beginning of the perimeter. Under the US "50 percent rule", an entity owned, directly or indirectly, 50 percent or more in aggregate by listed persons is itself treated as blocked, whether or not it appears on any list; EU guidance extends the restrictions to entities *owned or controlled* by listed persons, with control judged on a range of indicators. A firm must therefore screen not only its counterparties but their owners, and their owners' owners, through chains of holding companies that may be designed to obscure them.

Compliance perimeter — the set of counterparties, transactions, goods and jurisdictions a firm must treat as restricted in order to comply with every sanctions regime that applies to it. It is wider than any list: it includes entities owned or controlled by listed persons, goods restricted by end use, and the jurisdictions whose rules reach the firm through its nationality, its currency, its banks or its inputs.

10.9.2 Diversion and the duty to look downstream

Sanctions on goods have always leaked through third countries. After 2022 the leak ran through distributors in countries that did not apply the measures, which re-exported to Russia goods that had once been shipped there directly. The senders' response moved the perimeter towards the firm. The United States, the European Union, the United Kingdom and Japan published common lists of high-priority items — mostly electronic components found in Russian weapons — whose sale to any destination at risk of diversion called for particular scrutiny. The EU's twelfth package, of December 2023, obliged exporters of the most sensitive goods to write into their contracts with third-country buyers a clause prohibiting re-export to Russia, and in the same month the United States extended its secondary reach to foreign banks serving Russia's military-industrial base.

The trade statistics show the leak plainly. Figure 10.3 indexes the exports of the 27 member states of the European Union to five of Russia's neighbours against their 2021 level, as reported by the exporting countries to the IMF (International Monetary Fund 2026b). EU exports to Russia fell from 105 billion US dollars in 2021 to 41 billion in 2023. Over the same two years EU exports to Kyrgyzstan rose almost tenfold, from 0.31 to 2.95 billion dollars; exports to Armenia and Kazakhstan more than doubled, and those to Georgia and Uzbekistan rose by more than half (Table 10.2). German exports to Kyrgyzstan, a country of seven million people with no large manufacturing base, went from 59 million dollars to 772 million. Nothing in these economies explains such growth; their position between the European Union and Russia — inside a customs union with Russia, for three of them — does. Maxim Chupilkin, Beata Javorcik and Alexander Plekhanov trace the same "Eurasian roundabout" at the level of products: the surge is concentrated in the goods the European Union stopped sending to Russia directly, and the neighbours' exports to Russia rose in the same goods (Chupilkin, Javorcik, and Plekhanov 2026).

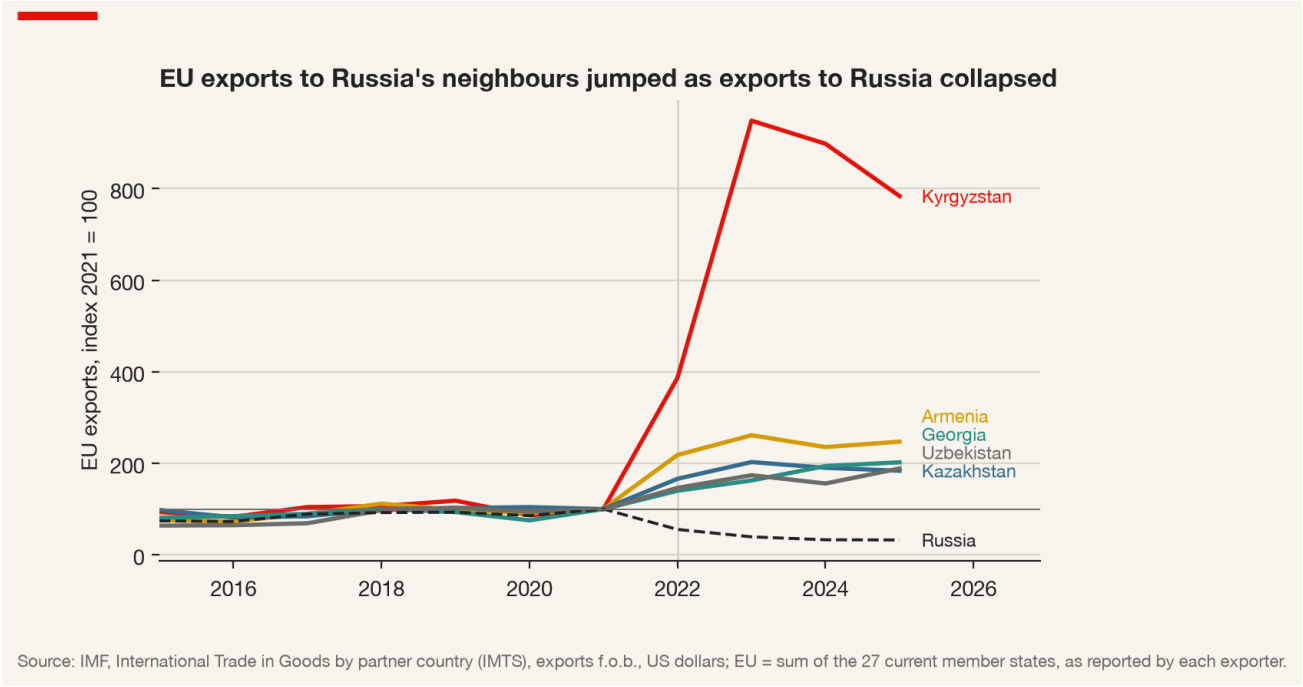


Figure 10.3: EU exports to Russia's neighbours jumped as exports to Russia collapsed. Exports of the 27 EU member states to five neighbours of Russia and to Russia itself, index 2021 = 100. Source: IMF, International Trade in Goods by partner country (IMTS).

Table 10.2: EU exports to Russia and five of its neighbours, billions of US dollars. EU = the 27 current member states, as reported by each exporter. Source: IMF, International Trade in Goods by partner country (IMTS); computed by the author.

EU exports to	2021	2022	2023	2025	2023 / 2021
Russia	105.4	58.4	41.4	34.2	0.4
Kazakhstan	6.48	10.77	13.14	11.91	2.0
Uzbekistan	2.71	3.96	4.71	5.11	1.7
Georgia	2.41	3.38	3.92	4.87	1.6
Armenia	0.89	1.95	2.34	2.21	2.6
Kyrgyzstan	0.31	1.20	2.95	2.44	9.5

The numbers also put the leak in proportion. EU exports to the five neighbours rose by 14 billion dollars between 2021 and 2023, while exports to Russia fell by 64 billion: even if every additional dollar had been re-exported, the roundabout replaced about a fifth of the lost trade. The sanctions worked on volume and leaked at the margin — and the margin is where enforcement, and the exporter's exposure, now sit.

The compliance question is therefore no longer only "is my customer listed?" but "where do my goods end up?". A manufacturer's exposure runs through its *downstream* channels — distributors, resellers, freight forwarders — as well as through its own counterparties.

10.9.3 Secondary sanctions and blocking statutes

Primary sanctions bind the sender's own nationals and transactions on its territory. Secondary sanctions go further: they threaten third-country firms with penalties — exclusion from the sender's market, its financial system, or its government contracts — if they deal with the target, even where no person or transaction of the sender is involved.

Secondary sanctions — measures that penalise third-country persons for dealing with a sanctioned target, by restricting their own access to the sender's market or financial system. They make a sender's foreign policy binding on firms that are not its nationals, by forcing them to choose between the sender and the target.

The tension this creates between allies is old. When the United States legislated against foreign firms investing in Cuba, Iran and Libya in 1996, the European Union adopted a *Blocking Statute* forbidding EU firms to comply, and updated it in 2018 when the United States reimposed secondary sanctions on Iran. European firms nevertheless left Iran in large numbers. The choice a secondary sanction forces is not a legal one but a commercial one: the US market and the dollar are worth more to almost any multinational than the target's market is, and no blocking statute changes that arithmetic. The case of BNP Paribas, which pleaded guilty in the United States in 2014 to processing transactions for sanctioned parties and paid a penalty without precedent at the time, settled the question for many compliance departments: dollar transactions with sanctioned parties anywhere in the world fall within US reach.

The blocking statute has since acquired a mirror image. As Chapter 3 noted, China adopted instruments of its own in 2020 and 2021 against compliance with foreign sanctions it regards as discriminatory, so that a firm large in both the United States and China can find that obeying one state's law breaches the other's.

10.9.4 Overcompliance and de-risking

Penalties are severe, rules are complex, lists change without notice, and the cost of a mistake is asymmetric: a bank that refuses a lawful payment loses a fee, a bank that processes an unlawful one may lose its licence. The rational response is to go further than the law.

Overcompliance and de-risking — the withdrawal of firms, and above all banks, from whole countries, sectors or categories of client that the law does not close, because the expected cost of an error exceeds the revenue at stake. The effective scope of sanctions is set by compliance departments as much as by governments.

The evidence that this matters is now substantial. Matthieu Crozet and Julian Hinz, studying the sanctions on Russia after 2014 and the Russian counter-sanctions, find that the trade losses borne by the sending countries fell largely on products that were not embargoed at all — "friendly fire" that they attribute mainly to the financial measures and the uncertainty they created (Crozet and Hinz 2020). A firm selling unrestricted goods to an unrestricted customer may still be unable to get paid, insured or shipped. If its lawyers say the measures do not require it to exit and its bank says it will no longer process payments to that country, its sanctions policy has been set by its bank.

10.10 Russia 2022: the exit decision

The full-scale invasion of Ukraine in February 2022 produced the largest and fastest expansion of sanctions against a major economy since the Second World War, and it confronted hundreds of Western multinationals with a decision most had never modelled: whether to leave a large market that the law, for many of them, did not require them to leave. Public and employee pressure, the difficulty of making payments and moving goods, and the expectation of further measures pushed many to suspend, scale back or exit.

Exit turned out to be harder than announcing it. Russia's countermeasures made the sale of a foreign-owned business subject to approval by a government commission, required sellers to accept a steep discount on the valuation, and added a payment to the state budget; buyers were frequently local managers or firms close to the state. In 2023 a presidential decree allowed the temporary state management of foreign-owned assets, and the Russian subsidiaries of Danone and Carlsberg were placed under it within weeks. The firms that waited lost the option of selling on their own terms.

The four options — stay, suspend, sell, walk away — each have a cost and a residual risk. McDonald's sold its Russian restaurants to a local licensee, which reopened them under a new name. Renault sold its controlling stake in AvtoVAZ to a state entity for a symbolic rouble, keeping an option to buy it back within six years — a sale designed to preserve an option on a return. Others suspended operations and paid for idle assets; others stayed, citing obligations to employees or the essential nature of their products, and accepted the reputational cost and the risk of seizure.

The firms that left did not leave a vacuum. Figure 10.4 follows Russia's imports from its main suppliers, as reported by the exporters. EU exports to Russia, 152 billion dollars in 2013, fell after Crimea, recovered to 105 billion by 2021, and after the invasion fell to 34 billion by 2025. China's exports to Russia went the other way: from 68 billion dollars in 2021 to 111 billion in 2023 and 116 billion in 2024. Türkiye's almost doubled, from 5.8 to 10.9 billion, before falling back as Turkish banks, wary of US secondary sanctions, balked at Russian payments. This is Early's sanctions busting at the scale of a great power: the target found a supplier large enough, and indifferent enough to the senders' displeasure, to make the measures a cost rather than a constraint (Early 2015).

For a firm weighing exit, the implication is uncomfortable: its market share was not held in trust. Competitors from non-sending states took it within two years, so a return after sanctions would be a new entry against established rivals — which makes an option to buy back, like Renault's, worth less than it looks.

The right answer depends on the firm, not on the country. Put the same choice to a consumer-goods firm and to an energy firm, and the answers diverge: the first has brands, local staff and a domestic supply chain that a buyer can run; the second has assets that cannot be moved, partners that may be sanctioned, and a product the target state regards as strategic. Country-level exposure tells a firm that a decision must be taken. It does not tell it which one.

10.11 Through which channels sanctions reach the firm

Sanctions run through all five of the channels set out in Chapter 1. Through **market access**, export and import bans close markets directly, or indirectly when customers are listed or owned by listed persons, and dual-use export controls close specific product lines to specific end users. Through **inputs**, a sanctioned supplier can no longer be paid, and the target's counter-sanctions close supply from the other side, often deeper in the supply chain than the firm can see. **Finance and payments** is where financial sanctions bite first — the correspondent bank that will not process a payment, the letter of credit that cannot be confirmed, the insurer that will not cover a cargo — and the channel through which overcompliance operates. Through **assets and people**, assets can be frozen by the sender, or seized, placed under temporary management or sold at a forced discount by the target, and employees are part of the exit's cost. Through **data and operations**, software licences, cloud services and technical support may be restricted services in themselves.



Figure 10.4: After 2022, China replaced Europe as Russia's supplier. Exports to Russia by exporter, billions of US dollars. Source: IMF, International Trade in Goods by partner country (IMTS), as reported by the exporter.

Four quantities follow. The firm's **exposure** — revenues, assets and suppliers in sanctioned or at-risk jurisdictions — is the starting point. The **pipeline** of threats is the selection effect applied: the cases that matter most are those not yet imposed. **Propagation** runs through the banks, insurers and logistics providers whose own exposure could cut the firm off. And the target's **counter-measures** against foreign firms on its soil determine, as Russia showed, what an exit costs.

10.12 From episodes to a dataset

The Global Sanctions Data Base was built to answer, at scale, questions that case studies cannot: how the use of sanctions has changed over time, what types and objectives dominate, and — joined to trade data — what sanctions do to trade flows (Felbermayr et al. 2020). Its unit is the *case*: a sanctions episode defined by its senders, its target, its start and end, its types and its objectives, with an outcome coded for each objective. Successive releases have extended the period and revised earlier cases; the third added the measures following the pandemic and the escalating sanctions on Russia (Syropoulos et al. 2024), and the fifth codes 1,794 cases whose first years run from 1949 to 2025; it is distributed on request and for non-commercial use only, so the most complete public record of the instrument cannot be built into a commercial product.

A time series of "active cases" counts, for each year, the episodes whose interval from start to end contains it, and three decisions are buried in that construction. Cases that have not ended must be given an end — and the distributed file gives ongoing cases an end year that is the last year of the data, which is not what its codebook says, so that an analyst who follows the codebook and one who follows the file obtain different series in the final years. A case of several types appears once in each type, so a stacked chart of types overstates the number of cases. And a case is a case whether it consists of a single listing or of a comprehensive embargo, so the number of active cases measures the frequency of the instrument's use, not its intensity.

The outcome coding carries the success debate into the data. Each objective of each case receives an outcome — total success, partial success, a negotiated settlement, failure, or ongoing — and the share judged successful depends on which objective a multi-objective case is counted under and on whether the ongoing cases, which by definition have not succeeded yet, enter the denominator. These are not technical details. They are Hufbauer's and Pape's disagreement reproduced in two lines of arithmetic.

TIES, set beside the GSDB, supplies what the GSDB cannot: the threats that were never imposed. The two differ in unit, period, coding of senders and targets and definition of outcome, so they cannot be merged case by case. What they can do is be compared, year by year, to give an order of magnitude to the selection effect: how much of economic coercion happens before any measure is imposed.

The end-year decision is not peculiar to the GSDB, and TIES shows how much it weighs. Of the 845 TIES cases in which sanctions were imposed, 337 have no end year. Some were still in force when coding stopped; others faded from the news unrecorded. With the date of the last news story the coders found, an analyst has three defensible rules: the case was still in force at the end of the data, it ended at the last news report, or — the implicit choice of anyone who drops missing values — it lasted only the year it began. Figure 10.5 draws the number of imposed sanctions in force each year under each rule. Until the late 1980s the three series barely differ. After 1990 they part company: in 2000 the count is 369 under the first rule, 91 under the second and 59 under the third; under the first it rises to a peak of 412 in 2003, under the other two it peaks in 1993 and then declines. Whether sanctions kept multiplying after the sanctions decade depends on a line of code the reader of the chart never sees.

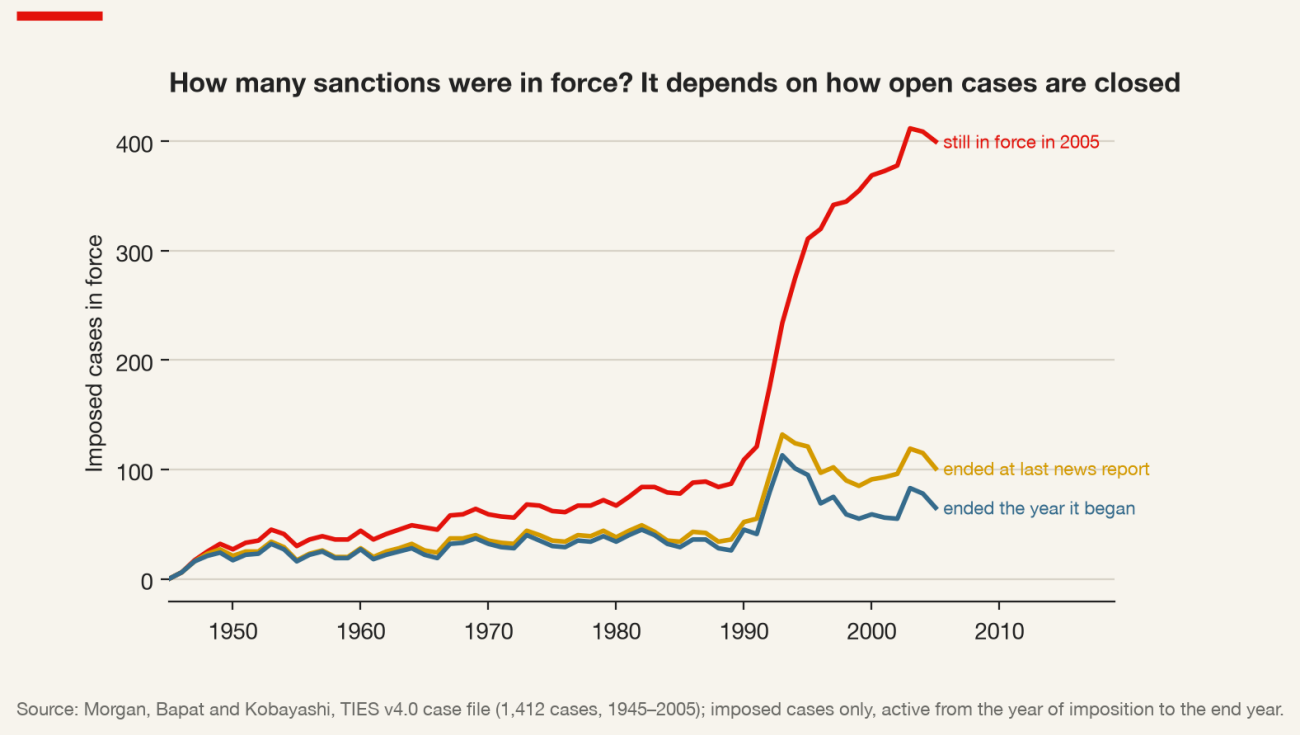


Figure 10.5: How many sanctions were in force? It depends on how open cases are closed. Imposed TIES cases in force each year, under three rules for the 337 cases without an end year. Source: Morgan, Bapat and Kobayashi, TIES v4.0.

The record also says who uses the instrument (Table 10.3). The United States is the primary sender in 681 of the 1,412 cases — 48 percent, about half in every decade but the 1970s. Less often noticed is the list of targets: the United States itself comes first, followed by Japan, South Korea, China, the European Community and Canada. In this record, economic coercion is largely a tool used among trading partners, many of them allies, over trade.

Table 10.3: Who sanctions whom: the most frequent primary senders and targets of economic sanctions, 1945–2005. "Imposed" is the share of the sender's cases in which sanctions took effect. Source: Morgan, Bapat and Kobayashi, TIES v4.0; computed by the author.

Primary sender	Cases	Share	Imposed	Most frequent targets	Cases
United States	681	48.2%	57%	United States	103
Canada	112	7.9%	80%	Japan	80

Primary sender	Cases	Share	Imposed	Most frequent targets	Cases
Russia / Soviet Union	38	2.7%	68%	South Korea	58
United Kingdom	38	2.7%	34%	China	45
India	34	2.4%	94%	European Community / Union	44
Mexico	25	1.8%	92%	Canada	41

10.13 A short review of the literature

The empirical study of sanctions begins with Hufbauer et al. (2007), whose three editions established the case list, the two-component coding of success and the "about a third" finding; Pape (1997) audited the claimed successes and found few that survived, Elliott (1998) replied, and Pape (1998) rejoined. D. A. Baldwin (1985) had already argued that effectiveness must be judged relative to the alternatives and their costs, which reframes the whole exchange. The mechanism literature runs from Galtung (1967), whose study of Rhodesia dismantled the "naive theory" that pain produces compliance, to Kirshner (1997), who located the effect of sanctions in the redistribution of power among groups within the target.

The strategic turn came with Drezner (1999), who explained why sanctions are imposed most on the adversaries least likely to concede, and Drezner (2003), who identified the selection effect of threats. The data response was TIES, which records threats as well as impositions (Morgan, Bapat, and Krustev 2009; Morgan, Bapat, and Kobayashi 2014); its outcome codes show complete acquiescence more often at the threat stage than after imposition, above all in economic disputes. Early (2015) showed how third-country sanctions busters sustain targets, which helps explain why multilateral sanctions do not always outperform unilateral ones.

On design, Biersteker, Eckert, and Tourinho (2016) evaluate UN targeted sanctions against the purposes of coercing, constraining and signalling, and Peksen (2009) documents the human-rights cost of sanctions to the populations of target states. Farrell and Newman (2019) explain the rise of financial sanctions through the chokepoints of global networks. The GSDB (Felbermayr et al. 2020) and its later releases (Syropoulos et al. 2024) made it possible to study sanctions at the scale of trade data, and Crozet and Hinz (2020) used that kind of data to show that much of the trade cost of the Russia sanctions after 2014 fell on products that were not sanctioned at all — the quantitative trace of overcompliance. Chupilkin, Javorcik, and Plekhanov (2026) trace the opposite leak after 2022: goods rerouted to Russia through the Caucasus and Central Asia.

10.14 What the case count does not license

The descriptive record of the GSDB is clear in its broad lines: the number of active sanctions cases has risen steeply since the end of the Cold War, and the mix of instruments has shifted from trade towards finance, travel and named persons. That picture is the right starting point for any view of sanctions risk. It is also easy to over-read, and four conclusions often drawn from it are not supported.

It does not license a conclusion about a firm's *exposure*: a count of cases naming a country says how often senders have used the instrument against it, not how much of a firm's revenue, supply or payments the measures touch, and a single financial sanction on a country's largest banks can matter more than a dozen travel bans. It does not license a conclusion about the *perimeter*, which is drawn by ownership rules, secondary sanctions, diversion clauses and the firm's own banks, and which reaches customers in countries that appear nowhere in the data as targets. It does not license a conclusion about the *pipeline*, because the cases that would matter most next year are threats today. And it says nothing about *duration*: whether a given set of measures will be lifted soon enough to make waiting the cheaper option, a cost the Russian exits showed to be set less by the senders than by the target's countermeasures.

The deeper conclusion is about the instrument itself. Sanctions were designed to regulate conflict between states without war, and on the evidence they do that job partially, at a cost that falls largely on people who did not choose the policy. For

firms the balance is less ambiguous. Each expansion of the instrument — from trade to finance, from lists to ownership, from targets to the neighbours through which goods are diverted — has moved more of the work of enforcement onto the private sector. Whatever sanctions do to their targets, they have made the multinational the place where foreign policy is executed, and its sanctions risk lives in its footprint, its counterparties' owners, its banks, its distributors and the countermeasures of its host states. None of those is in the case list.

Chapter 11

Space Geopolitics: The Orbital Commons and the New Space Race

11.1 Infrastructure overhead

Few multinationals own a satellite. Almost all of them depend on one, and most do not know which. A container line tracks its ships by satellite and talks to their crews by satellite when they are out of range of land. A mining company runs its remote sites over satellite links. An agribusiness forecasts harvests from imagery, an insurer settles flood claims from it, and a buyer of commodities checks the stockpiles of its suppliers with it. Above all, almost every firm depends on satellites for *time*. Telecommunications networks, electricity grids, data centres and trading venues synchronise their clocks to the signals of global navigation satellite systems, because those signals carry time from atomic clocks in orbit to any receiver on the ground, cheaply and everywhere.

Three services carry most of this dependence. **Positioning, navigation and timing** comes from four global systems, each run by a state: GPS by the United States, Galileo by the European Union, BeiDou by China and GLONASS by Russia. **Connectivity** comes from operators in geostationary orbit, 36,000 kilometres up, and since 2019 from constellations of thousands of satellites in low Earth orbit, a few hundred kilometres up. **Earth observation** comes from state agencies and from a growing number of commercial imaging firms.

Chapters 8 and 10 showed how networks become instruments of power: a network's hubs are under someone's jurisdiction, and whoever has jurisdiction can watch the traffic or cut it off. The network of this chapter is physically above the firm. Its hubs are a handful of states and, increasingly, a handful of companies, and its environment is degrading in a way no terrestrial network's does — through debris that stays where it was created for years or centuries. The scale of the change is recent. The European Space Agency's 2026 environment report notes that the exponential growth in the number of new payloads continued in 2025, driven by a higher launch frequency, rideshare launches and the deployment of large constellations by commercial operators (ESA Space Debris Office 2026).

The question of this chapter is who holds the risk when orbit is crowded, contested and dominated by one operator — and what the firm can do about it. The answer it develops is that the firm holds more of it than it thinks, that most of it arrives through the reversible and unspectacular threats rather than the destructive ones, and that dependence on a satellite service is, by the construction of space law, dependence on a state.

11.2 A domain written for two superpowers

The law of outer space was written at speed, during the Cold War, by and for two states that were then the only ones able to get there. Sputnik's launch in October 1957 settled one question before anyone had negotiated it: the Soviet satellite passed over every country on Earth, and no state protested that its airspace had been violated. The principle that satellites may overfly territory — which the United States, planning reconnaissance satellites of its own, had every interest in establishing — became the foundation of the domain.

The rest of the foundation was laid in the Outer Space Treaty, opened for signature in January 1967 and in force by October of that year (United Nations 1967). It remains the constitution of the domain, and the articles that matter to a firm can be read in two pages. Article II declares that outer space "is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means": no one can own an orbit. Article IV forbids placing nuclear weapons

or other weapons of mass destruction in orbit, and reserves the Moon and other celestial bodies for peaceful purposes — but it says nothing to prohibit conventional weapons in orbit, or the testing of weapons that destroy satellites from the ground. Article VI makes states internationally responsible for national activities in space "whether such activities are carried on by governmental agencies or by non-governmental entities", and requires those activities to be authorised and continually supervised by "the appropriate State". Article VII makes the launching state liable for damage caused by its objects. Article IX requires states to act with "due regard" to the interests of others and to consult before activities that could cause harmful interference — the closest the Treaty comes to traffic rules, without anyone to enforce them.

Orbital commons — Earth orbit treated as a resource that no one may appropriate (Outer Space Treaty, Article II) and from whose use no one can easily be excluded, but whose use by one actor degrades it for others — through congestion of orbits and spectrum and through debris. The commons is legal in origin and physical in its consequences.

Article IV's silence on conventional weapons was not an accident; neither superpower wanted to give up the option. Its prohibition on nuclear weapons in orbit, by contrast, had been learned the hard way.

From the archive · Starfish Prime, 1962

On 9 July 1962 the United States detonated a thermonuclear warhead of about 1.4 megatons some 400 kilometres above Johnston Island in the Pacific. The test, Starfish Prime, was intended to study the effects of a nuclear explosion at high altitude. Its most lasting effect was one its planners had underestimated: electrons from the explosion were trapped in the Earth's magnetic field, creating an artificial radiation belt that persisted for years.

The satellites then in low orbit were few, and several were damaged. Telstar 1, the first commercial communications satellite — built by AT&T's Bell Laboratories and launched the day after the test — suffered radiation damage to its electronics and failed within months; Britain's first satellite, Ariel 1, was also damaged. A single test by one state had degraded an orbital environment shared by everyone, and the first commercial satellite was among the casualties.

The Partial Test Ban Treaty of 1963 prohibited nuclear tests in outer space, and the Outer Space Treaty four years later prohibited nuclear weapons in orbit. Neither prohibited the next form of the same problem: a conventional weapon that destroys a satellite and fills its orbit with fragments.

Article II has also become the ground of a newer contest, over resources rather than weapons. The Treaty forbids national appropriation of outer space and celestial bodies; it does not say whether a firm may own what it extracts from them. The United States answered the question for its own citizens in the Commercial Space Launch Competitiveness Act of 2015, which recognises their property rights in resources obtained from asteroids and other bodies, and Luxembourg followed with a law of its own in 2017. In 2020 the United States and seven partners signed the Artemis Accords, a non-binding set of principles for the civil exploration of the Moon that treat resource extraction as consistent with the Treaty; China and Russia announced a joint lunar research station the following year and gathered partners of their own. The pattern resembles the contest over technical standards described in Chapter 6: where the treaty process has stalled, rules are written by coalitions, and a firm's access to a future market depends on which coalition its home state has joined.

11.3 Liability, registration and the right to transmit

The Outer Space Treaty is the constitution; three further layers turn it into rules a firm encounters.

The first is liability and registration. The Liability Convention of 1972 elaborated Article VII: a launching state is absolutely liable for damage its space objects cause on the surface of the Earth or to aircraft in flight, and liable for fault for damage elsewhere in space. The Registration Convention of 1975 requires launching states to register their objects with the United Nations. The two together mean that every object in orbit has, in law, a state that answers for it — long after it has stopped working.

From the archive · Cosmos 954, 1978

On 24 January 1978 a Soviet ocean-surveillance satellite, Cosmos 954, re-entered the atmosphere over the Northwest Territories of Canada. It was powered by a small nuclear reactor, and its re-entry scattered radioactive debris over a wide stretch of the Canadian north. Canada and the United States mounted a search and clean-up operation, Operation Morning Light, that lasted for months.

Canada presented a claim to the Soviet Union under the Liability Convention and general principles of international law, for the cost of locating, recovering and cleaning up the debris. The claim was settled in 1981, by protocol, for three million Canadian dollars — about half the amount claimed. It remains the only claim ever brought under the Convention.

The case shows both the strength and the limits of the regime. Liability attaches to a state, and a state paid. But it took three years, the settlement was negotiated rather than adjudicated, and the claimant was a state. A firm whose satellite is destroyed by a fragment from another state's anti-satellite test has no standing under the Convention: it must persuade its own state to claim, against a launching state that may deny responsibility, for damage whose cause it may be unable to prove.

The second layer is spectrum. A satellite is useless without radio frequencies to command it and to carry its service, and frequencies and orbital positions are coordinated by the International Telecommunication Union through its Radio Regulations. An operator's state files a satellite network with the ITU and must coordinate it with networks filed earlier, so priority goes, broadly, to the first to file. The incentive is to file early and large — sometimes for "paper satellites" that may never fly — and the result is that orbital slots and frequencies are a scarce asset allocated by a procedure most firms never see.

The third layer is national licensing. Article VI's "authorization and continuing supervision" becomes, in practice, a set of permits: to launch, to use spectrum, to operate a remote-sensing system, to export a satellite or its components, to land a service in a national market. This is where a firm meets space law directly, and where the balance of power enters most visibly — through the export-control lists discussed below and through decisions about which foreign constellations may sell service in which countries. It is also where the rules on debris acquire teeth. The international guidelines are voluntary, but a licence condition is not: in 2022 the US Federal Communications Commission adopted a rule requiring operators of satellites in low orbit that it licenses, or that serve the US market, to dispose of them within five years of the end of their missions, replacing a guideline of twenty-five years. A firm's satellite provider is therefore bound not by one law of space but by the licensing conditions of every state whose market it serves.

11.4 The physics of a commons

Law makes orbit a commons in one sense: no one may own it. Physics makes it a commons in the stricter sense economists use. A common-pool resource is one from which exclusion is difficult and whose use by one actor subtracts from what is left for others. Garrett Hardin's parable of the herdsmen on a common pasture, each adding animals because he captures the whole gain while bearing only a share of the cost, is the classic statement of why such resources tend to be overused (Hardin 1968). Elinor Ostrom showed that real communities often escape the tragedy through rules they design and enforce themselves — but under conditions of clear boundaries, monitoring and graduated sanctions that Earth orbit does not meet (Ostrom 1990). Brian Weeden and Tiffany Chow applied her framework to space and found most of its design principles missing (Weeden and Chow 2012).

What turns congestion into a tragedy in orbit is a mechanism identified by Donald Kessler and Burton Cour-Palais in 1978 (Kessler and Cour-Palais 1978). Objects in the same orbital shell occasionally collide; each collision produces fragments; the fragments raise the density of objects in the shell and therefore the rate of further collisions. Below a critical density, the atmosphere drags enough debris down to keep the population in check. Above it, collisions generate fragments faster than drag removes them, and the debris population grows on its own.

Kessler syndrome — the self-sustaining growth of the debris population in an orbital shell once the density of objects passes a threshold at which collisions create fragments faster than atmospheric drag removes them. Past the threshold, the population grows even if nobody launches again.

Altitude matters enormously, because the drag that cleans orbits weakens with height. Debris a few hundred kilometres up re-enters within years; debris above about 800 kilometres can remain for decades or centuries. A fragmentation in a high, busy shell is therefore a long-lived degradation of an asset shared by every operator in it. Two decades ago, simulations by NASA's debris specialists already suggested that the low-orbit debris population would keep growing through collisions even if all launches stopped (Liou and Johnson 2006).

The public catalogue shows where the objects are today (Figure 11.1). Of the 35,036 catalogued objects in Earth orbit in the October 2026 vintage used here, 29,200 have a mean altitude between 200 and 2,000 kilometres. Below 600 kilometres there are 14,568, and 95 percent of them are payloads — 11,153 of them Starlink satellites, stacked in a few narrow shells around 350, 480 and 550 kilometres. Between 700 and 900 kilometres the picture inverts: of 5,665 objects, only 12 percent

are payloads, and the rest are debris and spent rocket stages, many of them left by the events described below. The two populations sit in different places for the reason Kessler gave. Operators of large constellations fly low so that a failed satellite re-enters within a few years; the fragments of explosions and tests above 700 kilometres stay where they were made.

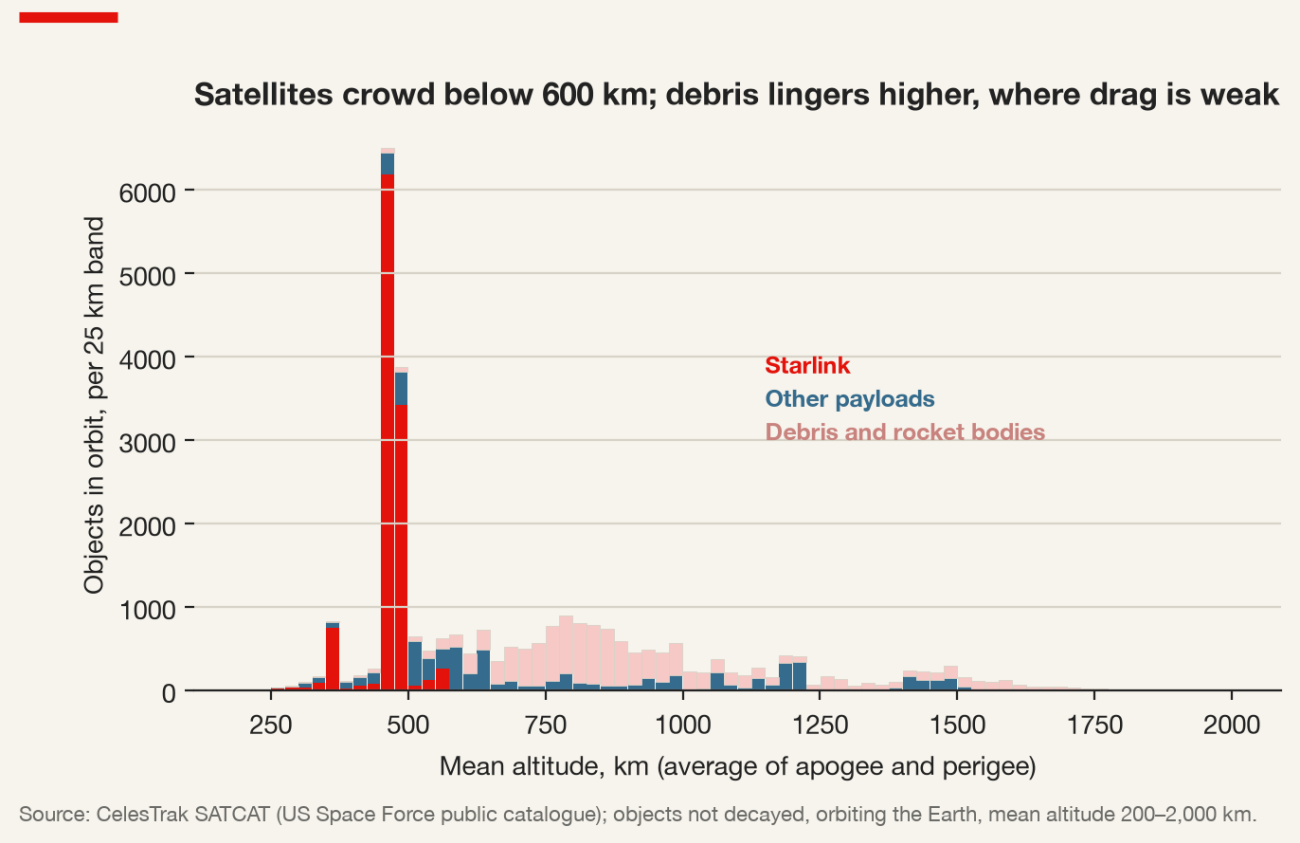


Figure 11.1: Satellites crowd below 600 km; debris lingers higher, where drag is weak. Catalogued objects in Earth orbit by mean altitude, in 25-kilometre bands. Source: CelesTrak SATCAT, October 2026 vintage.

For a firm the chart says that "orbit" is not one asset but several. The low shells, where the new broadband constellations fly, are crowded with working satellites that manoeuvre and are cleaned by drag; the risk there is congestion and the conduct of a few large operators. The shells near 800 kilometres, used by Earth-observation and weather satellites, carry the legacy of the twentieth century's tests and explosions, and that legacy will be there for decades whatever anyone does next.

Economists have since modelled the incentives. Nodir Adilov, Peter Alexander and Brendan Cunningham show that, because each operator ignores the collision risk its satellites and debris impose on others, open access leads to more launches and more debris than would be efficient (Adilov, Alexander, and Cunningham 2015). Akhil Rao, Matthew Burgess and Daniel Kaffine estimate that an internationally harmonised fee on each satellite in orbit — making operators pay for the risk they impose — would greatly increase the long-run value of the space industry (Rao, Burgess, and Kaffine 2020). No such fee exists. Mitigation guidelines adopted through the UN and by space agencies ask operators to limit debris and to remove satellites from busy orbits at the end of their lives; compliance is voluntary and uneven.

For the firm: an orbit is a shared infrastructure whose quality is set by its least careful user. A destructive test by one state, or a rocket stage that explodes years after launch, raises the collision risk — and the insurance price — for every operator in that shell, and therefore for every customer of those operators.

11.5 Three readings of the new space race

Who will govern this commons, and in whose interest? International relations offers three readings of what has come to be called the new space race, and they do not exclude one another.

The first is classical geopolitics carried upwards. Everett Dolman's *Astropolitik* takes Mackinder's logic — whoever controls the pivot controls the world — and applies it to orbit: whoever controls low Earth orbit controls access to the rest of space, and through it can dominate the Earth below (Dolman 2002). On this realist reading the United States should seize and hold that control while it can, and space is a territory to dominate, its orbits and transfer routes the chokepoints of a new strategic geography. It is the anarchy of Chapter 3, with orbits in place of straits.

The second reading is institutional and commercial. Saadia Pekkanen describes the new space race as driven by three intertwined trends (Pekkanen 2019). *Democratisation*: many more states, and many non-state actors, now operate in space. *Commercialisation*: firms no longer only supply governments but strike out on their own, with their own capital and customers. *Militarisation*: states increasingly treat space as a domain of warfare. Binding them together is the fact that about 95 percent of space technologies, on her account, are dual-use. With treaty-making in decline, the domain is governed less by new international law than by national legislation, contracts, and a few old principles. On this reading space is a market, and the firms in it are part of the security problem.

The third reading puts the Earth back at the centre. Bleddyn Bowen argues that spacepower matters for its effects on Earth: orbit is a "cosmic coastline" from which states support — and threaten — what happens on land, at sea and in the air (Bowen 2020). Satellites are valuable because they make terrestrial forces see, communicate and navigate; that is also why they are targets when terrestrial power is contested.

Read together, the three explain why the same constellation can be a consumer broadband business, a military communications network and an instrument of foreign policy all at once. Pekkanen supplies the dual use; Dolman and Bowen supply the strategic stakes. For a multinational the practical conclusion is the same under all three: a commercial satellite service is never only commercial.

11.6 Counterspace: what states can do to satellites

If satellites are targets, the question is what can be done to them, by whom, and with what consequences for the firms that use them. The Secure World Foundation's annual open-source assessment tracks the capabilities of thirteen countries to deceive, disrupt, deny, degrade or destroy space systems, in five categories (Samson and Brett 2026).

Counterspace capabilities — the means of deceiving, disrupting, denying, degrading or destroying space systems: direct-ascent missiles, co-orbital systems, electronic warfare, directed energy and cyber operations. They differ in whether their effects are reversible and in who bears them first — the satellite's owner, every operator in the orbit, or the users on the ground.

Direct-ascent weapons are missiles launched from the Earth to destroy a satellite. Their effects are irreversible and, at most altitudes, create debris that threatens every operator in the shell. **Co-orbital** systems are satellites that approach another to inspect it, interfere with it or attack it. **Electronic warfare** jams or spoofs signals — the uplink to a satellite, its downlink to users, or navigation signals on the ground — and its effects end when the jamming stops. **Directed-energy** weapons use lasers or microwaves to dazzle or damage sensors. **Cyber** operations attack satellites, ground stations or user terminals, and can disable a service without touching a satellite.

The assessment's central finding is the one that matters most for business: many states have researched and tested destructive capabilities, but only non-destructive ones are actively used against satellites in current military operations. Four states have destroyed one of their own satellites with a missile launched from the ground: the United States in 1985 — and, in 2008, by intercepting a failing satellite at low altitude — China in 2007, India in 2019 and Russia in 2021. In April 2022 the United States declared that it would not conduct destructive direct-ascent tests, and in December 2022 the UN General Assembly called on all states to make the same commitment.

What firms have actually suffered is the reversible kind. On 24 February 2022, the day Russia's full-scale invasion of Ukraine began, a cyberattack on the ground segment of Viasat's KA-SAT network disabled thousands of user modems in Ukraine and across Europe, including those used for the remote monitoring of wind farms in central Europe. The European

Union, the United Kingdom and the United States attributed the attack to Russia (UK Government 2022). The satellite was untouched; the service failed for its customers, many of them nowhere near the war. Since then, the jamming and spoofing of navigation signals around conflict zones has become routine. In 2024 Finnair suspended its flights to Tartu, in Estonia, for several weeks because interference made the airport's satellite-based approach procedures unusable.

The lesson for an analyst. Destructive tests make headlines and debris. Jamming, spoofing and cyberattacks on ground segments make *losses* — and the losses fall on the customers, not on the satellite. A firm's counterspace exposure is measured on the ground, in the regions where its ships, aircraft and sites depend on signals from orbit.

11.7 The debris record, and how it is counted

The most complete public account of the orbital environment comes from ESA's Space Debris Office, which has published an annual *Space Environment Report* since 2017. The 2026 edition, with data to the end of 2025, is the source of the figures that follow (ESA Space Debris Office 2026), and its method deserves reading before its numbers.

ESA counts two kinds of objects. *Catalogued* objects are those tracked and published in catalogues: the US public catalogue, the Russian Vimpel catalogue and, for the early decades, the United Kingdom's historical RAE tables. *Asserted* objects are known to exist by design — a satellite released from a dispenser, a component separated as planned — even if not yet catalogued. The two are cross-correlated in ESA's DISCOS database and classified into object classes. Objects too small or too erratic to be tracked are outside the report's scope.

On that basis, the report counts 44,964 objects in Earth orbit at the end of 2025, of which 16,946 are payloads, and 25,060 objects in low Earth orbit (ESA Space Debris Office 2026). It observes that, ever since the start of the space age, there has been more debris in orbit than operational satellites. And its most important finding for anyone who thinks the debris problem is the weapons tests is this: *fragmentations other than collisions* — explosions, mostly of old rocket stages and batteries — are the dominant source of debris. Non-deliberate fragmentations have averaged 9.8 a year over the last two decades, and 2024 alone added more than 3,000 catalogued fragments, mostly from propulsion-related break-ups.

The report's count of objects in geocentric orbit by object class since 1957 shows the cumulative result of seven decades of launches, explosions, collisions and re-entries, and three events stand out on it. On 11 January 2007 China destroyed its Fengyun-1C weather satellite with a ground-launched missile, at about 850 kilometres; the test remains the largest single source of catalogued debris in the public catalogue, and most of its thousands of fragments are still in orbit. On 10 February 2009 the active Iridium 33 and the defunct Cosmos 2251 collided at about 780 kilometres — the first accidental collision between two intact satellites, Kessler's mechanism observed. And on 15 November 2021 Russia destroyed Cosmos 1408 at roughly 470 to 490 kilometres, producing some 1,800 catalogued fragments; at that altitude almost all re-entered within three years, and ESA attributes the 2022 spike in re-entering debris to the test.

The contrast between the first and third events is a lesson in altitude. Both were deliberate destructions of a satellite by its own state; one polluted a high orbit for decades, the other a low orbit for a few years. A count of objects treats them alike.

The public catalogue lets the lesson be put in numbers, because every fragment is named after its parent. Table 11.1 counts the catalogued fragments of the main destructive tests and of three accidental break-ups, and how many of them are still in orbit. Of the 3,531 catalogued fragments of Fengyun-1C, 2,311 — almost two in three — are still in orbit nearly two decades later, at a median altitude of about 830 kilometres. Of the 1,806 fragments of Cosmos 1408, four remain. The American intercept of 2008 and India's test of 2019, both conducted at low altitude, have left nothing in the catalogue. The 2009 collision sits between the two: a third of the Cosmos 2251 fragments and a fifth of the Iridium 33 fragments are still up.

Table 11.1: Fragments of selected destructive events and how many remain in orbit. Fragments are objects named "DEB" after the parent; altitude is the mean of apogee and perigee. Source: CelesTrak SATCAT, October 2026 vintage; computed by the author.

Event	Fragments catalogued	Still in orbit	Share	Median altitude of survivors
Fengyun-1C, Chinese ASAT test, January 2007	3,531	2,311	65%	832 km
Cosmos 2251, collision, February 2009	1,714	608	35%	728 km
Iridium 33, collision, February 2009	656	115	18%	720 km
USA 193, US intercept, February 2008	174	0	0%	—
Microsat-R, Indian ASAT test, March 2019	129	0	0%	—
Cosmos 1408, Russian ASAT test, November 2021	1,806	4	0%	376 km
NOAA 16, break-up, November 2015	458	353	77%	776 km
DMSP 5D-2 F13, break-up, February 2015	241	122	51%	779 km
NOAA 17, break-up, March 2021	115	51	44%	691 km

The last three rows support ESA's finding that accidental fragmentations, not weapons, are the larger source of debris. Two ageing American weather satellites and one military weather satellite, each of which broke up years after its mission ended — the DMSP satellite through a battery failure — together left 526 fragments still in orbit between 700 and 800 kilometres, not far short of the 723 left by the 2009 collision. None of these break-ups made headlines; each degraded a shell that other operators use.

What the count does not measure. It counts objects, not risk: a ten-centimetre fragment and a five-tonne rocket stage each count once, and the objects too small to catalogue — the majority of the fragments able to disable a satellite — do not appear at all. It is drawn from catalogues that themselves improve: when sensors get better, the count rises with nothing new having happened in orbit. And it shows the *environment*, not the exposure of any operator or customer, which depends on which shells their satellites use.

11.8 From a catalogue to a stock

ESA's series can be approximated from public data, and the approximation shows how a concept — the number of objects in orbit — becomes a number. The public catalogue, maintained from US Space Force data and redistributed by CelesTrak, does not record a stock. It records objects: one row for each catalogued object over its whole life, with a launch date and, if it has re-entered, a decay date. There is no "year" in the file. An object is in orbit in a given year if it was launched by then and had not decayed by then, so the stock in each year is the running sum of launches minus the running sum of decays — a stock rebuilt from two flows. Much of what a risk analyst needs to know is a stock of this kind, which no one publishes and which has to be rebuilt from the events that change it.

The construction exposes the catalogue's decisions. Debris carries the launch date of its parent object, not the date of the event that created it: the fragments of China's 2007 test are dated to the launch of Fengyun-1C in 1999, so a debris series built on launch dates moves in the wrong year, and the published record and the reconstruction disagree precisely at the events that matter. "Not decayed" is not the same as "in Earth orbit", because some objects orbit the Sun, the Moon or a Lagrange point. The owner field records a country, organisation or company code, not the operator: every Starlink satellite is simply American, and the former Soviet Union and Russia share one code. Operational status describes today only, so the file cannot say how many satellites were active in any past year. And the public catalogue is not the environment: it omits classified objects and everything too small to track, which is one reason ESA's totals, drawn from more catalogues and asserted objects, are higher.

The dating decision can be measured, because a second public catalogue records what the first does not. Jonathan McDowell's General Catalog of Artificial Space Objects gives each fragment the date on which it separated from its parent as well as the parent's launch date (McDowell 2026). Figure 11.2 rebuilds the stock of objects in Earth orbit at the end of each year from that catalogue, with debris dated by the event that created it, and sets beside it the total obtained from CelesTrak's copy of the public catalogue (Kelso 2026) with debris dated by its parent's launch. The two agree on payloads and rocket bodies and disagree, sometimes by a factor of two, on debris. Dated by launch, the debris in orbit at the end of 2000 numbers 16,843 objects, because every fragment ever created from a satellite launched by then — including those of the 2007 test, the 2009 collision and every later break-up — is counted as if it already existed. Dated by the event, it numbers 6,530. The launch-dated series barely moves in 2007 (it rises by 39 objects) because the Fengyun-1C fragments were already in it from 1999; the event-dated series jumps by 3,765. The two series converge only after 2021, once most backdated fragments have either been created or re-entered.

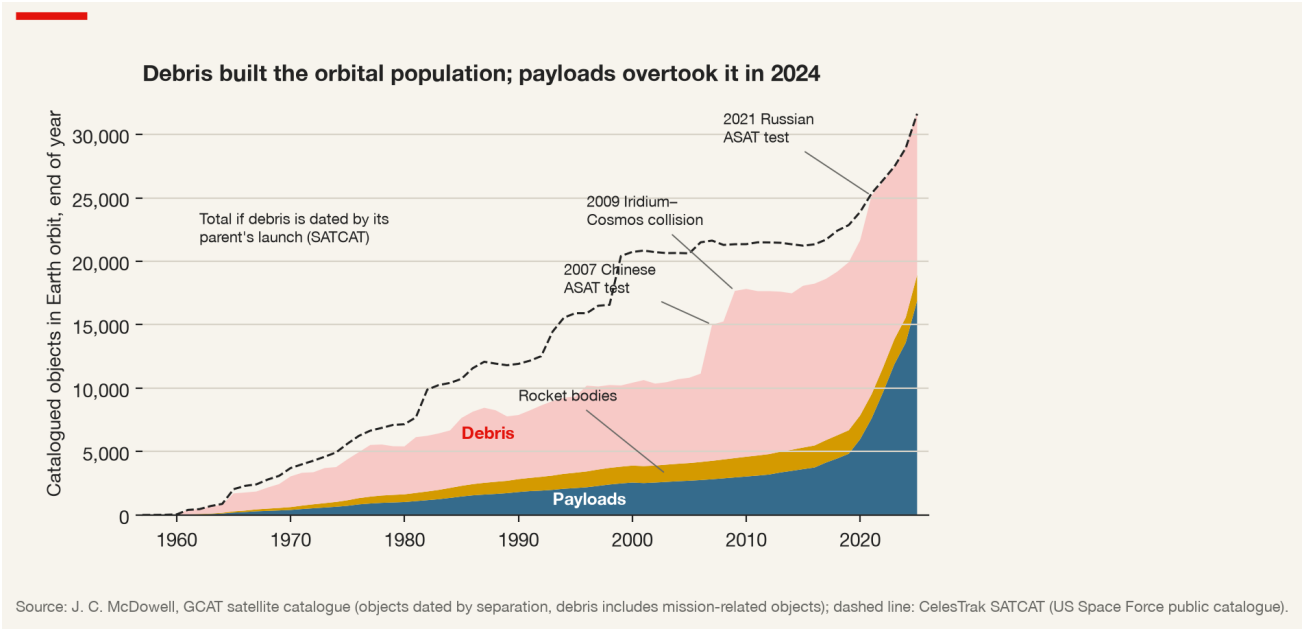


Figure 11.2: Debris built the orbital population; payloads overtook it in 2024. Catalogued objects in Earth orbit at the end of each year, with debris dated by the event that created it; the dashed line dates debris by its parent's launch. Source: GCAT (J. C. McDowell); CelesTrak SATCAT.

The event-dated series also shows the turn of the last few years. Debris outnumbered payloads in the catalogue from the 1960s onwards, as ESA notes. At the end of 2025 the reconstruction counts 16,899 payloads, 2,015 rocket bodies and 12,812 pieces of debris; payloads first outnumbered debris in 2024. The crossing is not a sign that the debris problem is solving itself. It is the arrival of the large constellations, whose satellites are counted as payloads while they work and after they fail, and the slow re-entry of the low fragments of the 2021 test.

Table 11.2: Three counts of the same orbital environment. The two public-catalogue counts use the vintages of October 2026 and the rule "launched by, and not decayed by, 31 December 2025". Sources: ESA Space Debris Office (2026); CelesTrak SATCAT; GCAT; computed by the author.

Count of objects in Earth orbit, end of 2025	Total	Payloads
ESA Space Environment Report 2026 (catalogued and asserted objects)	44,964	16,946
Public catalogue (CelesTrak SATCAT), debris dated by parent launch	32,460	17,414
GCAT, debris dated by fragmentation	31,726	16,899

Table 11.2 sets the three counts side by side. The two reconstructions from public catalogues agree within a few hundred objects on the total. ESA's is more than 12,000 higher, because it merges several catalogues and adds objects known to exist but not yet tracked. Its count of payloads is close to the reconstructions', so the difference is almost entirely debris: the environment a firm's satellite provider actually faces is more crowded than any public download shows.

11.9 One operator

The new space race has a feature the Cold War one did not. In the first years of the space age, concentration was extreme because only two states could launch anything; it fell for decades as Europe, Japan, China, India and many others acquired the capability, and as commercial operators multiplied. It has risen again — not because states have dropped out, but because one company has launched more satellites than everyone else together. SpaceX's Starlink, first launched in May 2019, grew in six years into the majority of the world's operating satellites. Measured on the public catalogue's owner codes, the concentration of payloads launched each year has returned to levels last seen in the Cold War, with a firm rather than a superpower at the top.

Concentration needs a measure, and the standard one is the index that antitrust authorities use. Its origins are in geopolitics: Albert Hirschman devised an index of the concentration of a country's trade in 1945, to show how Nazi Germany had used the structure of trade to make its smaller partners dependent on it (Hirschman 1945). Orris Herfindahl reinvented it a few years later for industrial concentration, and it carries both their names.

Herfindahl–Hirschman index (HHI) — the sum of the squared market shares of all participants, with shares in percent, so that it runs from near 0 (many small participants) to 10,000 (one participant with everything). It weights large shares heavily: one participant with 75 percent contributes 5,625 on its own. US merger guidelines treat markets above 2,500 (U.S. Department of Justice and Federal Trade Commission 2010) or, in the 2023 revision, above 1,800 (U.S. Department of Justice and Federal Trade Commission 2023) as highly concentrated.

The index, built from the catalogue, inherits the catalogue's owner-code problem twice over. Concentration by *state* hides the fact that one company dominates the American share; concentration by *operator* must be reconstructed from object names. Each choice yields a different index, and each answers a different question — about the balance of power among states, or about the dependence of customers on firms.

Figure 11.3 shows where the concentration comes from. In 2018, 448 payloads were launched, two of them SpaceX's. In 2020 the total was 1,242 and 833 were SpaceX's; in 2025 it was 4,495, of which 3,169 — 71 percent — were SpaceX's. China launched 368 payloads in 2025, Russia 88, and every other American operator together 530. The rest of the world, Europe, Japan and India included, launched 340.

Figure 11.4 computes the index on payloads launched each year, by owner state and by operator, from the GCAT codes. By state, the index stood at 7,450 in 1960 and 7,898 in 1980, when the Soviet Union launched most of the world's satellites; it fell to 2,559 in 2000 and a low of 1,312 in 2008, as Europe, China, India, Japan and many smaller states entered. Since 2019 it has risen again, to 6,845 in 2025 — Cold War levels — because the American share has grown with one firm. By operator, the index was below 1,000 for most of the space age, and as low as 142 in 2013, because even the Soviet programme was spread across many agencies and design bureaus; it reached 4,589 in 2020 and 5,007 in 2025. The two lines tell different

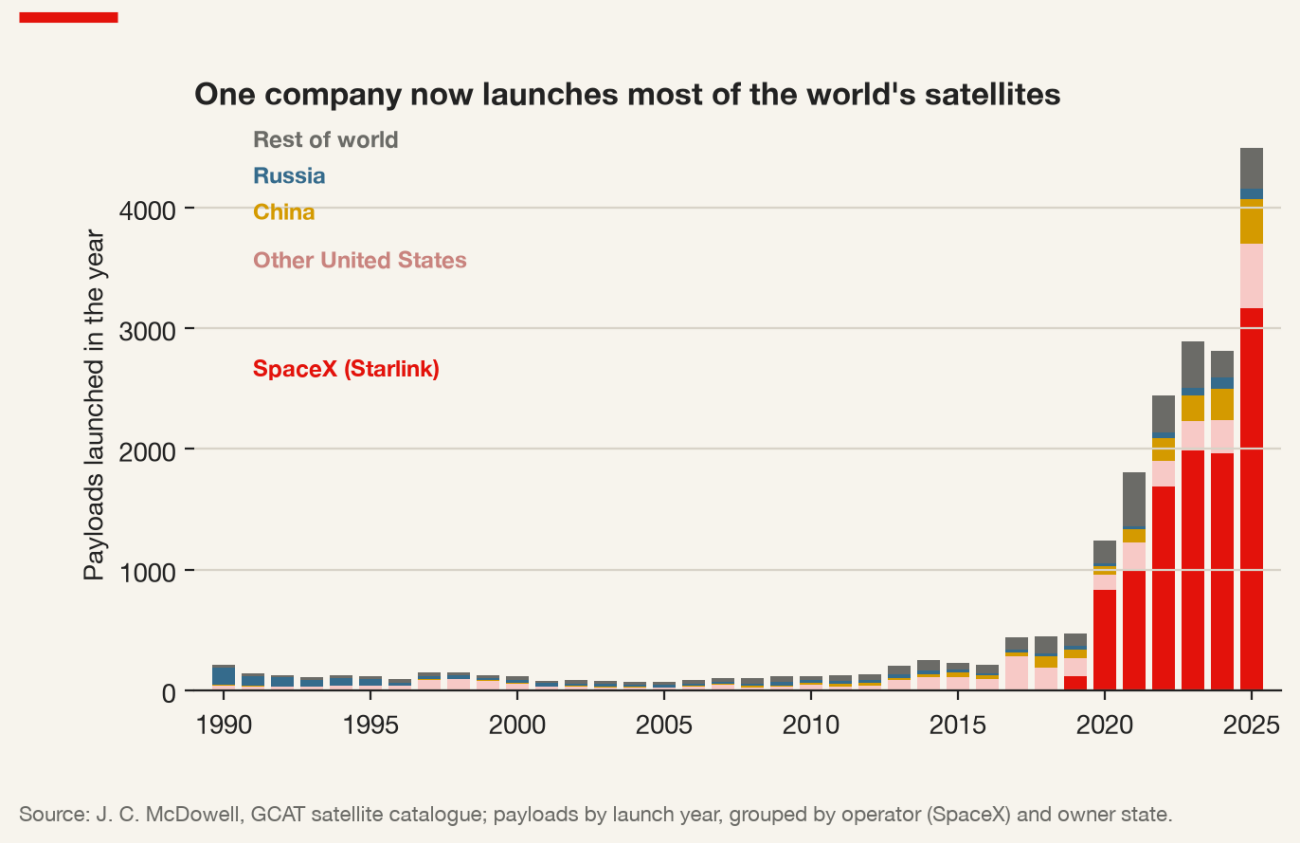


Figure 11.3: One company now launches most of the world's satellites. Payloads by launch year, grouped by operator (SpaceX) and owner state. Source: GCAT (J. C. McDowell).

stories about the past and the same one about the present. The balance among states looks like the 1980s; the balance among operators has no precedent.

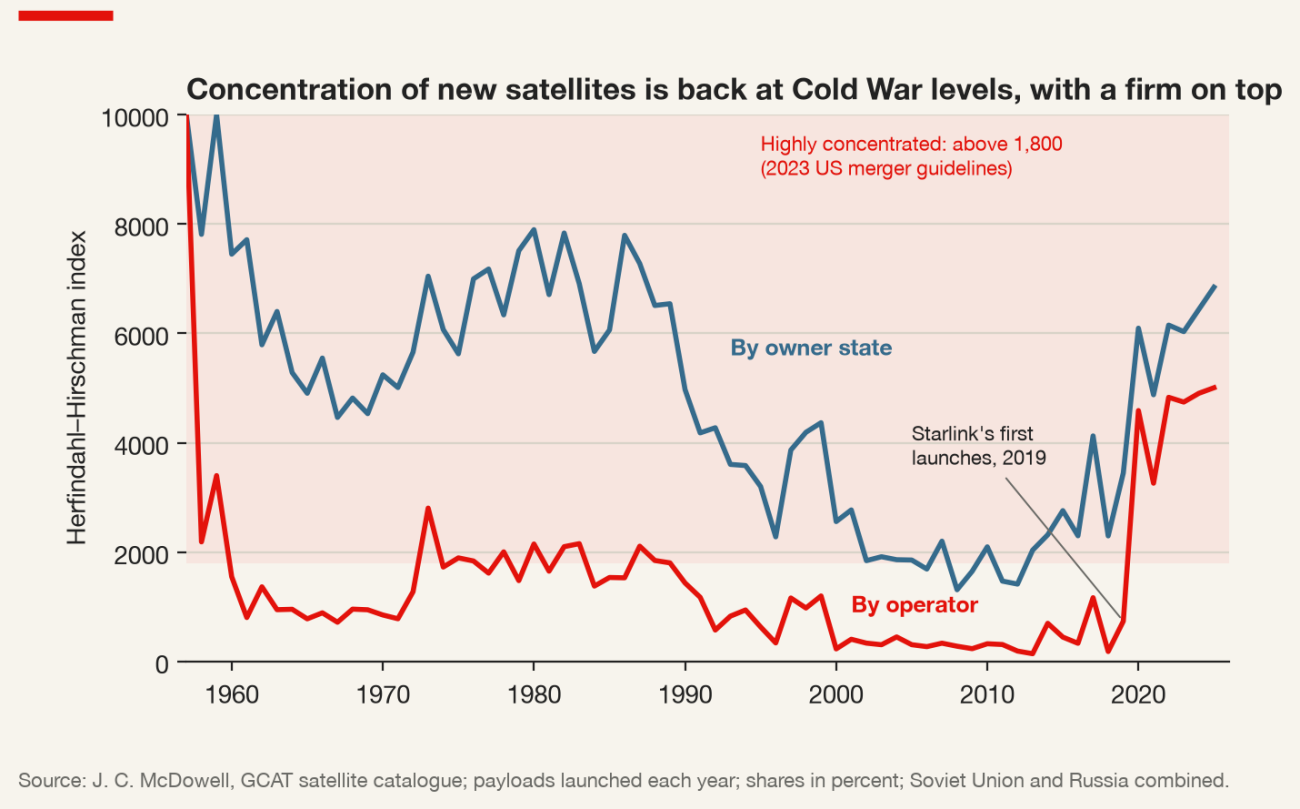


Figure 11.4: Concentration of new satellites is back at Cold War levels, with a firm on top. Herfindahl–Hirschman index of payloads launched each year, by owner state and by operator. Source: GCAT (J. C. McDowell).

Flows are not stocks, and launches overstate the weight of a constellation whose satellites last five years against geostationary satellites that last fifteen. On the stock, the conclusion is the same. Of the 17,247 catalogued objects marked as operational in the October 2026 vintage of the public catalogue, 11,151 — 65 percent — are Starlink satellites.

Concentration matters to a firm because of what it implies about dependence, and Starlink in Ukraine shows what that dependence means. Within days of the 2022 invasion, Starlink terminals became a backbone of Ukraine's military and civil communications. The company's decisions about where its service would work — including, by its owner's account, a refusal to enable it for an attack near Crimea in 2022 — became questions of war policy, and governments moved to contract for the service directly.

Farrell and Newman's framework, introduced in Chapter 8, applies without modification (Farrell and Newman 2019). A constellation is a hub. Its operator sees who connects and from where — a panopticon — and can switch off a user, a region or a country — a chokepoint. Weaponisation requires two conditions: that the network be concentrated, and that a state can direct the firm that runs the hub. Article VI of the Outer Space Treaty guarantees the second, because every operator has a licensing state that is internationally responsible for its activities and must supervise them. A firm that depends on one operator depends on that operator's state as well.

States are responding as the balance-of-power reasoning of Chapter 3 would predict: by building their own. The European Union's IRIS² secure-connectivity programme, established by Regulation (EU) 2023/588, China's Qianfan and Guowang constellations, and the many national launch programmes are all, in part, answers to dependence on one American firm. Export controls draw the same line from the other side. Spacecraft and their components sit either on the US Munitions List, under Category XV of the International Traffic in Arms Regulations, or — for most commercial satellites since 2014 — under the Export Administration Regulations, in the "9x515" entries; a non-US manufacturer whose satellite contains a controlled US component needs US permission to sell it to some customers. The balance of power reaches into the bill of materials.

Navigation is the older case of the same pattern, and the one on which almost every firm already depends. GPS was built by the US military. Its opening to the world followed a tragedy: after the Soviet Union shot down Korean Air Lines Flight 007, which had strayed into Soviet airspace, in September 1983, the United States announced that GPS would be made available for civilian use once it was operational. For years the civilian signal was deliberately degraded — "Selective Availability" — until the degradation was switched off in May 2000. The service on which the world's networks and logistics came to rely was, and remains, a gift that one state chose to give and could in principle withdraw or degrade, regionally or globally. Europe's decision to build Galileo, and China's to build BeiDou, were in large part refusals to accept that dependence. For a firm, the existence of four global systems is a genuine diversification — but only if its receivers use more than one, and only against the failure or withdrawal of a system, not against a jammer in the next port.

11.10 Through which channels, and what the firm can do

Space risk reaches the profit and loss account through the same five channels as every other risk in this book, and naming the channel is the first step towards a decision.

Through **market access**, export controls and licensing decide whether a satellite, a terminal or a service can be sold or switched on in a given market, and spectrum and landing rights decide whether a constellation may serve a country at all. Through **inputs**, a firm that buys connectivity for its ships or mines from one constellation, or that times its networks from one state's navigation system, has a single point of failure with a sovereign behind it. Through **finance and payments**, trading venues and payment systems time-stamp transactions from navigation signals, and the insurance capacity for space assets — a thin market — shrinks after large losses. Through **assets and people**, ships and aircraft operating in jammed or spoofed regions are rerouted, delayed, or misreport their positions, with consequences for safety and for sanctions compliance alike. Through **data and operations**, a cyberattack on a ground segment cuts the firm's links, as the KA-SAT modems showed, and the loss of an imagery feed leaves a crop forecast or a supply-chain monitor blind.

Four kinds of evidence bear on that exposure, and each can be tied in advance to a decision. *Dependence*: the share of a critical service — connectivity, imagery, timing — supplied by one operator or one state's system; above a share the firm has fixed beforehand, the decision is to contract a second. *Interference*: the official warnings of navigation-signal interference that aviation authorities publish for the regions where the firm's ships, aircraft or sites operate. *Environment*: destructive tests and fragmentation events in the orbital shells its providers use, and the yearly indicators in ESA's report. *Rules*: changes in export-control lists, licensing conditions and ITU filings that affect its suppliers or markets.

The strategies available each have a cost and a limit. A firm can buy **multi-orbit, multi-operator** connectivity — geostationary and low-orbit links from two providers — at the price of integration, but the second operator may share the first one's licensing state. It can use **multi-constellation receivers** that combine GPS, Galileo and others, which is cheap and protects against the failure of one system, but a local jammer can still blind them all. It can invest in **terrestrial back-ups** — holdover clocks and time delivered over fibre and cellular networks — which requires capital and is unavailable at sea or at remote sites. It can buy **insurance**, in a thin market whose capacity shrinks after large losses. And it can practise **compliance by design**, classifying every space-related component it buys or sells under the relevant export-control regimes, at the cost of staff and delay, knowing that the rules change with the balance of power.

A single case ties these together. A shipping subsidiary's only broadband at sea is one low-orbit constellation; the operator's government has been asked by an ally to switch the service off over a strait the firm's ships use. The firm's options this quarter turn on the distinction between dependence on a *service* and dependence on a *state*. A second operator under the same licensing state diversifies the first and not the second; a second operator under another state diversifies both, at the price of a second set of export controls and a second government's discretion.

11.11 A short review of the literature

The legal foundation is the Outer Space Treaty (United Nations 1967), whose Articles II, VI and VII make orbit unappropriable, place every private operator under a responsible state, and attach liability to the launching state. The governance problem it leaves open is that of a commons. Hardin (1968) states why open-access resources are overused and Ostrom (1990) the conditions under which communities escape the tragedy; Weeden and Chow (2012) apply Ostrom's design principles to orbit and find most of them absent. The physical mechanism is Kessler and Cour-Palais (1978), with Liou and

Johnson (2006) showing by simulation that the low-orbit debris population would grow even without new launches. The economics of the externality is set out by Adilov, Alexander, and Cunningham (2015), and Rao, Burgess, and Kaffine (2020) estimate the gains from orbital-use fees.

The strategic literature divides along the lines of this chapter's three readings. Dolman (2002) carries classical geopolitics into orbit; Bowen (2020) argues that spacepower matters for its terrestrial effects; Pekkanen (2019) describes the democratisation, commercialisation and militarisation of a domain in which most technology is dual-use and treaty-making has stalled. The empirical record of threats is the Secure World Foundation's annual assessment (Samson and Brett 2026), and that of the environment ESA's annual report (ESA Space Debris Office 2026); the public catalogue redistributed by CelesTrak (Kelso 2026) and McDowell's GCAT (McDowell 2026), which dates each fragment by the event that created it, allow the record to be rebuilt and its dating decisions tested.

For the firm, the bridge is Farrell and Newman (2019): a concentrated constellation under a state's jurisdiction meets both conditions for weaponised interdependence. The measure of concentration is Hirschman (1945)'s index, conceived to analyse how trade structures create political dependence, and its antitrust thresholds are those of the US merger guidelines (U.S. Department of Justice and Federal Trade Commission 2010, 2023).

11.12 What the evidence does not license

The public record of the orbital environment is unusually good. ESA's count of objects by class, the catalogue from which it can be approximated, and the concentration of who launches payloads from Sputnik to today together show the steady accumulation of debris, the steps that the 2007 test, the 2009 collision and the 2021 test cut into the record, the explosion of payloads since the large constellations began, and a concentration index that falls for decades and then climbs back — by owner state to 6,845 in 2025, by operator to 5,007, with two-thirds of the operational satellites in the catalogue belonging to one firm. They also show the decisions underneath: the dating of debris, the filtering of objects that do not orbit the Earth, the reading of owner codes, the definition of an active satellite.

Three conclusions do not follow. A rising count of objects does not tell a firm that its services are at greater risk, because risk depends on the size of objects, the shells its providers use and the manoeuvring capability of their satellites — none of which the count contains. The absence of a spike does not mean the absence of a threat: the counterspace effects firms have actually suffered — jamming, spoofing, cyberattacks on ground segments — leave no trace in a catalogue of objects. And concentration by owner code is not the firm's dependence. A firm may buy its connectivity from a small operator in an unconcentrated market and still depend on one state's navigation system for its time, or buy from the dominant operator through a reseller and not know it.

What the evidence does support is a conclusion about where the risk sits. Space was governed for half a century by a bargain between two states that could each destroy the other's satellites and chose, mostly, not to. That bargain has not been replaced; it has been overlaid by a market in which one firm carries much of the world's new capacity, by coalitions writing rules the treaty process cannot, and by an environment that the least careful user degrades for everyone. In each of those layers the risk falls less on the owner of the satellite than on the customer of the service, and the customer is usually the last to know. A multinational's exposure to space therefore begins not with the sky but with a list it rarely keeps: the satellite services it buys, the operators and states behind each, and the regions where its ships, aircraft and sites depend on signals that someone else can switch off.

Chapter 12

From Indicators to Decisions: What a Multinational Can Know About Geopolitical Risk

12.1 The question this book began with

This book opened with a question: how does a multinational firm know how much geopolitical risk it faces, and what should it do about it? Eleven chapters later the honest answer to the first half is that it never knows, in the sense of holding a number that settles the matter, and much of the book has been an account of why. The answer to the second half is more encouraging. A firm can organise what it does not know so that someone acts on it, at the right moment, for reasons that can be stated in advance and checked afterwards. That is what a risk function is for.

This concluding chapter draws the threads together. It takes two classic contributions — Stephen Kobrin's review of political risk, published in 1979, and Witold Henisz's measure of political constraint, published in 2000 — and uses them to frame what the preceding chapters established: that geopolitical risk is a contingency rather than an event; that it is measured through families of indicators that disagree; that every famous figure is a stack of decisions; that the system a firm operates in moves slowly and then suddenly; that the risk reaches the firm through identifiable channels, increasingly as the instrument or the collateral of state power; and that indicators decide nothing until someone attaches a threshold, a decision and an owner to them. It ends with the question with which any analyst should end: what would change your mind?

12.2 The event is not the risk

Frank Knight drew the distinction that every risk analyst still uses, even when they do not name it (Knight 1921). *Risk*, in his sense, is uncertainty that can be measured: the outcomes are known and their probabilities can be estimated from experience, so the uncertainty can be priced, insured or diversified. *Uncertainty* proper is the rest: situations unique enough, or new enough, that no frequency distribution exists. Most of what this book has called geopolitical risk is Knightian uncertainty in that sense. There is no actuarial table of invasions, no frequency distribution of secondary sanctions on a given country, no base rate for a constellation operator's government ordering it to switch off a region. The indices examined in the preceding chapters are attempts to make the second kind of uncertainty behave a little more like the first — to give it a scale, a history and a comparison — and they do not entirely succeed.

Kobrin's contribution, in a review written as the field of political risk assessment was taking shape, was to insist on a second distinction (Kobrin 1979). The literature he surveyed used "political risk" in two ways: some writers meant political *events* — instability, violence, changes of government — and others meant the *effects* of politics on business — expropriation, restrictions, changes in the rules. Kobrin argued that the first is not the risk. Political instability in a country is a property of that country; whether it affects a particular firm's operations depends on the firm's industry, its assets, its partners and its relationship with the government. A coup may leave a consumer-goods importer untouched and ruin a mining concession; a peaceful election may produce a government that rewrites the tax code for one sector. Risk lies in the *contingency* between the event and the firm.

Political risk (Kobrin) — not the political event itself, but the contingency that political events or conditions will affect a particular firm's operations — its revenues, costs, assets and freedom to act. The same event carries different risks for different firms; a measure of the event is not a measure of the risk.

This book has widened Kobrin's frame in one direction. His concern was largely with host-country risk: what the government of the country in which a firm invests might do to it. Geopolitical risk, as Chapter 1 defined it, includes the risk that originates *outside* the host country — in Washington, Beijing, Brussels or Moscow — and reaches the firm through its suppliers, customers, banks and networks. A sanction imposed by the firm's home state on the country where it has a subsidiary, an export control on a component it buys, a jammer operating in a strait its ships use: none of these is a decision of the host government, and all of them reach the firm through the contingencies Kobrin described.

From the archive · Iran, 1979

Kobrin's review appeared in the spring of 1979, in the first months of the revolution that ended the Shah's rule in Iran. Few events have taught multinationals more about the gap between country analysis and firm risk. Iran had been a favoured destination for Western investment and contracting through the 1970s, rich with oil revenue and governed by a regime regarded as a firm ally of the United States. Its collapse cancelled contracts, stranded assets and exposed staff to danger within months.

The year ended with a second lesson. After the seizure of the US embassy in Tehran in November 1979, President Carter used the International Emergency Economic Powers Act — the statute on which most of the American sanctions programmes discussed in Chapter 10 still rest — to freeze Iranian government assets under US jurisdiction. Many American firms had claims against Iran. They obtained redress only through the Algiers Accords of January 1981, which ended the hostage crisis and created the Iran–United States Claims Tribunal in The Hague to arbitrate them. The firms were paid, eventually, out of the frozen assets: their risk had been settled by a negotiation between states in which they had no seat.

The episode is commonly cited as a reason so many firms built or expanded formal political risk units in the years that followed. Kobrin's own study of how American firms organised political assessment appeared three years later (Kobrin 1982). The lesson both drew is the one this book has tried to establish: the analysis of a country is the beginning of the assessment of a firm's risk, not the end.

Every serious assessment of geopolitical risk is therefore a claim about Kobrin's contingency, and it starts from the firm's footprint — where its revenue, assets, suppliers and people are. A risk report that begins with a map of the world coloured by a country index has not yet begun to assess the firm's risk; one that begins with the firm's own disclosures, and asks which of its countries, suppliers and channels the index bears on, has. The finding of Chapter 7, that most of the variation in political risk measured from earnings calls is specific to the firm rather than common to its country or sector (Hassan et al. 2019), is the quantitative form of the same point.

12.3 Six families that disagree

Chapter 1 sorted the indicators of geopolitical risk into six families, and the sorting has carried the book. *Event-based* indicators count things that happened, coded from reports: conflicts, cyber incidents, sanctions episodes. *Text-based* indicators measure how much the press or firms talk about a risk. *Expert-based* indicators are scores assigned by analysts or by surveys, such as the Worldwide Governance Indicators (Kaufmann, Kraay, and Mastruzzi 2011). *Structural* indicators describe slow-moving features of the system: the distribution of material capabilities, the pattern of trade dependence. *Market-based* indicators are prices that embed risk, such as sovereign credit-default swap spreads. *Firm-level* indicators capture risk as firms themselves perceive and disclose it.

Family of indicators — a set of measures of geopolitical risk that share a source of information and therefore a characteristic strength and blind spot: event-based (concrete, but only what was reported and coded), text-based (timely, but measures attention rather than exposure), expert-based (covers latent concepts, but opaque and sticky), structural (explains why, rarely when), market-based (forward-looking, but mixes risk with risk appetite), firm-level (speaks to the decision, but self-reported).

The most cited text-based measure is the one Chapter 1 took apart. Dario Caldara and Matteo Iacoviello count the share of articles in leading newspapers that mention geopolitical tensions, wars and terrorism, and show that their index rises with the events that history books record and that high values are followed by lower investment and employment (Caldara and Iacoviello 2022). Its sibling, the economic policy uncertainty index of Scott Baker, Nicholas Bloom and Steven Davis, applies the same logic to economic policy (Baker, Bloom, and Davis 2016). Chapter 7 went down to the level of the firm with the measure of Tarek Hassan and his colleagues, built from the share of conversation in firms' earnings calls devoted to

political topics and risk (Hassan et al. 2019). Each is ingenious, carefully validated and limited by its source: newspapers in a few countries and one language; earnings calls of listed firms, speaking to investors.

The families disagree, and the disagreement is information. When the text-based indices are calm and the market-based ones signal danger, someone with money at stake knows or fears something the press has not yet written about. When an expert rating stays unchanged through a crisis that the event data record in detail, the rating is sticky or the raters see a resilience the events do not. When a structural indicator shows a power transition under way and every short-run indicator is quiet, the structural one is saying *why* something may happen and the others are saying, correctly, that it has not happened yet. An assessment that relies on one family inherits that family's blind spot. Triangulation across several families is not a matter of more being better; it is the only defence against a blind spot that cannot be seen from inside one family.

12.3.1 Five series, one quarter at a time

The claim can be tested with the series this book has used. Figure 12.1 sets five of them side by side, quarter by quarter, from 2002 to mid-2026: the geopolitical risk index built from newspapers (Caldara and Iacoviello 2022); the global index of economic policy uncertainty, also built from newspapers (Baker, Bloom, and Davis 2016); the average political risk that listed firms express in their earnings calls, available until the first quarter of 2022 (Hassan et al. 2019); the VIX, the volatility of the US stock market implied by option prices, the most watched market-based measure of fear (Whaley 2009); and the New York Fed's index of global supply-chain pressure, which combines shipping and airfreight costs with the delivery times and backlogs reported in manufacturing surveys (Benigno et al. 2022). Each is expressed in standard deviations from its own mean over the 81 quarters that all five cover, so that their peaks can be compared.

Each series has a different worst quarter. For the geopolitical risk index it is the first quarter of 2003, the invasion of Iraq, followed by the first quarter of 2022. For policy uncertainty it is the second quarter of 2025, the quarter of the American tariff announcements. For the political risk firms discuss with their investors it is the second quarter of 2020, the pandemic. For the VIX it is the last quarter of 2008, the financial crisis. For supply-chain pressure it is the last quarter of 2021, when ports were congested and container rates at their height. Five measures of risk, five crises at the top.

The disagreement is sharper than a list of peaks suggests. Ranked against the other 80 quarters of the common period, the second quarter of 2020 is the riskiest for policy uncertainty and for firms' political risk, the fourth for the VIX — and the least risky of all 81 for the geopolitical risk index, whose newspapers had stopped writing about war and terrorism to write about the virus. The last quarter of 2008 is the riskiest for the VIX and the third for firms, but the 71st for geopolitical risk and the 79th for supply-chain pressure, which collapsed with world trade. The first quarter of 2022, when Russia invaded Ukraine, is second for geopolitical risk and third for supply-chain pressure, but only 16th for the VIX and 34th for firms' political risk: equity markets and earnings calls, dominated by American firms, treated a war in Europe as a smaller event than Lehman or the pandemic.

Table 12.1: Correlations between five risk indicators, quarterly levels, 2002Q1–2022Q1 (81 quarters). Sources: as for Figure 12.1; computed by the author.

	Geopolitical risk	Policy uncertainty	Firms' political risk	VIX
Policy uncertainty (newspapers)	−0.19			
Political risk (earnings calls)	−0.19	0.72		
Market volatility (VIX)	0.03	0.27	0.54	
Supply-chain pressure (GSCPI)	−0.07	0.58	0.30	0.10

Table 12.1 puts the same point in numbers. Policy uncertainty and firms' political risk move together closely (a correlation of 0.72), and both move with the VIX: they are, in different ways, measures of the uncertainty that matters to investors in listed firms. The geopolitical risk index is correlated with none of the other four — its correlations range from −0.19 to 0.03 — and supply-chain pressure is correlated with policy uncertainty, through the pandemic and the tariffs, but barely with markets. Changes tell the same story as levels: from one quarter to the next, the geopolitical risk index moves independently

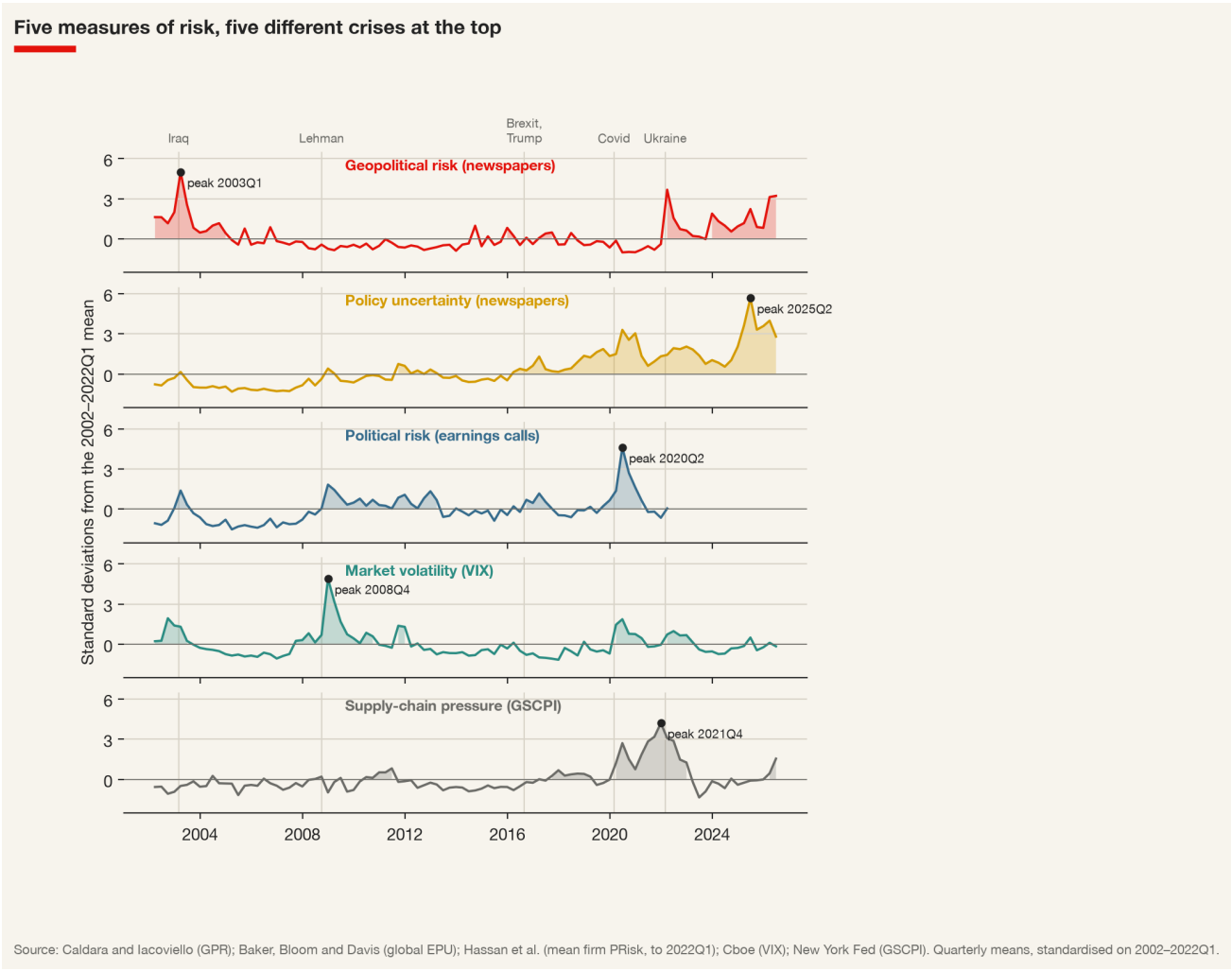


Figure 12.1: Five measures of risk, five different crises at the top. Quarterly means of each series, standardised on 2002–2022Q1; dots mark each series' highest quarter. Sources: Caldara and Iacoviello; Baker, Bloom and Davis; Hassan et al.; Cboe; Federal Reserve Bank of New York.

of policy uncertainty (a correlation of 0.00) and almost independently of everything else. A firm that watched only the VIX would have seen nothing unusual in the quarter Russia invaded Ukraine; one that watched only the geopolitical risk index would have seen nothing at all in the pandemic, which closed its plants and its customers' shops.

None of this means that four of the five measures are wrong. They measure different things — attention to war, attention to policy, the concerns of managers, the price of insurance against a fall in equities, the strain on logistics — and each was right about its own object. It means that "geopolitical risk is high" is never a complete statement until one says which family says so, and that an indicator chosen because it rose in the last crisis is likely to miss the next.

12.4 Every figure is a stack of decisions

The method of this book rested on a claim: that a chart in a famous work is a set of decisions, and that the decisions become visible only when the chart is taken apart and rebuilt from its sources. Each chapter applied the claim to one figure, and each found the decisions where the claim said they would be.

A share of world output, in Chapter 2, depends on whether output is valued at market exchange rates or at purchasing-power parity, and the date at which the G7 ceased to dominate the world economy moves by decades with the choice. A share of world capabilities, in Chapter 3, depends on which components enter the index and how they are weighted, and the composite index of national material capabilities rewards population and heavy industry because it was designed for the wars of the nineteenth and twentieth centuries (Singer, Bremer, and Stuckey 1972). Trade "within" and "between" blocs, in Chapter 4, depends on where the line between blocs is drawn, which is itself an inference from voting in the UN General Assembly (Gopinath et al. 2024). A map of certified management systems, in Chapter 6, counts certificates, not performance. A count of cyber incidents, in Chapter 8, counts what was attributed, between states coded as rivals (Valeriano and Maness 2015). A count of sanctions cases per year, in Chapter 10, depends on how a case with no end is ended and how a case of several types is counted. A count of objects in orbit, in Chapter 11, depends on whether debris is dated by the event that created it or by the launch of its parent. None of these decisions is wrong. Each is defensible, and each is a choice someone made, usually for reasons stated in a codebook that few readers open.

The places where the data disagree with the published account are where the decisions become visible. Series are revised; countries are recoded; a release adds cases the original paper did not have; a variable's meaning changes between the codebook and the distributed file. A reconstruction that matches the original perfectly is less informative than one that does not, if the difference is understood and explained. The analyst who notices and records the disagreement has learned something about the instrument that the analyst who merely reproduces the chart has not.

What each indicator cannot tell a firm. Read in sequence, the closing sections of the eleven preceding chapters form a list: attention is not exposure; a share is not a level; capabilities are not intentions; votes are not trade; a certificate is not performance; a composite is a choice of weights; an empty cell in an incident matrix is not safety; a correlation between embassies and exports is not a cause; a count of sanctions cases is not intensity; a catalogue of objects is not the environment. That list is the most useful thing a risk function can carry from the evidence reviewed in this book.

The same holds for any risk report a firm produces: it is an argument about the firm's risk, assembled from decisions of exactly this kind. Which countries enter the footprint, and on what measure of exposure — revenue, assets, employees, suppliers? Which indicator stands for which risk, and from which release? How are missing values treated, and how are countries named so that sources can be joined? An analyst who cannot list the data decisions behind a number is asking the board to trust the picture rather than the reasoning.

12.5 The system moves slowly, then suddenly

Chapters 2 to 6 described the system in which multinationals operate: an international economic order resting on a distribution of power among states that have no government above them, partly governed by public institutions and partly by private authority. The structural indicators of those chapters — shares of world output, shares of material capabilities, the alignment of states in UN votes, voting power inside the institutions, the diffusion of standards — share a characteristic. They move slowly. A power transition takes decades; the reorganisation of trade into blocs shows up in data over years; the authority of an institution erodes before anyone announces its collapse.

Slow-moving indicators are easy to ignore, because nothing in them demands a decision this quarter. Yet slow structural change is released in sudden events: the system bends for years and then breaks in a week. The collective-security order of the League of Nations, whose failure over Ethiopia opened Chapter 10, looked intact until it was tested; the integration of Russia into European energy and financial markets looked like interdependence until February 2022 revealed it as exposure. The structural indicator does not tell the firm *when*. It tells the firm *where* the break, when it comes, will be most costly — where interdependence has become dependence, where a bloc boundary runs through its supply chain, where an institution it relies on has lost the support of the states that made it.

Two findings from the middle of the book make the point concrete. Chapter 5 argued that an institution is a risk mitigator when the coalition that sustains it is stable and a risk source when it is not: a guarantee, a co-financing agreement or an investment treaty is a promise by a coalition of states, and should be read as such. Chapter 6 showed that the rules a firm meets in its markets are increasingly written by private bodies and by large markets whose regulation travels — the Brussels effect (Bradford 2020) — so that a firm's future requirements are visible in what its customers' home markets adopt today. Both are slow variables. Both decide, when they shift, what a firm may sell, to whom, and on what terms.

This is why a sound assessment combines families. The structural indicators locate the fault lines; the event-based and text-based ones signal movement along them; the market-based ones show whether anyone with money at stake has noticed. None of them alone is a warning system. Together, with thresholds, they can be.

12.6 Structure as well as events: Henisz on constraint

Most of the indicators reviewed in this book measure what is happening: events, attention, prices, perceptions. Henisz's measure of political constraint measures something different: how easily what is happening could change (Henisz 2000).

The idea comes from the theory of veto players, which George Tsebelis later developed into a general account of policy stability (Tsebelis 2002). A government that wants to change a policy — a tax, a regulation, a contract with a foreign investor — must secure the agreement of every actor with the power to block it: the executive, each chamber of the legislature, the courts, sub-national governments. The more such actors there are, and the more their preferences differ, the harder it is to change anything. Henisz built from this an index, POLCON, that combines the number of independent branches of government with veto power and the alignment of their partisan composition into a measure, between zero and one, of the feasibility of policy change. He found, across a large panel of countries, that greater political constraint was associated with higher economic growth (Henisz 2000), and in later work with more investment in infrastructure — the kind of long-lived, immobile asset most exposed to a government changing its mind (Henisz 2002).

Political constraint (POLCON) — a measure of the feasibility of policy change, built from the number of independent veto players in a political system and the alignment of their preferences. Low constraint means a government can change policy quickly, in either direction; high constraint means change is slow, which protects existing commitments and delays new ones.

The measure matters for the argument of this book in three ways. First, it is the structural complement to every event- and text-based indicator. An index of attention tells a firm that tension is rising; a measure of constraint tells it how fast that tension could become a change in the rules. Second, constraint cuts both ways, and a firm must ask which way it cuts *for it*. Low constraint means a government can expropriate quickly, but also that it can grant a licence, settle a dispute or lift a restriction quickly; high constraint protects an existing investment and slows the policy change a new entrant needs. Third, constraint connects the domestic politics of states to the weaponised interdependence of Chapter 8. The second condition for a network hub to become a chokepoint is that a state can direct the firms that run it (Farrell and Newman 2019). Whether a government can in practice order a bank, a cloud provider or a satellite operator to cut off a customer — quickly, without a court or a legislature stopping it — is a question about political constraint. The same measure that tells an investor how exposed it is to its host government tells a customer how exposed it is to its supplier's.

The structural measure behaves very differently from the event and expert measures, and a firm reading them together must know why. Figure 12.2 follows three countries at war — Russia, Ukraine and Israel — on three indicators from three families: the share of articles in the geopolitical risk index's newspapers that mention the country together with geopolitical risk, the World Bank's expert-based score of political stability and absence of violence (Kaufmann, Kraay, and Mastruzzi 2011), and Henisz's political constraint. The press reacted at once and by a large factor. Russia's share of articles rose from 0.82 percent in 2021 to 3.69 percent in 2022 and Ukraine's from 0.35 to 3.61; Israel's quadrupled in 2023. The expert score moved less, and not always in the expected direction. Russia's political stability score fell from 57.2 to 53.2 in the year of

the invasion and to 50.5 by 2024 — about where it had been in 2013. Ukraine's fell sharply in 2014, with the annexation of Crimea and the war in the Donbas, but in the years of full-scale war it *rose*, from 47.9 in 2022 to 58.5 in 2024, a reminder that the sources behind the score judge the state and its government, not only the fighting on its territory. Political constraint did not register the wars at all: Russia's rose slightly in 2022, from 0.23 to 0.30; Israel's has stayed between 0.54 and 0.60 for two decades.

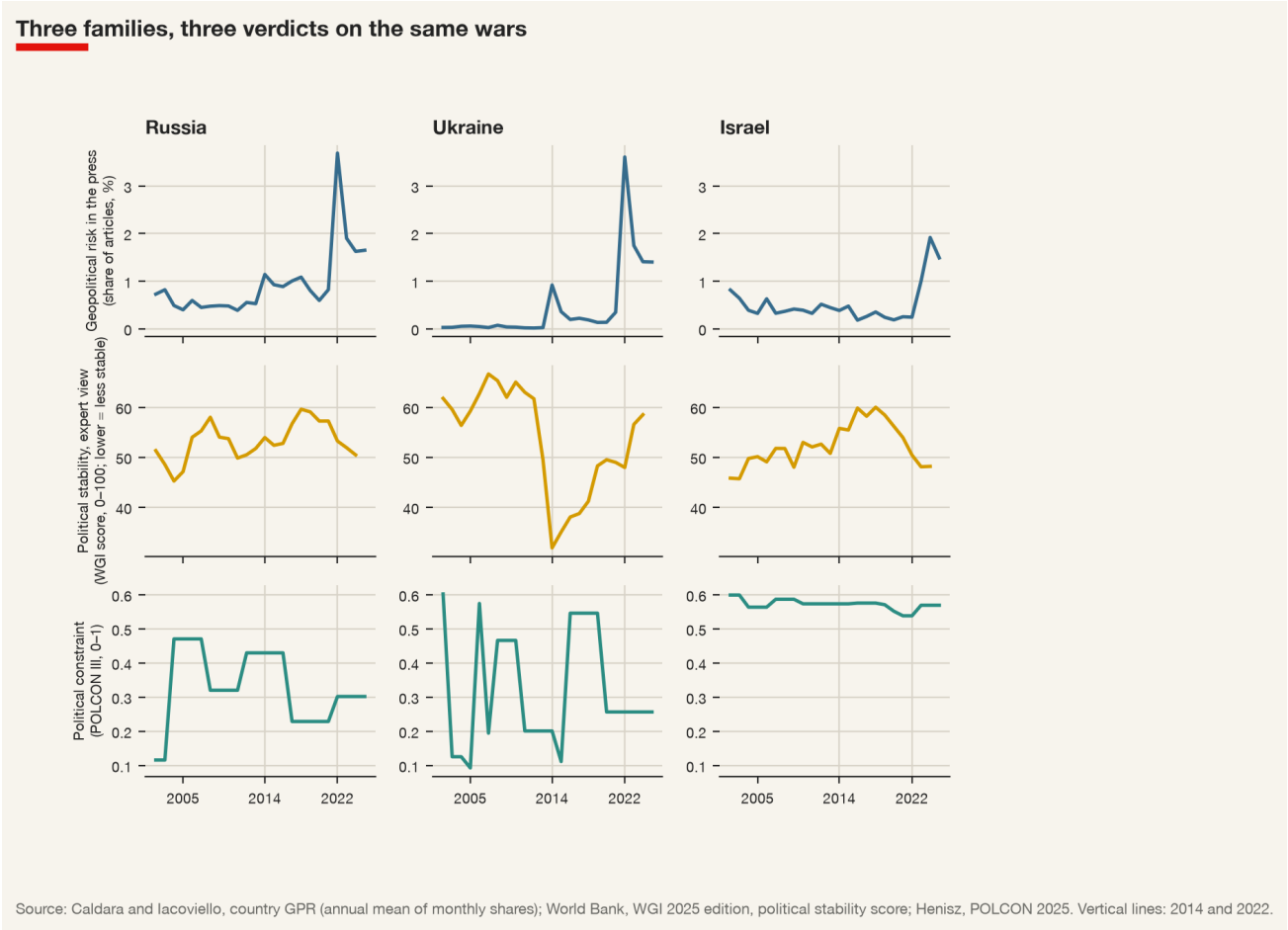


Figure 12.2: Three families, three verdicts on the same wars. Country geopolitical risk (share of articles, annual mean), WGI political stability (score 0–100) and POLCON III, 2002–2025. Sources: Caldara and Iacoviello; World Bank, WGI 2025 edition; Henisz, POLCON 2025.

None of the three is wrong. The press measures attention, and attention jumps. The expert score measures a latent property — the likelihood of political instability or violence — through surveys and ratings that update slowly and that judge a state at war on the cohesion of its government as well as on the fighting. The structural index measures the institutional feasibility of policy change, which a war need not alter. For a firm the useful reading combines them: the press says that something has happened, the expert score says how far informed observers have revised their view of the state, and the constraint index says how quickly the state could now change the rules the firm operates under.

The bargain between a firm and its host government also changes over time, in a direction that Raymond Vernon identified half a century ago and Chapter 9 examined (Vernon 1971). Before a multinational invests, the host government needs its capital, technology and market access, and the firm can extract favourable terms. Once the plant is built or the mine is sunk, the balance shifts: the firm's assets are hostage, its technology has been transferred or copied, and the government can revise the terms with less to lose. The wave of expropriations of the 1960s and 1970s that Kobrin documented (Kobrin 1980) gave the obsolescing bargain its evidence, and the forced, discounted exits from Russia after 2022 gave it a geopolitical form: when the home state sanctions the host, the host loses its reason to keep the investor content. Constraint slows the obsolescence; it does not stop it. The exposure of assets and people is not fixed at the time of investment. It grows as the firm's bargaining power declines, and the analysis that justified the investment must be redone as the bargain ages.

12.7 Instruments: where the risk meets the profit and loss account

Chapters 8 to 11 took four instruments through which geopolitical risk now reaches the firm most directly: cyber operations, economic diplomacy, sanctions and space. Read together, they share a pattern that the older political risk literature, focused on what host governments do to foreign investors, could not see. In each, the firm is not only the object of state action but part of its machinery. Firms run the networks that cyber operations traverse and the hubs that states weaponise. They are, in the two-level games of economic diplomacy (Putnam 1988), at once their home state's clients, its instruments and, abroad, its hostages. They execute sanctions, and set their effective scope through their own compliance departments. They operate the constellations whose service a government can switch off. Geopolitical risk, in this sense, is increasingly risk that the firm carries *on behalf of* states — including its own.

Chapter 1 proposed five channels to trace how such a risk becomes a cost, and the instruments run through all of them. **Market access** is the firm's ability to sell: closed by sanctions, export controls, licensing decisions, boycotts and the loss of a diplomatic relationship, opened by commercial diplomacy and trade agreements. **Inputs** are what the firm buys: components and raw materials whose supply is concentrated in one country, services — connectivity, timing, cloud computing — supplied by one operator under one state's jurisdiction. **Finance and payments** are the plumbing: correspondent banks, payment networks, insurance, the currency of settlement — the channel through which financial sanctions and overcompliance operate. **Assets and people** are what the firm has in place: plants, mines, ships, inventories and employees exposed to expropriation, temporary management, conflict or interference. **Data and operations** are the systems the firm runs on: networks open to cyberattack, software and services that can be withdrawn, imagery and signals that can be denied.

Risk channel — the route by which a geopolitical development reaches a firm's revenues, costs, assets or ability to operate: market access, inputs, finance and payments, assets and people, data and operations. Naming the channel converts a statement about the world into a statement about the firm, and points to the decision that could reduce the exposure.

The channels turn Kobrin's contingency into something a firm can analyse. A statement that "tension between the United States and China is rising" is about the world. A statement that "a large share of our revenue is in a market exposed to bloc realignment, our key component comes from a single supplier in a country subject to export controls, and our payments in that market clear through banks that a financial sanction would cut off" is about the firm — and each clause points to a decision.

The instruments also share an evidentiary problem. The most consequential uses of state power against firms are the least likely to appear in the data. A sanction that works at the threat stage is never imposed (Drezner 2003); a cyber operation never attributed is never coded; a jammed navigation signal leaves no trace in a catalogue of objects in orbit; a firm quietly told by its home government not to bid for a contract appears in no dataset at all. The datasets on which this book has relied are records of the coercion that failed to deter or that was made public. A firm that reads them as the full distribution of what states do will underestimate the risk it carries.

12.7.1 When risks compound

Indicators are built one at a time, and risk reports display them side by side, which invites the reader to add them up. Risks do not add. They interact.

They compound when one risk removes the remedy for another. A firm that depends on a single supplier in a country that is drifting out of its home state's bloc faces an input risk and an alignment risk; if the same country is a plausible target of sanctions, the sanction would cut the input at the moment the alignment drift made a replacement supplier in a friendly country most expensive. A firm whose ships depend on one satellite constellation faces a dependence risk; if the operator's state is a sender of sanctions against a country whose waters the ships cross, the dependence becomes an exposure to a foreign-policy decision. They offset when one exposure hedges another: a firm with plants on both sides of a bloc boundary loses one market in a rupture but may keep the other, and a firm whose suppliers are spread across states with different alignments can shift sourcing as the boundary moves.

Interactions are where the analyst's knowledge of the firm, rather than of the world, does the work. No index can be built for them in advance. They are also where the most expensive surprises lie, because a risk function organised by risk type — cyber, sanctions, country risk — has no one whose job is to see them.

12.8 The composite index, and who chooses the weights

Most practical risk scores, commercial or internal, are composites: select indicators, normalise them to a common scale, weight them, and aggregate. The handbook examined in Chapter 7 sets out each step and the choices it involves (OECD and European Commission Joint Research Centre 2008).

Composite index — a single measure built by combining several indicators after normalisation and weighting. Its ranking of units depends on which indicators are chosen, how they are normalised and how they are weighted — and the weights are not technical parameters but statements of what matters, and how much.

The composite is the least neutral number in any risk report. Normalisation decides whether a country's score is relative to the best and worst in the sample or to a fixed benchmark, and therefore whether adding a country changes everyone else's score. Weighting decides how much a conflict risk counts against a sanctions risk or a dependence risk, and no data can settle that, because the answer depends on the firm's exposure through each channel and on how much loss of each kind it can bear. Equal weights are a choice too: the choice to treat every indicator as equally important, which is rarely what anyone believes.

The remedy is not to abandon composites but to expose their weights. A ranking that does not change when the weights move within a plausible range is robust, and a board can rely on it. A ranking that reverses when the weight on one indicator moves from a third to a half is telling the board that the decision depends on a judgement about priorities, not on the data — and that judgement belongs to the board, not to the analyst. Showing the sensitivity is not a confession of weakness. It is the analyst handing the decision to the person who must own it.

Figure 12.3 shows how much the weights matter with three indicators already discussed: the press-based country geopolitical risk index, the expert-based political stability score and the structural political constraint index, for the 39 countries covered by all three in 2024. Each is rescaled from 0 to 1 across these countries, with 1 for the most risk — most press attention, least stability, least constraint — and combined with four sets of weights: equal, or 60 percent on one indicator and 20 on each of the others. Seven countries are among the ten riskiest whatever the weights: China, Egypt, the Philippines, Russia, Ukraine, the United States and Venezuela. Below them the ranking is fluid. The median country moves seven places between the most and least favourable weighting, and sixteen of the 39 move ten places or more.

Table 12.2: Rank of selected countries on the composite under four weightings (1 = riskiest of 39). Sources: Caldara and Iacoviello; World Bank, WGI 2025 edition; Henisz, POLCON 2025; computed by the author.

Country	Equal weights	Press-heavy	Expert-heavy	Constraint-heavy
Venezuela	1	8	1	1
United States	3	1	6	10
Russia	4	2	3	7
Israel	7	3	4	16
United Kingdom	11	7	20	13
Mexico	24	24	14	33
Germany	27	16	25	31
Brazil	31	31	21	39

The table shows where the fluidity comes from, and it is not noise. The United States is the riskiest of the 39 countries when the press is given most weight, because the newspapers behind the index are American and British and write about American wars, alliances and threats more than about anyone else's; it falls to tenth when constraint is given most weight, because the American system of checks and balances places it among the more constrained countries in the sample. Israel moves from third to sixteenth for the same reason. Brazil and Mexico, rarely mentioned in Anglo-American newspapers' war coverage, rise sharply when the expert score dominates. Each weighting embodies a view of what geopolitical risk is — the attention of a particular press, the judgement of a particular set of experts, the institutional ease of changing the rules — and the ranking a board sees is that view, not the world.

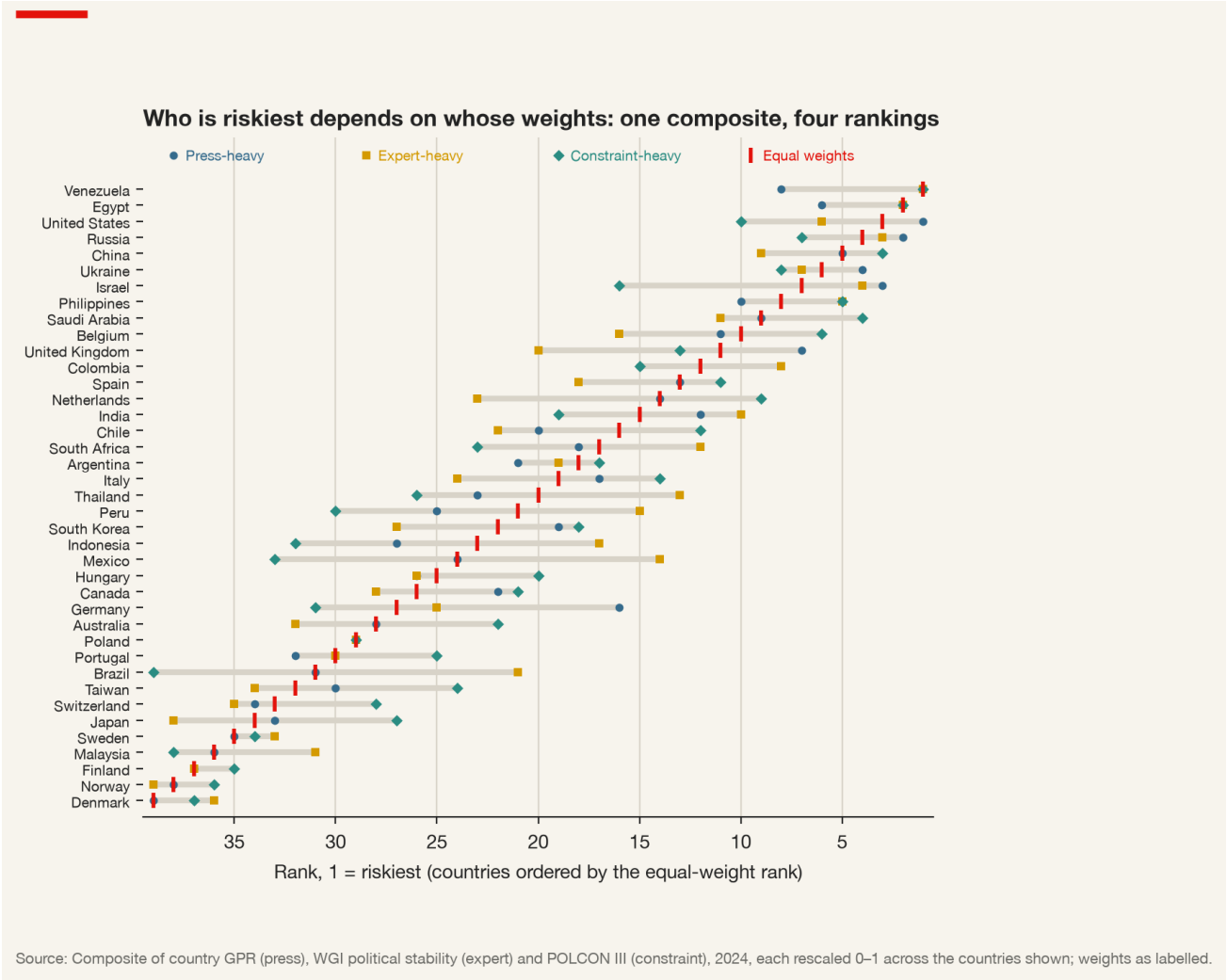


Figure 12.3: Who is riskiest depends on whose weights. Rank of 39 countries on a composite of press attention, political stability and political constraint, 2024, under four weightings; grey bars show the range. Sources: Caldara and Iacoviello; World Bank, WGI; Henisz, POLCON; computed by the author.

12.9 Indicators do not decide

An indicator without a threshold and an owner is decoration. A risk report can display every indicator in this book, beautifully, and change nothing in the firm, because nothing in it tells anyone what to do or when.

Watchlist — a short list of indicators, each paired with a threshold that triggers a named decision and with the person who owns that decision. It converts monitoring into commitment: the decision is made in advance, conditionally, so that when the threshold is crossed the question is no longer whether to act but whether the threshold was right.

A watchlist has four parts, and each does work. The indicator must be one the firm can actually observe, at the frequency it needs, from a source whose version it has recorded. The threshold must be a number, set before the event, so that it cannot be rationalised away when the moment comes; it may be revised, but revising it should be a decision with a record. The decision must be specific — contract a second operator, cap new commitments in a market, pre-position inventory, start the approval process for a sale — not "monitor the situation" or "review our exposure". And the owner must be a person with the authority to take the decision, because a threshold crossed without an owner is crossed in silence.

From the archive · Shell's scenarios, 1972–1973

The most famous case for deciding in advance comes from the oil industry. In the early 1970s the planning group of Royal Dutch Shell, led by Pierre Wack, concluded that the forecasting methods on which the company based its investment decisions assumed a continuation of the conditions of the previous two decades — cheap oil, controlled by the companies — that the politics of the producing countries no longer guaranteed. Instead of a single forecast, the group produced scenarios, among them one in which producing governments took control of supply and prices rose sharply.

Wack's later account emphasised that the scenarios at first changed little: managers read them as forecasts to accept or reject, not as alternative worlds whose signposts they should watch. The group's work became effective only when it was recast to change how managers thought about their business, and to make them identify in advance what they would do in each world. When the producers' embargo and the price rises of 1973 arrived, Shell's managers had, in Wack's telling, already thought through what such a world would require of them (Wack 1985).

The lesson is not that scenarios predict. It is that a risk analysis changes behaviour only when it is tied, in advance, to the signals that would reveal which world is arriving and to the decisions each world calls for. A scenario without signposts and a risk report without thresholds fail in the same way.

The most useful risk report is therefore not the one with the most indicators but the one that would make someone act on a given Monday.

12.9.1 Why firms wait, and when they should not

There is a rational reason firms delay decisions in the face of geopolitical risk, and a watchlist must be designed against it. Nicholas Bloom showed that a rise in uncertainty leads firms to pause investment and hiring: when the future is less predictable, the option to wait and see becomes more valuable, and the economy slows as many firms exercise it at once (Bloom 2009). Waiting is often right. But Chapters 10 and 11 described cases where waiting destroyed options rather than preserving them — the firms that delayed their exit from Russia until the target's countermeasures had closed the door on a sale on their own terms; the firm with one connectivity provider that waits until the service is switched off to contract a second. The value of waiting depends on whether the options remain open while one waits. A good threshold is set at the point where they begin to close.

12.10 Where the risk function sits

A risk report is an instrument; a risk function is an organisation that uses one. The argument of this book implies three things about how it should be organised.

The first is that geopolitical risk assessment cannot live only in a central unit. Kobrin's contingency is discovered in the firm's operations — in procurement, which knows where the inputs come from; in treasury, which knows which banks clear the payments; in the information-security function, which knows which networks and providers the firm runs on; in the country managers, who know what the host government is likely to do. A central function can build the indicators and the

method; the exposure must be mapped with the people who run the business, from the firm's own records, before reaching for any index.

The second is that the function must report to people who can decide. Ian Bremmer and Preston Keat argued that the most damaging political risks are in the "fat tail" — rare, high-impact events that standard models underweight — and that firms need political knowledge integrated into strategic decisions, not appended to them (Bremmer and Keat 2009). A brief to the board that names a small number of risks, the channels through which they would reach the firm and what to do about each is that integration in its simplest form: a document addressed to the people who own the decisions, written so that they can act on it.

The third is that the function must keep a record. It must be able to say, a year later, which version of which dataset produced the number on which the firm acted, which decisions were taken in building it, and what the threshold was at the time. Without that record a firm cannot learn whether the number, the threshold or the decision was wrong, because it cannot reconstruct what it knew. A risk function that cannot reconstruct its own past judgements cannot improve them.

12.11 What would change your mind?

Of all the questions that can be asked of a risk assessment, the most revealing is the last: what would change your recommendation? It is the question that separates an assessment from an opinion.

An assessment that nothing could change is not an assessment. An analyst who recommends reducing exposure to a market should be able to say which indicator, moving by how much, would lead to the opposite recommendation — and should be able to say it before the indicator moves. This is the same discipline as the threshold, applied to the analyst rather than to the firm. It protects against the most common failure of risk analysis, which is not ignorance but commitment: the analyst who has argued for a view reads every new piece of evidence as confirmation, and the firm that has invested on a view finds reasons to stay.

The evidence on expert forecasting supports the discipline. Philip Tetlock's long study of political forecasts found that experts' predictions were, on average, barely better than simple baselines, and that the forecasters who did better were those who held their views provisionally, drew on many sources and revised when the evidence moved, rather than those who explained events through one big idea (P. E. Tetlock 2005). Geopolitical risk is a field that rewards the single big idea — the coming war, the end of globalisation, the inevitable transition — with attention. A risk function should reward the opposite habit.

Every chapter of this book ended by asking what a famous figure did not license a multinational to conclude. The same question, asked of a firm's own analysis, tells a board how much weight to put on its recommendation, and tells the reader where to look for the next surprise.

12.12 A short review of the literature

The study of political risk in international business took shape in the 1970s, in the shadow of the wave of expropriations that Kobrin (1980) documented. Kobrin (1979) reviewed the early literature and argued that risk lies in the contingency between political events and a firm's operations, not in the events themselves; Fitzpatrick (1983) surveyed the same field and found its definitions of risk equally unsettled; and Kobrin (1982) studied how firms actually organised the assessment. The distinction between measurable risk and unmeasurable uncertainty on which the field rests is Knight (1921)'s, and Vernon (1971) supplied the dynamic of the obsolescing bargain.

The institutional strand begins from veto players (Tsebelis 2002). Henisz (2000) turned the idea into the POLCON index of political constraint and related it to growth, and Henisz (2002) to investment in infrastructure, the kind of asset most exposed to policy reversal. The measurement strand is more recent and largely text-based: Baker, Bloom, and Davis (2016) on economic policy uncertainty, Caldara and Iacoviello (2022) on geopolitical risk, and Hassan et al. (2019) on political risk as firms themselves discuss it, alongside expert-based aggregates such as Kaufmann, Kraay, and Mastruzzi (2011), the market-based VIX (Whaley 2009) and composite measures of supply-chain strain such as Benigno et al. (2022) — series which, set side by side, peak in different crises and are barely correlated with one another. Bloom (2009) established

the mechanism by which uncertainty, once measured, depresses investment, and OECD and European Commission Joint Research Centre (2008) codified the construction of the composite indicators on which most practical risk scores rely.

The geopolitical turn of the last decade reframed the field around networks and dependence: Farrell and Newman (2019) explain how states weaponise the hubs of global networks, which turns a firm's suppliers and service providers into channels of state power, and Drezner (2003) shows why the coercion that works is the coercion least likely to be observed. On deciding under such uncertainty, Wack (1985) remains the classic account of scenarios tied to decisions, Bremmer and Keat (2009) the practitioner's case for integrating political knowledge into strategy, and P. E. Tetlock (2005) the evidence on which habits of judgement forecast better.

12.13 Conclusion: decisions inside uncertainty

Geopolitical risk will never be known with the confidence of a price or the regularity of an actuarial table. The world in which multinationals now operate makes that truer, not less true. The order that allowed firms to treat geopolitics as background — a single dominant power, institutions that refereed disputes, networks that nobody used as weapons — has given way to one in which power is contested, institutions compete, standards and supply chains divide along political lines, and the infrastructure firms depend on, from payment systems to satellites, is an instrument of state policy. In that world the question a firm faces is not whether it is exposed but how, through which channels, and to whose decisions.

The evidence reviewed in this book does not answer that question for any particular firm, and it cannot. What it offers is a discipline for answering it. Begin with the firm, not the country. Use indicators from several families, and treat their disagreement as information. Know the decisions inside every number, and record where the data depart from the published account. Name the channel through which each risk would reach the firm. Watch the slow structural variables as well as the fast ones, because the slow ones say where the break will be costly. Attach to each indicator a threshold, a decision and an owner. And say in advance what would change your mind.

The analyst's job is not to remove uncertainty but to make the decisions inside it visible — the decisions in the data, the decisions in the method, and the decisions the firm will take when a threshold is crossed — to the person who must act. An assessment that does that has done what an assessment can. One that does not, however rich its indicators, is only a picture of the world.

References

Abbott, Kenneth W., and Duncan Snidal. 2009. "The Governance Triangle: Regulatory Standards Institutions and the Shadow of the State." In *The Politics of Global Regulation*, edited by Walter Mattli and Ngaire Woods, 44–88. Princeton: Princeton University Press. https://doi.org/10.1515/9781400830732.44.

Adilov, Nodir, Peter J. Alexander, and Brendan M. Cunningham. 2015. "An Economic Analysis of Earth Orbit Pollution." *Environmental and Resource Economics* 60 (1): 81–98. https://doi.org/10.1007/s10640-013-9758-4.

AidData. 2023. "AidData's Global Chinese Development Finance Dataset, Version 3.0." AidData at William & Mary. https://www.aiddata.org/data/aiddatas-global-chinese-development-finance-dataset-version-3-0.

Aiyar, Shekhar, Jiaqian Chen, Christian Ebeke, Roberto Garcia-Saltos, Tryggvi Gudmundsson, Anna Ilyina, Alvar Kangur, et al. 2023. "Geoeconomic Fragmentation and the Future of Multilateralism." *IMF Staff Discussion Notes*, no. SDN/2023/001. https://doi.org/10.5089/9798400229046.006.

Alfaro, Laura, and Davin Chor. 2023. "Global Supply Chains: The Looming 'Great Reallocation'." NBER Working Paper 31661. National Bureau of Economic Research. https://doi.org/10.3386/w31661.

Allee, Todd, and Clint Peinhardt. 2011. "Contingent Credibility: The Impact of Investment Treaty Violations on Foreign Direct Investment." *International Organization* 65 (3): 401–32. https://doi.org/10.1017/S0020818311000099.

Allison, Graham. 2017. *Destined for War: Can America and China Escape Thucydides's Trap?* Boston: Houghton Mifflin Harcourt.

Amiti, Mary, Stephen J. Redding, and David E. Weinstein. 2019. "The Impact of the 2018 Tariffs on Prices and Welfare." *Journal of Economic Perspectives* 33 (4): 187–210. https://doi.org/10.1257/jep.33.4.187.

Anderson, James E., and Eric van Wincoop. 2003. "Gravity with Gravitas: A Solution to the Border Puzzle." *American Economic Review* 93 (1): 170–92. https://doi.org/10.1257/000282803321455214.

Angell, Norman. 1910. *The Great Illusion: A Study of the Relation of Military Power in Nations to Their Economic and Social Advantage*. London: Heinemann.

Asian Infrastructure Investment Bank. 2026. "Members and Prospective Members of the Bank: Subscriptions and Voting Power." Asian Infrastructure Investment Bank. https://www.aiib.org/en/about-aiib/governance/members-of-bank/index.html.

Autor, David H., David Dorn, and Gordon H. Hanson. 2013. "The China Syndrome: Local Labor Market Effects of Import Competition in the United States." *American Economic Review* 103 (6): 2121–68. https://doi.org/10.1257/aer.103.6.2121.

Bailey, Michael A., Anton Strezhnev, and Erik Voeten. 2017. "Estimating Dynamic State Preferences from United Nations Voting Data." *Journal of Conflict Resolution* 61 (2): 430–56. https://doi.org/10.1177/0022002715595700.

Baker, Scott R., Nicholas Bloom, and Steven J. Davis. 2016. "Measuring Economic Policy Uncertainty." *Quarterly Journal of Economics* 131 (4): 1593–1636. https://doi.org/10.1093/qje/qjw024.

Baldwin, David A. 1985. *Economic Statecraft*. Princeton, NJ: Princeton University Press.

Baldwin, Richard. 2016. *The Great Convergence: Information Technology and the New Globalization*. Cambridge, MA: Belknap Press of Harvard University Press. https://doi.org/10.4159/9780674972667.

Baldwin, Richard, and Javier Lopez-Gonzalez. 2015. "Supply-Chain Trade: A Portrait of Global Patterns and Several Testable Hypotheses." *The World Economy* 38 (11): 1682–1721. https://doi.org/10.1111/twec.12189.

Barbieri, Katherine. 1996. "Economic Interdependence: A Path to Peace or a Source of Interstate Conflict?" *Journal of Peace Research* 33 (1): 29–49. https://doi.org/10.1177/0022343396033001003.

———. 2002. *The Liberal Illusion: Does Trade Promote Peace?* Ann Arbor: University of Michigan Press.

Beckley, Michael. 2018. "The Power of Nations: Measuring What Matters." *International Security* 43 (2): 7–44. https://doi.org/10.1162/isec_a_00328.

Benigno, Gianluca, Julian di Giovanni, Jan J. Groen, and Adam I. Noble. 2022. "The GSCPI: A New Barometer of Global Supply Chain Pressures." Staff Reports 1017. Federal Reserve Bank of New York. https://doi.org/10.2139/ssrn.4114973.

Berelson, Bernard. 1952. *Content Analysis in Communication Research*. Glencoe, IL: Free Press.

Bernanke, Ben S. 1983. "Irreversibility, Uncertainty, and Cyclical Investment." *Quarterly Journal of Economics* 98 (1): 85–106. https://doi.org/10.2307/1885568.

Biersteker, Thomas J., Sue E. Eckert, and Marcos Tourinho, eds. 2016. *Targeted Sanctions: The Impacts and Effectiveness of United Nations Action*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781316460290.

Bjola, Corneliu, and Ilan Manor. 2018. "Revisiting Putnam's Two-Level Game Theory in the Digital Age: Domestic Digital Diplomacy and the Iran Nuclear Deal." *Cambridge Review of International Affairs* 31 (1): 3–32. https://doi.org/10.1080/09557571.2018.1476836.

Bjørn, Anders, H. Damon Matthews, Maida Hadziosmanovic, Nicolas Desmoitier, Amr Addas, and Shannon M. Lloyd. 2023. "Increased Transparency Is Needed for Corporate Science-Based Targets to Be Effective." *Nature Climate Change* 13 (8): 756–59. https://doi.org/10.1038/s41558-023-01727-z.

Blackwill, Robert D., and Jennifer M. Harris. 2016. *War by Other Means: Geoeconomics and Statecraft*. Cambridge, MA: Belknap Press of Harvard University Press. https://doi.org/10.4159/9780674545960.

Bloom, Nicholas. 2009. "The Impact of Uncertainty Shocks." *Econometrica* 77 (3): 623–85. https://doi.org/10.3982/ECTA6248.

Bowen, Bleddyn E. 2020. *War in Space: Strategy, Spacepower, Geopolitics*. Edinburgh: Edinburgh University Press. https://doi.org/10.3366/edinburgh/9781474450485.001.0001.

Bradford, Anu. 2020. *The Brussels Effect: How the European Union Rules the World*. New York: Oxford University Press. https://doi.org/10.1093/oso/9780190088583.001.0001.

———. 2023. *Digital Empires: The Global Battle to Regulate Technology*. New York: Oxford University Press. https://doi.org/10.1093/oso/9780197649268.001.0001.

Bremmer, Ian, and Preston Keat. 2009. *The Fat Tail: The Power of Political Knowledge for Strategic Investing*. New York: Oxford University Press.

BRICS Governments. 2014. "Agreement on the New Development Bank." Signed at Fortaleza, 15 July 2014. https://www.ndb.int/wp-content/uploads/2022/11/Agreement-on-the-New-Development-Bank.pdf.

Brooks, Stephen G., and William C. Wohlforth. 2016. "The Rise and Fall of the Great Powers in the Twenty-First Century: China's Rise and the Fate of America's Global Position." *International Security* 40 (3): 7–53. https://doi.org/10.1162/ISEC_a_00225.

Brunsson, Nils, Andreas Rasche, and David Seidl. 2012. "The Dynamics of Standardization: Three Perspectives on Standards in Organization Studies." *Organization Studies* 33 (5–6): 613–32. https://doi.org/10.1177/0170840612450120.

Buchanan, Ben. 2017. *The Cybersecurity Dilemma: Hacking, Trust and Fear Between Nations*. New York: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190665012.001.0001.

Büthe, Tim, and Walter Mattli. 2011. *The New Global Rulers: The Privatization of Regulation in the World Economy*. Princeton: Princeton University Press. https://doi.org/10.1515/9781400838790.

Büthe, Tim, and Helen V. Milner. 2008. "The Politics of Foreign Direct Investment into Developing Countries: Increasing FDI Through International Trade Agreements?" *American Journal of Political Science* 52 (4): 741–62. https://doi.org/10.1111/j.1540-5907.2008.00340.x.

Caldara, Dario, and Matteo Iacoviello. 2022. "Measuring Geopolitical Risk." *American Economic Review* 112 (4): 1194–1225. https://doi.org/10.1257/aer.20191823.

Cashore, Benjamin. 2002. "Legitimacy and the Privatization of Environmental Governance: How Non-State Market-Driven (NSMD) Governance Systems Gain Rule-Making Authority." *Governance* 15 (4): 503–29. https://doi.org/10.1111/1468-0491.00199.

Central Committee of the Communist Party of China and State Council. 2021. "National Standardization Development Outline." Issued 10 October 2021; Chinese text on the Central Government Portal. https://www.gov.cn/zhengce/2021-10/10/content_5641727.htm.

Christmann, Petra, and Glen Taylor. 2001. "Globalization and the Environment: Determinants of Firm Self-Regulation in China." *Journal of International Business Studies* 32 (3): 439–58. https://doi.org/10.1057/palgrave.jibs.8490976.

———. 2006. "Firm Self-Regulation Through International Certifiable Standards: Determinants of Symbolic Versus Substantive Implementation." *Journal of International Business Studies* 37 (6): 863–78. https://doi.org/10.1057/palgrave.jibs.8400231.

Chupilkin, Maxim, Beata Javorcik, and Alexander Plekhanov. 2026. "The Eurasian Roundabout: Trade Flows into Russia Through the Caucasus and Central Asia." *European Economic Review* 187: 105340. https://doi.org/10.1016/j.euroecorev.2026.105340.

Cohen, Jacob. 1960. "A Coefficient of Agreement for Nominal Scales." *Educational and Psychological Measurement* 20 (1): 37–46. https://doi.org/10.1177/001316446002000104.

Copeland, Dale C. 1996. "Economic Interdependence and War: A Theory of Trade Expectations." *International Security* 20 (4): 5–41. https://doi.org/10.1162/isec.20.4.5.

———. 2015. *Economic Interdependence and War*. Princeton: Princeton University Press. https://doi.org/10.23943/princ

eton/9780691161587.001.0001.
Crozet, Matthieu, and Julian Hinz. 2020. "Friendly Fire: The Trade Impact of the Russia Sanctions and Counter-Sanctions." *Economic Policy* 35 (101): 97–146. https://doi.org/10.1093/epolic/eiaa006.
Cutler, A. Claire, Virginia Haufler, and Tony Porter, eds. 1999. *Private Authority and International Affairs*. Albany: State University of New York Press.
Cybersecurity and Infrastructure Security Agency. 2023. "The Attack on Colonial Pipeline: What We've Learned & What We've Done over the Past Two Years." https://www.cisa.gov/news-events/news/attack-colonial-pipeline-what-weve-learned-what-weve-done-over-past-two-years.
Damgaard, Jannick, Thomas Elkjaer, and Niels Johannesen. 2024. "What Is Real and What Is Not in the Global FDI Network?" *Journal of International Money and Finance* 140: 102971. https://doi.org/10.1016/j.jimonfin.2023.102971.
Davis, Steven J. 2016. "An Index of Global Economic Policy Uncertainty." Working Paper 22740. National Bureau of Economic Research. https://doi.org/10.3386/w22740.
Delmas, Magali A. 2002. "The Diffusion of Environmental Management Standards in Europe and in the United States: An Institutional Perspective." *Policy Sciences* 35 (1): 91–119. https://doi.org/10.1023/A:1016108804453.
Dixit, Avinash K., and Robert S. Pindyck. 1994. *Investment Under Uncertainty*. Princeton, NJ: Princeton University Press. https://doi.org/10.1515/9781400830176.
Dolman, Everett C. 2002. *Astropolitik: Classical Geopolitics in the Space Age*. London: Frank Cass. https://doi.org/10.4324/9780203016640.
Dreher, Axel, Andreas Fuchs, Bradley Parks, Austin Strange, and Michael J. Tierney. 2021. "Aid, China, and Growth: Evidence from a New Global Development Finance Dataset." *American Economic Journal: Economic Policy* 13 (2): 135–74. https://doi.org/10.1257/pol.20180631.
———. 2022. *Banking on Beijing: The Aims and Impacts of China's Overseas Development Program*. Cambridge: Cambridge University Press. https://doi.org/10.1017/9781108564496.
Dreher, Axel, Jan-Egbert Sturm, and James Raymond Vreeland. 2009a. "Development Aid and International Politics: Does Membership on the UN Security Council Influence World Bank Decisions?" *Journal of Development Economics* 88 (1): 1–18. https://doi.org/10.1016/j.jdeveco.2008.02.003.
———. 2009b. "Global Horse Trading: IMF Loans for Votes in the United Nations Security Council." *European Economic Review* 53 (7): 742–57. https://doi.org/10.1016/j.euroecorev.2009.03.002.
Drezner, Daniel W. 1999. *The Sanctions Paradox: Economic Statecraft and International Relations*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511549366.
———. 2003. "The Hidden Hand of Economic Coercion." *International Organization* 57 (3): 643–59. https://doi.org/10.1017/S0020818303573052.
Early, Bryan R. 2015. *Busted Sanctions: Explaining Why Economic Sanctions Fail*. Stanford, CA: Stanford University Press. https://doi.org/10.1515/9780804794329.
Eichengreen, Barry. 2019. *Globalizing Capital: A History of the International Monetary System*. 3rd ed. Princeton, NJ: Princeton University Press. https://doi.org/10.1515/9780691194585.
Elliott, Kimberly Ann. 1998. "The Sanctions Glass: Half Full or Completely Empty?" *International Security* 23 (1): 50–65. https://doi.org/10.1162/isec.23.1.50.
ESA Space Debris Office. 2026. "ESA's Annual Space Environment Report." GEN-DB-LOG-00288-OPS-SD, Issue 10.0, 1 May 2026. Darmstadt: European Space Agency, European Space Operations Centre. https://www.sdo.esoc.esa.int/publications/Space_Environment_Report_I10R0_20260501.pdf.
European Bank for Reconstruction and Development. 2022. "Board of Governors Vote for Firm Action Against Russia and Belarus." News release, 4 April 2022. https://www.ebrd.com/home/news-and-events/news/2022/board-of-governors-vote-for-firm-action-against-russia-and-belarus.html.
European Commission. 2022. "An EU Strategy on Standardisation: Setting Global Standards in Support of a Resilient, Green and Digital EU Single Market." Communication from the Commission, COM(2022) 31 final. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52022DC0031.
European Parliament and Council of the European Union. 2022a. "Directive (EU) 2022/2380 Amending Directive 2014/53/EU on the Harmonisation of the Laws of the Member States Relating to the Making Available on the Market of Radio Equipment." Official Journal of the European Union. https://eur-lex.europa.eu/eli/dir/2022/2380/oj.
———. 2022b. "Regulation (EU) 2022/2480 Amending Regulation (EU) No 1025/2012 as Regards Decisions of European Standardisation Organisations Concerning European Standards and European Standardisation Deliverables." Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2022/2480/oj.
Fajgelbaum, Pablo D., Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal. 2020. "The Return to Protectionism." *The Quarterly Journal of Economics* 135 (1): 1–55. https://doi.org/10.1093/qje/qjz036.

Farrell, Henry, and Abraham L. Newman. 2019. "Weaponized Interdependence: How Global Economic Networks Shape State Coercion." *International Security* 44 (1): 42–79. https://doi.org/10.1162/isec_a_00351.

Felbermayr, Gabriel, Aleksandra Kirilakha, Constantinos Syropoulos, Erdal Yalcin, and Yoto V. Yotov. 2020. "The Global Sanctions Data Base." *European Economic Review* 129: 103561. https://doi.org/10.1016/j.euroecorev.2020.103561.

Fitzpatrick, Mark. 1983. "The Definition and Assessment of Political Risk in International Business: A Review of the Literature." *Academy of Management Review* 8 (2): 249–54. https://doi.org/10.2307/257752.

Freund, Caroline, Aaditya Mattoo, Alen Mulabdic, and Michele Ruta. 2024. "Is US Trade Policy Reshaping Global Supply Chains?" *Journal of International Economics* 152: 104011. https://doi.org/10.1016/j.jinteco.2024.104011.

Galtung, Johan. 1967. "On the Effects of International Economic Sanctions, with Examples from the Case of Rhodesia." *World Politics* 19 (3): 378–416. https://doi.org/10.2307/2009785.

Gartzke, Erik. 2013. "The Myth of Cyberwar: Bringing War in Cyberspace Back down to Earth." *International Security* 38 (2): 41–73. https://doi.org/10.1162/ISEC_a_00136.

Gelpern, Anna, Sebastian Horn, Scott Morris, Brad Parks, and Christoph Trebesch. 2023. "How China Lends: A Rare Look into 100 Debt Contracts with Foreign Governments." *Economic Policy* 38 (114): 345–416. https://doi.org/10.1093/epolic/eiac054.

Gentzkow, Matthew, Bryan Kelly, and Matt Taddy. 2019. "Text as Data." *Journal of Economic Literature* 57 (3): 535–74. https://doi.org/10.1257/jel.20181020.

George, Alexander L. 1959. *Propaganda Analysis: A Study of Inferences Made from Nazi Propaganda in World War II*. Evanston, IL: Row, Peterson.

Gereffi, Gary, John Humphrey, and Timothy Sturgeon. 2005. "The Governance of Global Value Chains." *Review of International Political Economy* 12 (1): 78–104. https://doi.org/10.1080/09692290500049805.

Gertz, Geoffrey. 2018. "Commercial Diplomacy and Political Risk." *International Studies Quarterly* 62 (1): 94–107. https://doi.org/10.1093/isq/sqx079.

Gilpin, Robert. 1981. *War and Change in World Politics*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511664267.

Gopinath, Gita, Pierre-Olivier Gourinchas, Andrea F. Presbitero, and Petia Topalova. 2024. "Changing Global Linkages: A New Cold War?" IMF Working Paper WP/24/76. International Monetary Fund. https://doi.org/10.5089/9798400272745.001.

———. 2025. "Changing Global Linkages: A New Cold War?" *Journal of International Economics* 153: 104042. https://doi.org/10.1016/j.jinteco.2024.104042.

Gowa, Joanne. 1994. *Allies, Adversaries, and International Trade*. Princeton: Princeton University Press.

Gowa, Joanne, and Edward D. Mansfield. 1993. "Power Politics and International Trade." *American Political Science Review* 87 (2): 408–20. https://doi.org/10.2307/2939050.

Green, Jessica F. 2013. *Rethinking Private Authority: Agents and Entrepreneurs in Global Environmental Governance*. Princeton: Princeton University Press. https://doi.org/10.23943/princeton/9780691157580.001.0001.

Greenberg, Andy. 2019. *Sandworm: A New Era of Cyberwar and the Hunt for the Kremlin's Most Dangerous Hackers*. New York: Doubleday.

Grieco, Joseph M. 1988. "Anarchy and the Limits of Cooperation: A Realist Critique of the Newest Liberal Institutionalism." *International Organization* 42 (3): 485–507. https://doi.org/10.1017/S0020818300027715.

Grimmer, Justin, and Brandon M. Stewart. 2013. "Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts." *Political Analysis* 21 (3): 267–97. https://doi.org/10.1093/pan/mps028.

Guler, Isin, Mauro F. Guillén, and John Muir Macpherson. 2002. "Global Competition, Institutions, and the Diffusion of Organizational Practices: The International Spread of ISO 9000 Quality Certificates." *Administrative Science Quarterly* 47 (2): 207–32. https://doi.org/10.2307/3094804.

Hall, Rodney Bruce, and Thomas J. Biersteker, eds. 2002. *The Emergence of Private Authority in Global Governance*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511491238.

Hardin, Garrett. 1968. "The Tragedy of the Commons." *Science* 162 (3859): 1243–48. https://doi.org/10.1126/science.162.3859.1243.

Hassan, Tarek A., Stephan Hollander, Laurence van Lent, and Ahmed Tahoun. 2019. "Firm-Level Political Risk: Measurement and Effects." *Quarterly Journal of Economics* 134 (4): 2135–2202. https://doi.org/10.1093/qje/qjz021.

Hayes, Andrew F., and Klaus Krippendorff. 2007. "Answering the Call for a Standard Reliability Measure for Coding Data." *Communication Methods and Measures* 1 (1): 77–89. https://doi.org/10.1080/19312450709336664.

Head, Keith, and Thierry Mayer. 2014. "Gravity Equations: Workhorse, Toolkit, and Cookbook." In *Handbook of International Economics*, edited by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, 4:131–95. Amsterdam: Elsevier. https://doi.org/10.1016/B978-0-444-54314-1.00003-3.

Head, Keith, and John Ries. 2010. "Do Trade Missions Increase Trade?" *Canadian Journal of Economics* 43 (3): 754–75. https://doi.org/10.1111/j.1540-5982.2010.01593.x.

Henisz, Witold J. 2000. "The Institutional Environment for Economic Growth." *Economics & Politics* 12 (1): 1–31. https://doi.org/10.1111/1468-0343.00066.

———. 2002. "The Institutional Environment for Infrastructure Investment." *Industrial and Corporate Change* 11 (2): 355–89. https://doi.org/10.1093/icc/11.2.355.

Herz, John H. 1950. "Idealist Internationalism and the Security Dilemma." *World Politics* 2 (2): 157–80. https://doi.org/10.2307/2009187.

Hirschman, Albert O. 1945. *National Power and the Structure of Foreign Trade*. Berkeley: University of California Press. https://doi.org/10.1525/9780520378179.

Horn, Sebastian, Carmen M. Reinhart, and Christoph Trebesch. 2021. "China's Overseas Lending." *Journal of International Economics* 133: 103539. https://doi.org/10.1016/j.jinteco.2021.103539.

Hufbauer, Gary Clyde, Jeffrey J. Schott, Kimberly Ann Elliott, and Barbara Oegg. 2007. *Economic Sanctions Reconsidered*. 3rd ed. Washington, DC: Peterson Institute for International Economics.

Humphrey, Chris. 2022. *Financing the Future: Multilateral Development Banks in the Changing World Order of the 21st Century*. Oxford: Oxford University Press. https://doi.org/10.1093/oso/9780192871503.001.0001.

Humphrey, Chris, and Katharina Michaelowa. 2013. "Shopping for Development: Multilateral Lending, Shareholder Composition and Borrower Preferences." *World Development* 44: 142–55. https://doi.org/10.1016/j.worlddev.2012.12.007.

Ikenberry, G. John. 2011. *Liberal Leviathan: The Origins, Crisis, and Transformation of the American World Order*. Princeton, NJ: Princeton University Press. https://doi.org/10.1515/9781400838196.

Ikenberry, G. John, and Darren J. Lim. 2017. "China's Emerging Institutional Statecraft: The Asian Infrastructure Investment Bank and the Prospects for Counter-Hegemony." Washington, DC: Brookings Institution. https://www.brookings.edu/wp-content/uploads/2017/04/chinas-emerging-institutional-statecraft.pdf.

International Centre for Settlement of Investment Disputes. 2026. "ICSID Releases Caseload Statistics for the 2026 Fiscal Year." News release, 14 August 2026. https://icsid.worldbank.org/news-and-events/news-releases/icsid-releases-caseload-statistics-2026-fiscal-year.

International Monetary Fund. 2026a. "Annual Report 2026: Appendices (Appendix IV, Executive Directors and Voting Power as of April 30, 2026)." Washington, DC: International Monetary Fund. https://www.imf.org/annual-report/2026/pdfs/imf-annual-report-2026-appendices.pdf.

———. 2026b. "International Trade in Goods (by Partner Country) (IMTS), Formerly Direction of Trade Statistics (DOTS)." IMF Data. https://data.imf.org/en/datasets/IMF.STA:IMTS.

International Organization for Standardization. 2026. "ISO Open Data: Technical Committees and Deliverables Metadata." ISO. https://www.iso.org/open-data.html.

Jensen, Nathan M. 2003. "Democratic Governance and Multinational Corporations: Political Regimes and Inflows of Foreign Direct Investment." *International Organization* 57 (3): 587–616. https://doi.org/10.1017/S0020818303573040.

Jervis, Robert. 1978. "Cooperation Under the Security Dilemma." *World Politics* 30 (2): 167–214. https://doi.org/10.2307/2009958.

Julio, Brandon, and Youngsuk Yook. 2012. "Political Uncertainty and Corporate Investment Cycles." *Journal of Finance* 67 (1): 45–83. https://doi.org/10.1111/j.1540-6261.2011.01707.x.

Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi. 2011. "The Worldwide Governance Indicators: Methodology and Analytical Issues." *Hague Journal on the Rule of Law* 3 (2): 220–46. https://doi.org/10.1017/S1876404511200046.

Kello, Lucas. 2013. "The Meaning of the Cyber Revolution: Perils to Theory and Statecraft." *International Security* 38 (2): 7–40. https://doi.org/10.1162/ISEC_a_00138.

———. 2017. *The Virtual Weapon and International Order*. New Haven, CT: Yale University Press.

Kelso, T. S. 2026. "SATCAT: Satellite Catalog." CelesTrak. https://celestrak.org/satcat/.

Kennedy, Paul. 1987. *The Rise and Fall of the Great Powers: Economic Change and Military Conflict from 1500 to 2000*. New York: Random House.

Keohane, Robert O. 1984. *After Hegemony: Cooperation and Discord in the World Political Economy*. Princeton, NJ: Princeton University Press. https://doi.org/10.1515/9781400820269.

Kessler, Donald J., and Burton G. Cour-Palais. 1978. "Collision Frequency of Artificial Satellites: The Creation of a Debris Belt." *Journal of Geophysical Research* 83 (A6): 2637–46. https://doi.org/10.1029/JA083iA06p02637.

Kilby, Christopher. 2009. "The Political Economy of Conditionality: An Empirical Analysis of World Bank Loan Disbursements." *Journal of Development Economics* 89 (1): 51–61. https://doi.org/10.1016/j.jdeveco.2008.06.014.

Kindleberger, Charles P. 1973. *The World in Depression, 1929–1939*. Berkeley: University of California Press.
Kirshner, Jonathan. 1997. "The Microfoundations of Economic Sanctions." *Security Studies* 6 (3): 32–64. https://doi.org/10.1080/09636419708429314.
Knight, Frank H. 1921. *Risk, Uncertainty and Profit*. Boston: Houghton Mifflin.
Kobrin, Stephen J. 1979. "Political Risk: A Review and Reconsideration." *Journal of International Business Studies* 10 (1): 67–80. https://doi.org/10.1057/palgrave.jibs.8490631.
———. 1980. "Foreign Enterprise and Forced Divestment in LDCs." *International Organization* 34 (1): 65–88. https://doi.org/10.1017/S0020818300003970.
———. 1982. *Managing Political Risk Assessment: Strategic Response to Environmental Change*. Berkeley: University of California Press. https://doi.org/10.1525/9780520310094.
———. 1984. "Expropriation as an Attempt to Control Foreign Firms in LDCs: Trends from 1960 to 1979." *International Studies Quarterly* 28 (3). https://doi.org/10.2307/2600634.
Kostyuk, Nadiya, and Yuri M. Zhukov. 2019. "Invisible Digital Front: Can Cyber Attacks Shape Battlefield Events?" *Journal of Conflict Resolution* 63 (2): 317–47. https://doi.org/10.1177/0022002717737138.
Krasner, Stephen D. 1976. "State Power and the Structure of International Trade." *World Politics* 28 (3): 317–47. https://doi.org/10.2307/2009974.
Krippendorff, Klaus. 2018. *Content Analysis: An Introduction to Its Methodology*. 4th ed. Thousand Oaks, CA: SAGE.
Kugler, Jacek, and Douglas Lemke, eds. 1996. *Parity and War: Evaluations and Extensions of the War Ledger*. Ann Arbor: University of Michigan Press. https://doi.org/10.3998/mpub.10422.
Kuziemko, Ilyana, and Eric Werker. 2006. "How Much Is a Seat on the Security Council Worth? Foreign Aid and Bribery at the United Nations." *Journal of Political Economy* 114 (5): 905–30. https://doi.org/10.1086/507155.
Landis, J. Richard, and Gary G. Koch. 1977. "The Measurement of Observer Agreement for Categorical Data." *Biometrics* 33 (1): 159–74. https://doi.org/10.2307/2529310.
Lazer, David, Ryan Kennedy, Gary King, and Alessandro Vespignani. 2014. "The Parable of Google Flu: Traps in Big Data Analysis." *Science* 343 (6176): 1203–5. https://doi.org/10.1126/science.1248506.
Le Billon, Philippe, and Lauren Shykora. 2020. "Conflicts, Commodities and the Environmental Geopolitics of Supply Chains." In *A Research Agenda for Environmental Geopolitics*, edited by Shannon O'Lear, 59–73. Cheltenham: Edward Elgar. https://doi.org/10.4337/9781788971249.00012.
Lederman, Daniel, Marcelo Olarreaga, and Lucy Payton. 2010. "Export Promotion Agencies: Do They Work?" *Journal of Development Economics* 91 (2): 257–65. https://doi.org/10.1016/j.jdeveco.2009.09.003.
Lindsay, Jon R. 2015. "Tipping the Scales: The Attribution Problem and the Feasibility of Deterrence Against Cyberattack." *Journal of Cybersecurity* 1 (1). https://doi.org/10.1093/cybsec/tyv003.
Liou, J.-C., and N. L. Johnson. 2006. "Risks in Space from Orbiting Debris." *Science* 311 (5759): 340–41. https://doi.org/10.1126/science.1121337.
Loughran, Tim, and Bill McDonald. 2011. "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks." *Journal of Finance* 66 (1): 35–65. https://doi.org/10.1111/j.1540-6261.2010.01625.x.
Lowy Institute. 2024. "Global Diplomacy Index 2024." Lowy Institute, Sydney. https://globaldiplomacyindex.lowyinstitute.org/.
Maness, Ryan C., and Brandon Valeriano. 2016. "The Impact of Cyber Conflict on International Interactions." *Armed Forces & Society* 42 (2): 301–23. https://doi.org/10.1177/0095327X15572997.
Maness, Ryan C., Brandon Valeriano, Kathryn Hedgecock, Jose M. Macias, and Benjamin Jensen. 2022. "Dyadic Cyber Incident and Campaign Data (DCID), Version 2.0." Harvard Dataverse. https://doi.org/10.7910/DVN/CQOMYV.
Mansfield, Edward D. 2021. "International Trade and Conflict." In *The Oxford Handbook of International Political Economy*, edited by Jon C. W. Pevehouse and Leonard Seabrooke, 660–71. Oxford: Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198793519.013.27.
Mansfield, Edward D., and Jon C. Pevehouse. 2000. "Trade Blocs, Trade Flows, and International Conflict." *International Organization* 54 (4): 775–808. https://doi.org/10.1162/002081800551361.
Martin, Philippe, Thierry Mayer, and Mathias Thoenig. 2008. "Make Trade Not War?" *Review of Economic Studies* 75 (3): 865–900. https://doi.org/10.1111/j.1467-937X.2008.00492.x.
Mattli, Walter, and Tim Büthe. 2003. "Setting International Standards: Technological Rationality or Primacy of Power?" *World Politics* 56 (1): 1–42. https://doi.org/10.1353/wp.2004.0006.
McDowell, Jonathan C. 2026. "General Catalog of Artificial Space Objects (GCAT)." planet4589.org. https://planet4589.org/space/gcat/.
Mearsheimer, John J. 1994. "The False Promise of International Institutions." *International Security* 19 (3): 5–49. https://doi.org/10.2307/2539078.

———. 2001. *The Tragedy of Great Power Politics*. New York: W. W. Norton.
Meng, Xiao-Li. 2018. "Statistical Paradises and Paradoxes in Big Data (I): Law of Large Populations, Big Data Paradox, and the 2016 US Presidential Election." *Annals of Applied Statistics* 12 (2): 685–726. https://doi.org/10.1214/18-AOAS1161SF.
Meyer, John W., and Brian Rowan. 1977. "Institutionalized Organizations: Formal Structure as Myth and Ceremony." *American Journal of Sociology* 83 (2): 340–63. https://doi.org/10.1086/226550.
Moons, Selwyn J. V., and Peter A. G. van Bergeijk. 2017. "Does Economic Diplomacy Work? A Meta-Analysis of Its Impact on Trade and Investment." *The World Economy* 40 (2): 336–68. https://doi.org/10.1111/twec.12392.
Morgan, T. Clifton, Navin Bapat, and Yoshiharu Kobayashi. 2014. "Threat and Imposition of Economic Sanctions 1945–2005: Updating the TIES Dataset." *Conflict Management and Peace Science* 31 (5): 541–58. https://doi.org/10.1177/0738894213520379.
Morgan, T. Clifton, Navin Bapat, and Valentin Krustev. 2009. "The Threat and Imposition of Economic Sanctions, 1971–2000." *Conflict Management and Peace Science* 26 (1): 92–110. https://doi.org/10.1177/0738894208097668.
Morse, Julia C., and Robert O. Keohane. 2014. "Contested Multilateralism." *The Review of International Organizations* 9 (4): 385–412. https://doi.org/10.1007/s11558-014-9188-2.
Multilateral Investment Guarantee Agency. 2026. "Annual Report 2026: MIGA Appendix." Washington, DC: World Bank Group. https://www.miga.org/sites/default/files/2026-10/WB_2026AR_MIGA_Appendix_v1e_0.pdf.
Nitsch, Volker. 2007. "State Visits and International Trade." *The World Economy* 30 (12): 1797–1816. https://doi.org/10.1111/j.1467-9701.2007.01062.x.
OECD, and European Commission Joint Research Centre. 2008. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. Paris: OECD Publishing. https://doi.org/10.1787/9789264043466-en.
Oneal, John R., and Bruce M. Russett. 1997. "The Classical Liberals Were Right: Democracy, Interdependence, and Conflict, 1950–1985." *International Studies Quarterly* 41 (2): 267–94. https://doi.org/10.1111/1468-2478.00042.
Organski, A. F. K. 1958. *World Politics*. New York: Knopf.
Organski, A. F. K., and Jacek Kugler. 1980. *The War Ledger*. Chicago: University of Chicago Press. https://doi.org/10.7208/chicago/9780226351841.001.0001.
Ostrom, Elinor. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511807763.
Pape, Robert A. 1997. "Why Economic Sanctions Do Not Work." *International Security* 22 (2): 90–136. https://doi.org/10.1162/isec.22.2.90.
———. 1998. "Why Economic Sanctions Still Do Not Work." *International Security* 23 (1): 66–77. https://doi.org/10.1162/isec.23.1.66.
———. 2005. "Soft Balancing Against the United States." *International Security* 30 (1): 7–45. https://doi.org/10.1162/0162288054894607.
Pekkanen, Saadia M. 2019. "Governing the New Space Race." *AJIL Unbound* 113: 92–97. https://doi.org/10.1017/aju.2019.16.
Peksen, Dursun. 2009. "Better or Worse? The Effect of Economic Sanctions on Human Rights." *Journal of Peace Research* 46 (1): 59–77. https://doi.org/10.1177/0022343308098404.
Pelc, Krzysztof J. 2017. "What Explains the Low Success Rate of Investor-State Disputes?" *International Organization* 71 (3): 559–83. https://doi.org/10.1017/S0020818317000212.
Polachek, Solomon William. 1980. "Conflict and Trade." *Journal of Conflict Resolution* 24 (1): 55–78. https://doi.org/10.1177/002200278002400103.
Potoski, Matthew, and Aseem Prakash. 2005. "Covenants with Weak Swords: ISO 14001 and Facilities' Environmental Performance." *Journal of Policy Analysis and Management* 24 (4): 745–69. https://doi.org/10.1002/pam.20136.
Powell, Robert. 1991. "Absolute and Relative Gains in International Relations Theory." *American Political Science Review* 85 (4): 1303–20. https://doi.org/10.2307/1963947.
Prakash, Aseem, and Matthew Potoski. 2006a. "Racing to the Bottom? Trade, Environmental Governance, and ISO 14001." *American Journal of Political Science* 50 (2): 350–64. https://doi.org/10.1111/j.1540-5907.2006.00188.x.
———. 2006b. *The Voluntary Environmentalists: Green Clubs, ISO 14001, and Voluntary Environmental Regulations*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511617683.
Putnam, Robert D. 1988. "Diplomacy and Domestic Politics: The Logic of Two-Level Games." *International Organization* 42 (3): 427–60. https://doi.org/10.1017/S0020818300027697.
Rao, Akhil, Matthew G. Burgess, and Daniel Kaffine. 2020. "Orbital-Use Fees Could More Than Quadruple the Value of the Space Industry." *Proceedings of the National Academy of Sciences* 117 (23): 12756–62. https://doi.org/10.1073/pnas.1921260117.

Ravallion, Martin. 2012. "Troubling Tradeoffs in the Human Development Index." *Journal of Development Economics* 99 (2): 201–9. https://doi.org/10.1016/j.jdeveco.2012.01.003.
Ray, James Lee, and J. David Singer. 1973. "Measuring the Concentration of Power in the International System." *Sociological Methods & Research* 1 (4): 403–37. https://doi.org/10.1177/004912417300100401.
Rid, Thomas. 2012. "Cyber War Will Not Take Place." *Journal of Strategic Studies* 35 (1): 5–32. https://doi.org/10.1080/01402390.2011.608939.
———. 2013. *Cyber War Will Not Take Place*. New York: Oxford University Press.
Rid, Thomas, and Ben Buchanan. 2015. "Attributing Cyber Attacks." *Journal of Strategic Studies* 38 (1–2): 4–37. https://doi.org/10.1080/01402390.2014.977382.
Rodrik, Dani. 2011. *The Globalization Paradox: Democracy and the Future of the World Economy*. New York: W. W. Norton.
Rose, Andrew K. 2007. "The Foreign Service and Foreign Trade: Embassies as Export Promotion." *The World Economy* 30 (1): 22–38. https://doi.org/10.1111/j.1467-9701.2007.00870.x.
Ruggie, John Gerard. 1982. "International Regimes, Transactions, and Change: Embedded Liberalism in the Postwar Economic Order." *International Organization* 36 (2): 379–415. https://doi.org/10.1017/S0020818300018993.
Rühlig, Tim. 2023. "Chinese Influence Through Technical Standardization Power." *Journal of Contemporary China* 32 (139): 54–72. https://doi.org/10.1080/10670564.2022.2052439.
Russett, Bruce, and John R. Oneal. 2001. *Triangulating Peace: Democracy, Interdependence, and International Organizations*. New York: W. W. Norton.
Saisana, Michaela, Andrea Saltelli, and Stefano Tarantola. 2005. "Uncertainty and Sensitivity Analysis Techniques as Tools for the Quality Assessment of Composite Indicators." *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 168 (2): 307–23. https://doi.org/10.1111/j.1467-985X.2005.00350.x.
Samson, Victoria, and Kristin Brett, eds. 2026. *Global Counterspace Capabilities: An Open Source Assessment*. Secure World Foundation. https://www.swfound.org/publications-and-reports/2026-global-counterspace-capabilities-report.
Santos Silva, J. M. C., and Silvana Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88 (4): 641–58. https://doi.org/10.1162/rest.88.4.641.
Schmitt, Michael N., ed. 2013. *Tallinn Manual on the International Law Applicable to Cyber Warfare*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781139169288.
Science Based Targets initiative. 2024. "Statement from the SBTi Board of Trustees on Use of Environmental Attribute Certificates, Including but Not Limited to Voluntary Carbon Markets, for Abatement Purposes Limited to Scope 3." https://sciencebasedtargets.org/news/statement-from-the-sbti-board-of-trustees-on-use-of-environmental-attribute-certificates-including-but-not-limited-to-voluntary-carbon-markets-for-abatement-purposes-limited-to-scope-3.
———. 2026. "Target Dashboard: Companies and Targets Data." Science Based Targets initiative. https://sciencebasedtargets.org/target-dashboard.
Sell, Susan K. 2003. *Private Power, Public Law: The Globalization of Intellectual Property Rights*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511491665.
Signorino, Curtis S., and Jeffrey M. Ritter. 1999. "Tau-b or Not Tau-b: Measuring the Similarity of Foreign Policy Positions." *International Studies Quarterly* 43 (1): 115–44. https://doi.org/10.1111/0020-8833.00113.
Simmons, Beth A. 2014. "Bargaining over BITs, Arbitrating Awards: The Regime for Protection and Promotion of International Investment." *World Politics* 66 (1): 12–46. https://doi.org/10.1017/S0043887113000312.
Singer, J. David, Stuart Bremer, and John Stuckey. 1972. "Capability Distribution, Uncertainty, and Major Power War, 1820–1965." In *Peace, War, and Numbers*, edited by Bruce Russett, 19–48. Beverly Hills, CA: Sage.
Stone, Randall W. 2011. *Controlling Institutions: International Organizations and the Global Economy*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511793943.
Stopford, John M., Susan Strange, and John S. Henley. 1991. *Rival States, Rival Firms: Competition for World Market Shares*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511549830.
Strange, Susan. 1992. "States, Firms and Diplomacy." *International Affairs* 68 (1): 1–15. https://doi.org/10.2307/2620458.
———. 1996. *The Retreat of the State: The Diffusion of Power in the World Economy*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511559143.
Syropoulos, Constantinos, Gabriel Felbermayr, Aleksandra Kirilakha, Erdal Yalcin, and Yoto V. Yotov. 2024. "The Global Sanctions Data Base–Release 3: COVID-19, Russia, and Multilateral Sanctions." *Review of International Economics* 32 (1): 12–48. https://doi.org/10.1111/roie.12691.
Tetlock, Paul C. 2007. "Giving Content to Investor Sentiment: The Role of Media in the Stock Market." *Journal of Finance* 62 (3): 1139–68. https://doi.org/10.1111/j.1540-6261.2007.01232.x.
Tetlock, Philip E. 2005. *Expert Political Judgment: How Good Is It? How Can We Know?* Princeton, NJ: Princeton

University Press. https://doi.org/10.1515/9781400830312.
The White House. 2023. "United States Government National Standards Strategy for Critical and Emerging Technology." https://bidenwhitehouse.archives.gov/wp-content/uploads/2023/05/US-Gov-National-Standards-Strategy-2023.pdf.
Tinbergen, Jan. 1962. *Shaping the World Economy: Suggestions for an International Economic Policy*. New York: Twentieth Century Fund.
Tsebelis, George. 2002. *Veto Players: How Political Institutions Work*. Princeton, NJ: Princeton University Press. https://doi.org/10.1515/9781400831456.
UK Government. 2022. "Russia Behind Cyber-Attack with Europe-Wide Impact an Hour Before Ukraine Invasion." Press release, 10 May 2022. https://www.gov.uk/government/news/russia-behind-cyber-attack-with-europe-wide-impact-an-hour-before-ukraine-invasion.
UNCTAD. 2019. *World Investment Report 2019: Special Economic Zones*. New York; Geneva: United Nations. https://doi.org/10.18356/8a8d05f9-en.
United Nations. 1967. "Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies." Opened for signature 27 January 1967, entered into force 10 October 1967, 610 U.N.T.S. 205. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.html.
United Nations Development Programme. 1990. *Human Development Report 1990*. New York: Oxford University Press. https://hdr.undp.org/content/human-development-report-1990.
———. 2010. *Human Development Report 2010. The Real Wealth of Nations: Pathways to Human Development*. New York: Palgrave Macmillan. https://hdr.undp.org/content/human-development-report-2010.
U.S. Department of Justice and Federal Trade Commission. 2010. "Horizontal Merger Guidelines." Washington, DC: U.S. Department of Justice; Federal Trade Commission. https://www.justice.gov/atr/horizontal-merger-guidelines-08192010.
———. 2023. "Merger Guidelines." Washington, DC: U.S. Department of Justice; Federal Trade Commission. https://www.justice.gov/atr/2023-merger-guidelines.
U.S. Securities and Exchange Commission. 2023. "Cybersecurity Risk Management, Strategy, Governance, and Incident Disclosure." Final rule, Release No. 33-11216. Washington, DC: U.S. Securities; Exchange Commission. https://www.sec.gov/files/rules/final/2023/33-11216.pdf.
Valeriano, Brandon, Benjamin Jensen, and Ryan C. Maness. 2018. *Cyber Strategy: The Evolving Character of Power and Coercion*. New York: Oxford University Press. https://doi.org/10.1093/oso/9780190618094.001.0001.
Valeriano, Brandon, and Ryan C. Maness. 2015. *Cyber War Versus Cyber Realities: Cyber Conflict in the International System*. New York: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190204792.001.0001.
Vernon, Raymond. 1971. *Sovereignty at Bay: The Multinational Spread of U.S. Enterprises*. New York: Basic Books.
Voeten, Erik. 2000. "Clashes in the Assembly." *International Organization* 54 (2): 185–215. https://doi.org/10.1162/002081800551154.
———. 2026. "United Nations General Assembly Ideal Points." Harvard Dataverse. https://doi.org/10.7910/DVN/LEJUQZ.
Vogel, David. 1995. *Trading up: Consumer and Environmental Regulation in a Global Economy*. Cambridge, MA: Harvard University Press.
Vreeland, James Raymond, and Axel Dreher. 2014. *The Political Economy of the United Nations Security Council: Money and Influence*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781139027755.
Wack, Pierre. 1985. "Scenarios: Uncharted Waters Ahead." *Harvard Business Review* 63 (5).
Waltz, Kenneth N. 1979. *Theory of International Politics*. Reading, MA: Addison-Wesley.
Wang, Hongying. 2017. "New Multilateral Development Banks: Opportunities and Challenges for Global Governance." *Global Policy* 8 (1): 113–18. https://doi.org/10.1111/1758-5899.12396.
Weeden, Brian C., and Tiffany Chow. 2012. "Taking a Common-Pool Resources Approach to Space Sustainability: A Framework and Potential Policies." *Space Policy* 28 (3): 166–72. https://doi.org/10.1016/j.spacepol.2012.06.004.
Whaley, Robert E. 2009. "Understanding the VIX." *The Journal of Portfolio Management* 35 (3): 98–105. https://doi.org/10.3905/jpm.2009.35.3.098.
Wohlforth, William C. 1999. "The Stability of a Unipolar World." *International Security* 24 (1): 5–41. https://doi.org/10.1162/016228899560031.
Woods, Ngaire. 2006. *The Globalizers: The IMF, the World Bank, and Their Borrowers*. Ithaca, NY: Cornell University Press. https://doi.org/10.7591/9780801456022.
World Bank. 2026. "World Development Indicators." https://datatopics.worldbank.org/world-development-indicators/.
World Bank Group. 2026. "Voting Powers: IBRD, IDA, IFC and MIGA Country Voting Tables (Data as of June 30, 2026)." World Bank Group. https://www.worldbank.org/en/about/leadership/votingpowers.

World Trade Organization. n.d. “Understanding the WTO: The GATT Years: From Havana to Marrakesh.” https://www.wto.org/english/thewto_e/whatis_e/tif_e/fact4_e.htm.

Zettl-Schabath, Kerstin, Jakob Bund, Martin Müller, Camille Borrett, Jonas Hemmelskamp, Asaf Alibegovic, Enis Bajra, et al. 2025. “Global Dataset of Cyber Incidents (European Repository of Cyber Incidents), Version 1.3.” Zenodo. https://doi.org/10.5281/zenodo.14965395.